Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-10 22:25:28
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

-- Concat Chains --
```

```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_64x128/std_params.nc
  trend_params_path..... data/processed_64x128/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
======================================================================

METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

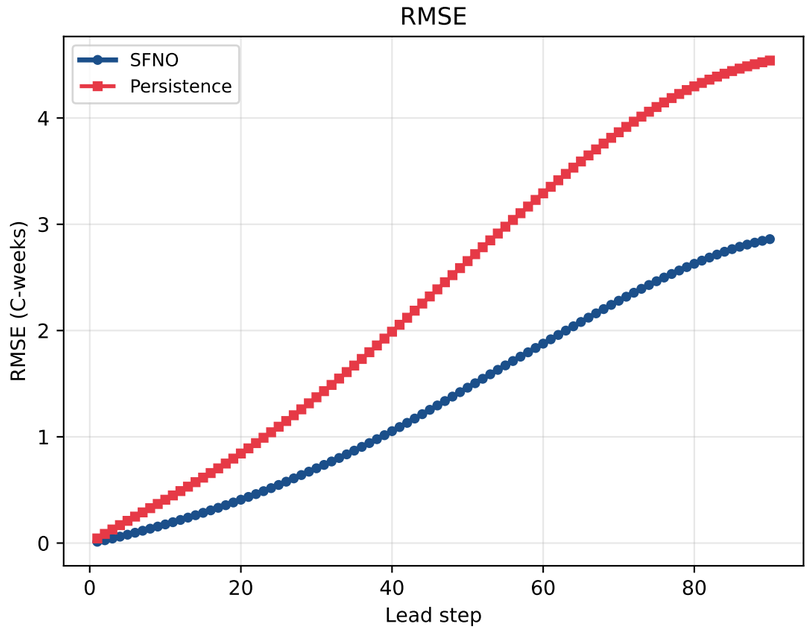
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

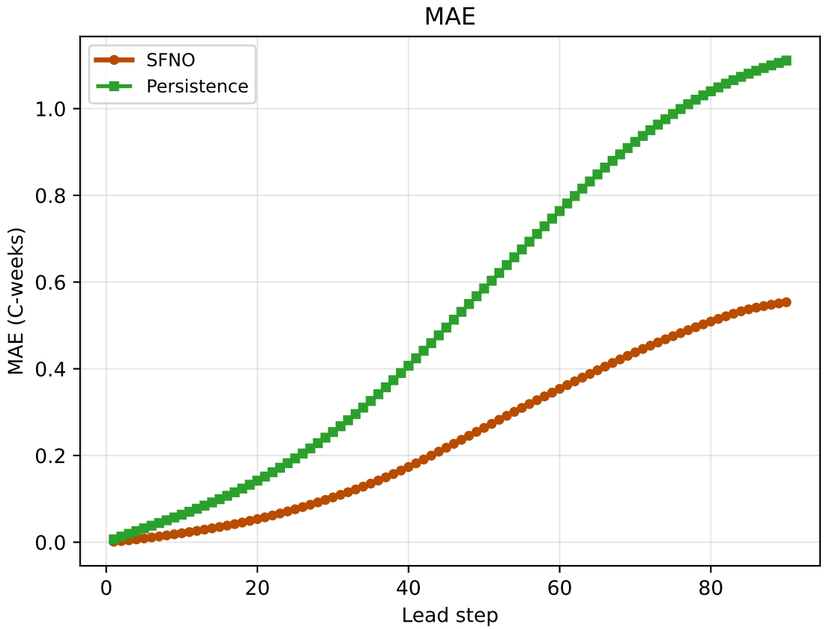
DATASETS
  Test          19 chains x 90 leads x 64 lat x 128 lon
  Val           16 chains x 90 leads x 64 lat x 128 lon
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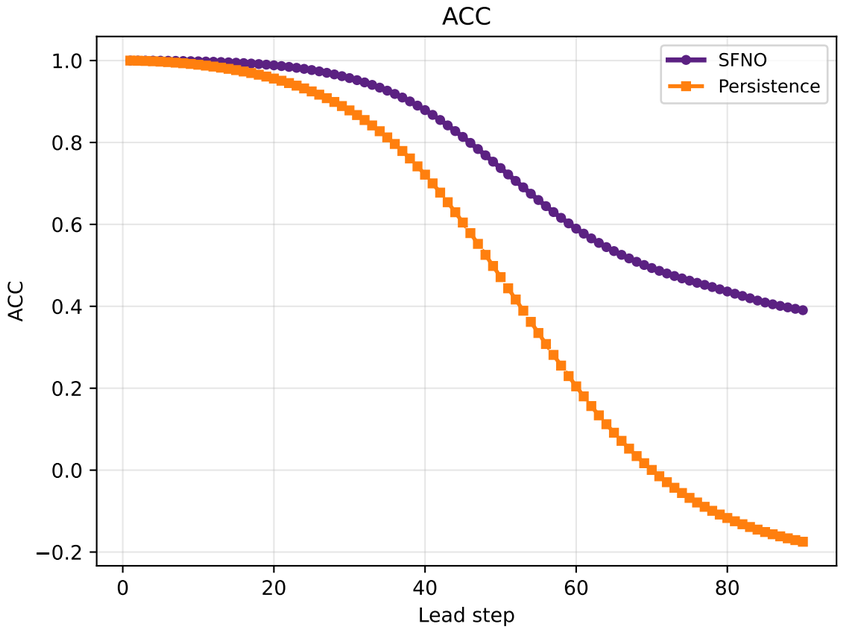
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

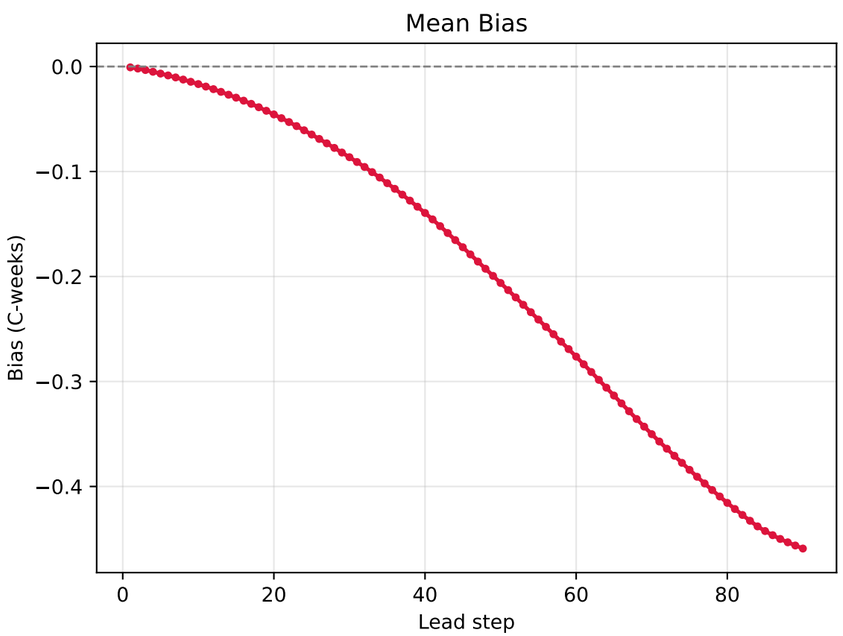
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0126	0.0437	71.2%	0.0011	0.0065	-0.0008	1.0000	0.9999	1.0000
2	0.0266	0.0862	69.1%	0.0027	0.0130	-0.0019	1.0000	0.9996	0.9999
3	0.0428	0.1278	66.5%	0.0045	0.0193	-0.0033	0.9999	0.9990	0.9998
4	0.0604	0.1687	64.2%	0.0066	0.0257	-0.0050	0.9998	0.9983	0.9996
5	0.0788	0.2091	62.3%	0.0088	0.0319	-0.0067	0.9996	0.9973	0.9994
6	0.0972	0.2491	61.0%	0.0110	0.0381	-0.0085	0.9994	0.9962	0.9990
7	0.1160	0.2888	59.8%	0.0133	0.0443	-0.0104	0.9992	0.9949	0.9986
8	0.1354	0.3285	58.8%	0.0157	0.0507	-0.0124	0.9989	0.9934	0.9981
9	0.1554	0.3680	57.8%	0.0182	0.0571	-0.0145	0.9985	0.9917	0.9975
10	0.1756	0.4079	57.0%	0.0207	0.0635	-0.0167	0.9981	0.9898	0.9968
11	0.1964	0.4484	56.2%	0.0234	0.0702	-0.0191	0.9976	0.9877	0.9959
12	0.2177	0.4893	55.5%	0.0262	0.0772	-0.0215	0.9970	0.9853	0.9950
13	0.2396	0.5310	54.9%	0.0290	0.0844	-0.0241	0.9964	0.9827	0.9939
14	0.2622	0.5733	54.3%	0.0320	0.0918	-0.0269	0.9956	0.9798	0.9926
15	0.2848	0.6158	53.8%	0.0351	0.0993	-0.0297	0.9948	0.9767	0.9912
16	0.3080	0.6590	53.3%	0.0383	0.1071	-0.0325	0.9938	0.9733	0.9895
17	0.3319	0.7031	52.8%	0.0417	0.1152	-0.0356	0.9927	0.9695	0.9877
18	0.3565	0.7484	52.4%	0.0454	0.1237	-0.0388	0.9914	0.9654	0.9856
19	0.3820	0.7951	52.0%	0.0492	0.1326	-0.0422	0.9899	0.9609	0.9833
20	0.4081	0.8429	51.6%	0.0533	0.1419	-0.0456	0.9883	0.9560	0.9806
21	0.4343	0.8913	51.3%	0.0574	0.1515	-0.0492	0.9865	0.9507	0.9778
22	0.4610	0.9406	51.0%	0.0617	0.1614	-0.0529	0.9845	0.9450	0.9746
23	0.4886	0.9912	50.7%	0.0663	0.1717	-0.0567	0.9822	0.9387	0.9710
24	0.5172	1.0427	50.4%	0.0710	0.1824	-0.0607	0.9797	0.9319	0.9670
25	0.5469	1.0951	50.1%	0.0760	0.1933	-0.0648	0.9768	0.9246	0.9625
26	0.5777	1.1486	49.7%	0.0811	0.2046	-0.0690	0.9736	0.9167	0.9575
27	0.6089	1.2029	49.4%	0.0864	0.2163	-0.0732	0.9700	0.9081	0.9521
28	0.6401	1.2585	49.1%	0.0919	0.2285	-0.0775	0.9661	0.8989	0.9461
29	0.6719	1.3152	48.9%	0.0975	0.2411	-0.0819	0.9619	0.8889	0.9397
30	0.7037	1.3721	48.7%	0.1033	0.2541	-0.0864	0.9572	0.8783	0.9327
31	0.7355	1.4298	48.6%	0.1092	0.2675	-0.0909	0.9521	0.8669	0.9252
32	0.7678	1.4883	48.4%	0.1153	0.2813	-0.0957	0.9466	0.8547	0.9172
33	0.8007	1.5482	48.3%	0.1216	0.2956	-0.1006	0.9406	0.8416	0.9084
34	0.8348	1.6090	48.1%	0.1282	0.3104	-0.1057	0.9338	0.8274	0.8989
35	0.8699	1.6707	47.9%	0.1352	0.3257	-0.1111	0.9265	0.8123	0.8885
36	0.9052	1.7328	47.8%	0.1423	0.3414	-0.1165	0.9184	0.7963	0.8773
37	0.9411	1.7960	47.6%	0.1497	0.3573	-0.1220	0.9097	0.7792	0.8654
38	0.9777	1.8599	47.4%	0.1573	0.3737	-0.1277	0.9002	0.7609	0.8526
39	1.0153	1.9244	47.2%	0.1652	0.3902	-0.1335	0.8900	0.7417	0.8389
40	1.0533	1.9890	47.0%	0.1734	0.4070	-0.1395	0.8790	0.7214	0.8246
41	1.0918	2.0544	46.9%	0.1819	0.4241	-0.1456	0.8673	0.7001	0.8095
42	1.1313	2.1205	46.6%	0.1907	0.4416	-0.1520	0.8549	0.6776	0.7936
43	1.1718	2.1870	46.4%	0.1997	0.4592	-0.1586	0.8416	0.6541	0.7770
44	1.2130	2.2540	46.2%	0.2088	0.4771	-0.1653	0.8278	0.6297	0.7600
45	1.2540	2.3209	46.0%	0.2179	0.4951	-0.1721	0.8135	0.6045	0.7427
46	1.2950	2.3878	45.8%	0.2270	0.5132	-0.1790	0.7989	0.5787	0.7253
47	1.3365	2.4546	45.6%	0.2362	0.5313	-0.1858	0.7840	0.5524	0.7077
48	1.3785	2.5213	45.3%	0.2454	0.5494	-0.1926	0.7686	0.5255	0.6898
49	1.4204	2.5876	45.1%	0.2546	0.5673	-0.1994	0.7531	0.4985	0.6719
50	1.4624	2.6535	44.9%	0.2637	0.5852	-0.2062	0.7375	0.4712	0.6541
51	1.5041	2.7188	44.7%	0.2729	0.6031	-0.2130	0.7219	0.4439	0.6366
52	1.5463	2.7840	44.5%	0.2823	0.6211	-0.2199	0.7061	0.4166	0.6191
53	1.5884	2.8487	44.2%	0.2916	0.6391	-0.2270	0.6904	0.3892	0.6019
54	1.6302	2.9131	44.0%	0.3008	0.6572	-0.2340	0.6748	0.3620	0.5850
55	1.6720	2.9772	43.8%	0.3098	0.6752	-0.2410	0.6595	0.3348	0.5685
56	1.7133	3.0409	43.7%	0.3187	0.6931	-0.2480	0.6446	0.3079	0.5525
57	1.7543	3.1042	43.5%	0.3275	0.7110	-0.2550	0.6300	0.2813	0.5371
58	1.7952	3.1668	43.3%	0.3363	0.7287	-0.2620	0.6159	0.2552	0.5221
59	1.8359	3.2292	43.1%	0.3450	0.7463	-0.2691	0.6024	0.2296	0.5078
60	1.8766	3.2912	43.0%	0.3537	0.7638	-0.2763	0.5896	0.2045	0.4942
61	1.9176	3.3528	42.8%	0.3624	0.7813	-0.2836	0.5774	0.1801	0.4811
62	1.9586	3.4139	42.6%	0.3710	0.7985	-0.2910	0.5658	0.1565	0.4687
63	1.9995	3.4741	42.4%	0.3796	0.8155	-0.2984	0.5548	0.1338	0.4569
64	2.0400	3.5329	42.3%	0.3881	0.8320	-0.3059	0.5445	0.1121	0.4459
65	2.0807	3.5910	42.1%	0.3966	0.8483	-0.3134	0.5348	0.0913	0.4353
66	2.1214	3.6481	41.9%	0.4050	0.8641	-0.3209	0.5256	0.0715	0.4254
67	2.1618	3.7045	41.6%	0.4134	0.8795	-0.3283	0.5169	0.0524	0.4158
68	2.2020	3.7596	41.4%	0.4217	0.8945	-0.3357	0.5087	0.0343	0.4067
69	2.2416	3.8136	41.2%	0.4299	0.9091	-0.3430	0.5008	0.0169	0.3980
70	2.2804	3.8659	41.0%	0.4379	0.9233	-0.3502	0.4934	0.0005	0.3898
71	2.3185	3.9166	40.8%	0.4457	0.9370	-0.3572	0.4866	-0.0149	0.3822
72	2.3559	3.9656	40.6%	0.4533	0.9503	-0.3640	0.4801	-0.0294	0.3749
73	2.3926	4.0132	40.4%	0.4607	0.9631	-0.3708	0.4740	-0.0431	0.3681
74	2.4286	4.0591	40.2%	0.4680	0.9755	-0.3775	0.4681	-0.0558	0.3616
75	2.4641	4.1036	40.0%	0.4752	0.9875	-0.3842	0.4626	-0.0678	0.3555
76	2.4988	4.1466	39.7%	0.4823	0.9991	-0.3907	0.4573	-0.0790	0.3496
77	2.5322	4.1876	39.5%	0.4892	1.0101	-0.3971	0.4523	-0.0895	0.3440
78	2.5649	4.2264	39.3%	0.4959	1.0206	-0.4034	0.4470	-0.0993	0.3383
79	2.5970	4.2630	39.1%	0.5024	1.0305	-0.4095	0.4416	-0.1084	0.3324
80	2.6282	4.2978	38.8%	0.5089	1.0400	-0.4156	0.4361	-0.1168	0.3267
81	2.6580	4.3305	38.6%	0.5151	1.0491	-0.4215	0.4307	-0.1248	0.3209
82	2.6869	4.3612	38.4%	0.5211	1.0576	-0.4271	0.4252	-0.1321	0.3153
83	2.7148	4.3898	38.2%	0.5269	1.0657	-0.4327	0.4197	-0.1388	0.3095
84	2.7417	4.4167	37.9%	0.5326	1.0733	-0.4380	0.4141	-0.1452	0.3039
85	2.7662	4.4418	37.7%	0.5371	1.0805	-0.4425	0.4092	-0.1511	0.2990
86	2.7884	4.4649	37.5%	0.5411	1.0873	-0.4464	0.4049	-0.1565	0.2947
87	2.8087	4.4865	37.4%	0.5446	1.0937	-0.4500	0.4010	-0.1616	0.2908
88	2.8271	4.5062	37.3%	0.5478	1.0998	-0.4532	0.3974	-0.1664	0.2874
89	2.8444	4.5244	37.1%	0.5507	1.1055	-0.4562	0.3939	-0.1708	0.2841
90	2.8603	4.5408	37.0%	0.5535	1.1109	-0.4591	0.3905	-0.1748	0.2810

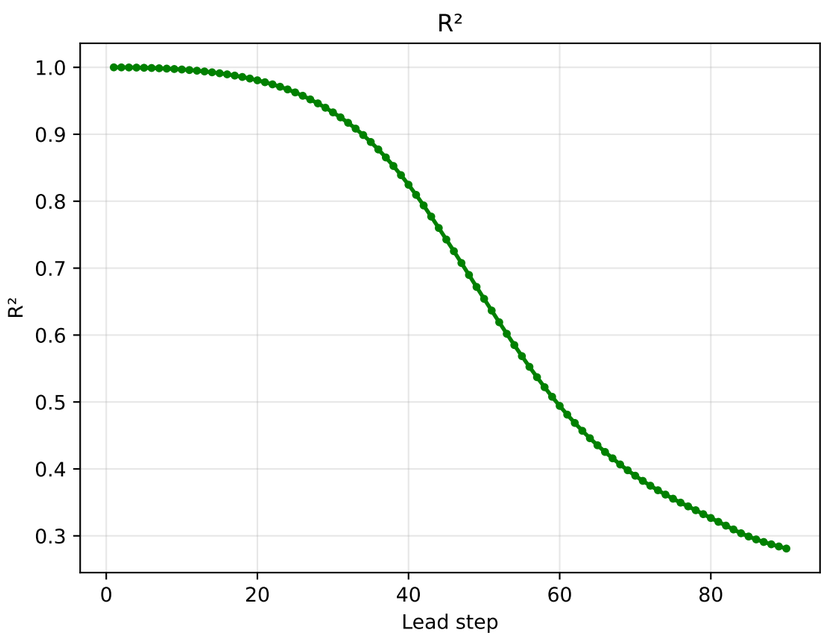
DHW Forecast Skill vs Lead Step - Test

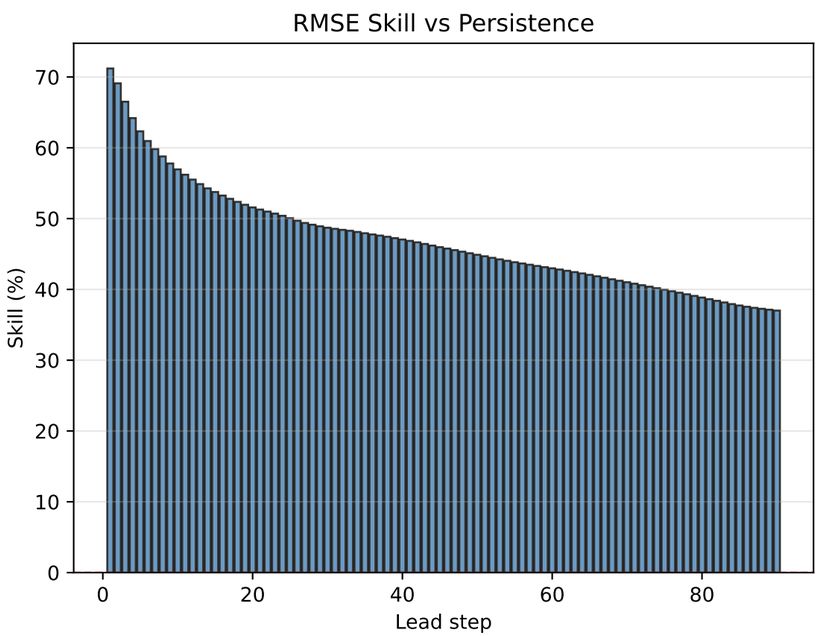


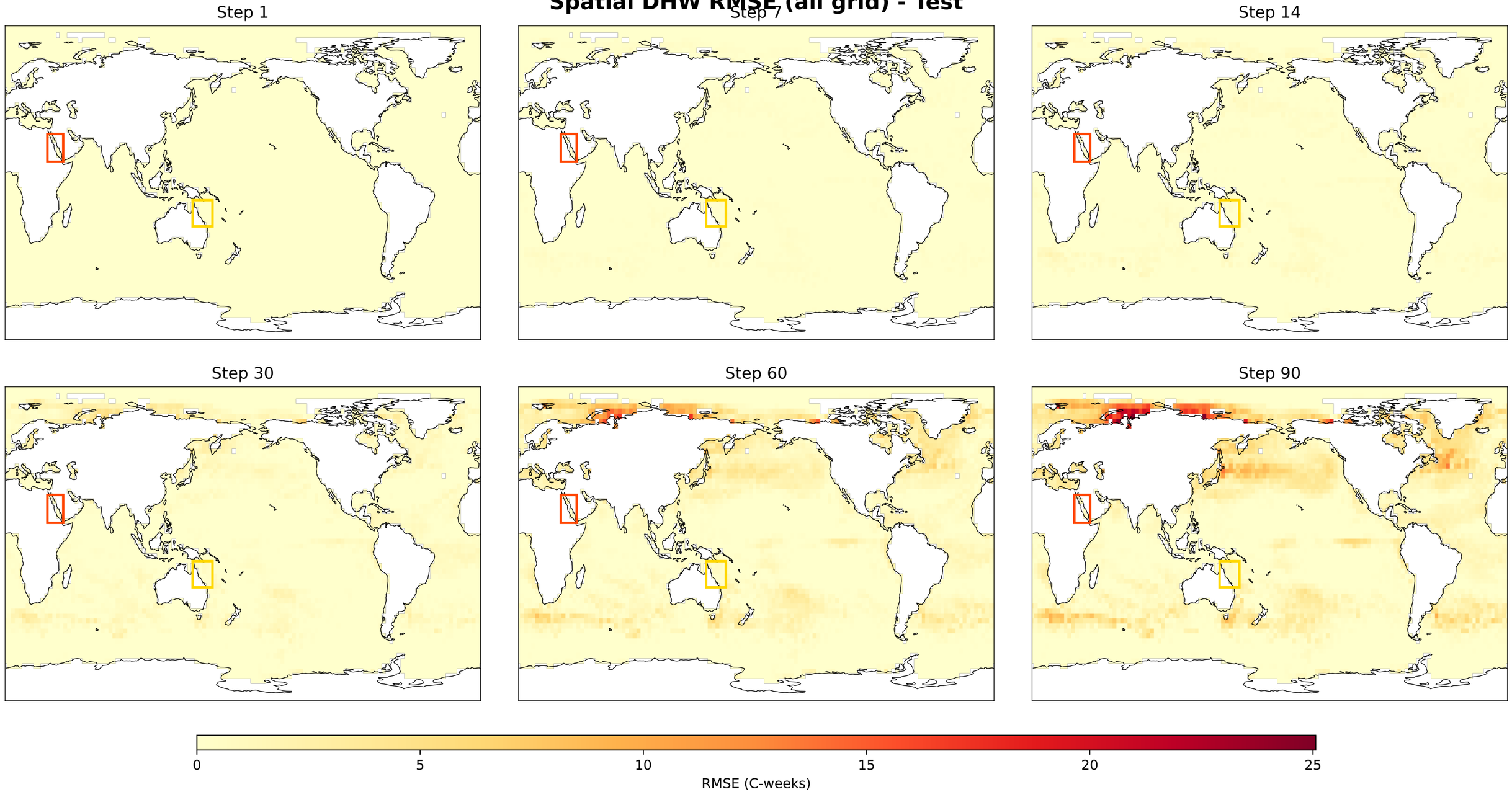


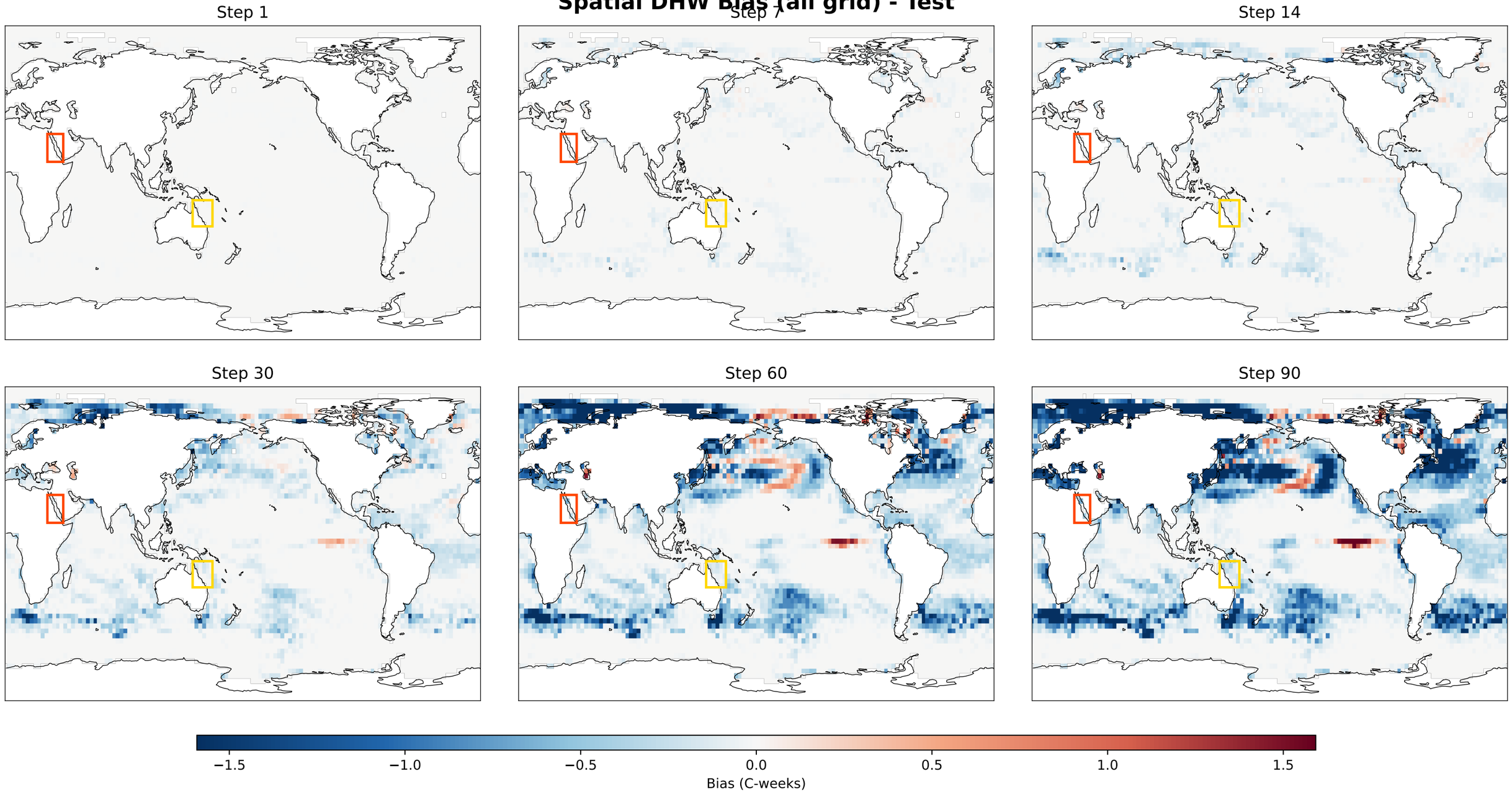


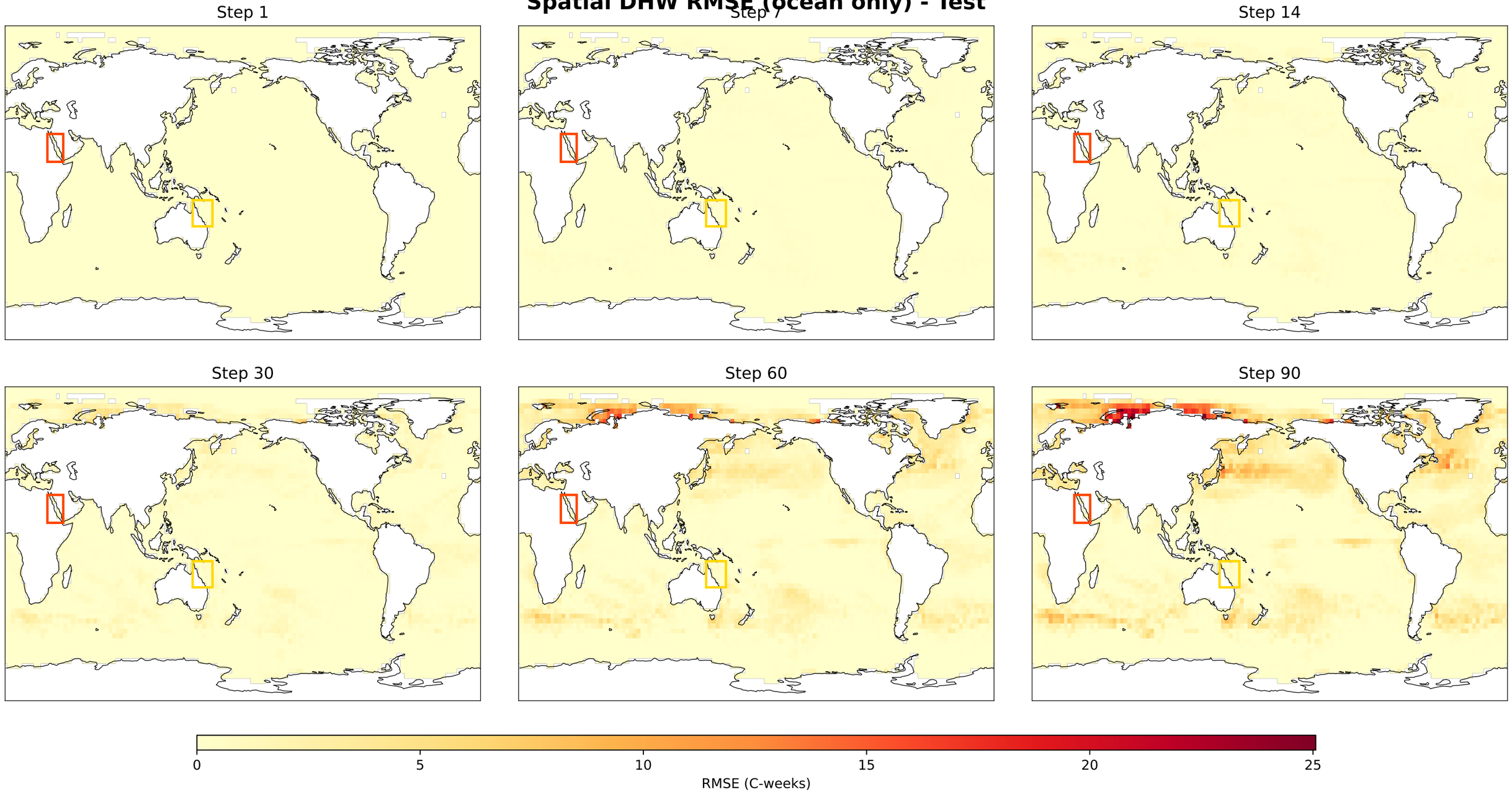


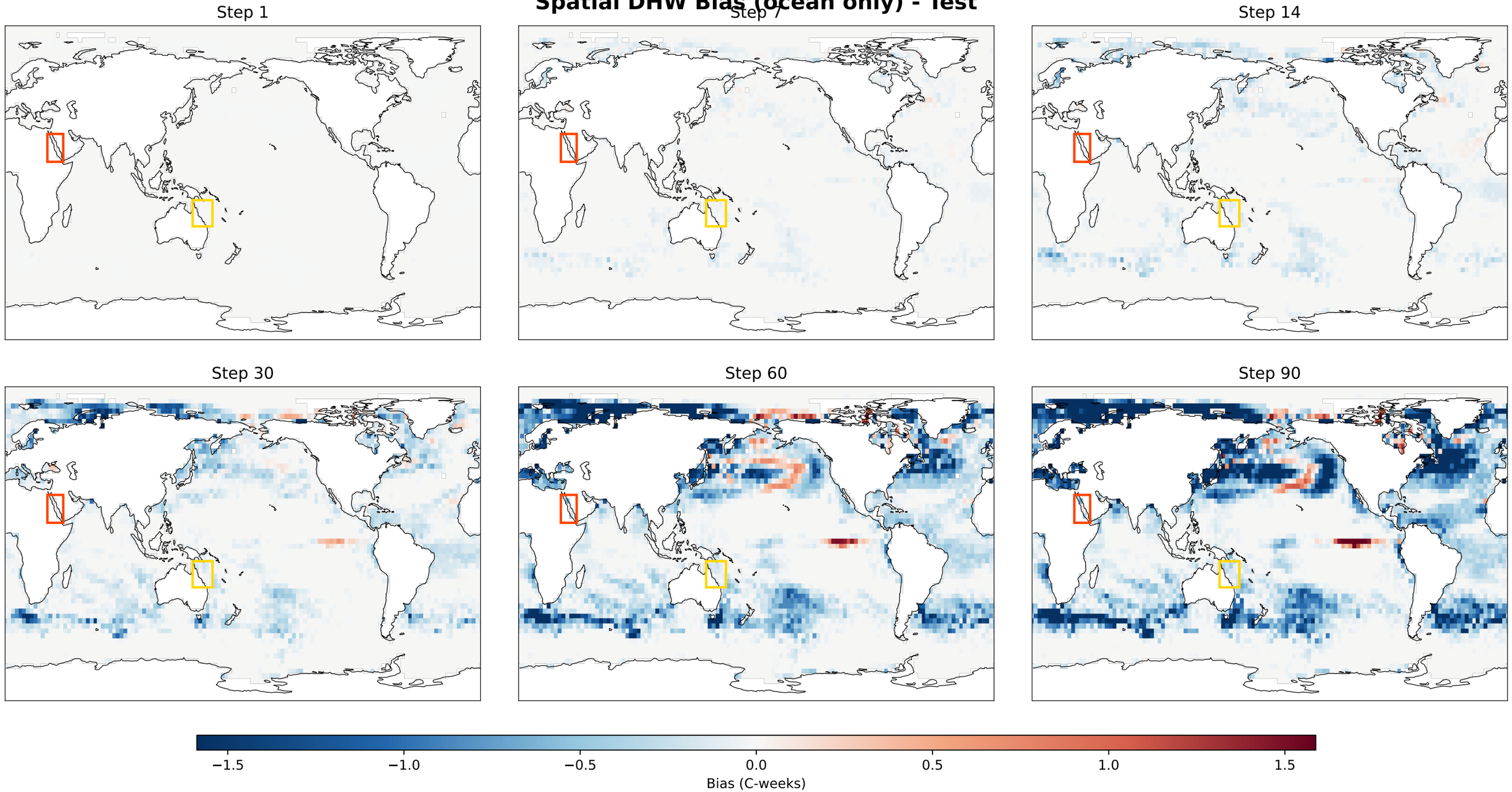





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

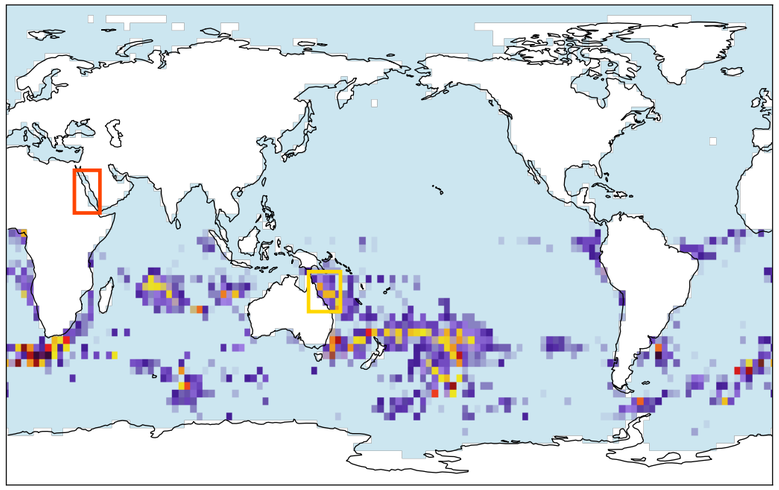

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

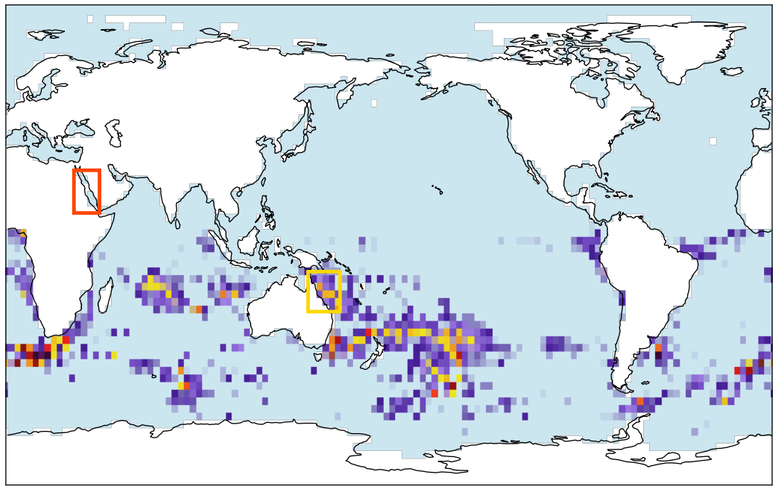

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

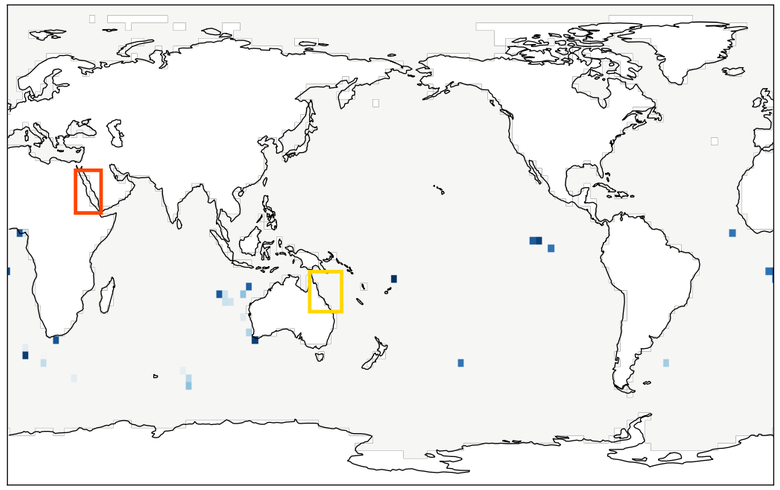
Actual DHW - Step 1



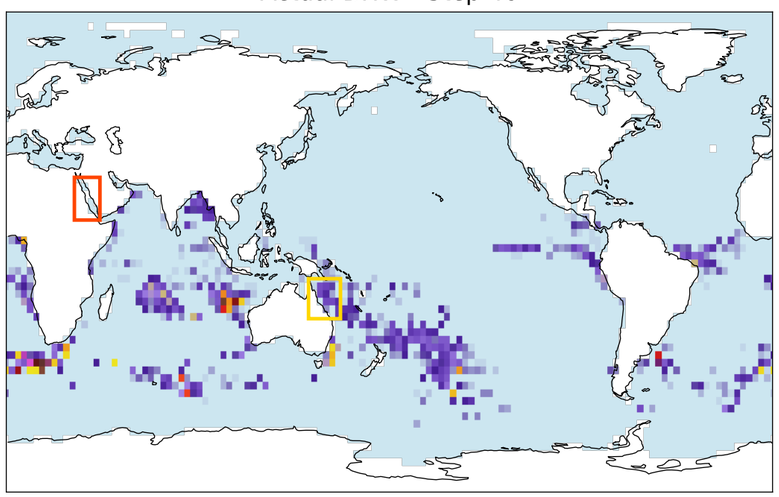
Predicted DHW - Step 1



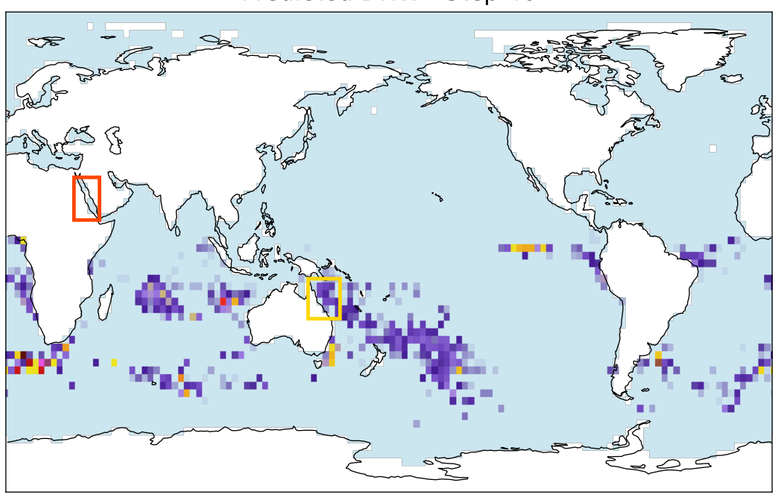
Error (Pred - Actual) - Step 1 [scale ±0.24]



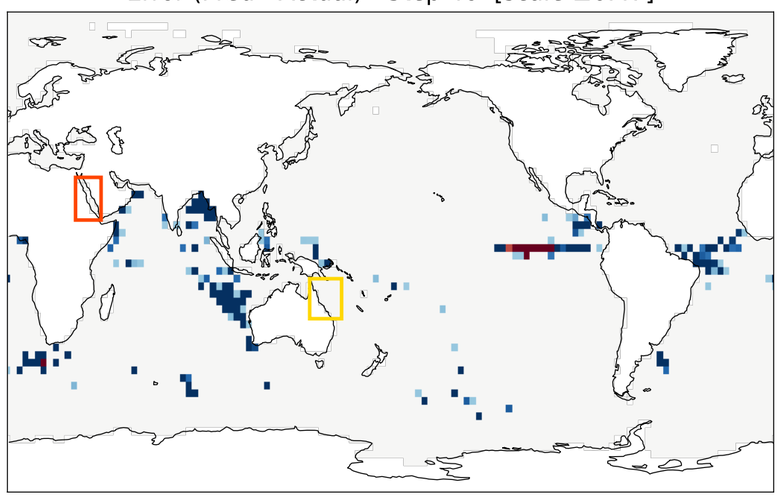
Actual DHW - Step 46



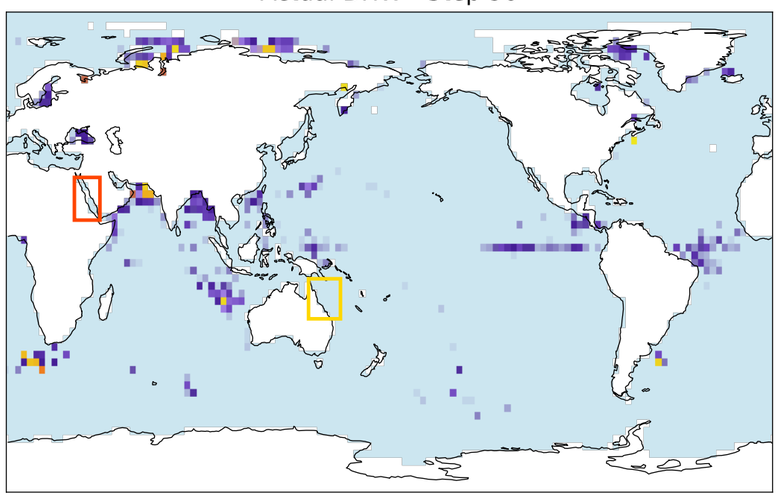
Predicted DHW - Step 46



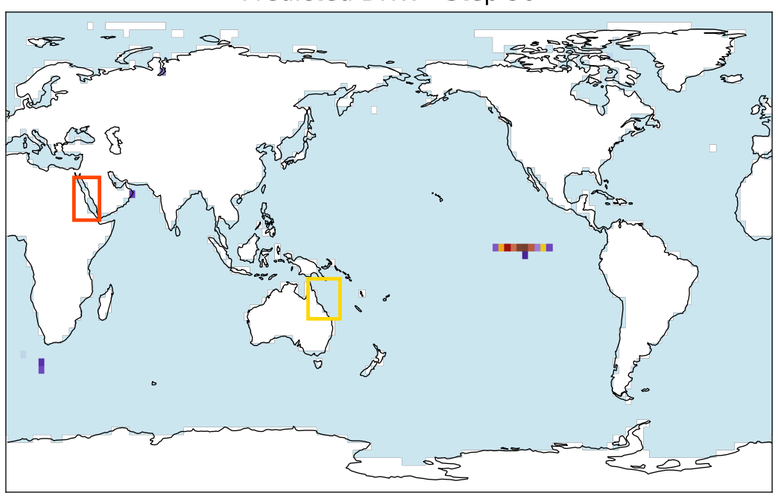
Error (Pred - Actual) - Step 46 [scale ±0.47]



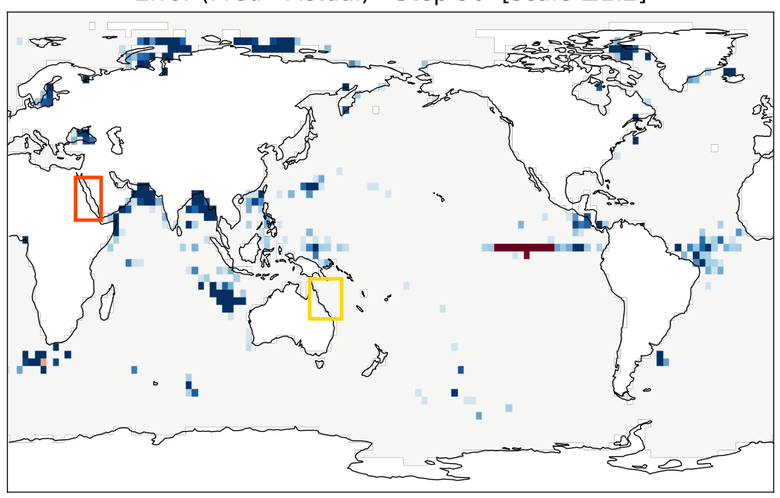
Actual DHW - Step 90

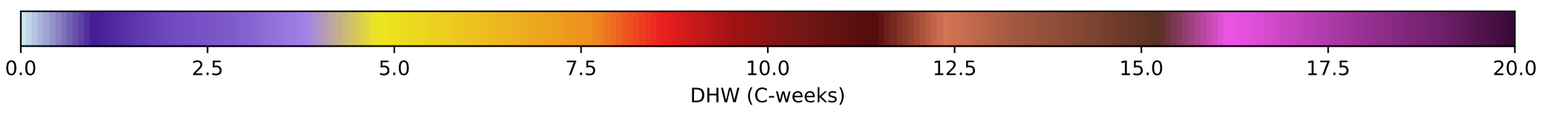


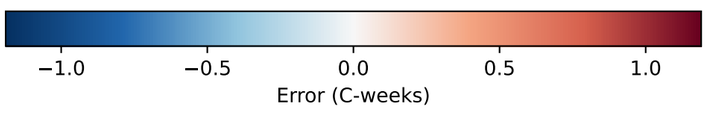
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.2]

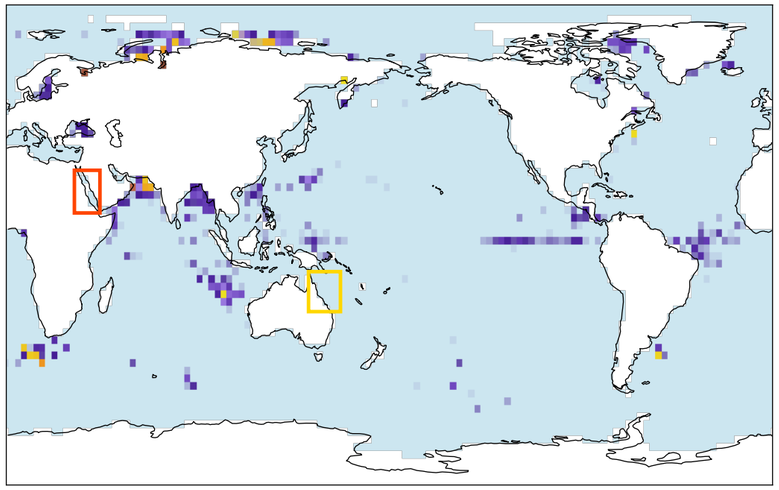




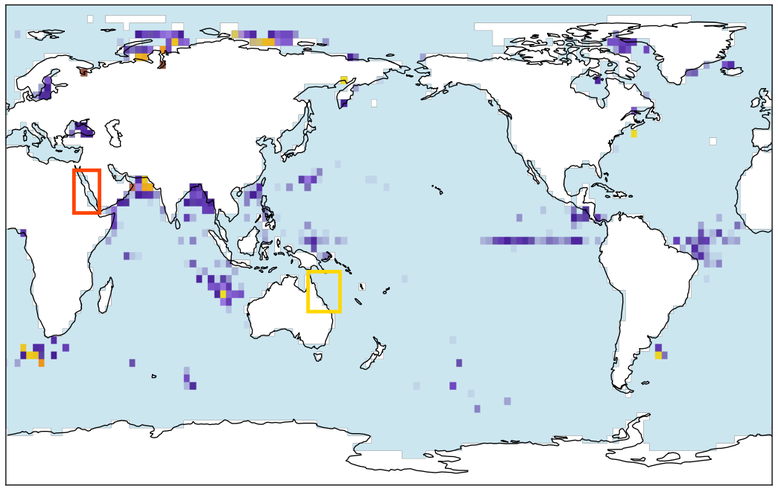


DHW Chain init: 2020-07-09 - Test (C-weeks)

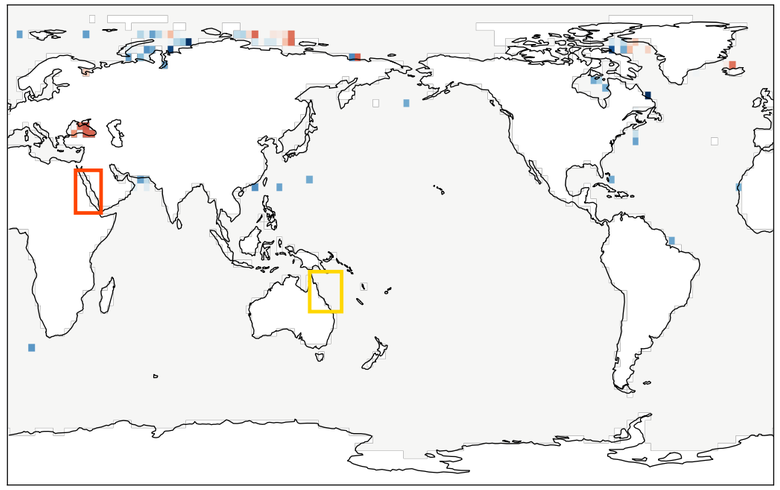
Actual DHW - Step 1



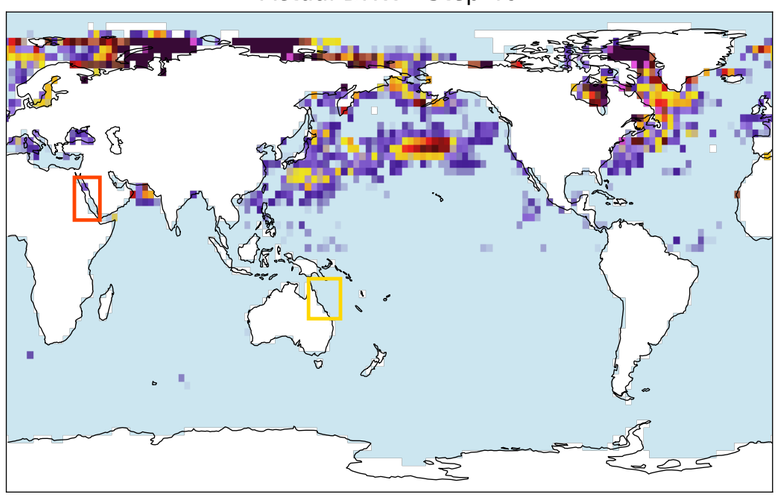
Predicted DHW - Step 1



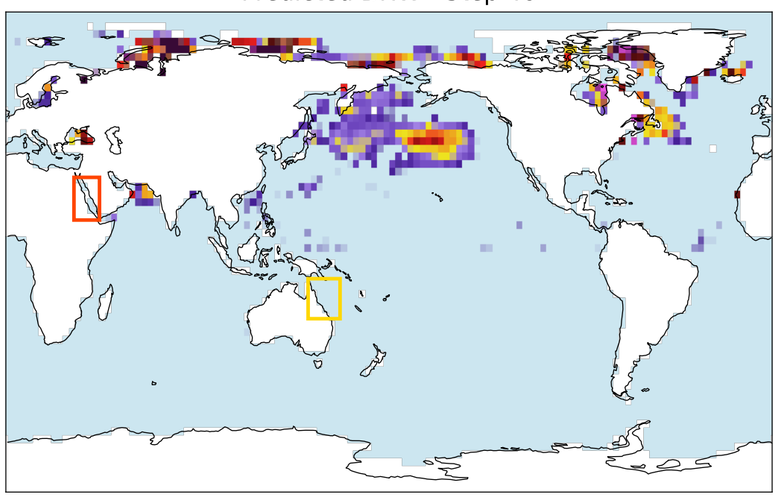
Error (Pred - Actual) - Step 1 [scale ±0.36]



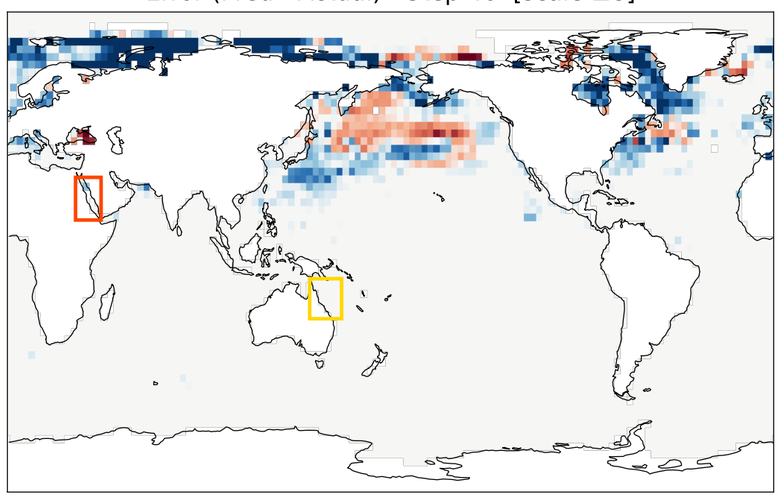
Actual DHW - Step 46



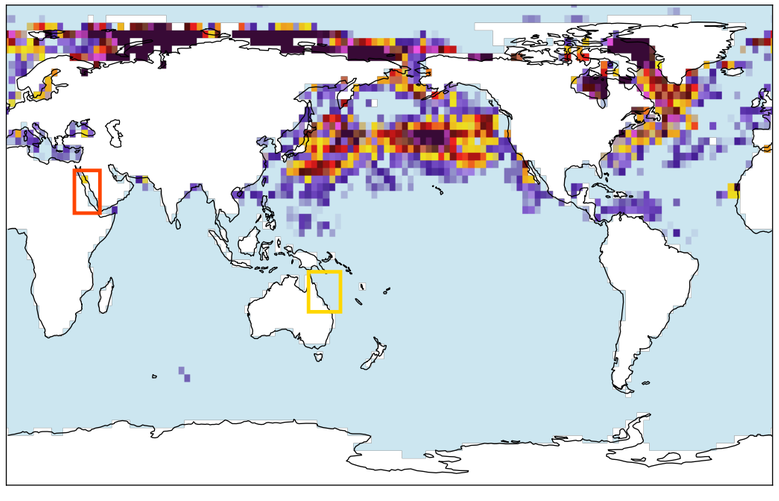
Predicted DHW - Step 46



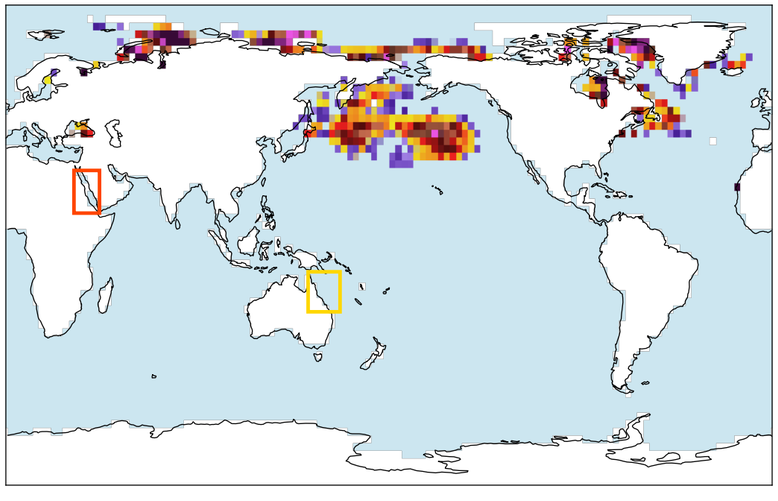
Error (Pred - Actual) - Step 46 [scale ±8]



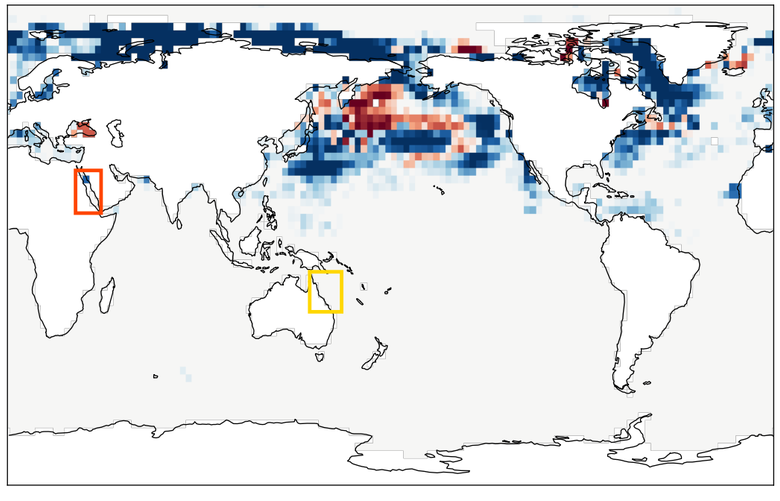
Actual DHW - Step 90

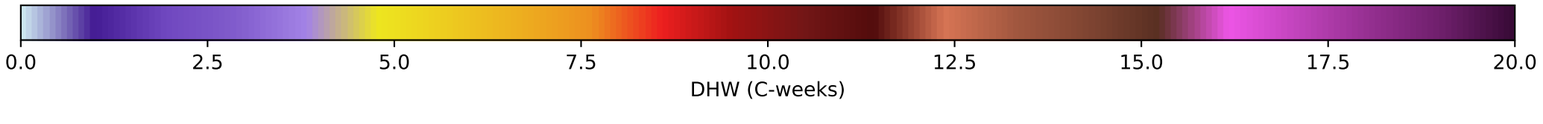


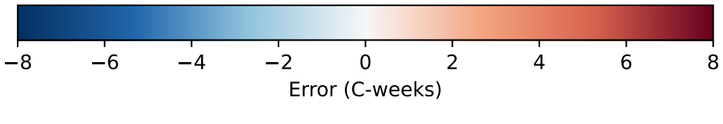
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

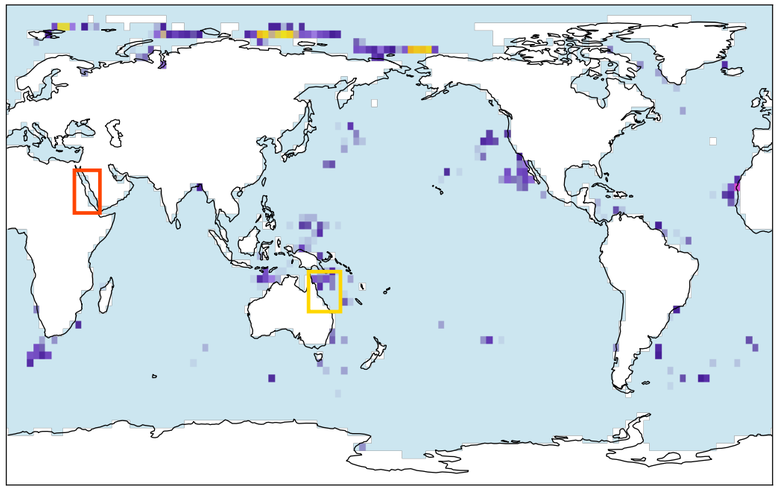




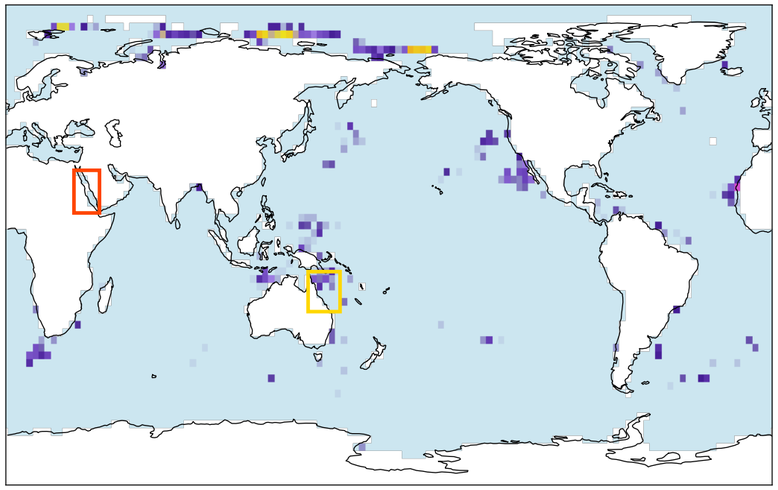


DHW Chain init: 2021-01-06 - Test (C-weeks)

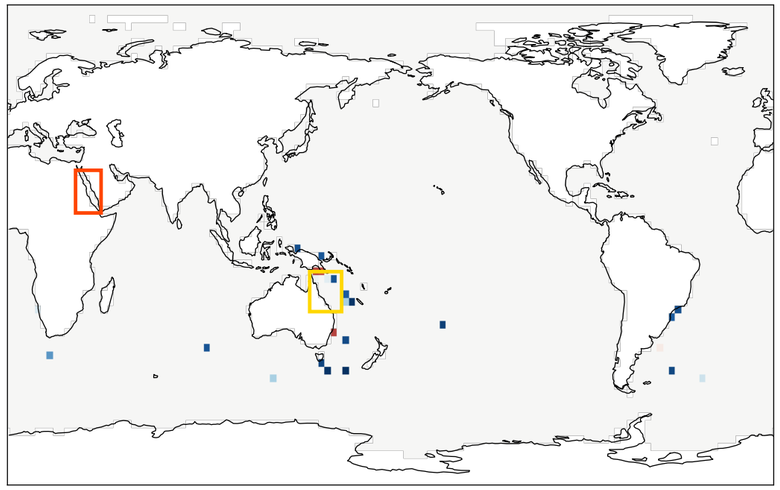
Actual DHW - Step 1



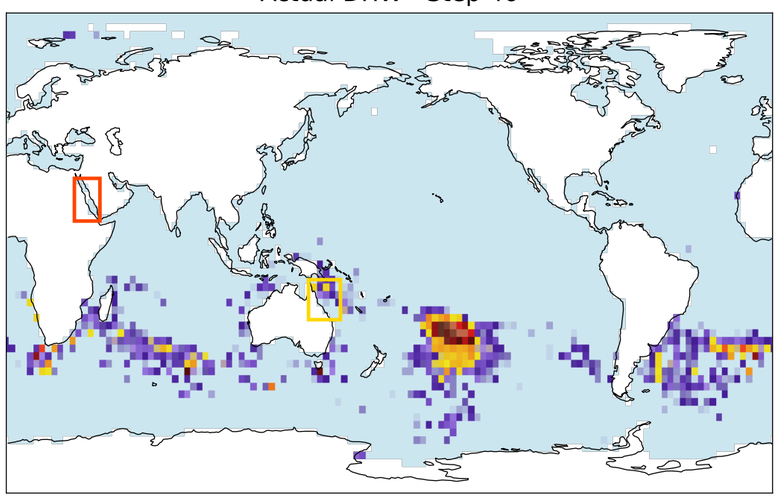
Predicted DHW - Step 1



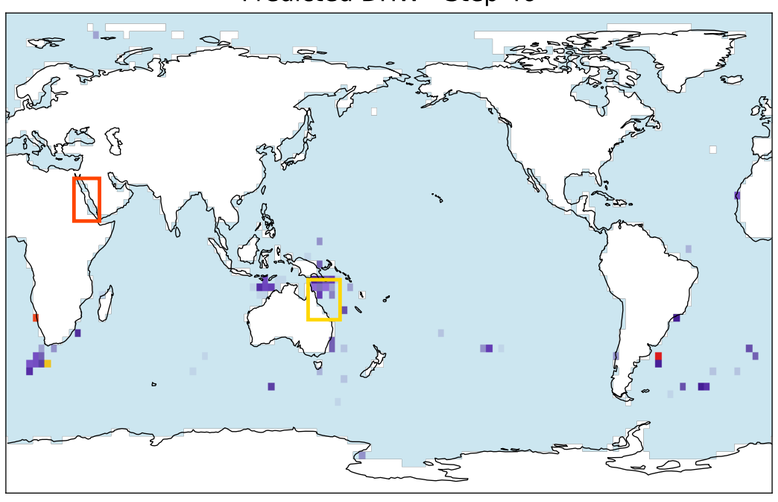
Error (Pred - Actual) - Step 1 [scale ±0.2]



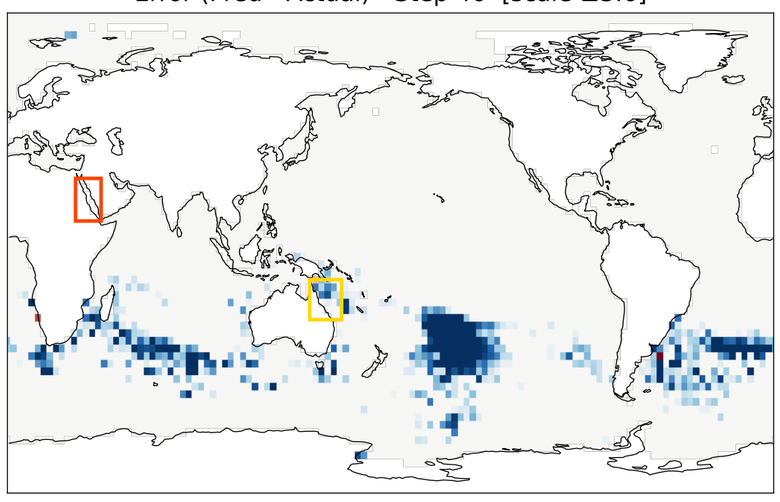
Actual DHW - Step 46



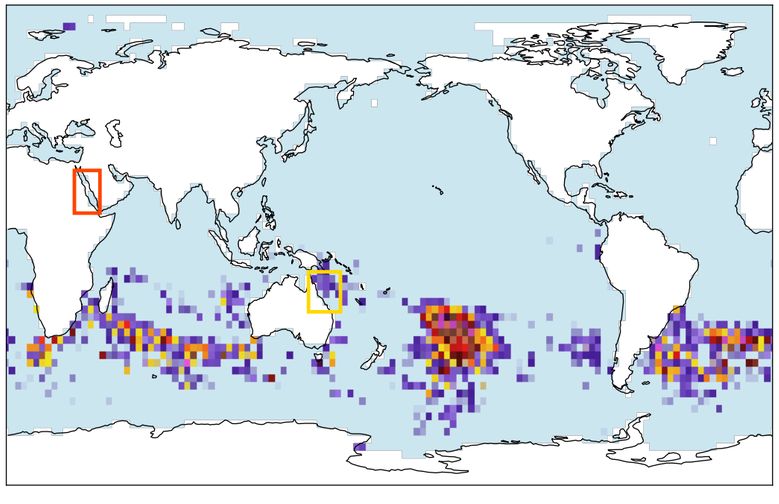
Predicted DHW - Step 46



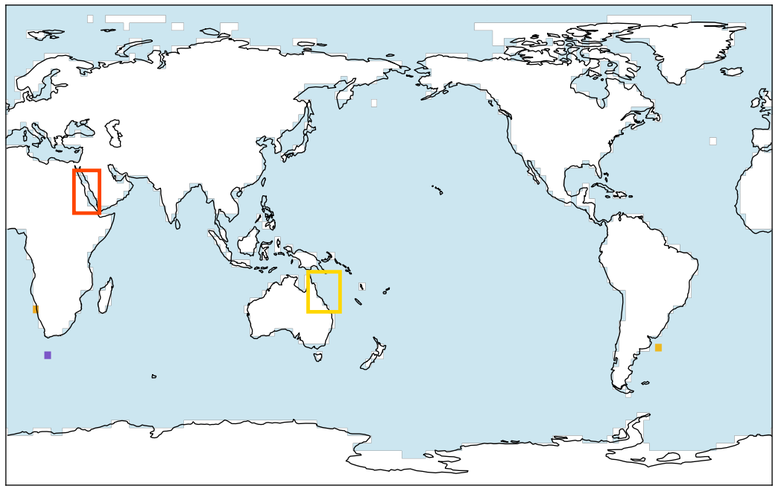
Error (Pred - Actual) - Step 46 [scale ±3.6]



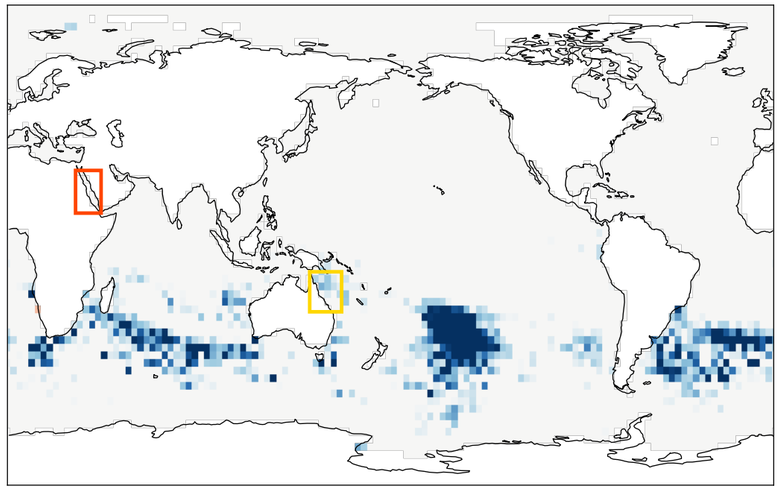
Actual DHW - Step 90

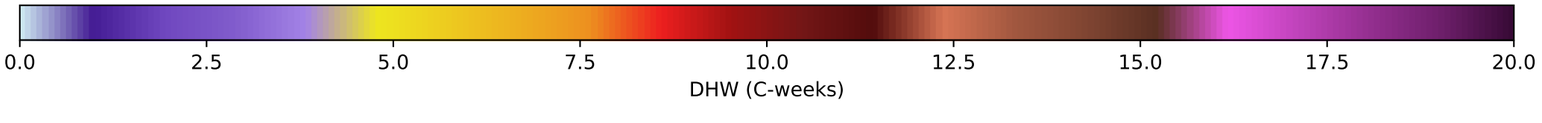


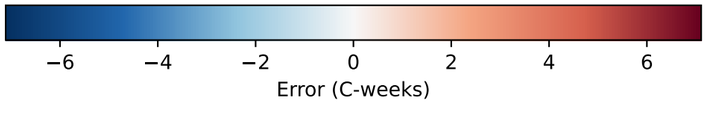
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7.1]

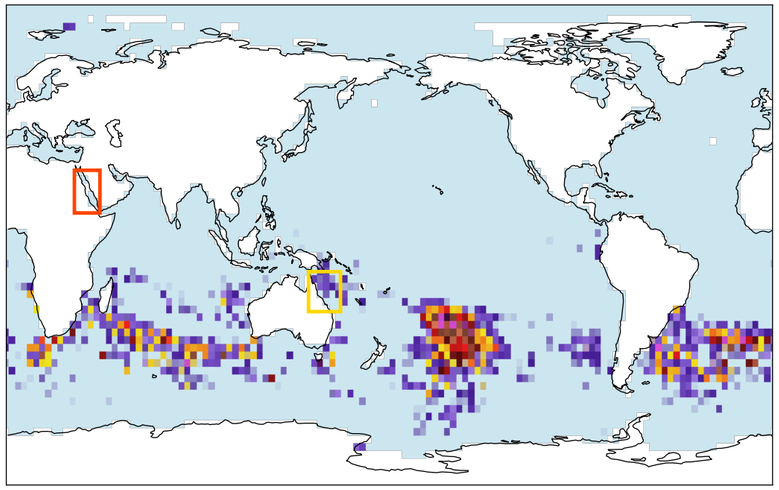




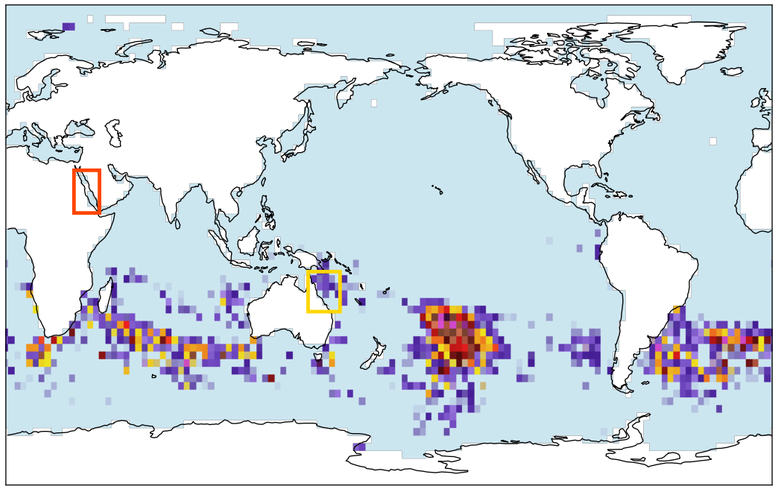


DHW Chain init: 2021-04-06 - Test (C-weeks)

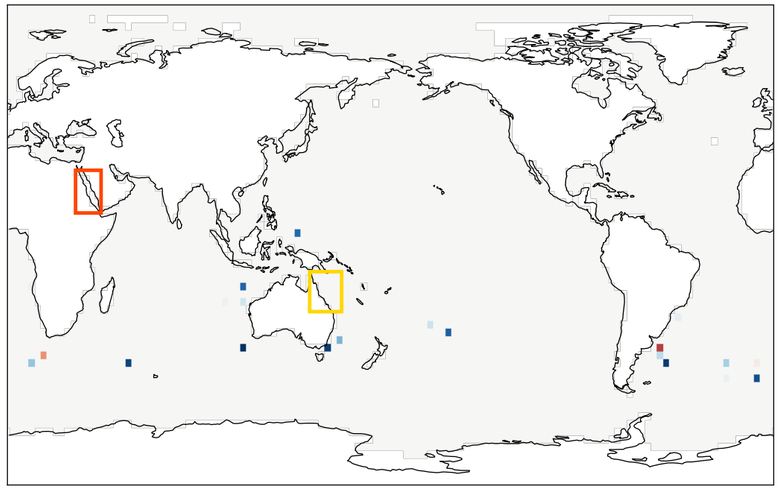
Actual DHW - Step 1



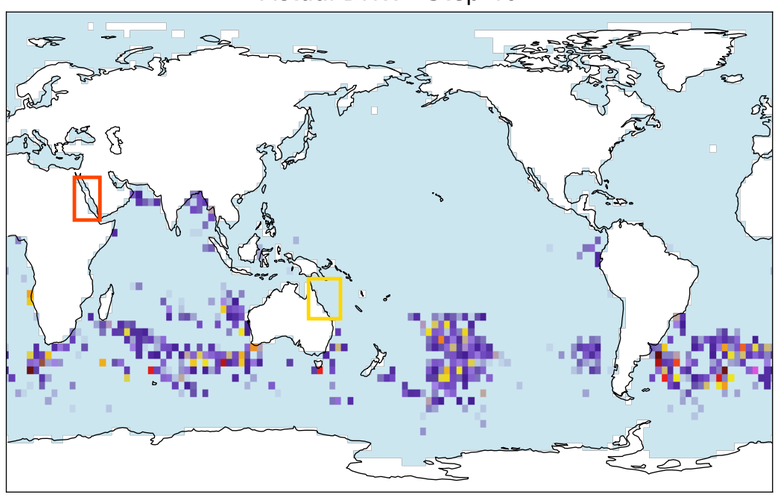
Predicted DHW - Step 1



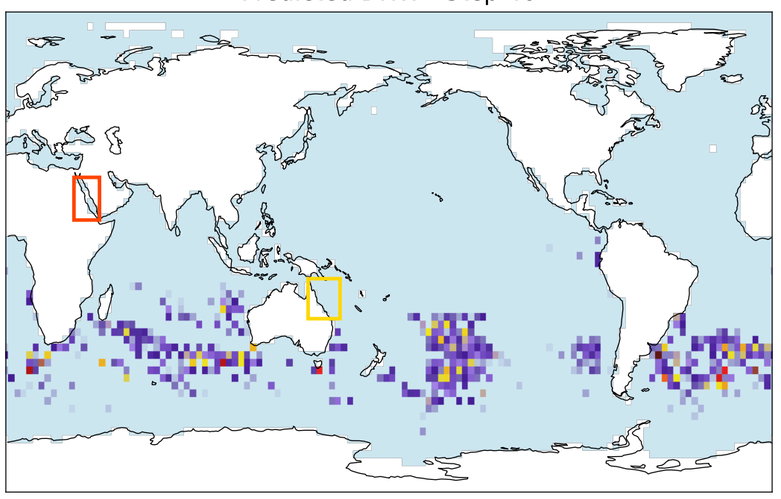
Error (Pred - Actual) - Step 1 [scale ±0.23]



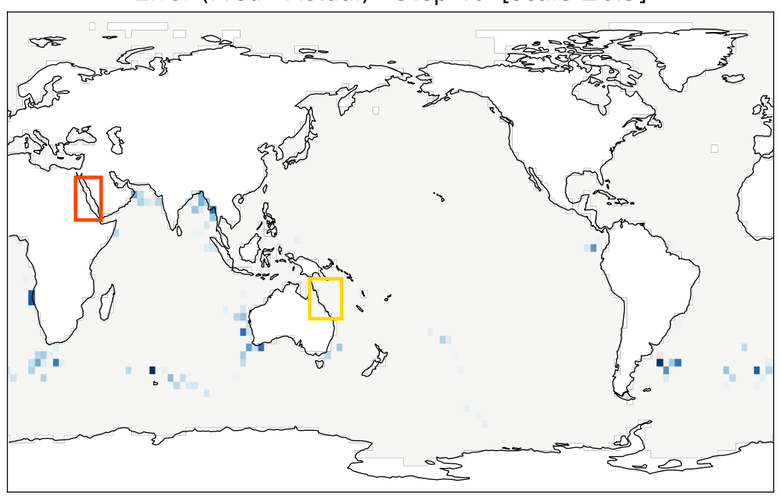
Actual DHW - Step 46



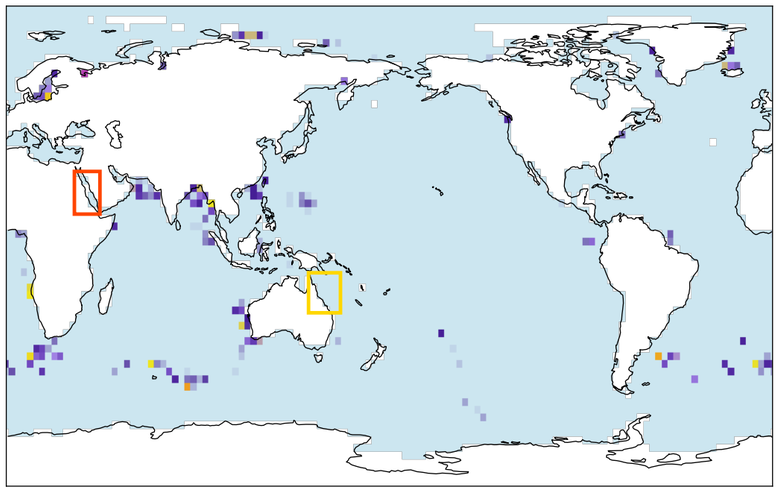
Predicted DHW - Step 46



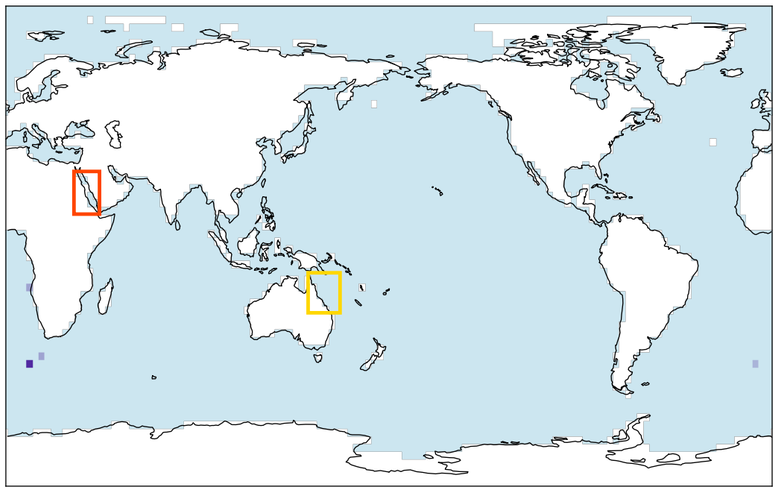
Error (Pred - Actual) - Step 46 [scale ±6.5]



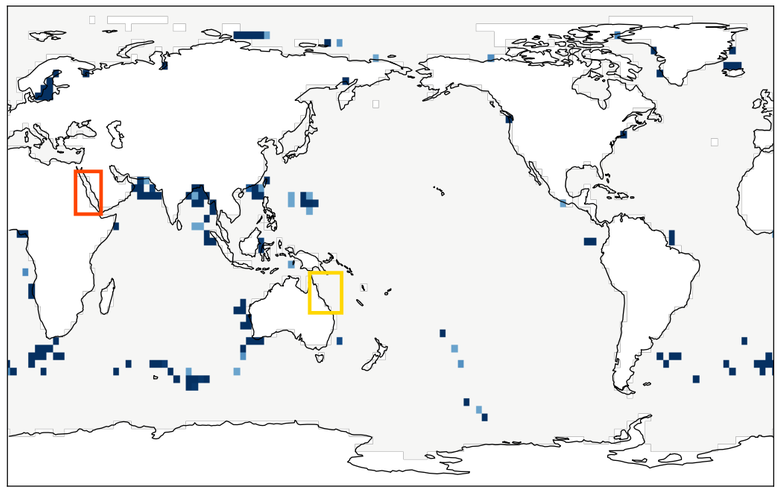
Actual DHW - Step 90

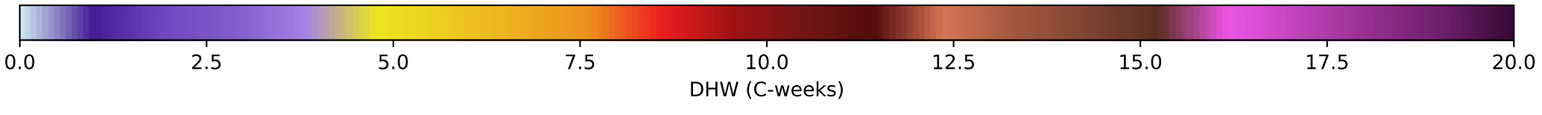


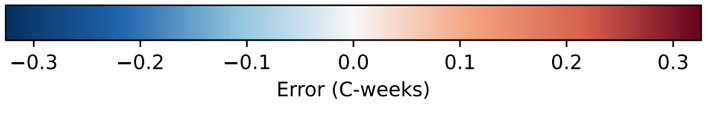
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

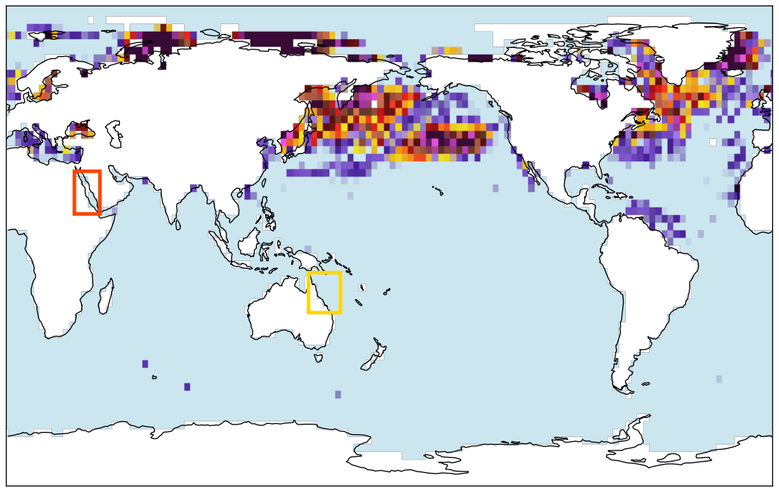




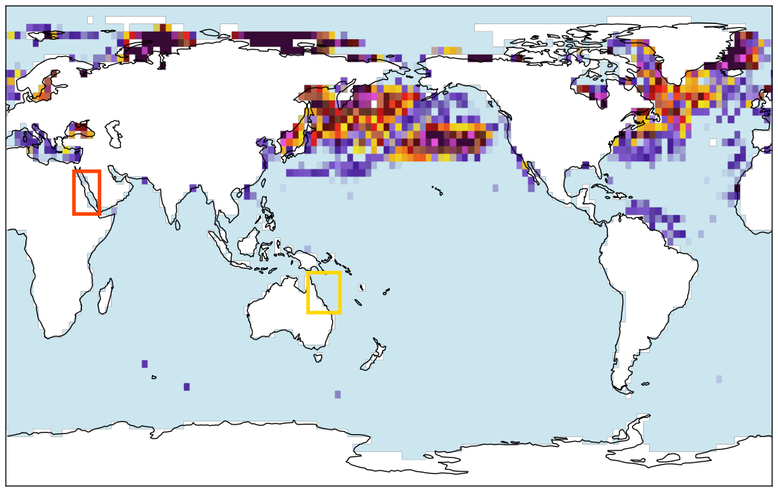


DHW Chain init: 2021-10-03 - Test (C-weeks)

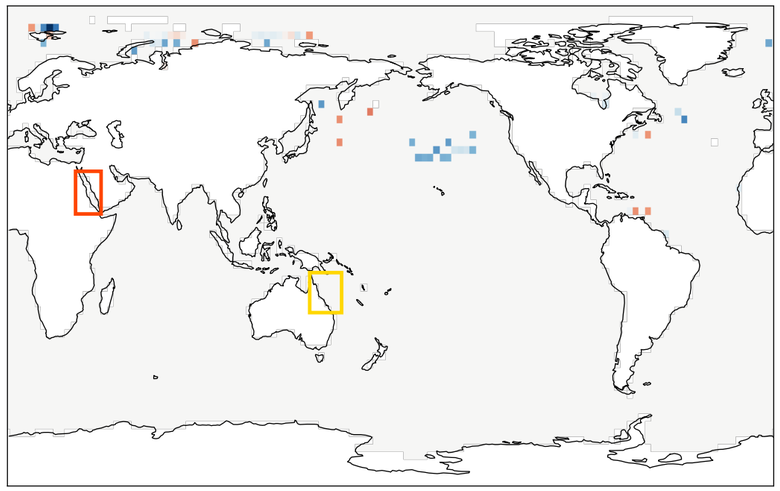
Actual DHW - Step 1



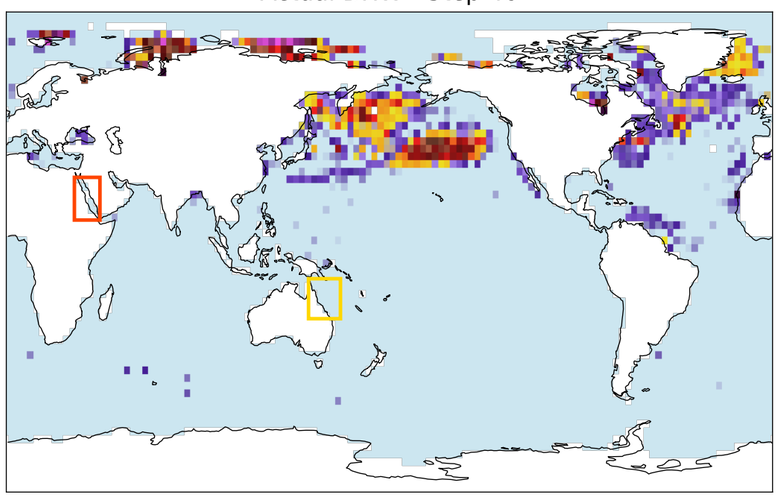
Predicted DHW - Step 1



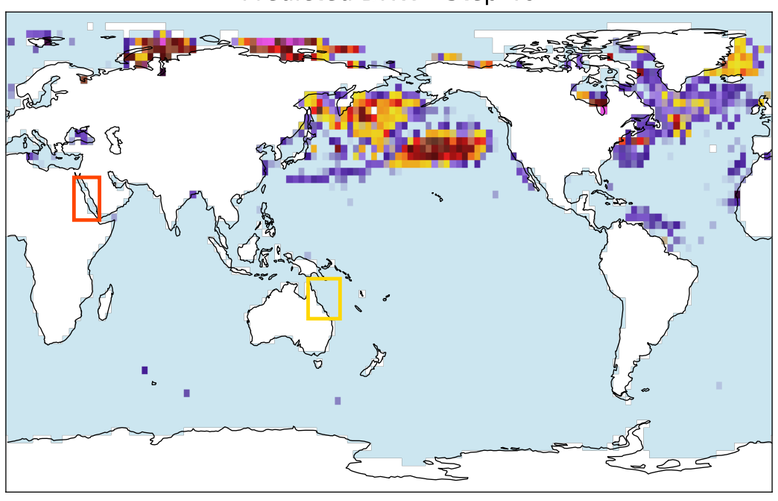
Error (Pred - Actual) - Step 1 [scale ±0.37]



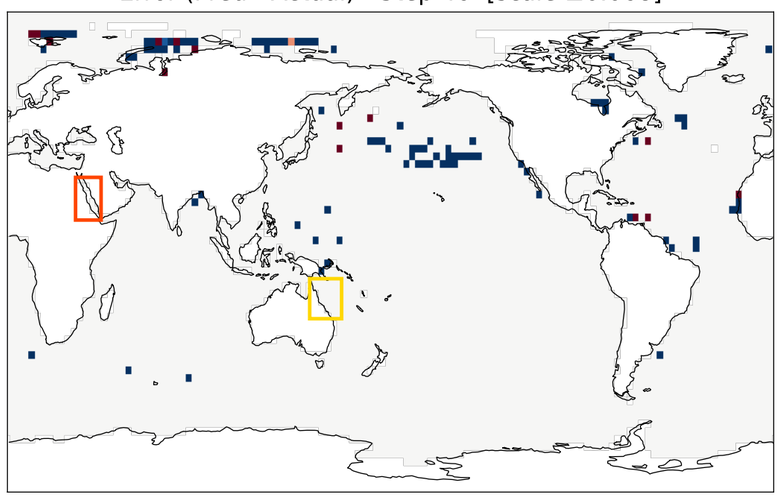
Actual DHW - Step 46



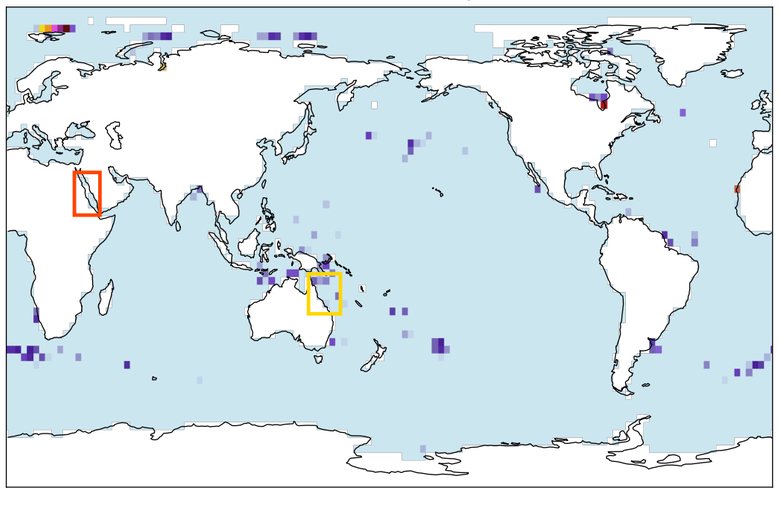
Predicted DHW - Step 46



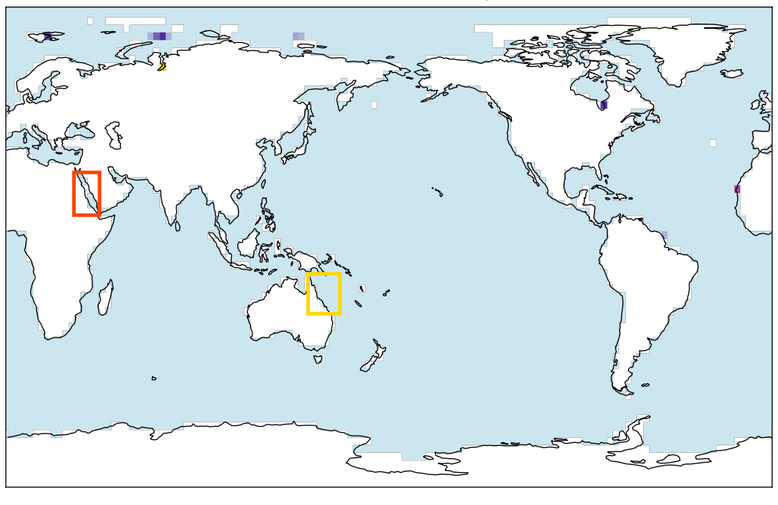
Error (Pred - Actual) - Step 46 [scale ±0.068]



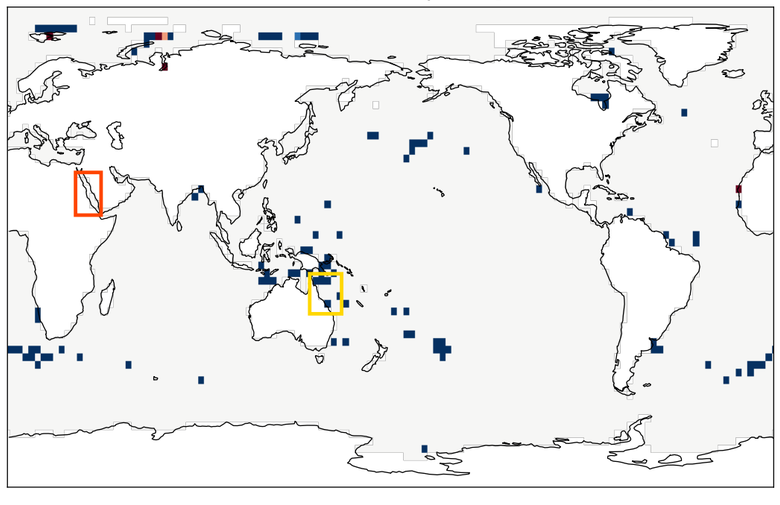
Actual DHW - Step 90

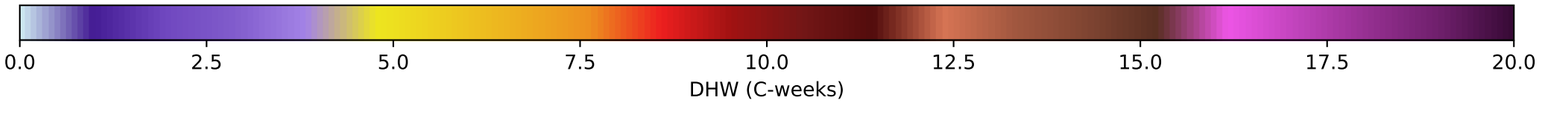


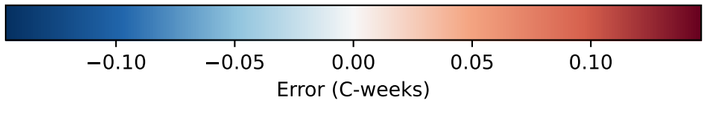
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

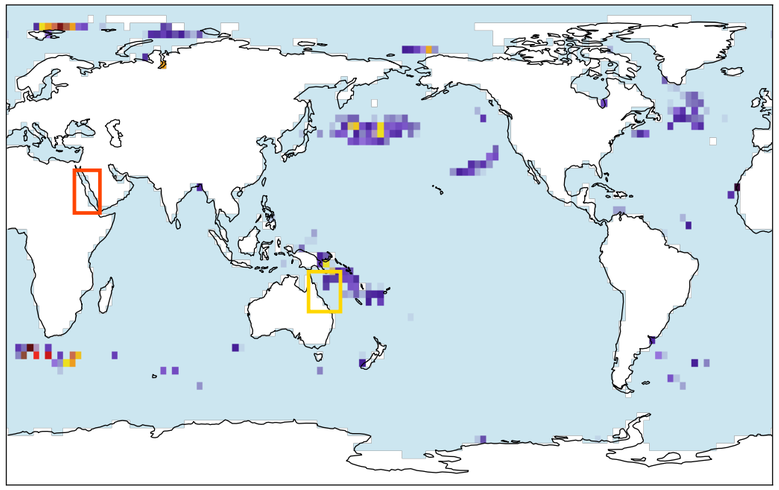




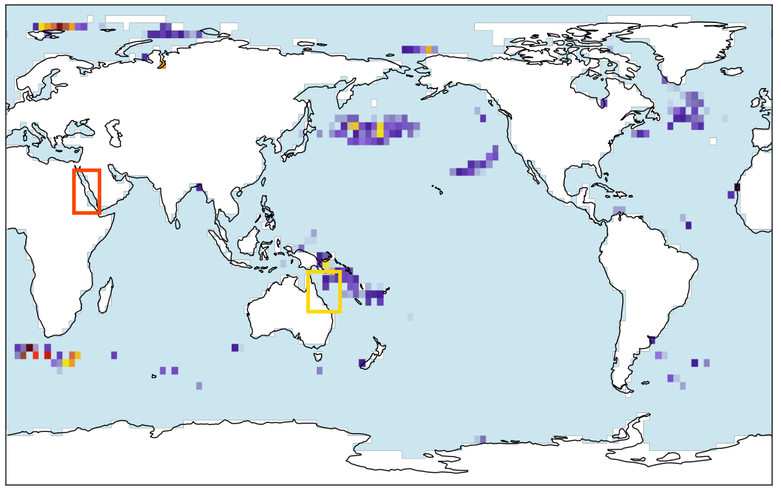


DHW Chain init: 2022-12-27 - Test (C-weeks)

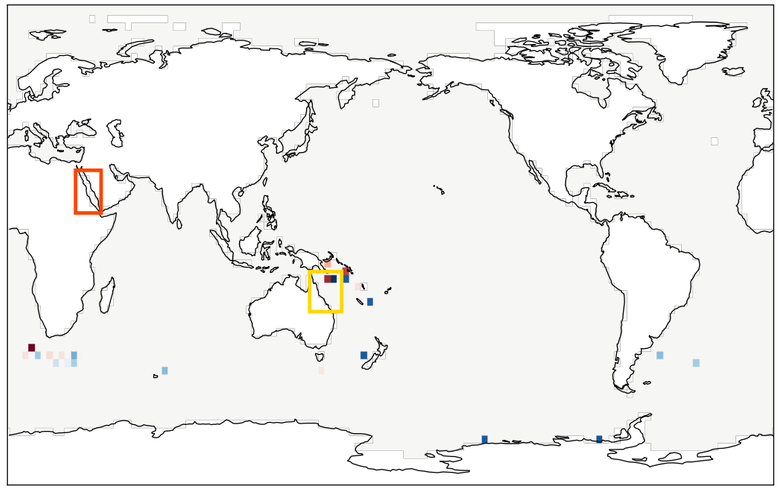
Actual DHW - Step 1



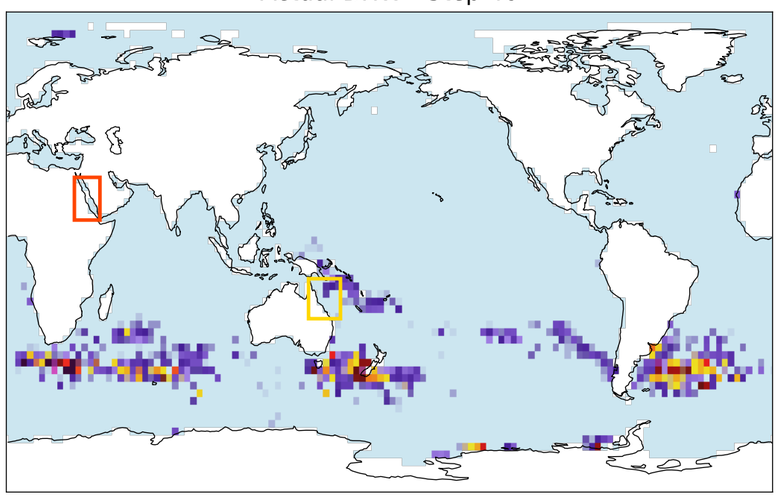
Predicted DHW - Step 1



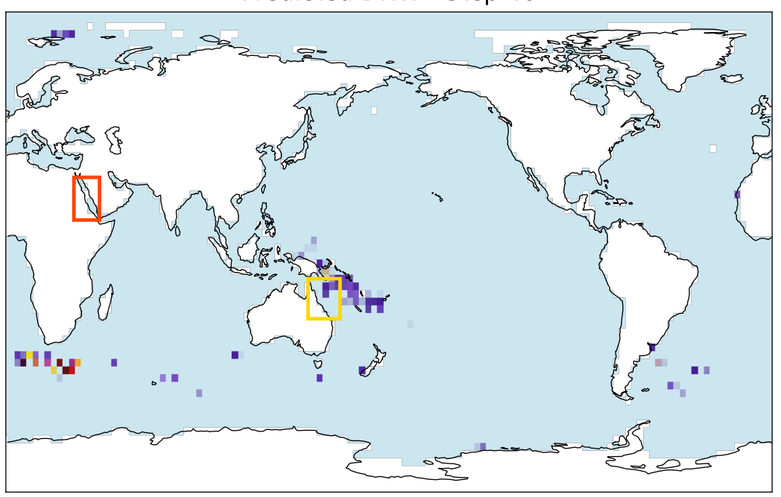
Error (Pred - Actual) - Step 1 [scale ±0.2]



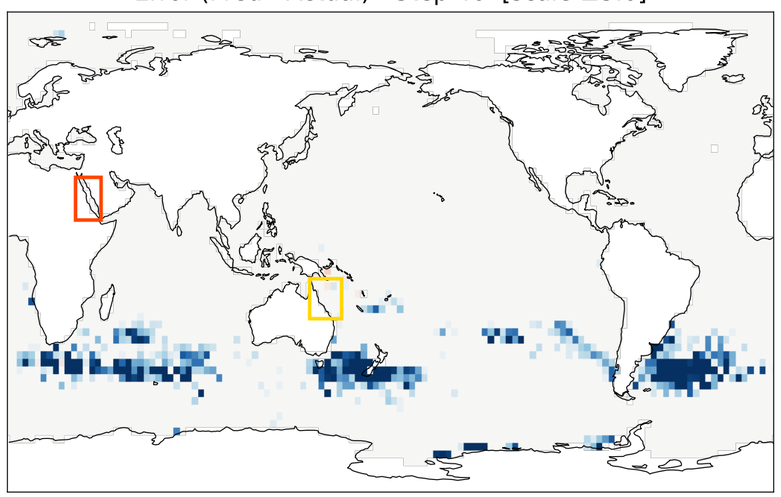
Actual DHW - Step 46



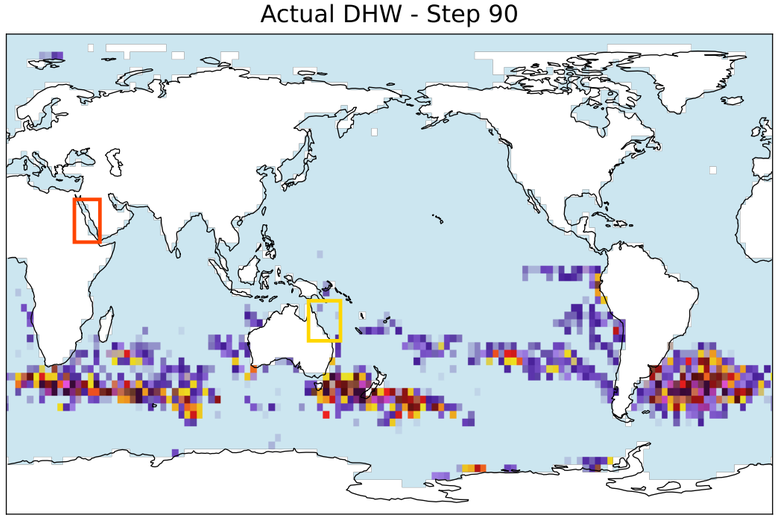
Predicted DHW - Step 46



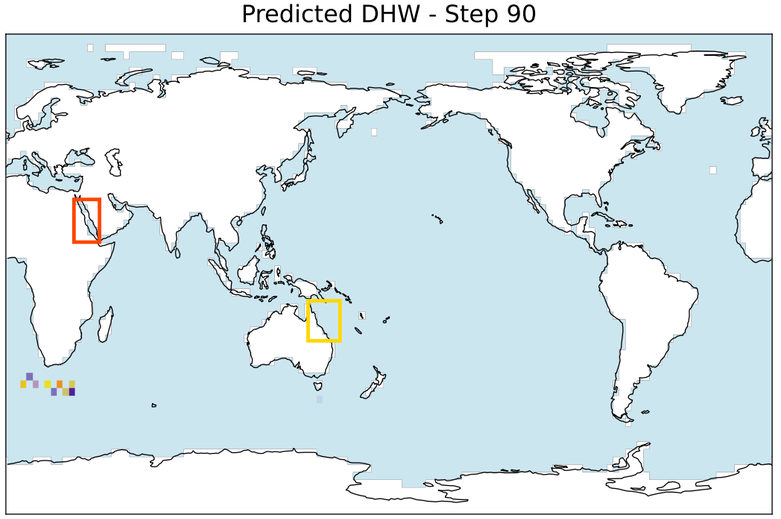
Error (Pred - Actual) - Step 46 [scale ±3.6]



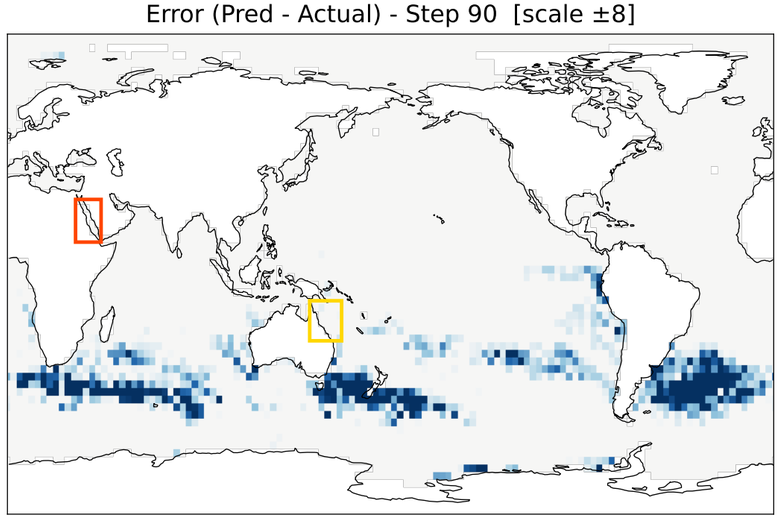
Actual DHW - Step 90

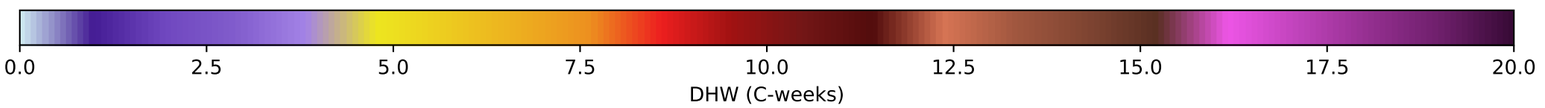


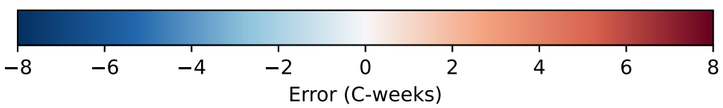
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

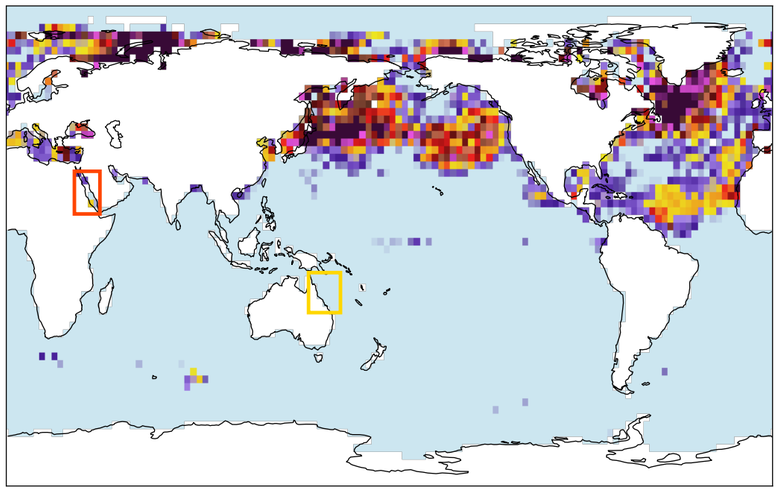




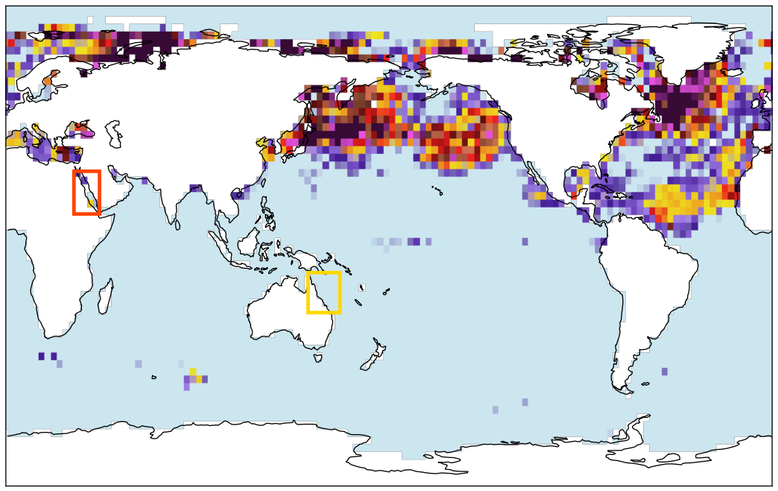


DHW Chain init: 2023-09-23 - Test (C-weeks)

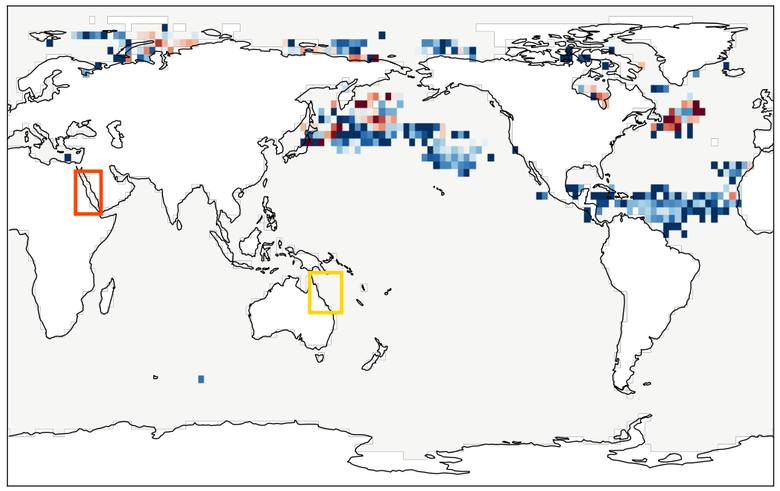
Actual DHW - Step 1



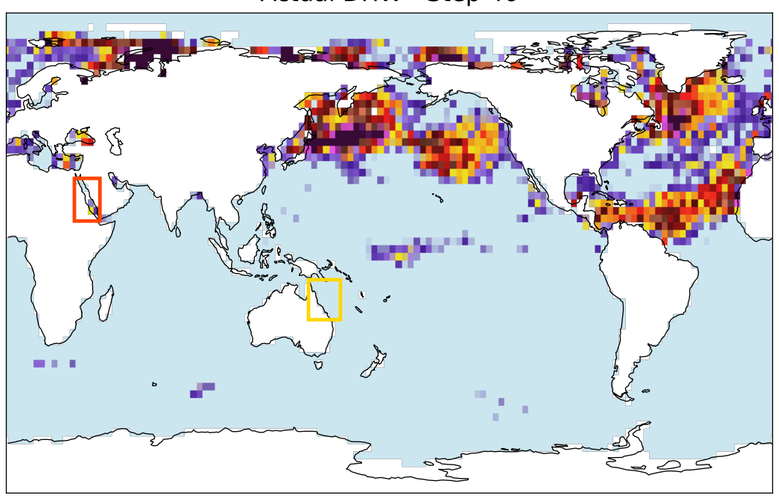
Predicted DHW - Step 1



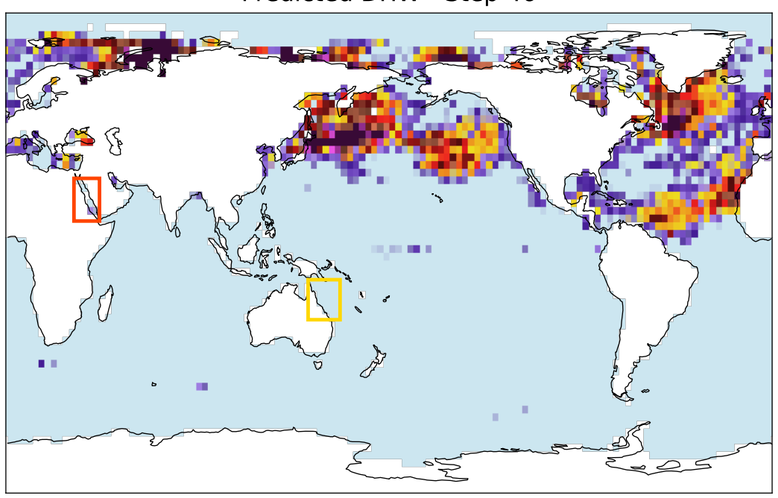
Error (Pred - Actual) - Step 1 [scale ±0.044]



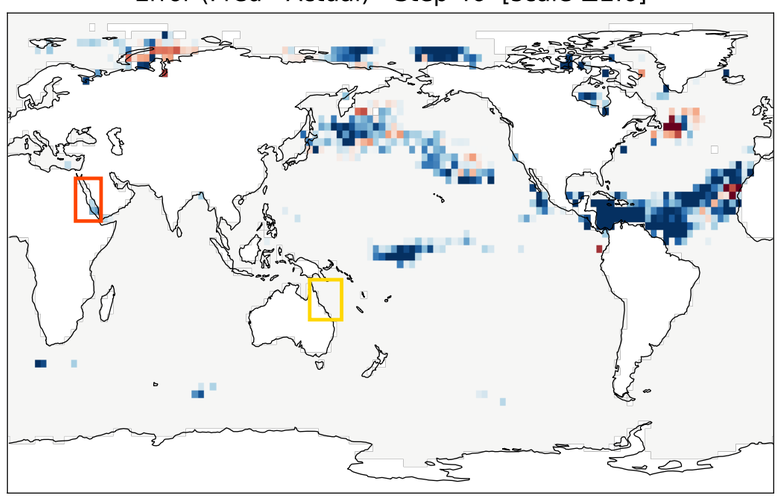
Actual DHW - Step 46



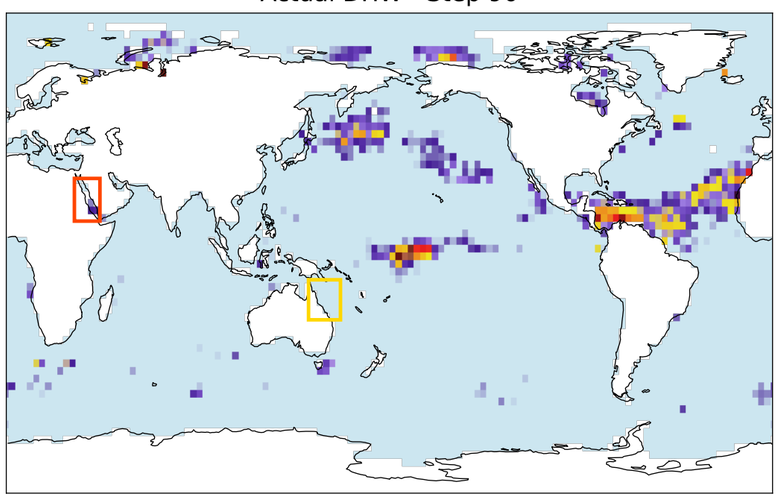
Predicted DHW - Step 46



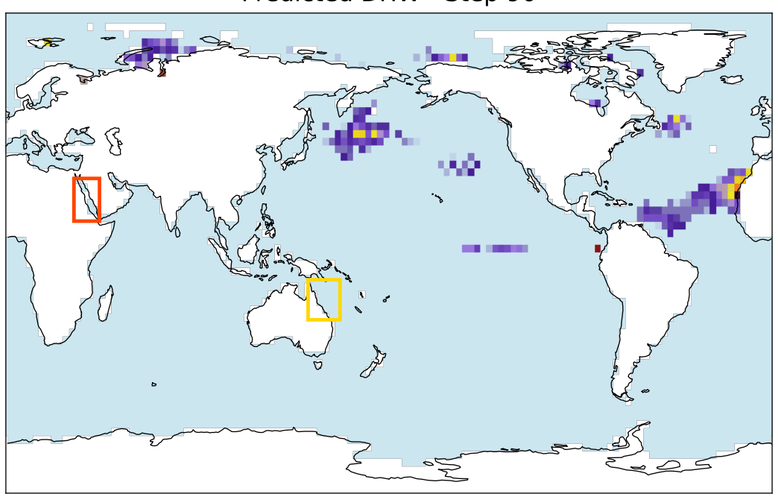
Error (Pred - Actual) - Step 46 [scale ±2.6]



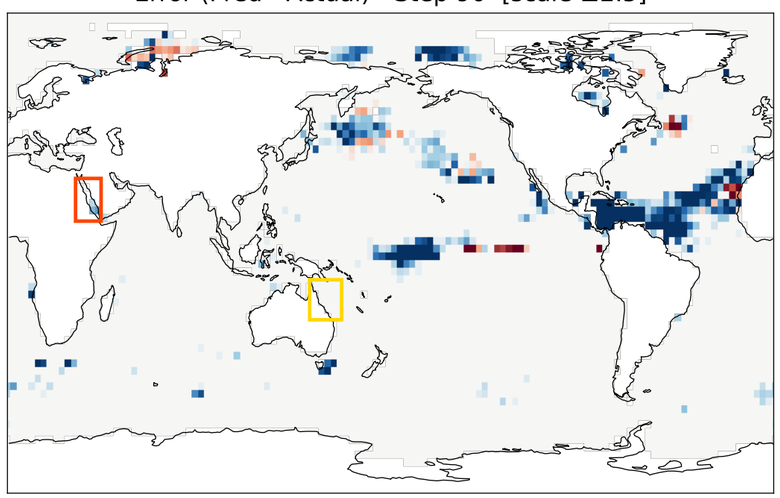
Actual DHW - Step 90

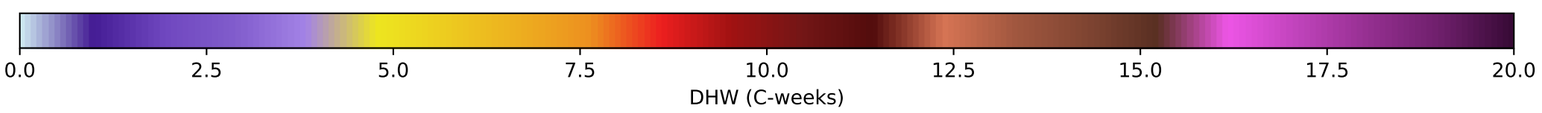


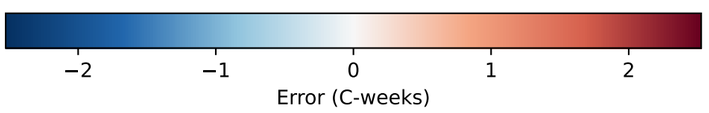
Predicted DHW - Step 90



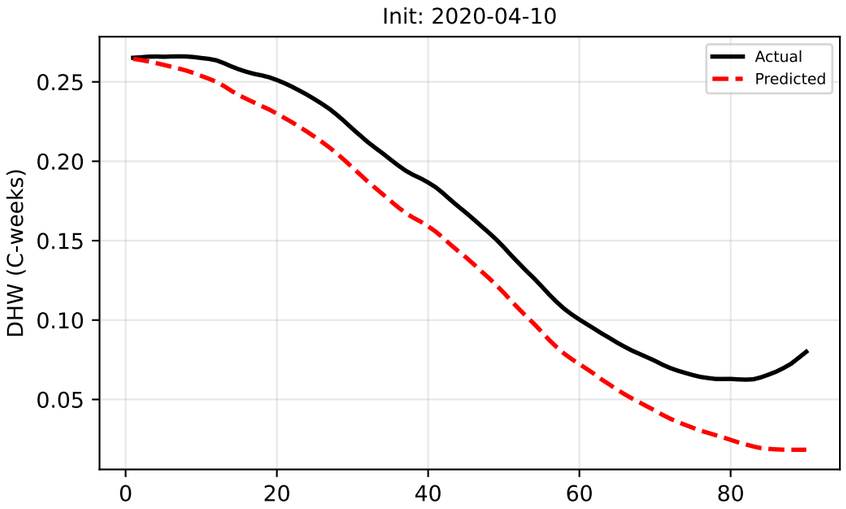
Error (Pred - Actual) - Step 90 [scale ±2.5]

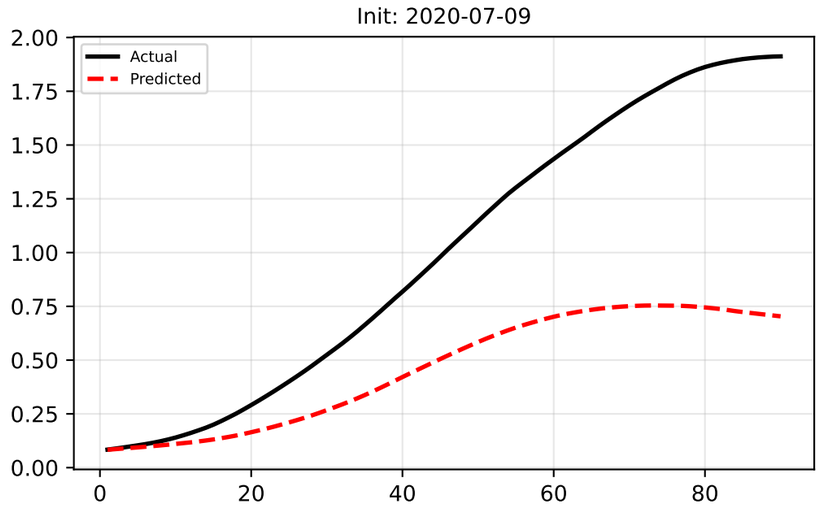


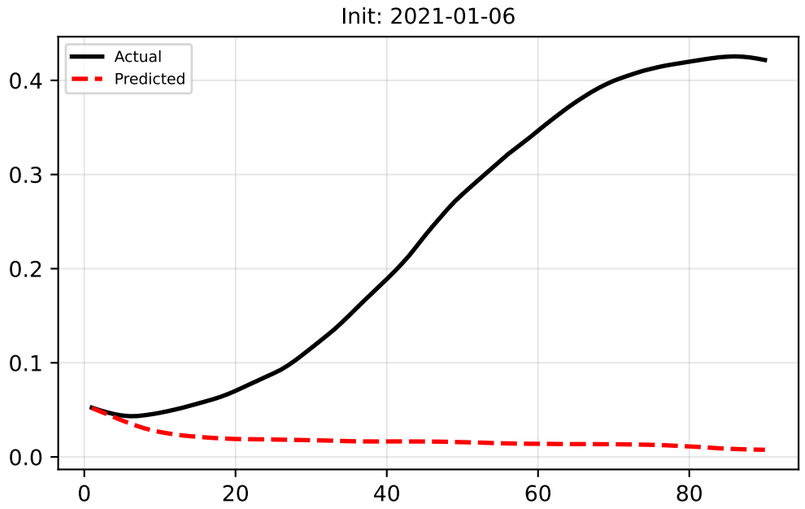


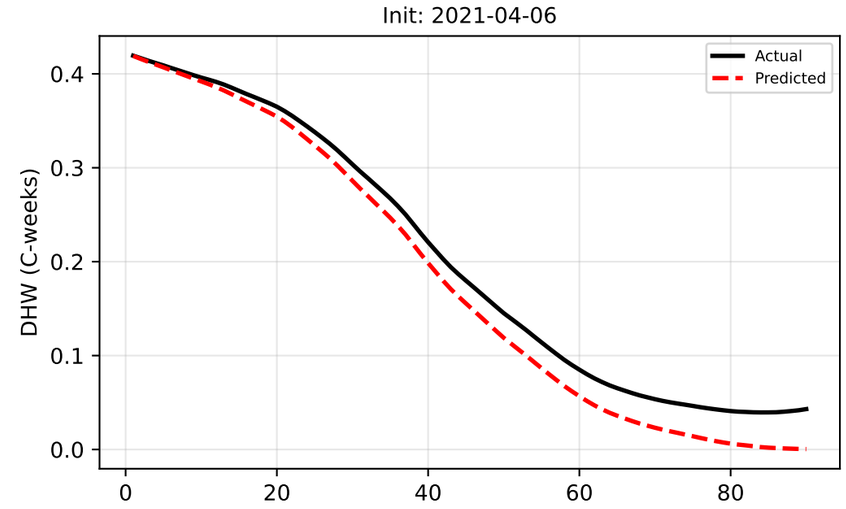


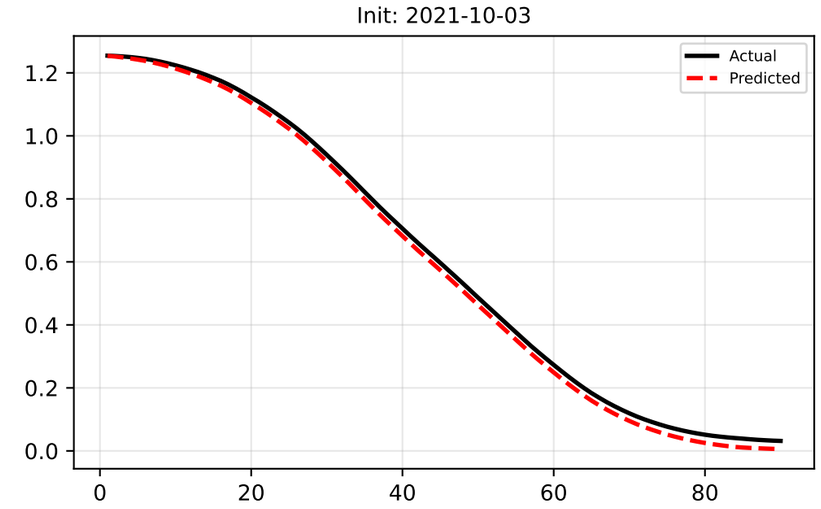
Global-Mean DHW - Test (C-weeks)

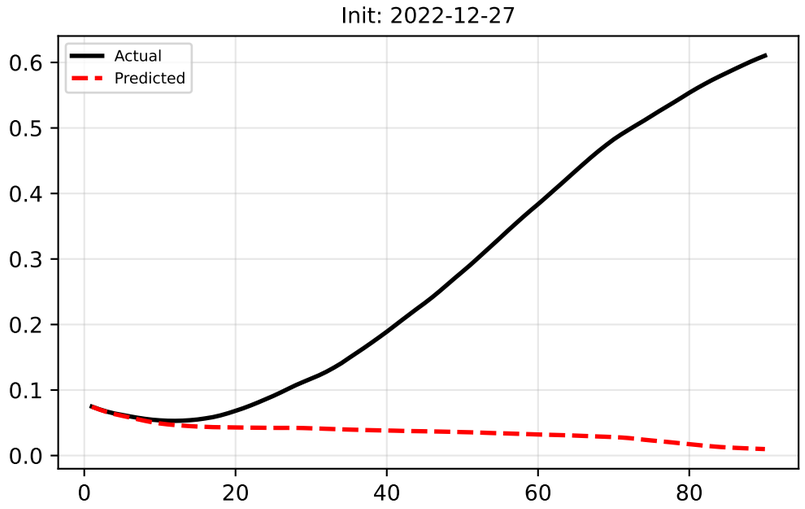


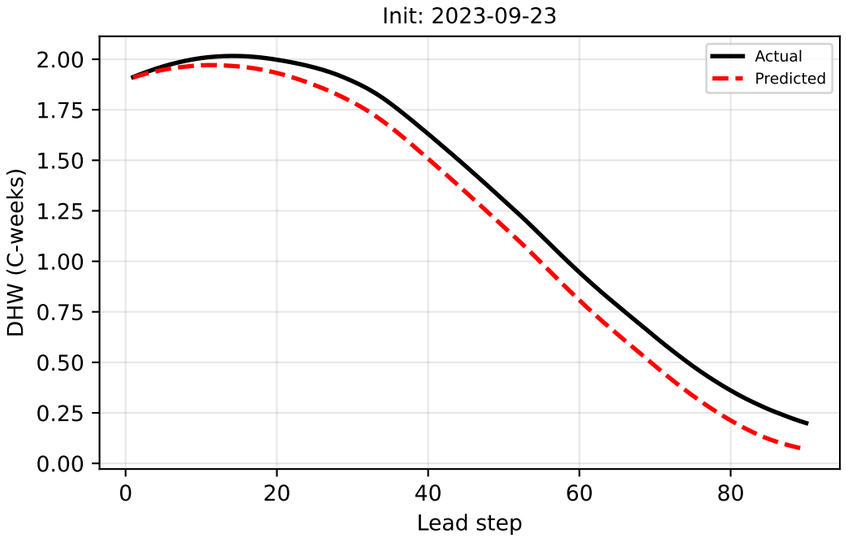




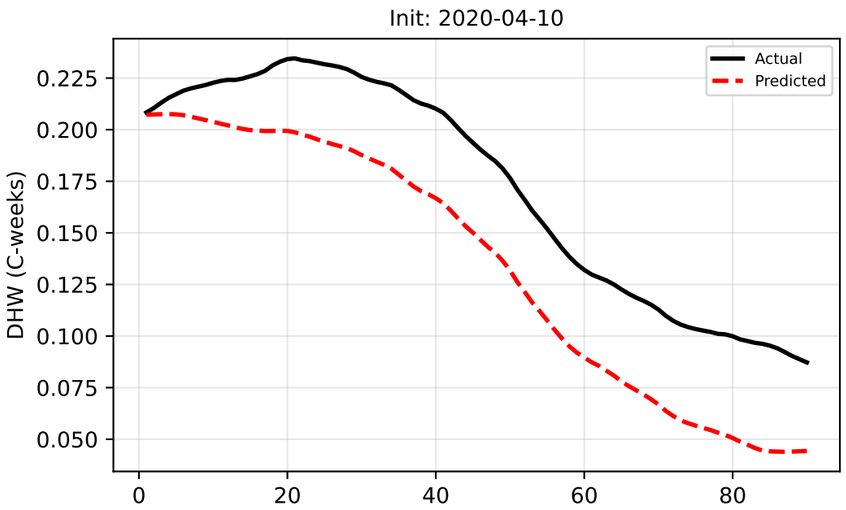


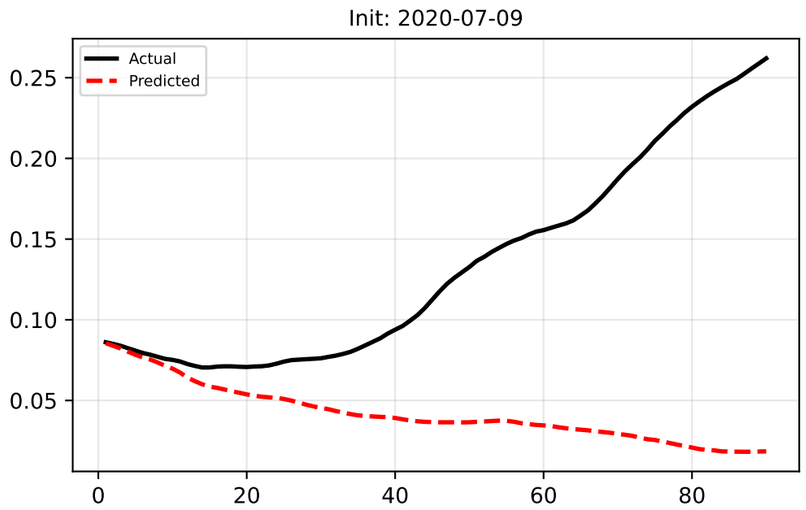


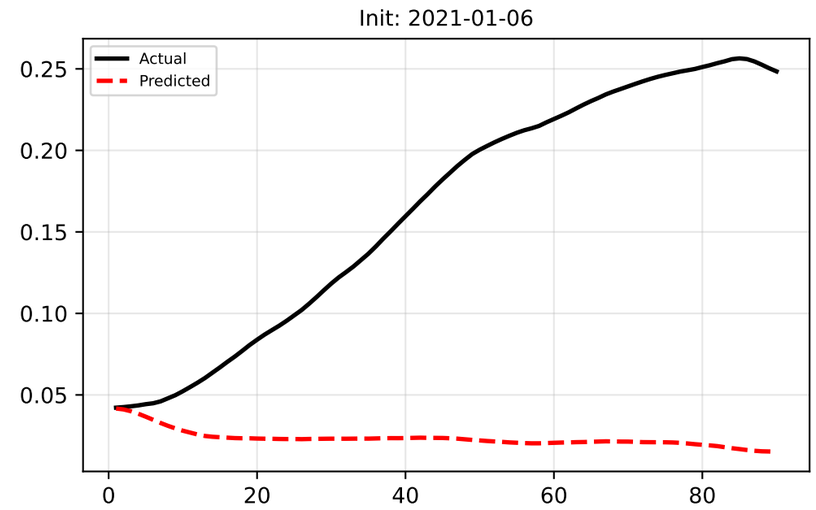


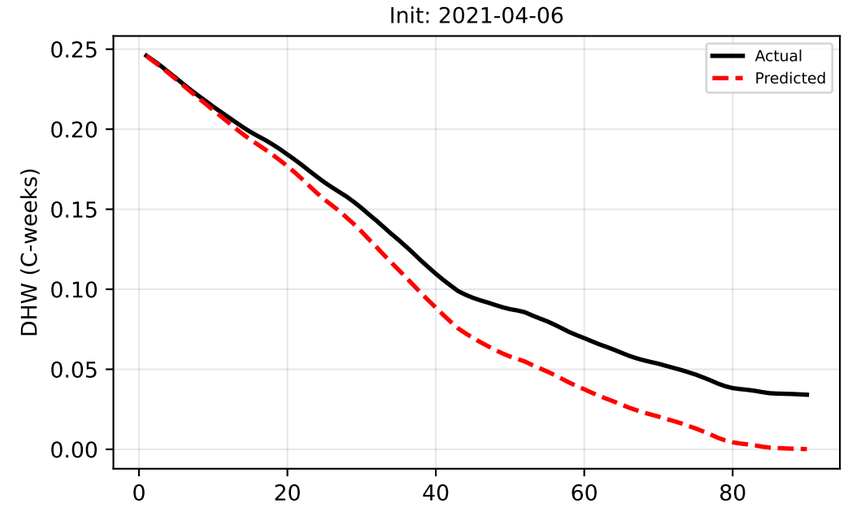


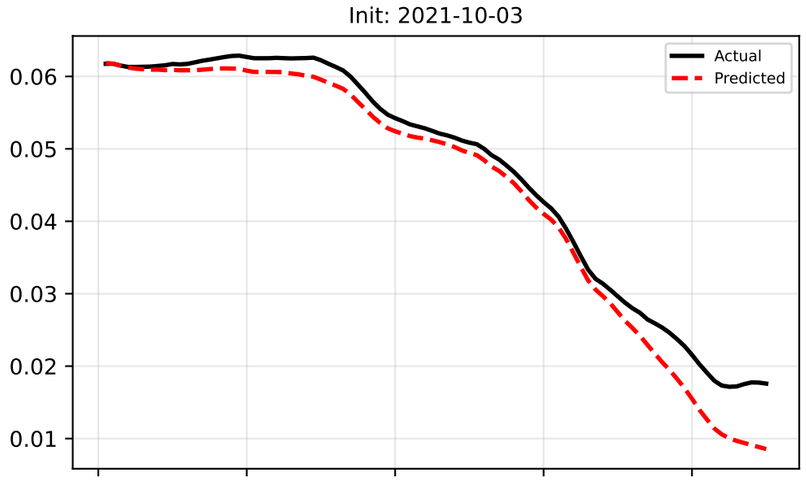
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

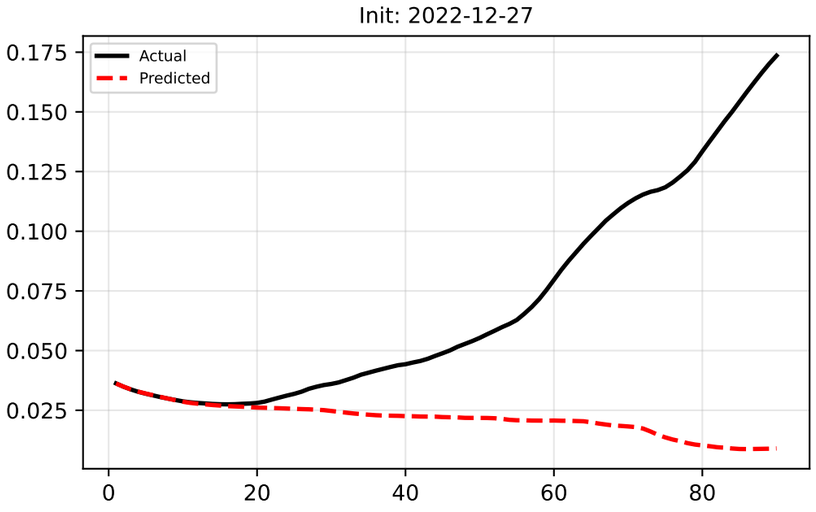


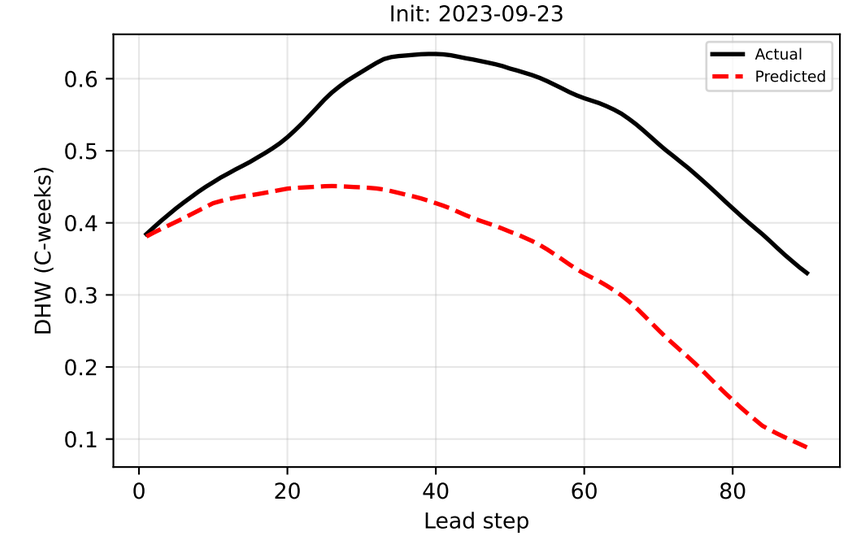




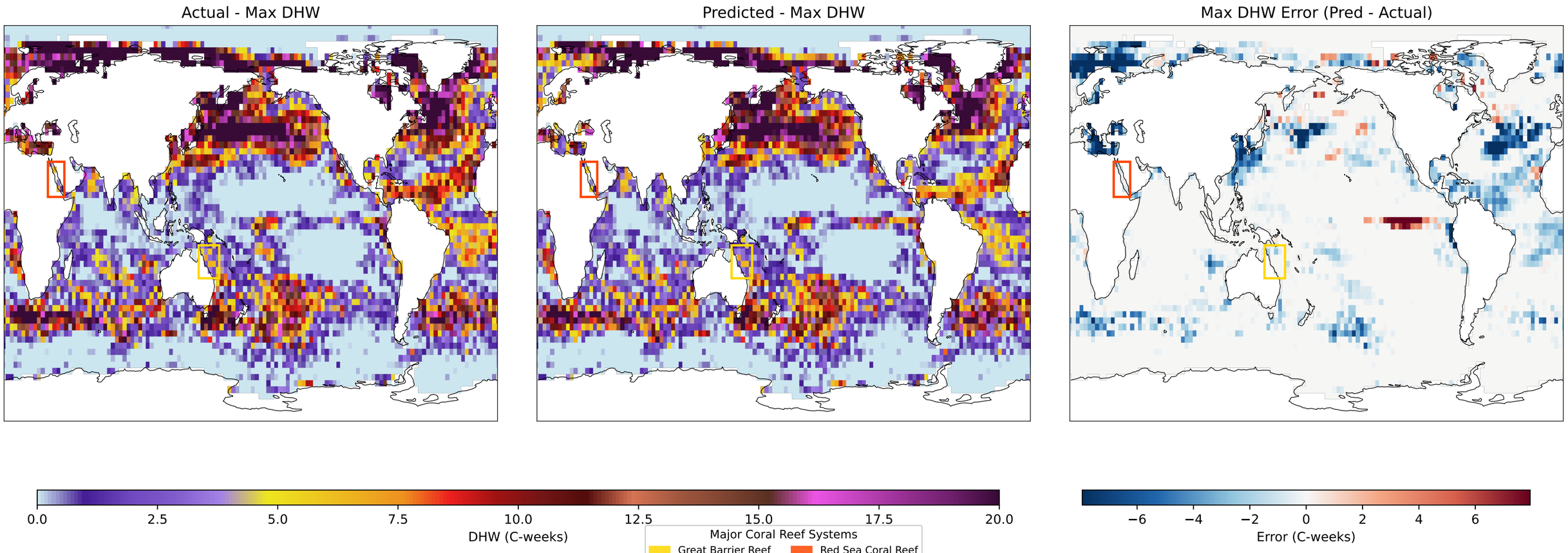


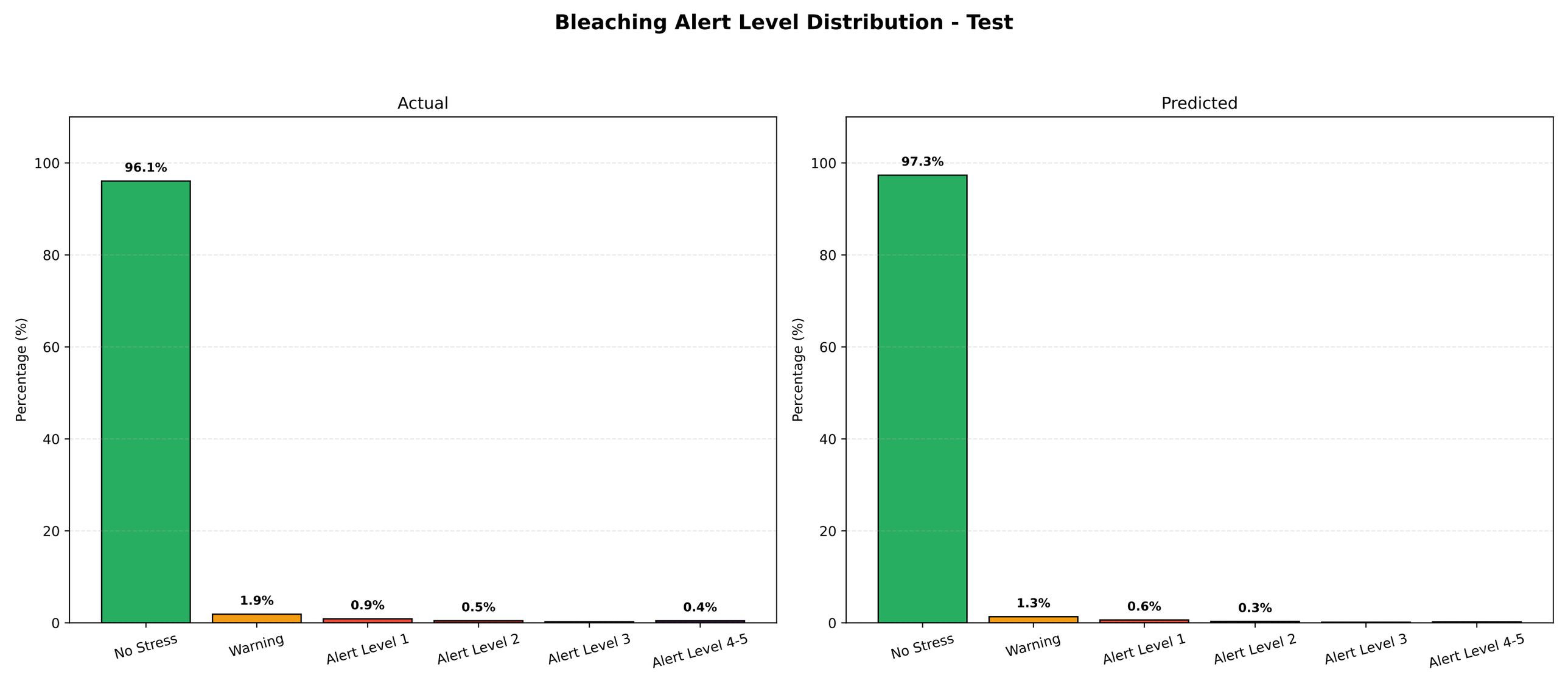






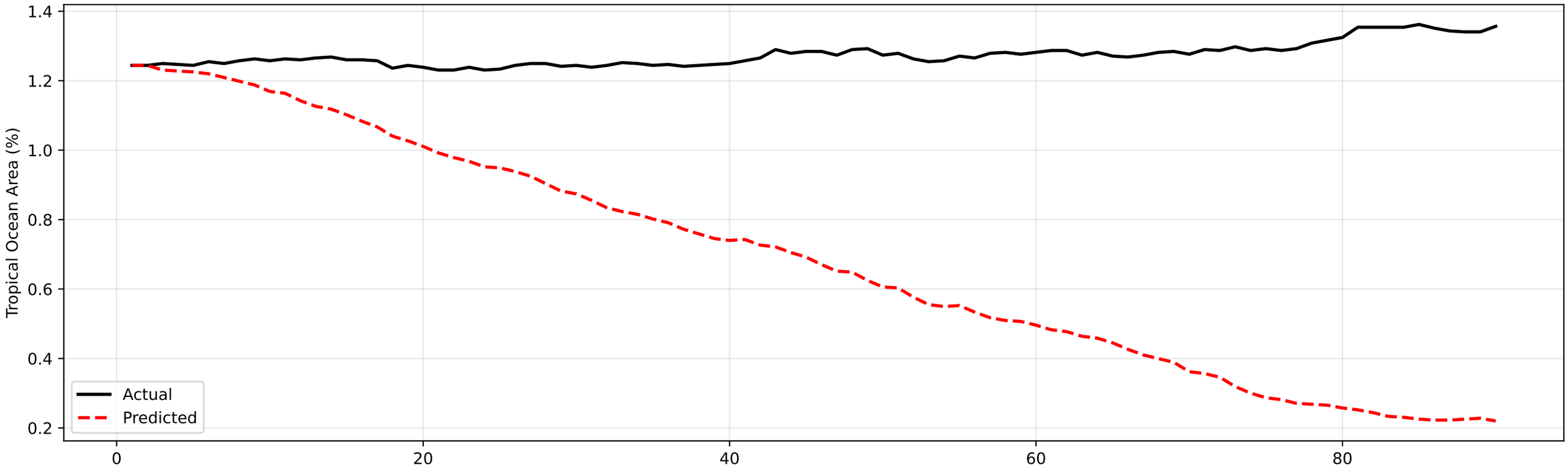
Peak DHW (max over all chains & leads) - Test (C-weeks)



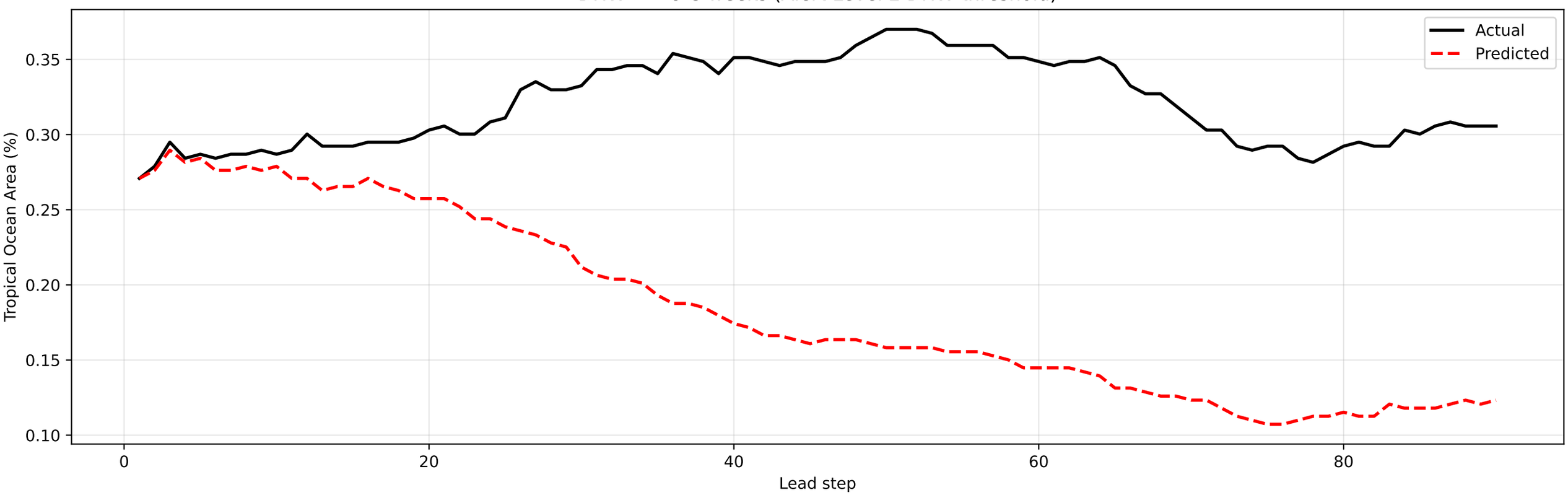

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.1%
1.9%
0.9%
0.5%
0.4%
97.3%
1.3%
0.6%
0.3%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



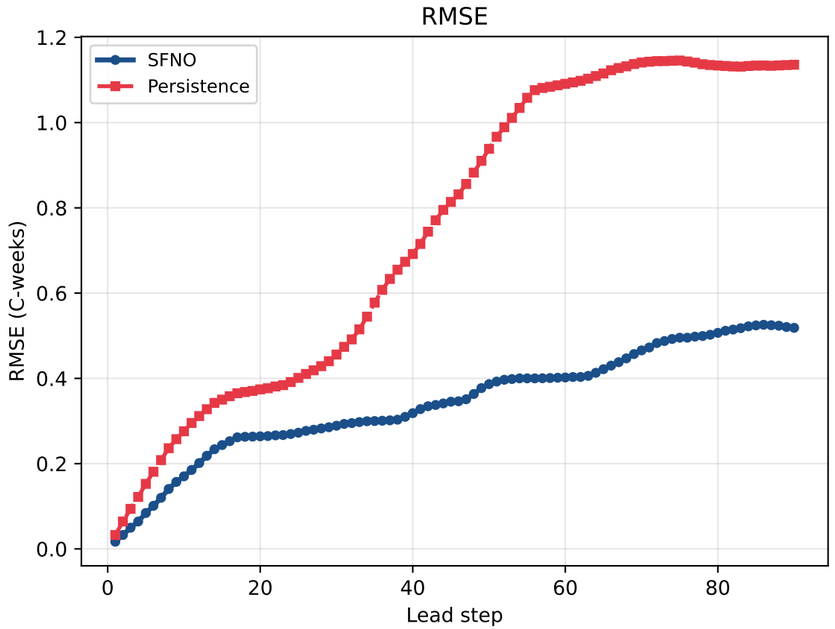
Great Barrier Reef Analysis - Test

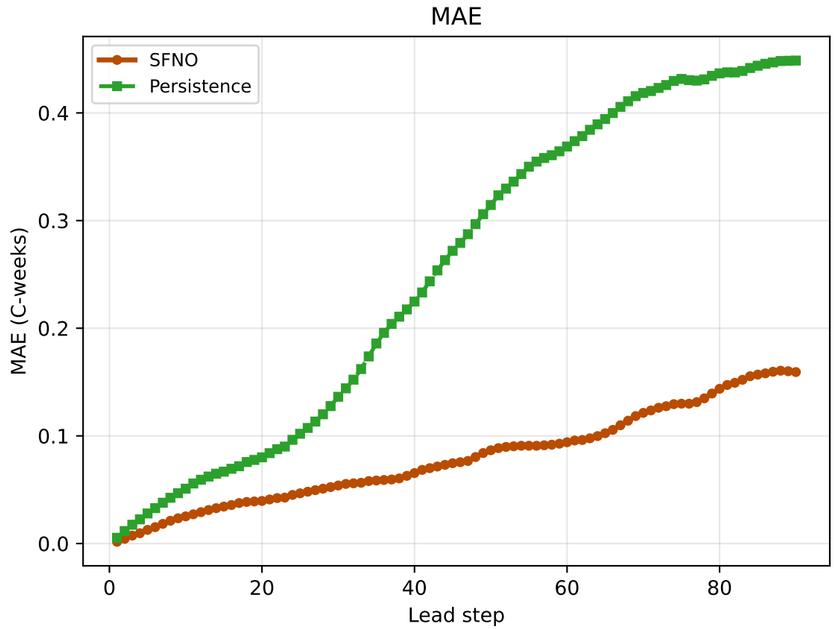
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                      Grid: 5 lat x 5 lon points
```

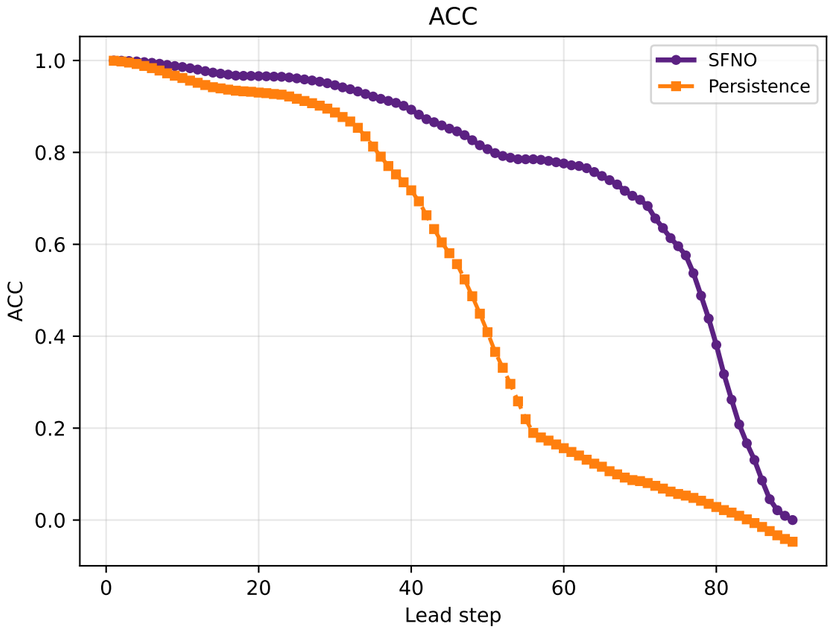
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

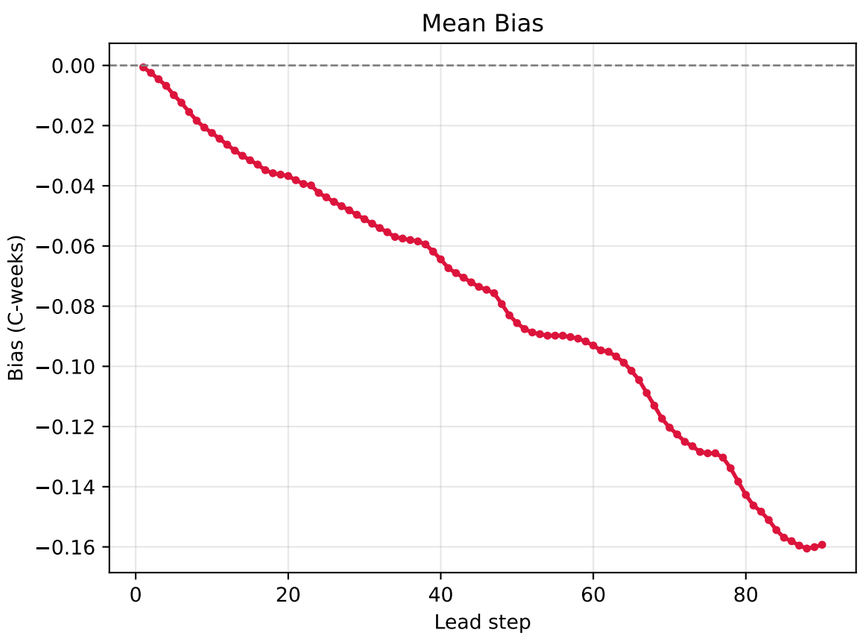
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0168	0.0325	48.3%	0.0017	0.0054	-0.0006	0.9999	0.9995	0.9997
2	0.0327	0.0643	49.2%	0.0044	0.0116	-0.0025	0.9994	0.9979	0.9989
3	0.0497	0.0939	47.1%	0.0074	0.0175	-0.0046	0.9987	0.9955	0.9976
4	0.0643	0.1219	47.3%	0.0096	0.0224	-0.0068	0.9979	0.9924	0.9959
5	0.0842	0.1525	44.8%	0.0127	0.0279	-0.0099	0.9964	0.9881	0.9931
6	0.1011	0.1810	44.1%	0.0152	0.0329	-0.0124	0.9949	0.9834	0.9901
7	0.1199	0.2083	42.4%	0.0183	0.0379	-0.0155	0.9928	0.9781	0.9861
8	0.1405	0.2362	40.5%	0.0212	0.0425	-0.0184	0.9903	0.9721	0.9812
9	0.1572	0.2574	38.9%	0.0235	0.0468	-0.0206	0.9879	0.9669	0.9765
10	0.1702	0.2759	38.3%	0.0253	0.0510	-0.0224	0.9858	0.9619	0.9724
11	0.1852	0.2955	37.3%	0.0272	0.0558	-0.0244	0.9832	0.9562	0.9674
12	0.2014	0.3116	35.4%	0.0292	0.0593	-0.0263	0.9802	0.9515	0.9616
13	0.2185	0.3277	33.3%	0.0311	0.0622	-0.0283	0.9769	0.9466	0.9551
14	0.2338	0.3424	31.7%	0.0328	0.0649	-0.0300	0.9736	0.9419	0.9489
15	0.2436	0.3502	30.4%	0.0343	0.0668	-0.0315	0.9714	0.9393	0.9446
16	0.2526	0.3582	29.5%	0.0358	0.0694	-0.0329	0.9693	0.9366	0.9406
17	0.2617	0.3652	28.3%	0.0376	0.0718	-0.0348	0.9671	0.9343	0.9364
18	0.2630	0.3680	28.5%	0.0386	0.0757	-0.0358	0.9665	0.9329	0.9354
19	0.2634	0.3705	28.9%	0.0391	0.0777	-0.0363	0.9664	0.9319	0.9351
20	0.2641	0.3742	29.4%	0.0396	0.0801	-0.0367	0.9659	0.9302	0.9342
21	0.2647	0.3768	29.8%	0.0410	0.0840	-0.0381	0.9656	0.9289	0.9336
22	0.2664	0.3810	30.1%	0.0422	0.0876	-0.0394	0.9649	0.9271	0.9324
23	0.2670	0.3841	30.5%	0.0427	0.0901	-0.0399	0.9645	0.9256	0.9317
24	0.2695	0.3914	31.2%	0.0452	0.0963	-0.0423	0.9630	0.9217	0.9288
25	0.2726	0.4011	32.0%	0.0466	0.1020	-0.0438	0.9611	0.9167	0.9253
26	0.2769	0.4104	32.5%	0.0481	0.1074	-0.0453	0.9588	0.9118	0.9211
27	0.2795	0.4189	33.3%	0.0496	0.1133	-0.0468	0.9565	0.9070	0.9169
28	0.2826	0.4287	34.1%	0.0510	0.1201	-0.0481	0.9539	0.9017	0.9121
29	0.2854	0.4402	35.2%	0.0524	0.1277	-0.0496	0.9505	0.8952	0.9061
30	0.2889	0.4558	36.6%	0.0539	0.1363	-0.0511	0.9463	0.8868	0.8984
31	0.2932	0.4739	38.1%	0.0554	0.1442	-0.0526	0.9415	0.8770	0.8900
32	0.2951	0.4915	40.0%	0.0560	0.1522	-0.0540	0.9375	0.8672	0.8827
33	0.2975	0.5150	42.2%	0.0565	0.1621	-0.0554	0.9326	0.8535	0.8740
34	0.2995	0.5448	45.0%	0.0580	0.1738	-0.0570	0.9270	0.8349	0.8641
35	0.3000	0.5777	48.1%	0.0586	0.1858	-0.0575	0.9216	0.8127	0.8543
36	0.3007	0.6076	50.5%	0.0590	0.1958	-0.0580	0.9166	0.7908	0.8455
37	0.3016	0.6333	52.4%	0.0595	0.2040	-0.0585	0.9120	0.7703	0.8376
38	0.3031	0.6548	53.7%	0.0605	0.2107	-0.0595	0.9077	0.7521	0.8302
39	0.3098	0.6735	54.0%	0.0629	0.2175	-0.0619	0.9014	0.7349	0.8194
40	0.3184	0.6918	54.0%	0.0655	0.2248	-0.0644	0.8929	0.7175	0.8049
41	0.3279	0.7155	54.2%	0.0684	0.2332	-0.0674	0.8821	0.6935	0.7864
42	0.3347	0.7441	55.0%	0.0700	0.2436	-0.0690	0.8722	0.6632	0.7692
43	0.3376	0.7709	56.2%	0.0716	0.2537	-0.0705	0.8656	0.6331	0.7571
44	0.3412	0.7952	57.1%	0.0731	0.2632	-0.0721	0.8587	0.6042	0.7443
45	0.3453	0.8139	57.6%	0.0746	0.2720	-0.0736	0.8517	0.5806	0.7307
46	0.3463	0.8315	58.4%	0.0756	0.2793	-0.0745	0.8457	0.5570	0.7192
47	0.3510	0.8560	59.0%	0.0767	0.2874	-0.0757	0.8375	0.5236	0.7033
48	0.3634	0.8827	58.8%	0.0803	0.2967	-0.0793	0.8265	0.4870	0.6821
49	0.3772	0.9103	58.6%	0.0841	0.3059	-0.0831	0.8155	0.4490	0.6611
50	0.3864	0.9384	58.8%	0.0866	0.3145	-0.0856	0.8071	0.4087	0.6443
51	0.3925	0.9668	59.4%	0.0886	0.3234	-0.0876	0.7984	0.3658	0.6288
52	0.3968	0.9892	59.9%	0.0898	0.3297	-0.0887	0.7922	0.3313	0.6182
53	0.3984	1.0111	60.6%	0.0903	0.3362	-0.0893	0.7884	0.2963	0.6105
54	0.4000	1.0344	61.3%	0.0908	0.3432	-0.0898	0.7851	0.2584	0.6034
55	0.4000	1.0582	62.2%	0.0908	0.3501	-0.0898	0.7850	0.2195	0.6011
56	0.4000	1.0761	62.8%	0.0908	0.3548	-0.0898	0.7851	0.1896	0.5994
57	0.4003	1.0812	63.0%	0.0913	0.3581	-0.0902	0.7839	0.1797	0.5952
58	0.4008	1.0841	63.0%	0.0918	0.3607	-0.0908	0.7815	0.1728	0.5897
59	0.4015	1.0875	63.1%	0.0928	0.3643	-0.0917	0.7786	0.1645	0.5832
60	0.4021	1.0909	63.1%	0.0941	0.3688	-0.0931	0.7759	0.1564	0.5770
61	0.4032	1.0943	63.2%	0.0957	0.3738	-0.0947	0.7720	0.1481	0.5679
62	0.4033	1.0980	63.3%	0.0962	0.3785	-0.0951	0.7704	0.1404	0.5621
63	0.4059	1.1030	63.2%	0.0978	0.3843	-0.0967	0.7657	0.1313	0.5544
64	0.4131	1.1092	62.8%	0.0998	0.3894	-0.0988	0.7572	0.1227	0.5430
65	0.4216	1.1152	62.2%	0.1025	0.3943	-0.1015	0.7486	0.1160	0.5326
66	0.4300	1.1228	61.7%	0.1056	0.3999	-0.1046	0.7396	0.1063	0.5216
67	0.4380	1.1280	61.2%	0.1099	0.4056	-0.1088	0.7302	0.0994	0.5107
68	0.4469	1.1321	60.5%	0.1141	0.4108	-0.1131	0.7163	0.0924	0.4935
69	0.4570	1.1374	59.8%	0.1184	0.4156	-0.1174	0.7056	0.0870	0.4818
70	0.4657	1.1409	59.2%	0.1214	0.4186	-0.1204	0.6968	0.0847	0.4722
71	0.4724	1.1428	58.7%	0.1236	0.4203	-0.1226	0.6833	0.0805	0.4564
72	0.4828	1.1442	57.8%	0.1261	0.4232	-0.1251	0.6562	0.0744	0.4253
73	0.4876	1.1443	57.4%	0.1276	0.4259	-0.1265	0.6353	0.0684	0.3997
74	0.4924	1.1451	57.0%	0.1295	0.4297	-0.1284	0.6135	0.0619	0.3756
75	0.4954	1.1456	56.8%	0.1299	0.4316	-0.1289	0.5959	0.0567	0.3561
76	0.4954	1.1433	56.7%	0.1299	0.4305	-0.1289	0.5758	0.0532	0.3364
77	0.4980	1.1405	56.3%	0.1313	0.4300	-0.1303	0.5371	0.0482	0.3018
78	0.4993	1.1370	56.1%	0.1349	0.4312	-0.1338	0.4881	0.0419	0.2619
79	0.5023	1.1351	55.7%	0.1393	0.4344	-0.1383	0.4383	0.0353	0.2244
80	0.5068	1.1339	55.3%	0.1437	0.4368	-0.1427	0.3811	0.0284	0.1824
81	0.5117	1.1328	54.8%	0.1473	0.4377	-0.1463	0.3174	0.0216	0.1366
82	0.5145	1.1317	54.5%	0.1493	0.4376	-0.1483	0.2620	0.0162	0.0969
83	0.5178	1.1313	54.2%	0.1521	0.4390	-0.1511	0.2080	0.0092	0.0585
84	0.5221	1.1328	53.9%	0.1554	0.4417	-0.1544	0.1668	0.0015	0.0293
85	0.5243	1.1337	53.8%	0.1569	0.4438	-0.1569	0.1307	-0.0065	0.0030
86	0.5257	1.1340	53.6%	0.1581	0.4457	-0.1581	0.0862	-0.0150	-0.0299
87	0.5247	1.1333	53.7%	0.1596	0.4469	-0.1596	0.0453	-0.0240	-0.0634
88	0.5235	1.1341	53.8%	0.1605	0.4482	-0.1605	0.0213	-0.0332	-0.0850
89	0.5205	1.1349	54.1%	0.1601	0.4484	-0.1601	0.0093	-0.0412	-0.0958
90	0.5186	1.1357	54.3%	0.1593	0.4487	-0.1593	0.0000	-0.0472	-0.1042

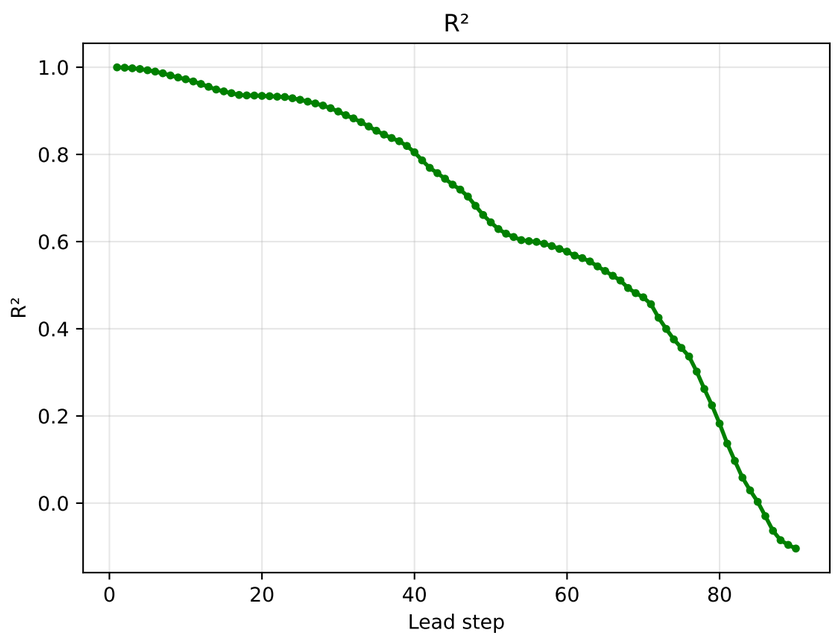
DHW Forecast Skill vs Lead Step - Test - GBR

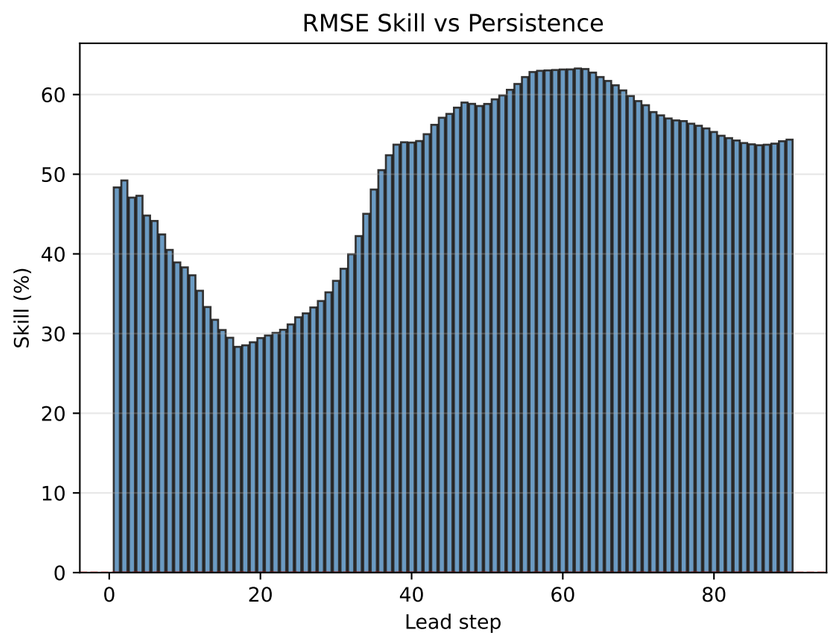


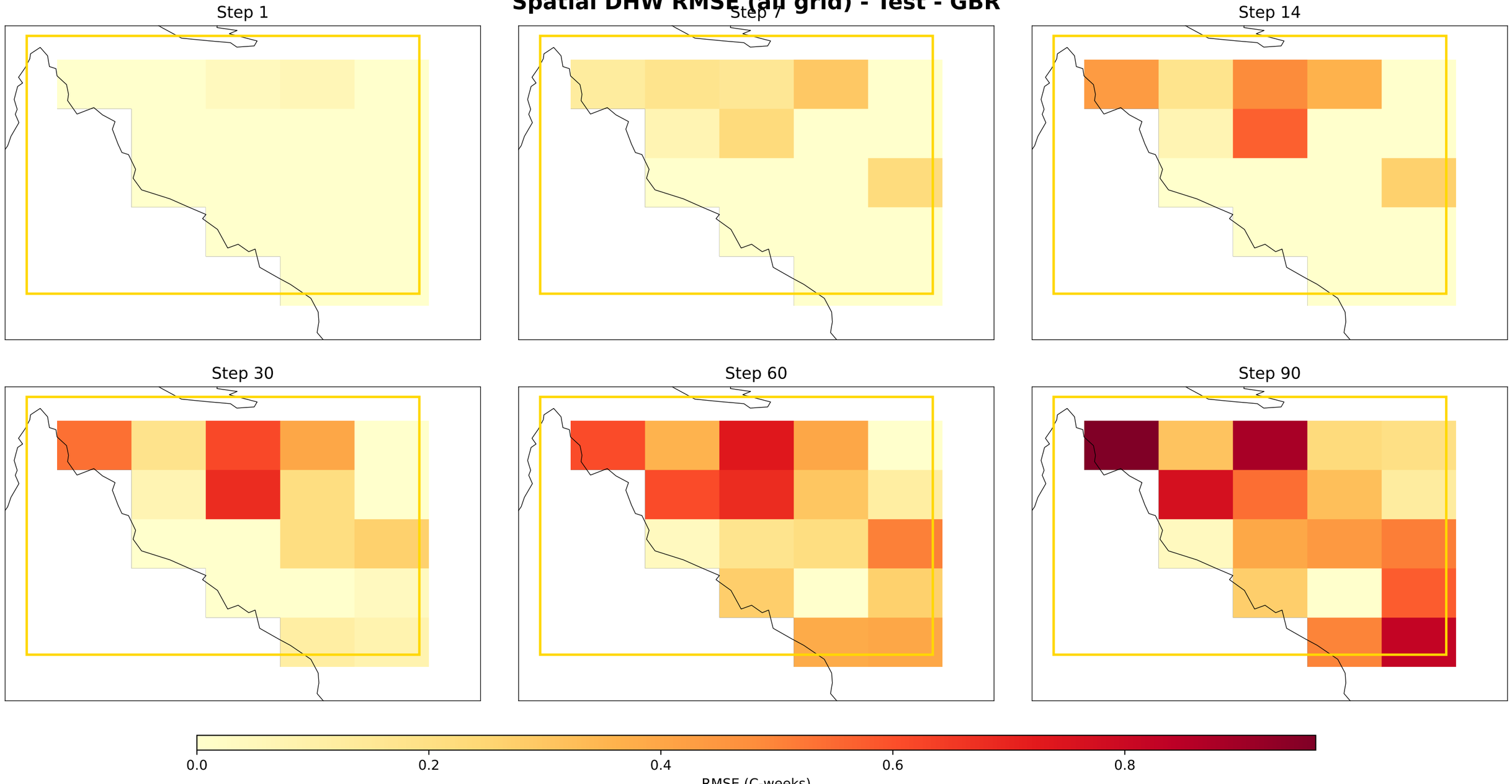


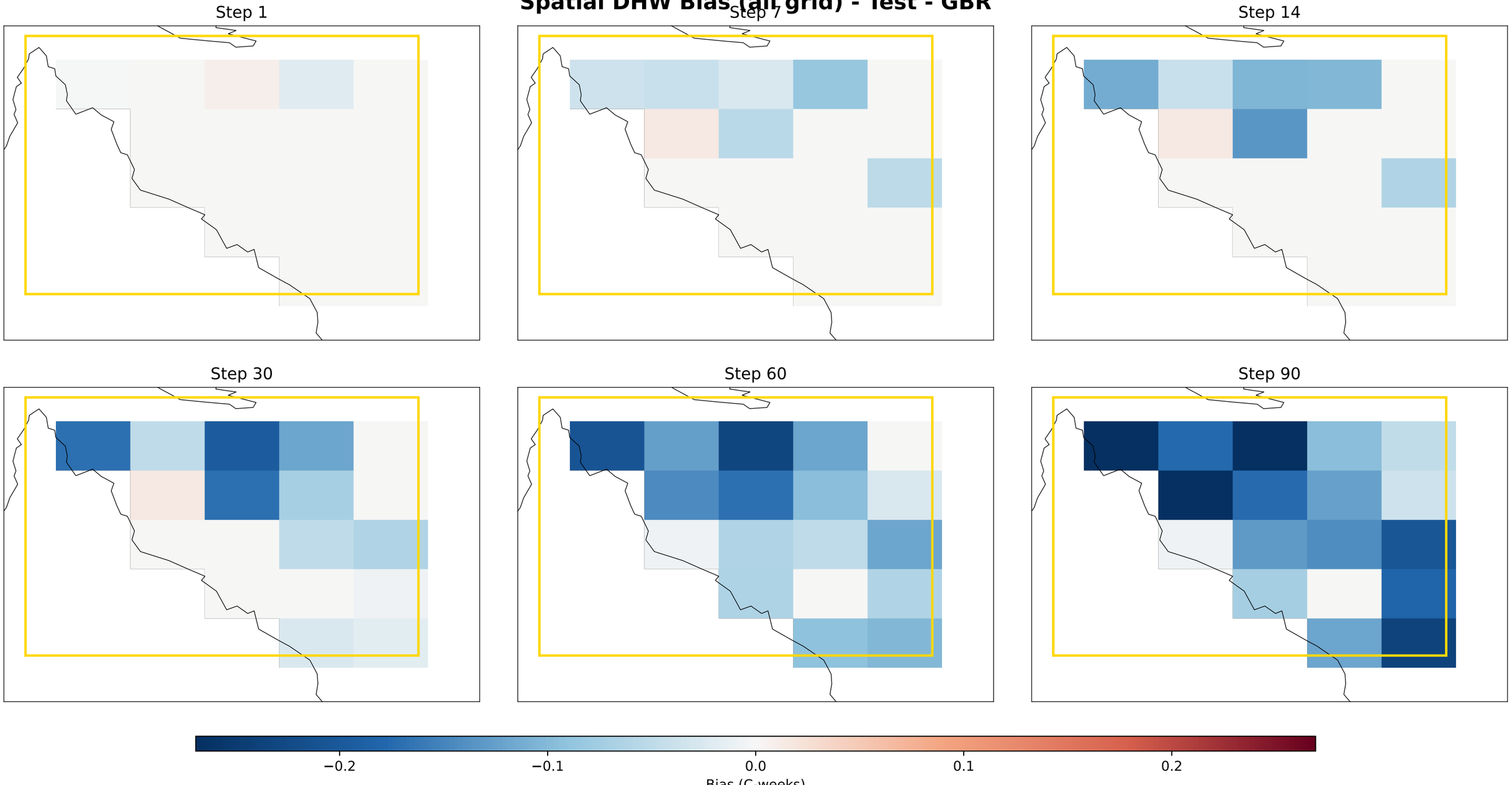


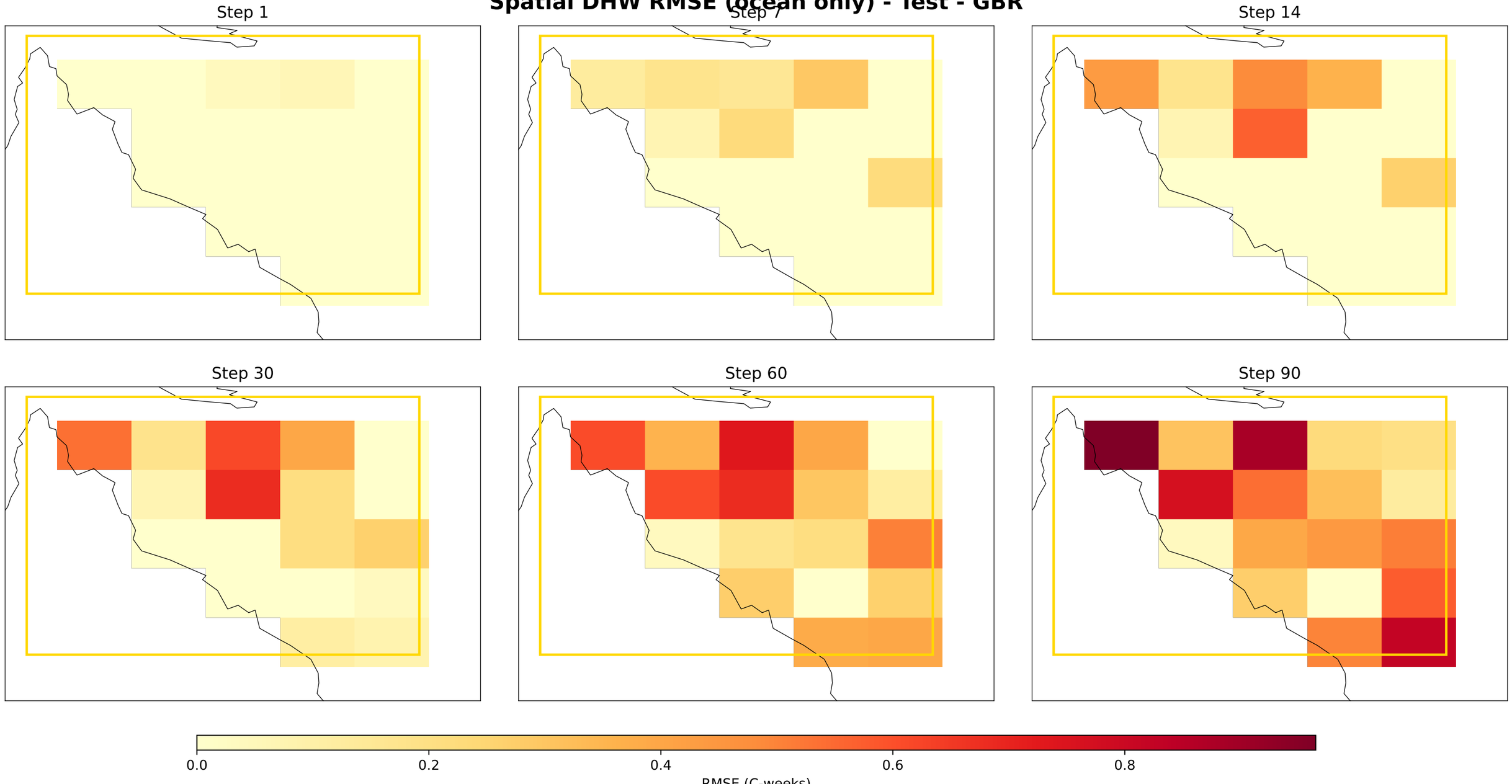


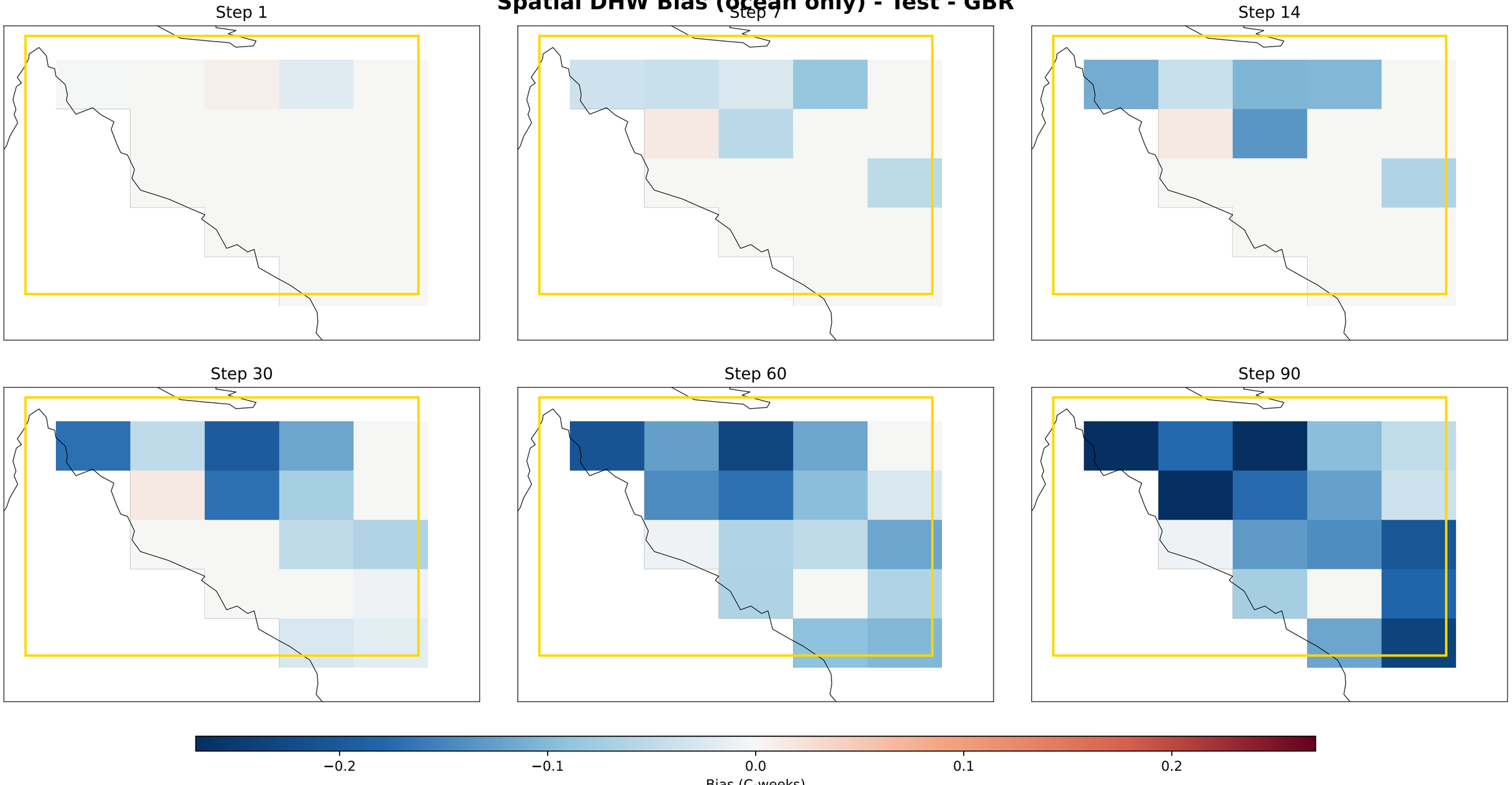





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

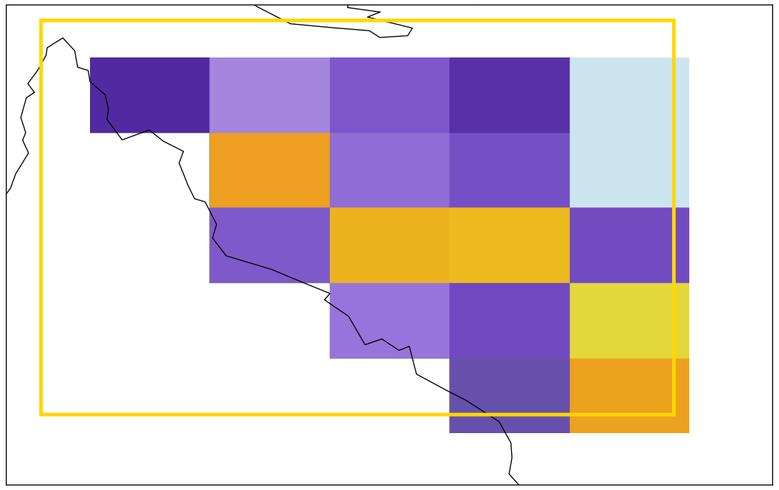

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

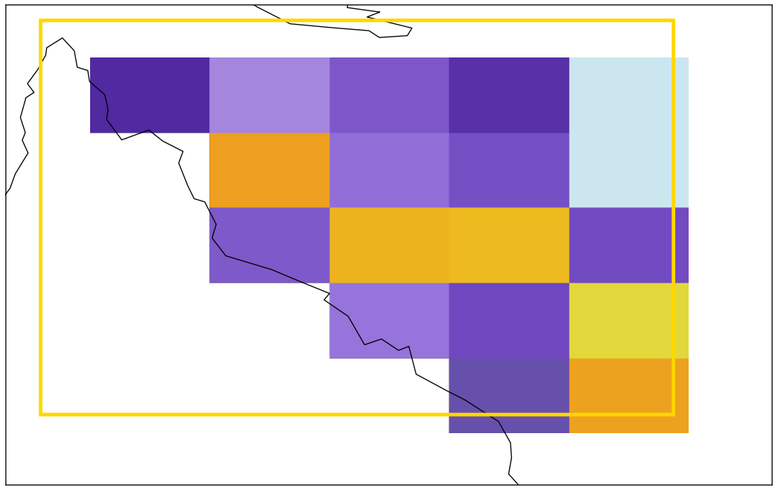

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

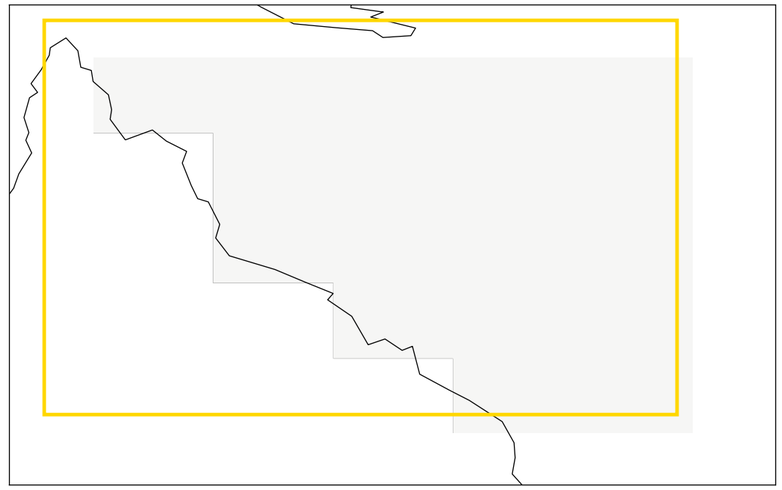
Actual DHW - Step 1



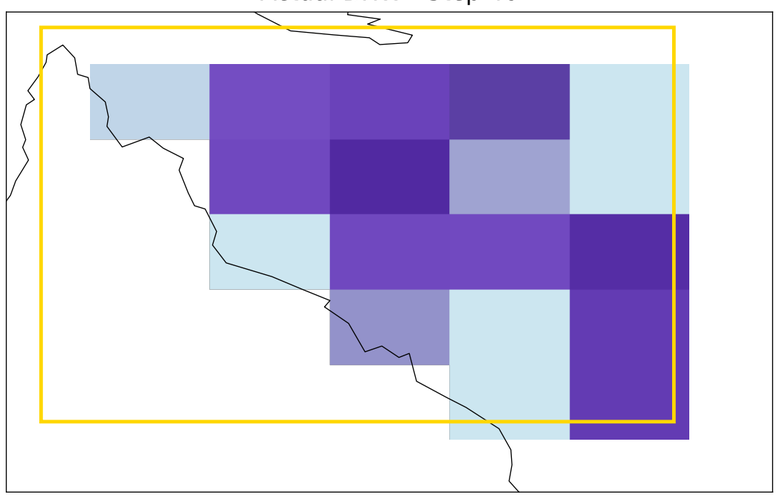
Predicted DHW - Step 1



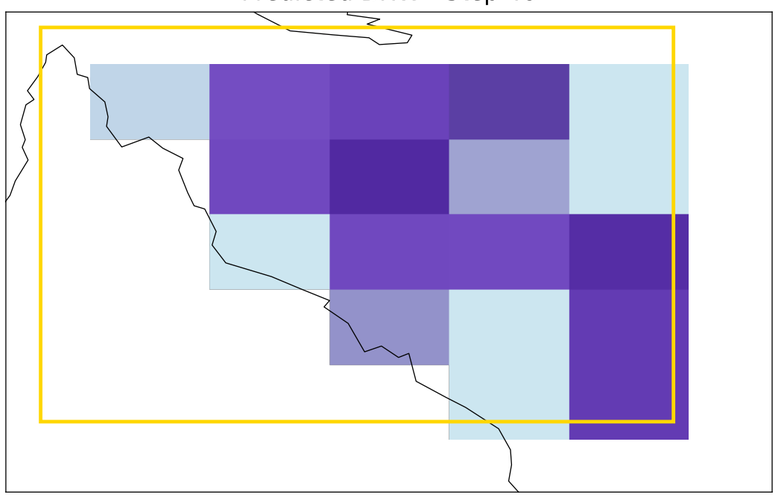
Error (Pred - Actual) - Step 1 [scale ±8]



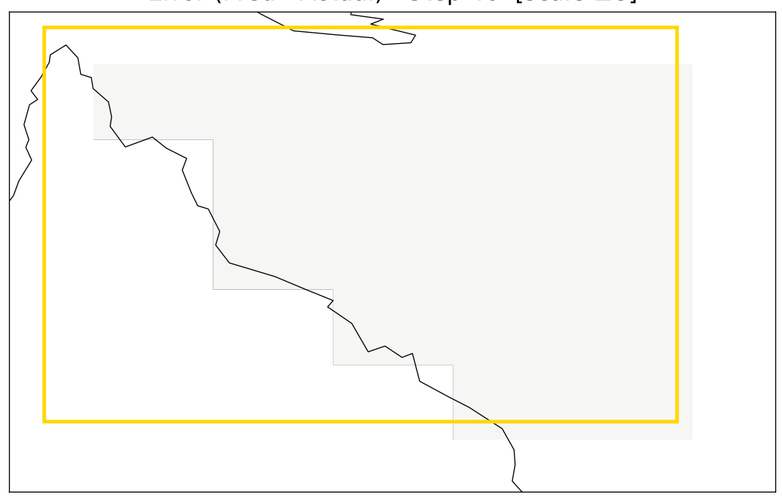
Actual DHW - Step 46



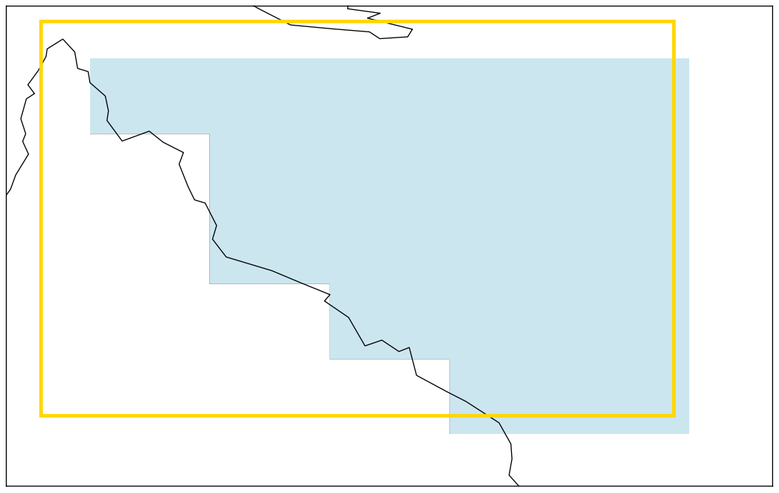
Predicted DHW - Step 46



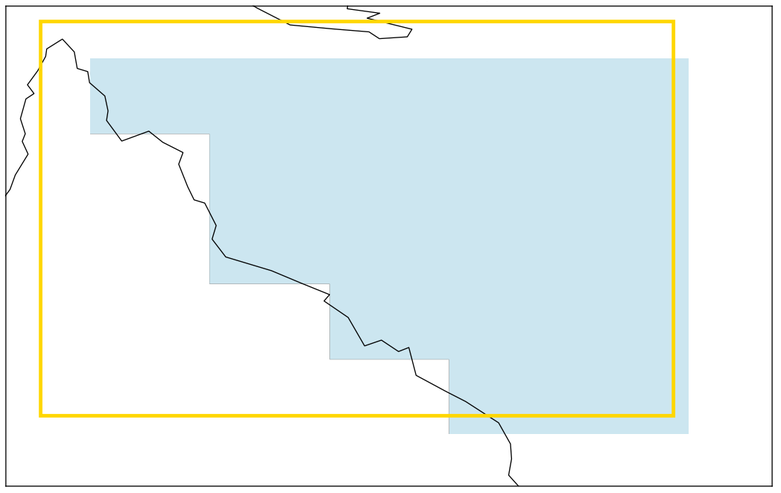
Error (Pred - Actual) - Step 46 [scale ±8]



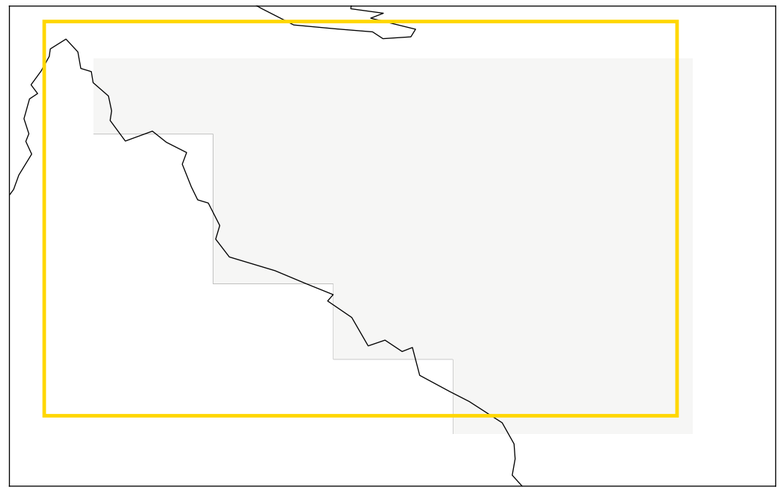
Actual DHW - Step 90

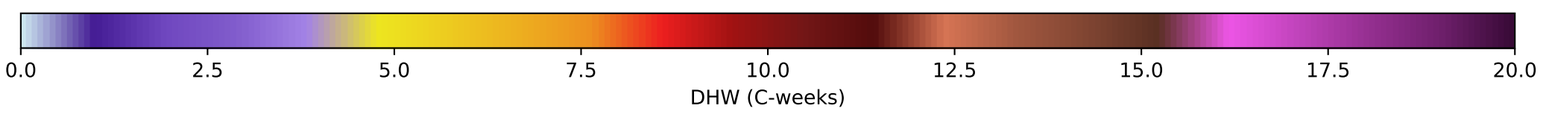


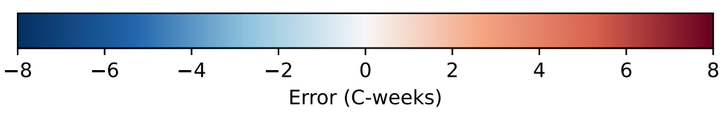
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

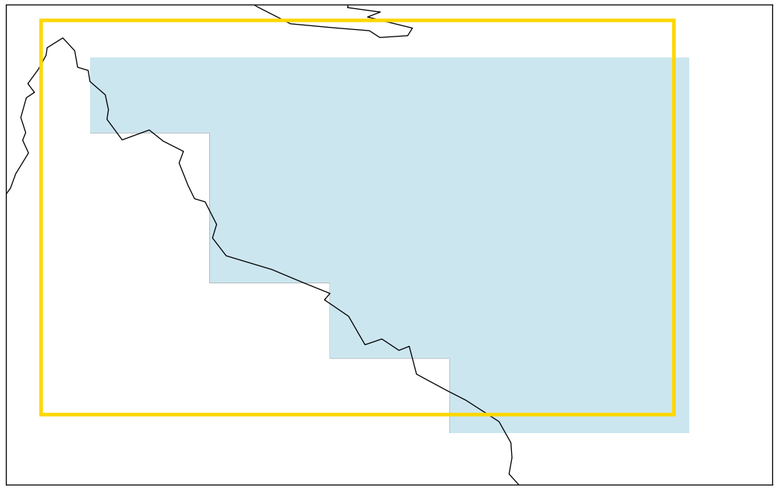




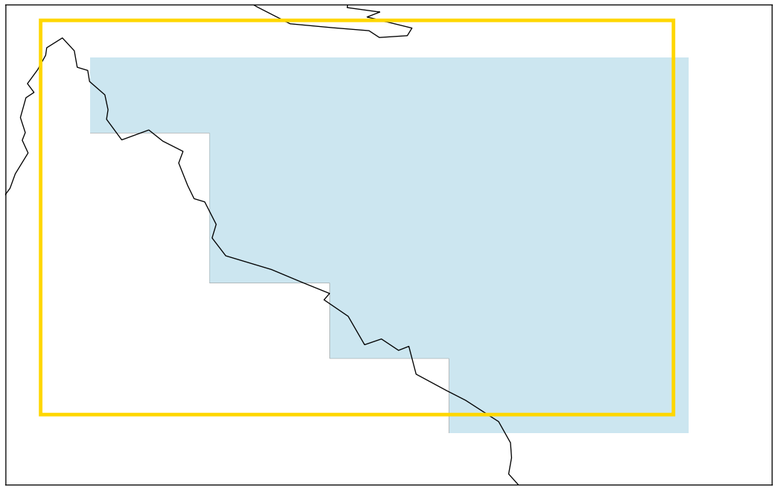


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

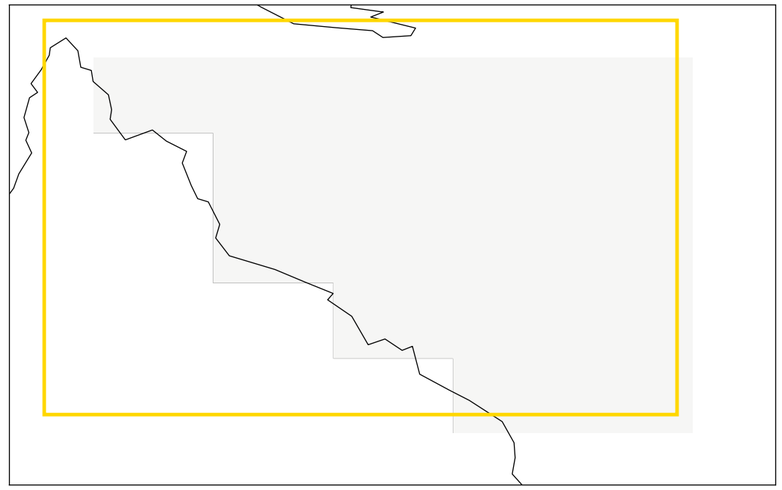
Actual DHW - Step 1



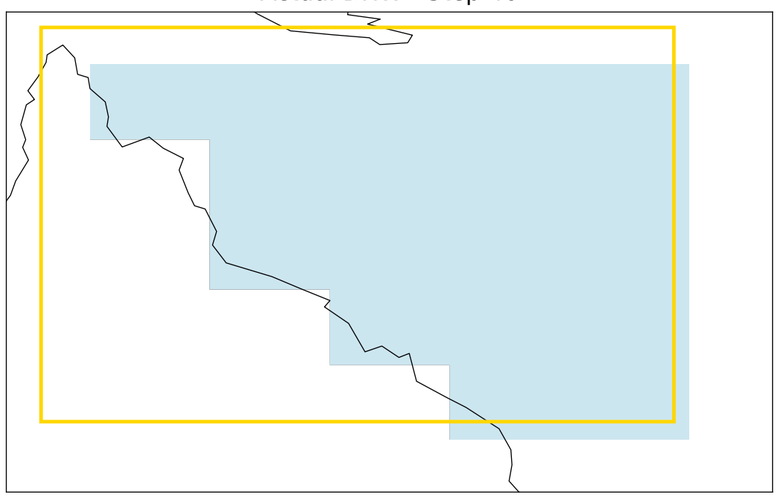
Predicted DHW - Step 1



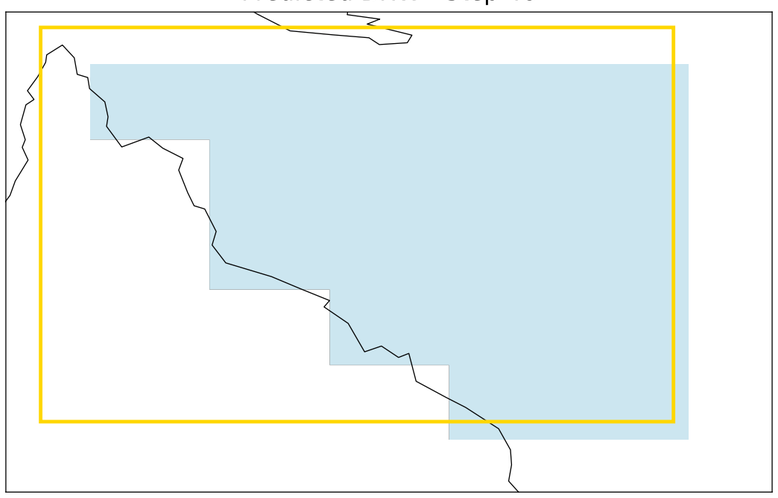
Error (Pred - Actual) - Step 1 [scale ±8]



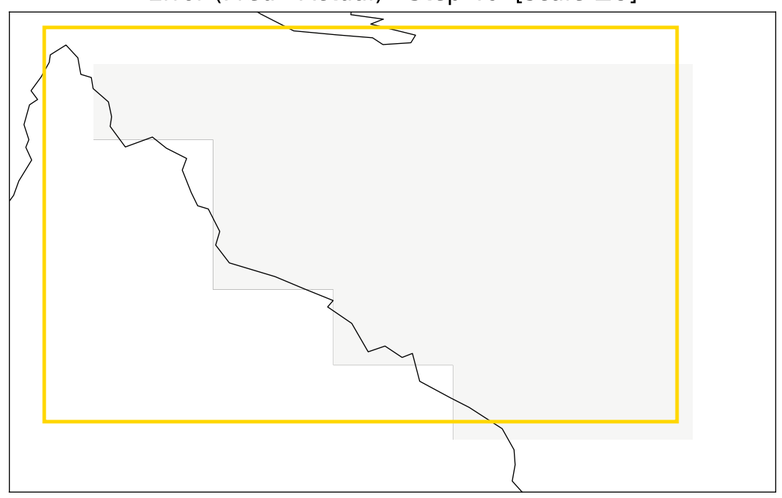
Actual DHW - Step 46



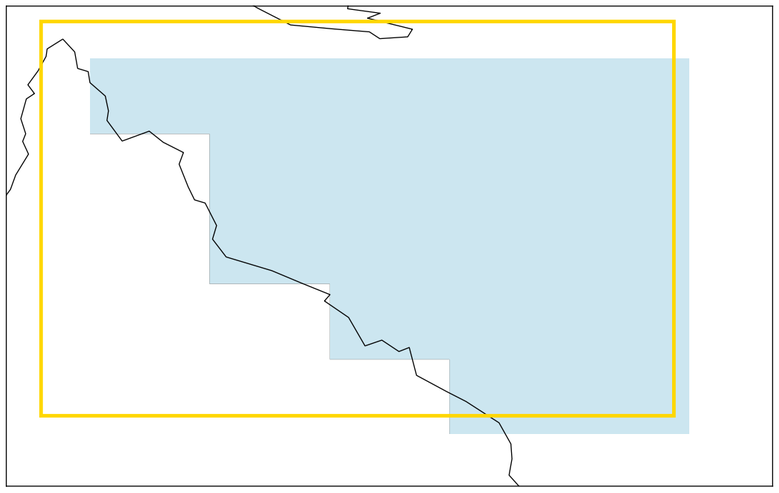
Predicted DHW - Step 46



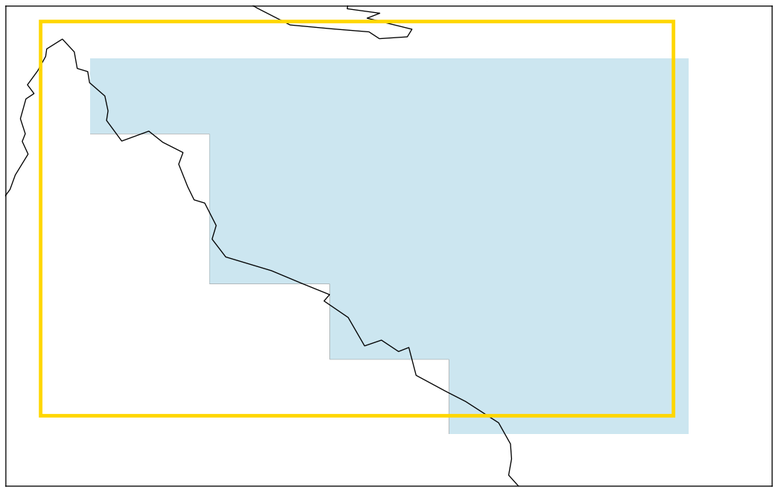
Error (Pred - Actual) - Step 46 [scale ±8]



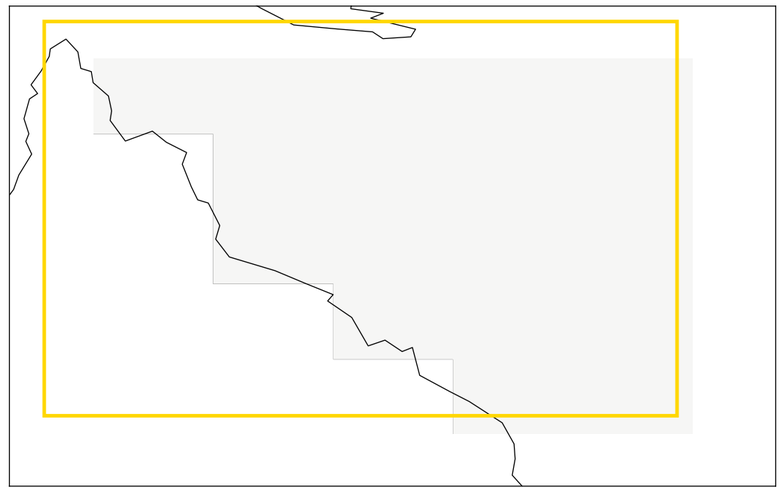
Actual DHW - Step 90

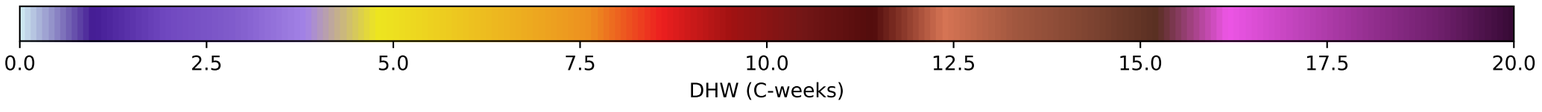


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

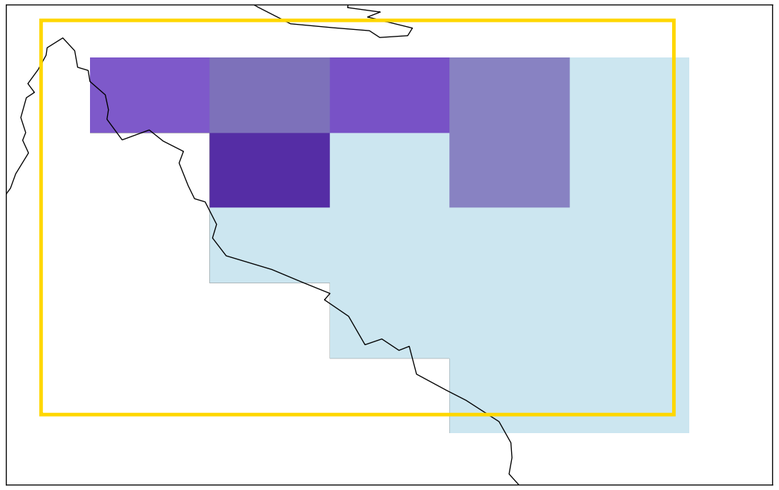




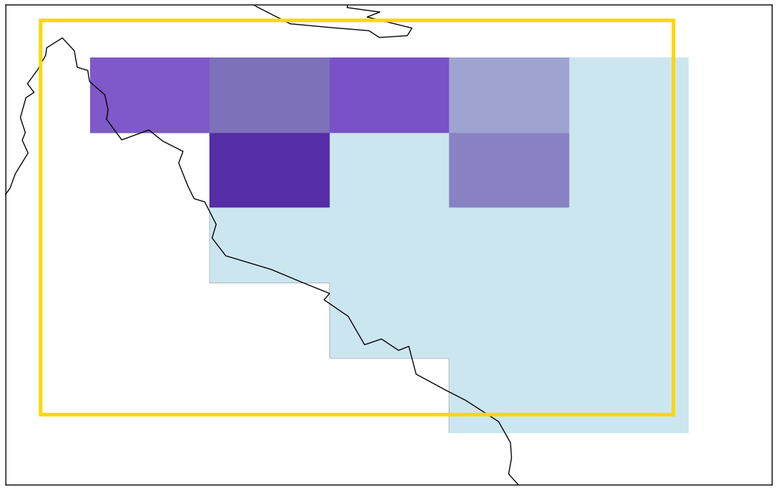


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

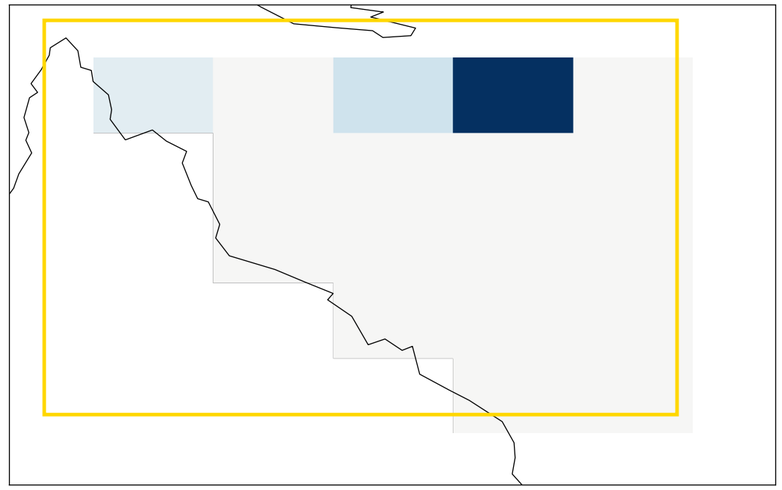
Actual DHW - Step 1



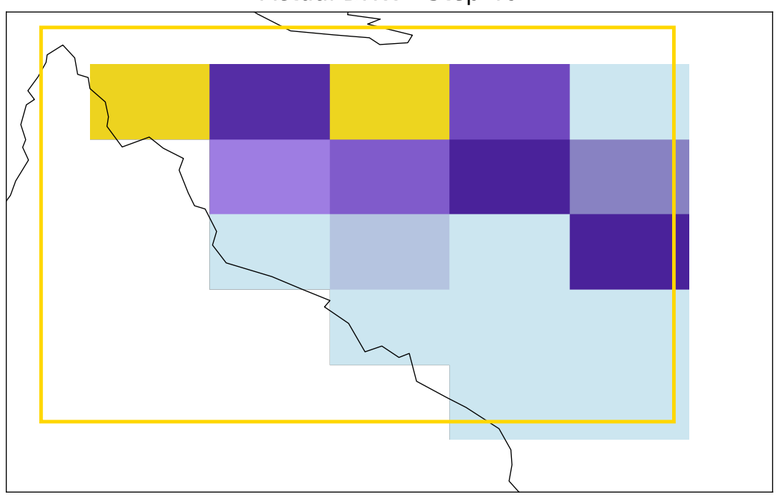
Predicted DHW - Step 1



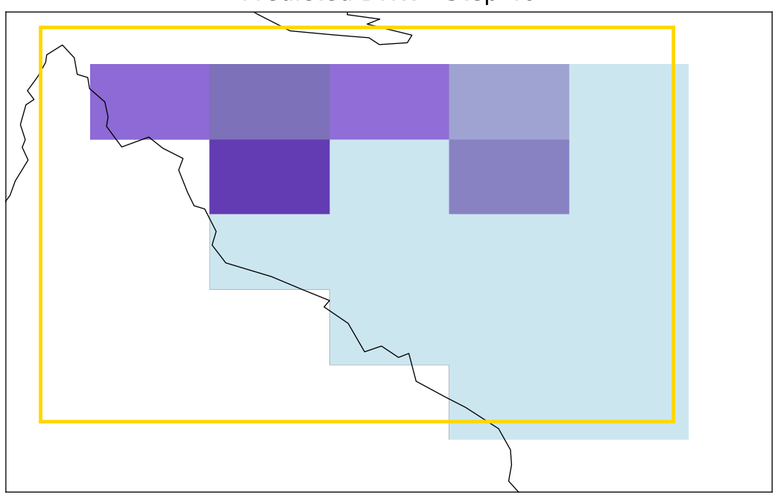
Error (Pred - Actual) - Step 1 [scale ±0.11]



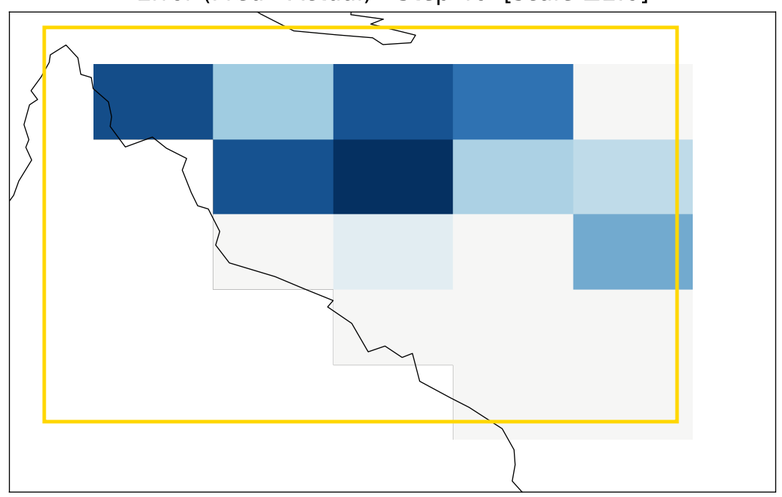
Actual DHW - Step 46



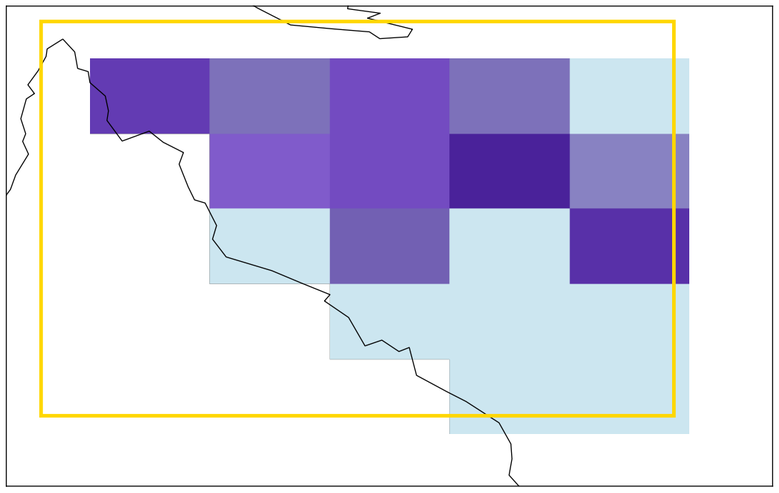
Predicted DHW - Step 46



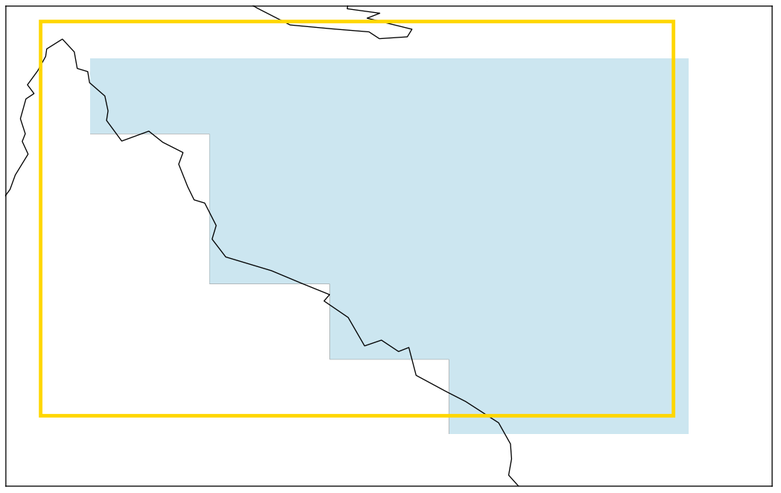
Error (Pred - Actual) - Step 46 [scale ±2.6]



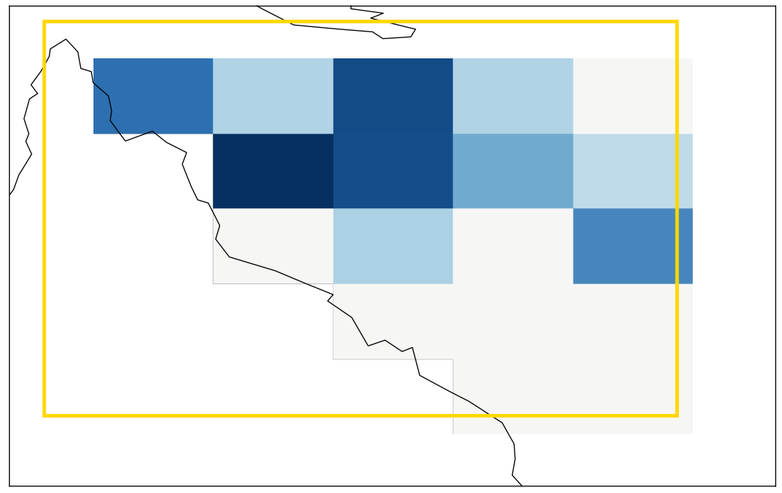
Actual DHW - Step 90

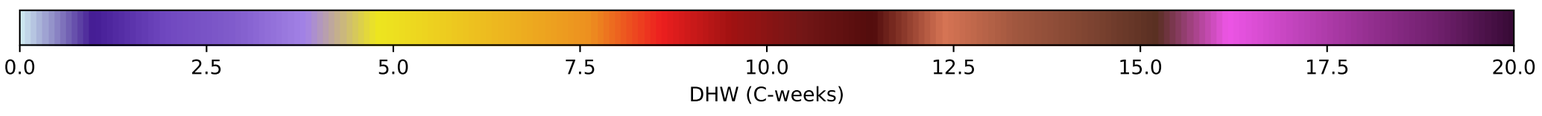


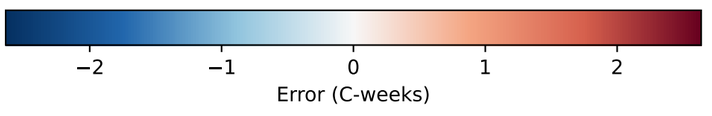
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

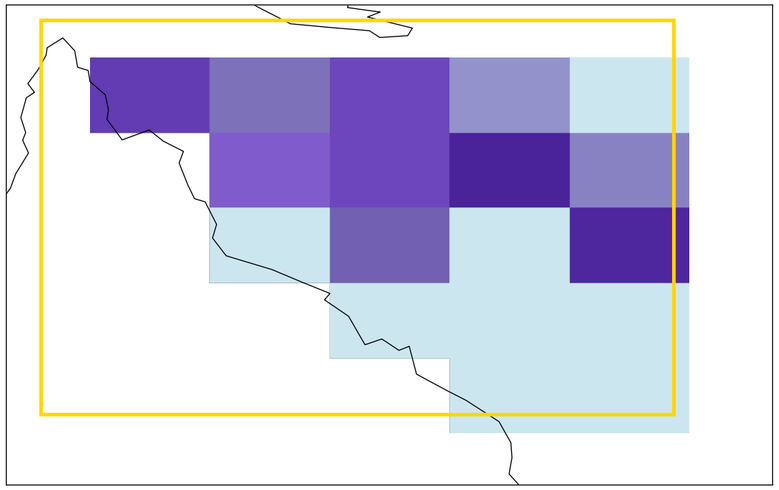




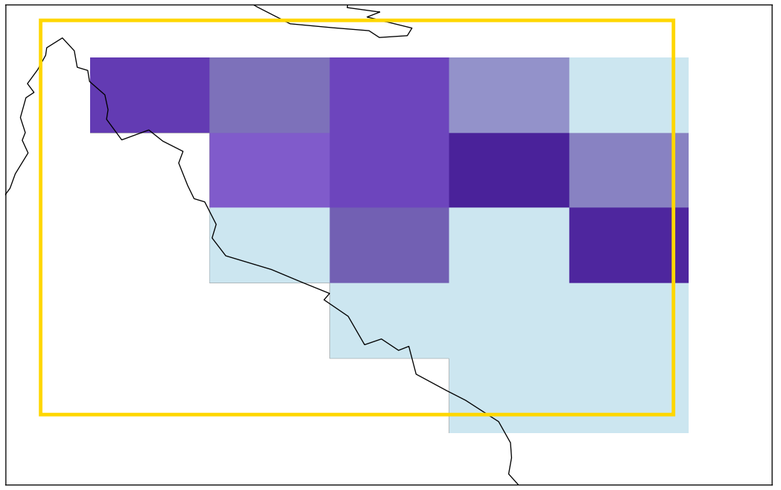


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

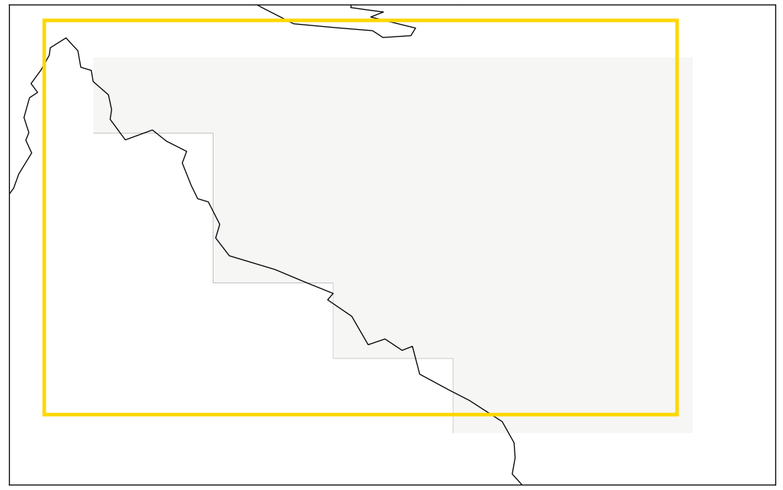
Actual DHW - Step 1



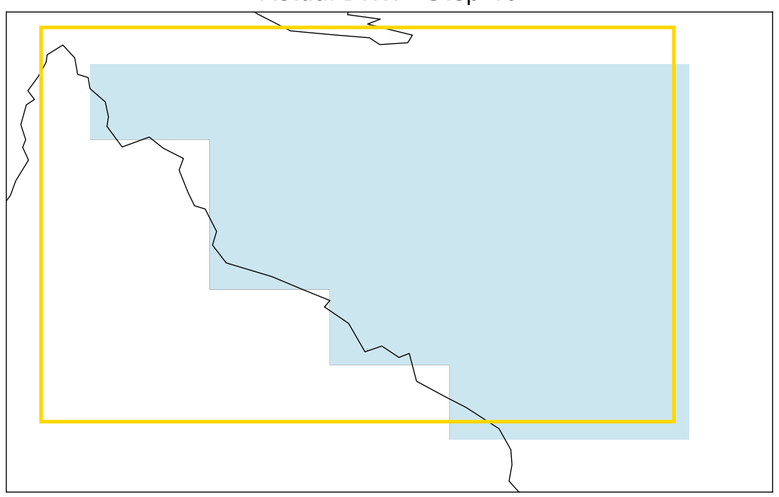
Predicted DHW - Step 1



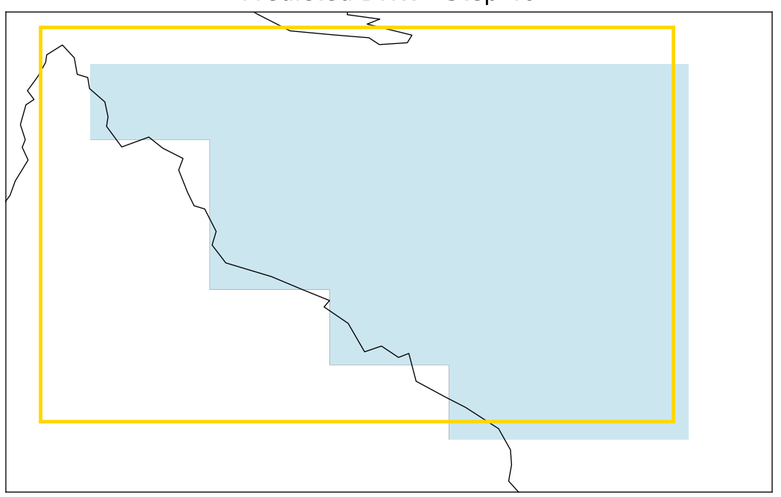
Error (Pred - Actual) - Step 1 [scale ±8]



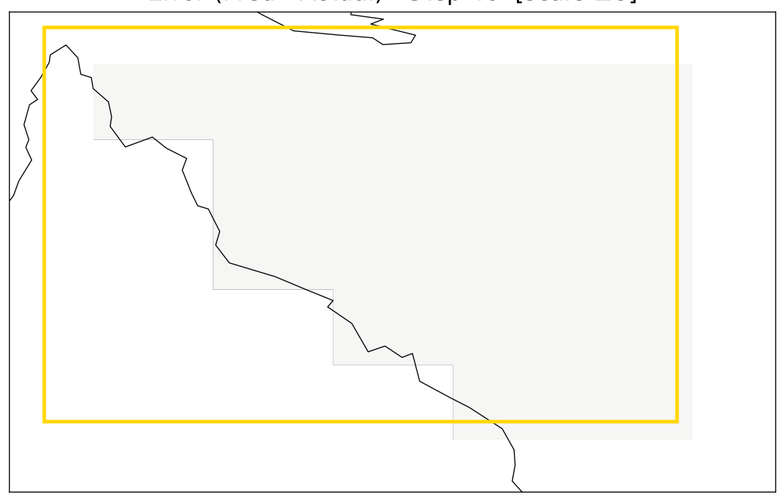
Actual DHW - Step 46



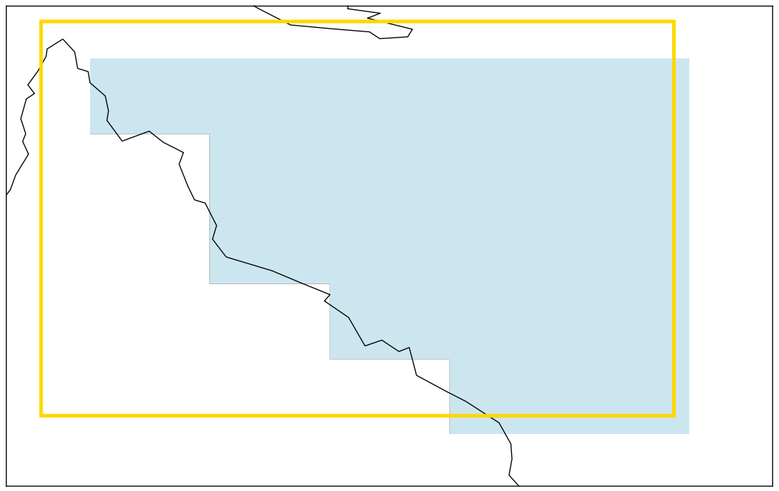
Predicted DHW - Step 46



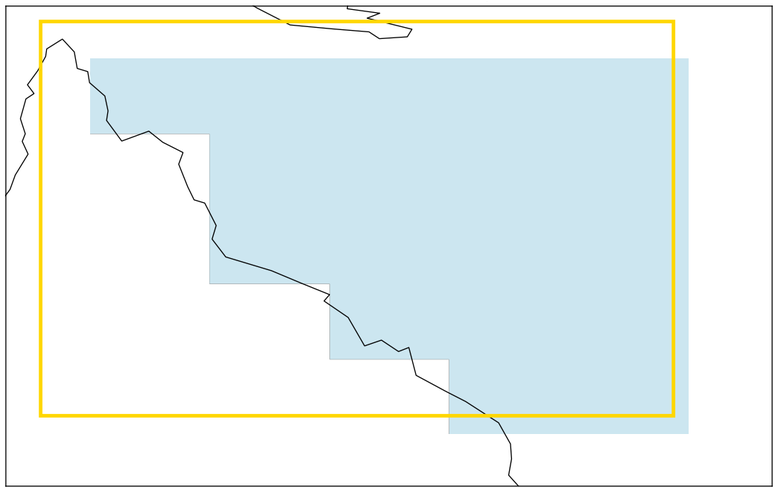
Error (Pred - Actual) - Step 46 [scale ±8]



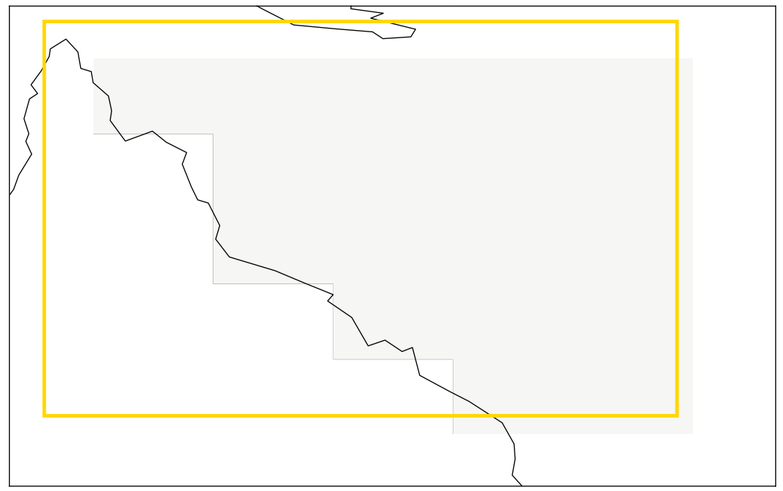
Actual DHW - Step 90

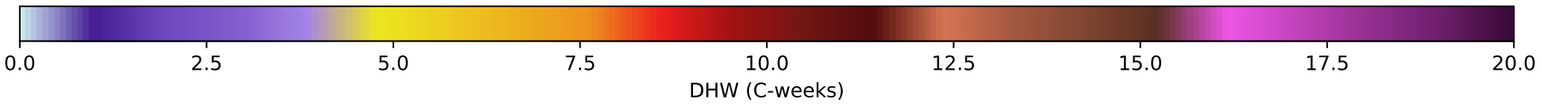


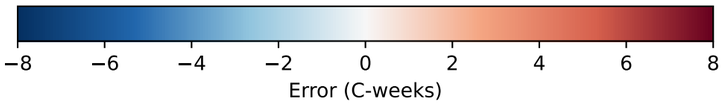
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

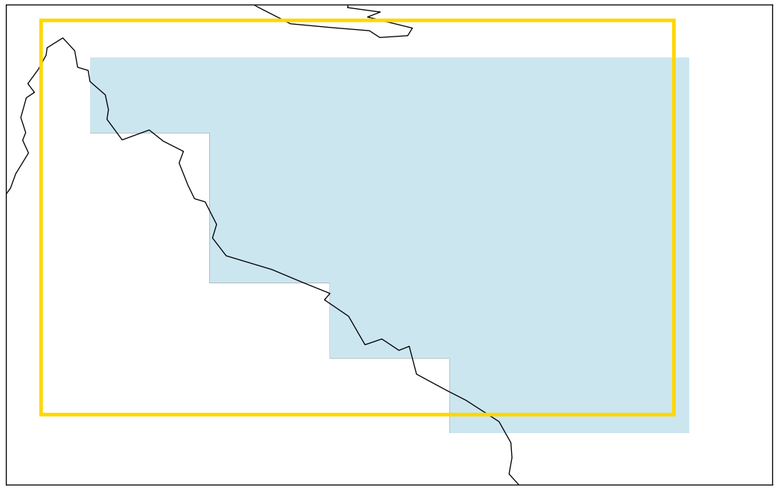




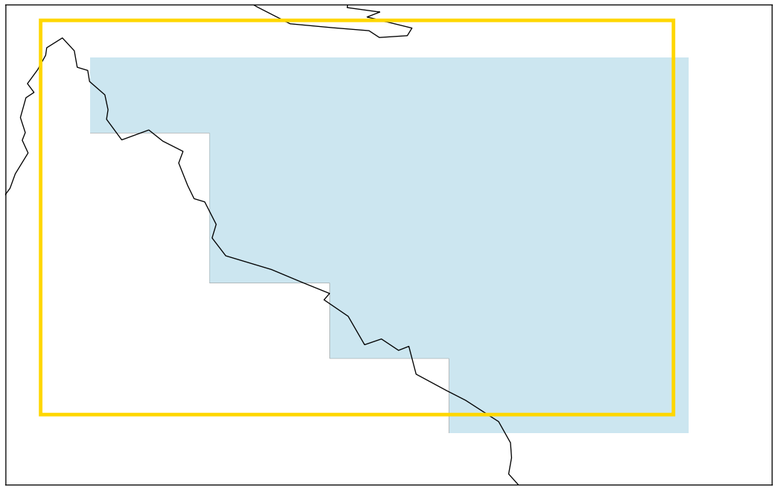


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

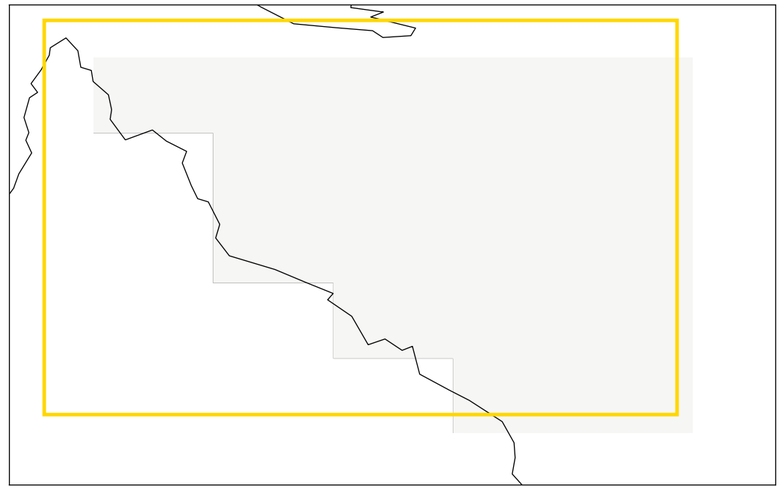
Actual DHW - Step 1



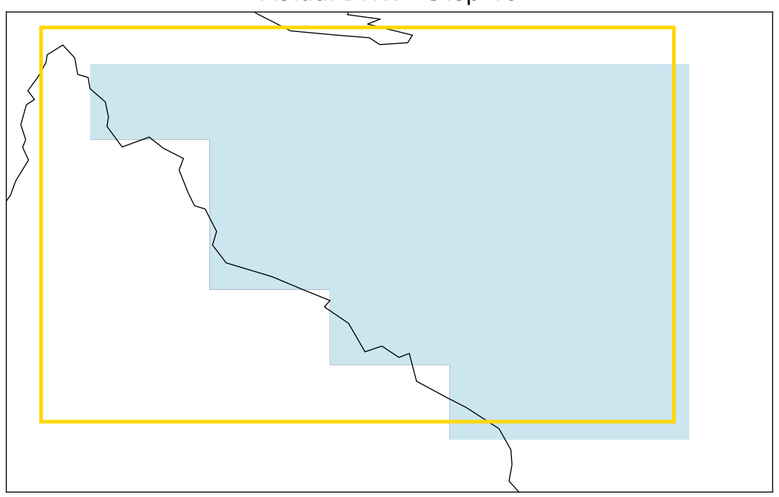
Predicted DHW - Step 1



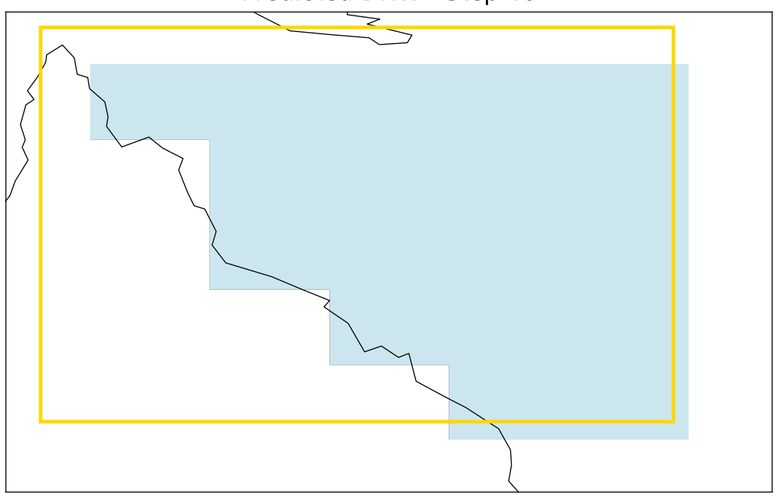
Error (Pred - Actual) - Step 1 [scale ±8]



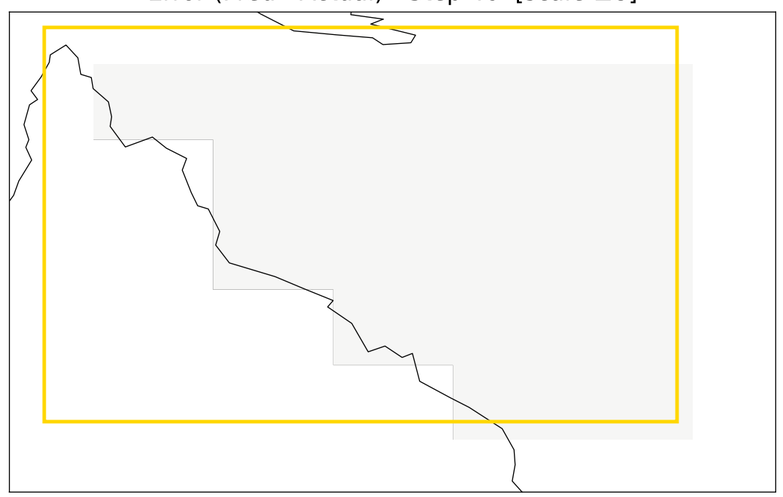
Actual DHW - Step 46



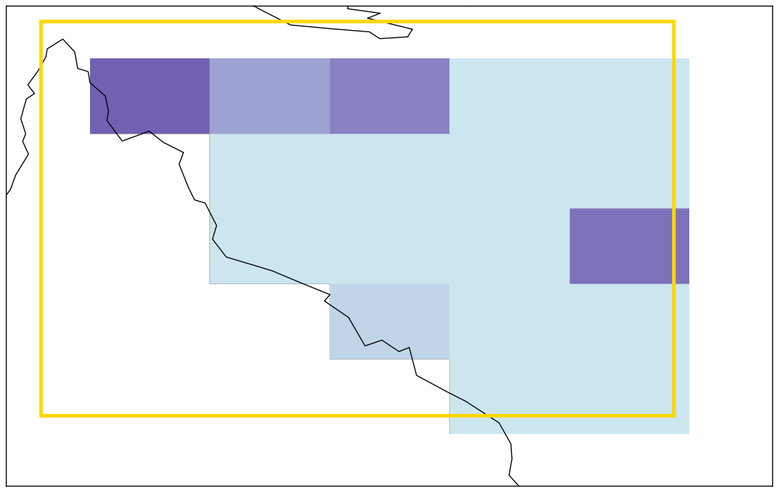
Predicted DHW - Step 46



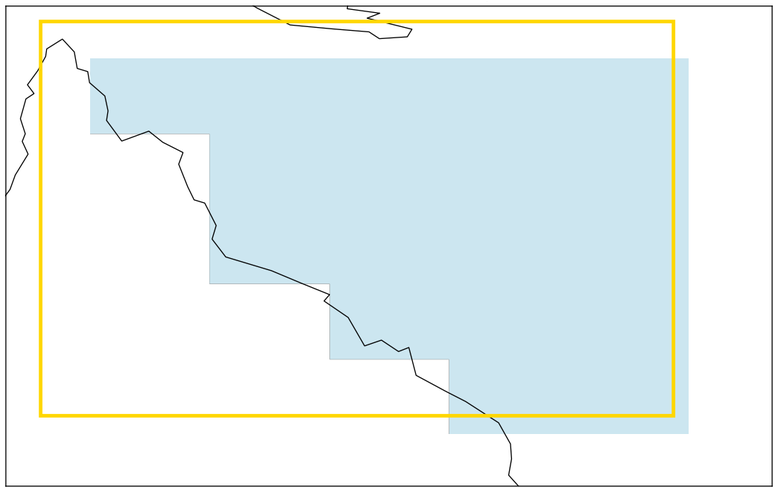
Error (Pred - Actual) - Step 46 [scale ±8]



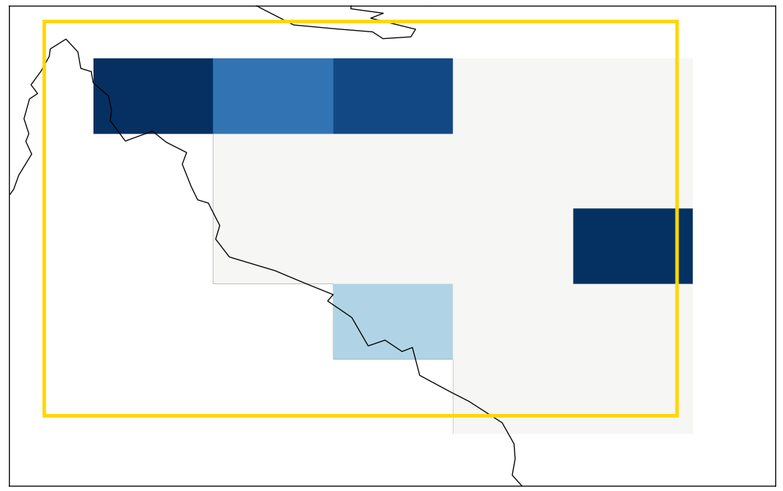
Actual DHW - Step 90

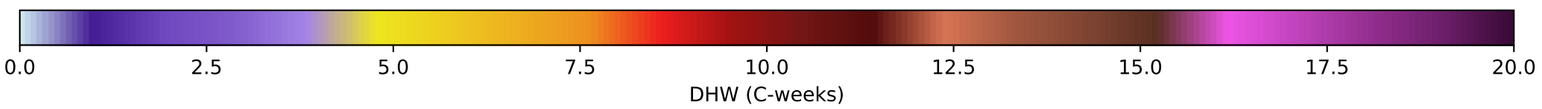


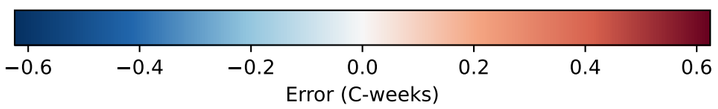
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

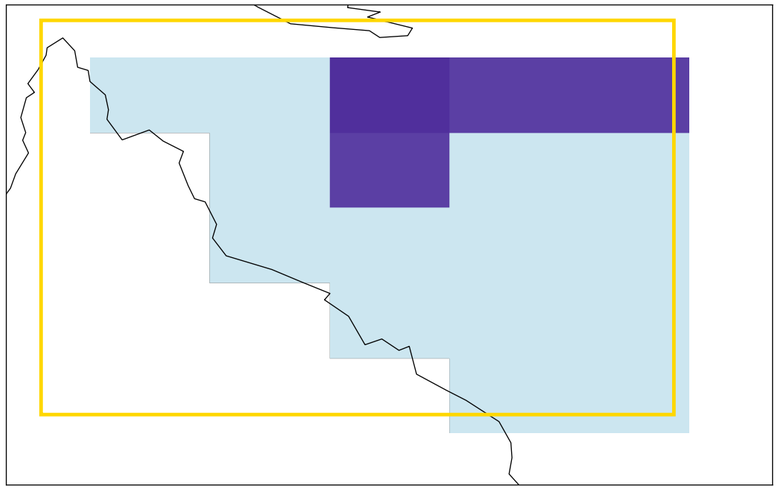




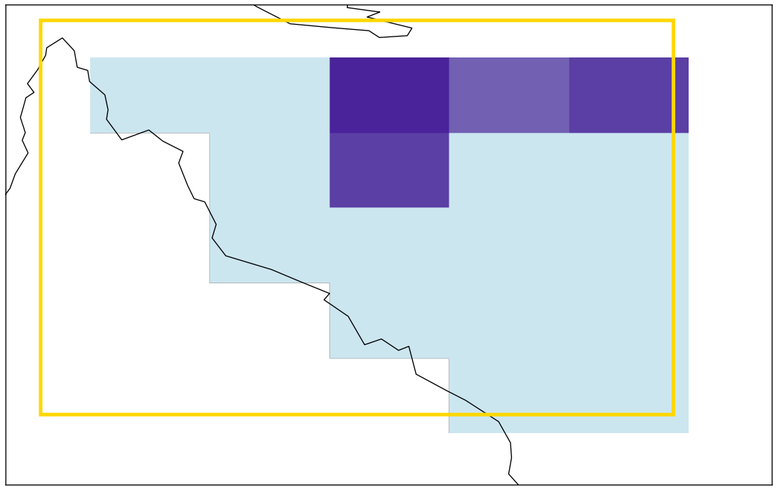


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

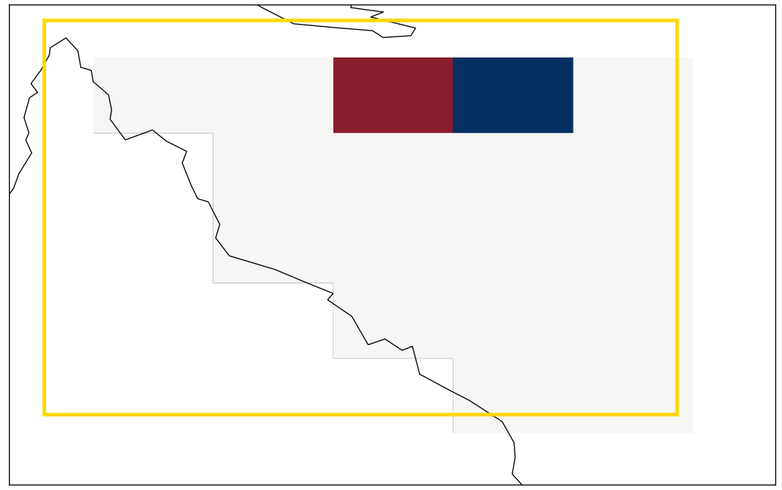
Actual DHW - Step 1



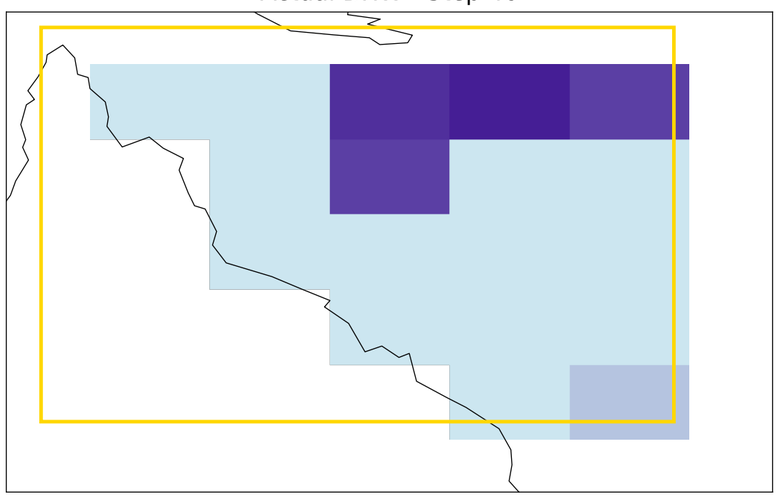
Predicted DHW - Step 1



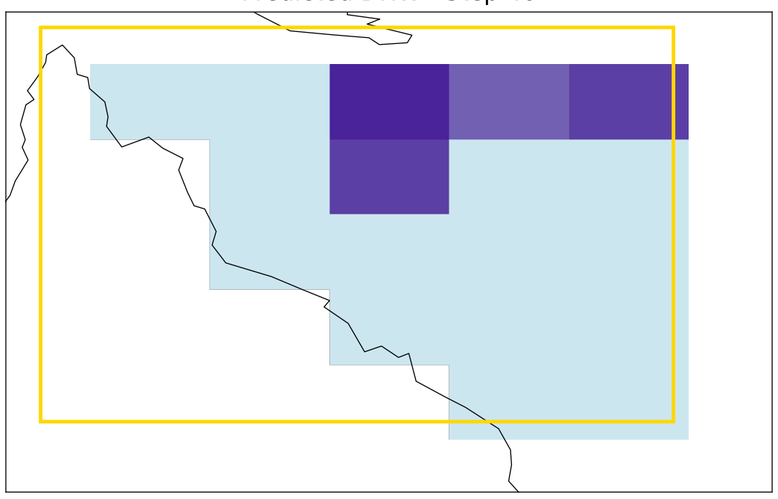
Error (Pred - Actual) - Step 1 [scale ±0.19]



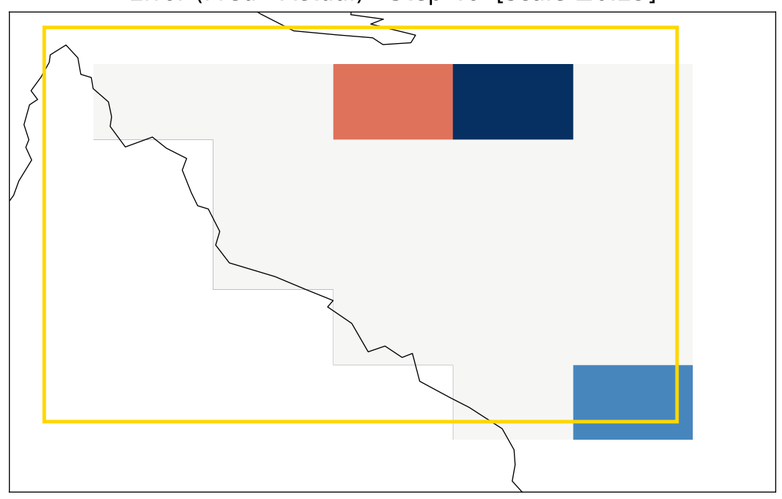
Actual DHW - Step 46



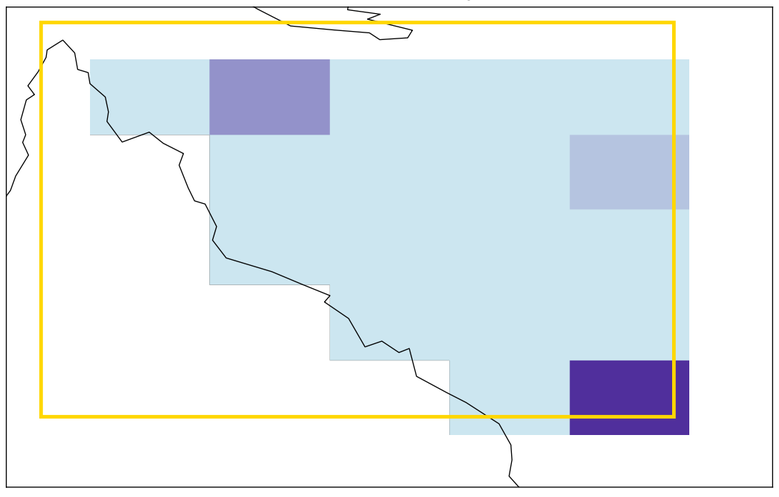
Predicted DHW - Step 46



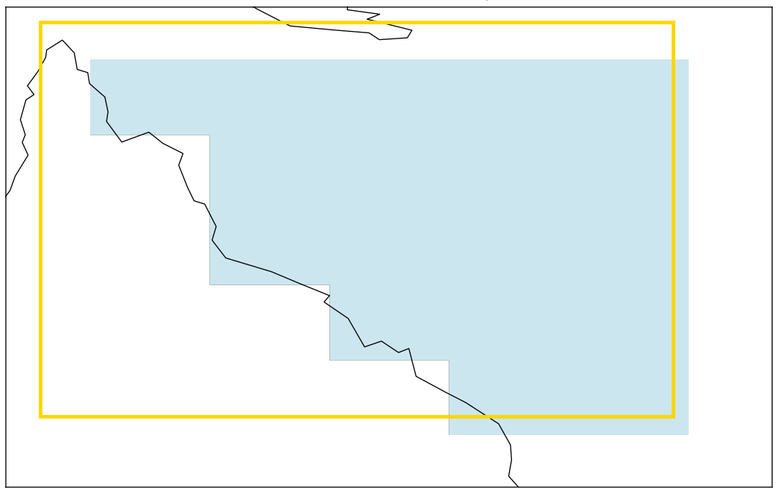
Error (Pred - Actual) - Step 46 [scale ±0.29]



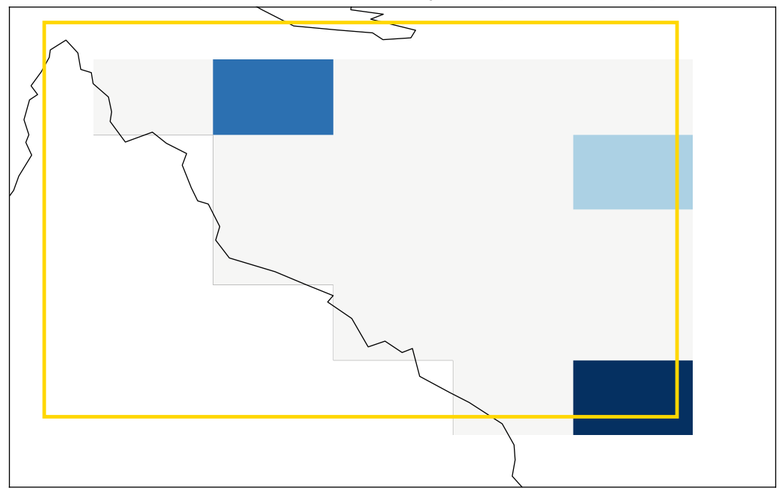
Actual DHW - Step 90

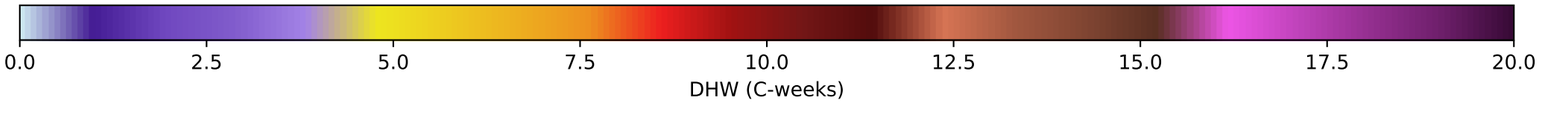


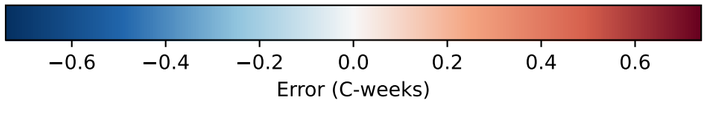
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.74]

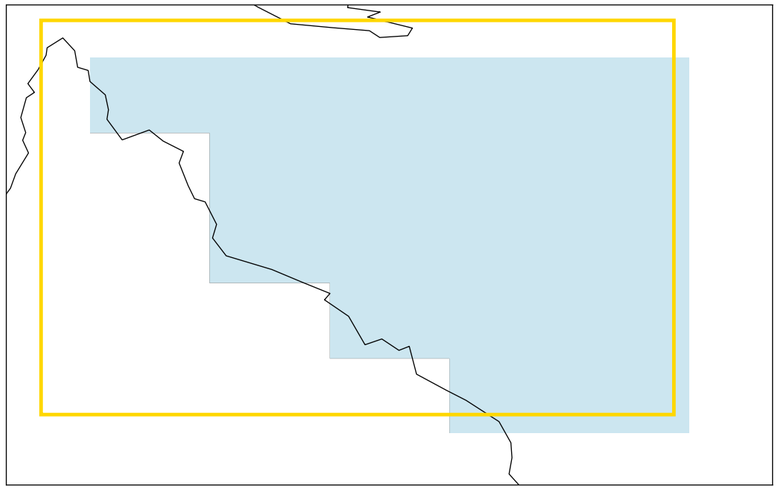




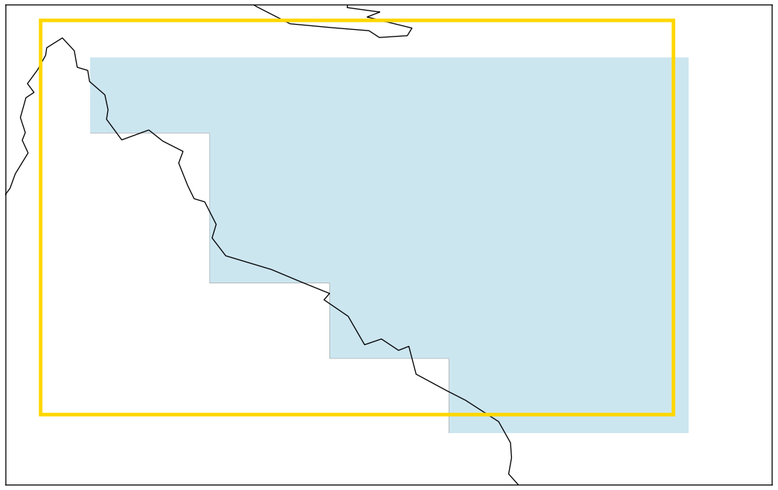


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

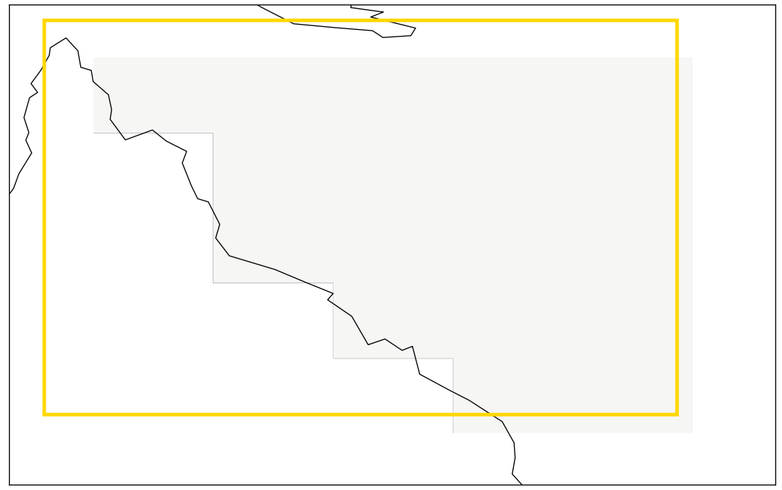
Actual DHW - Step 1



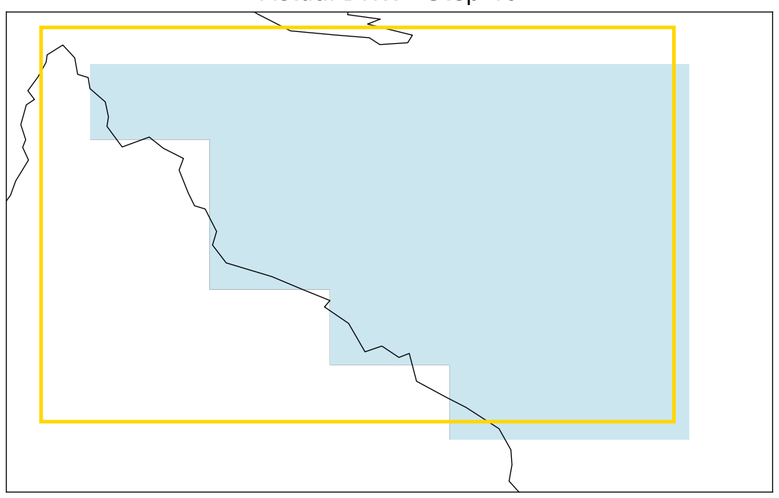
Predicted DHW - Step 1



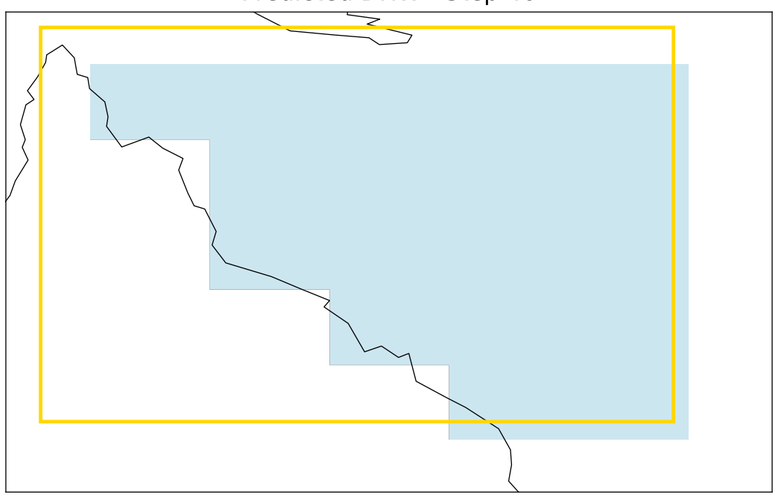
Error (Pred - Actual) - Step 1 [scale ±8]



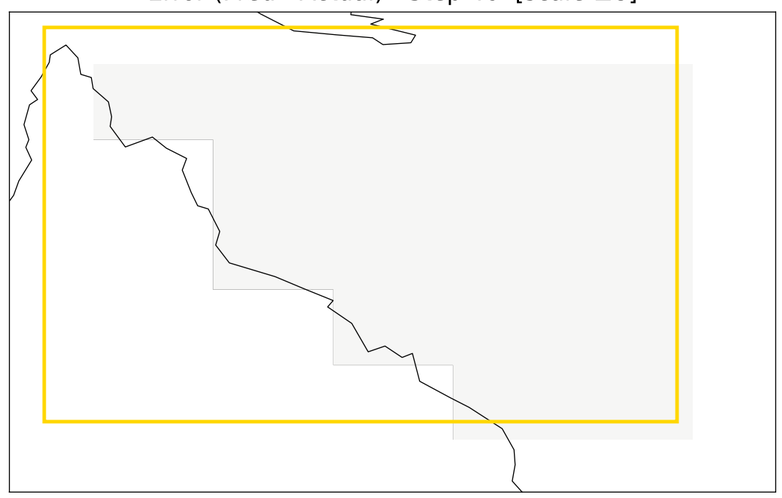
Actual DHW - Step 46



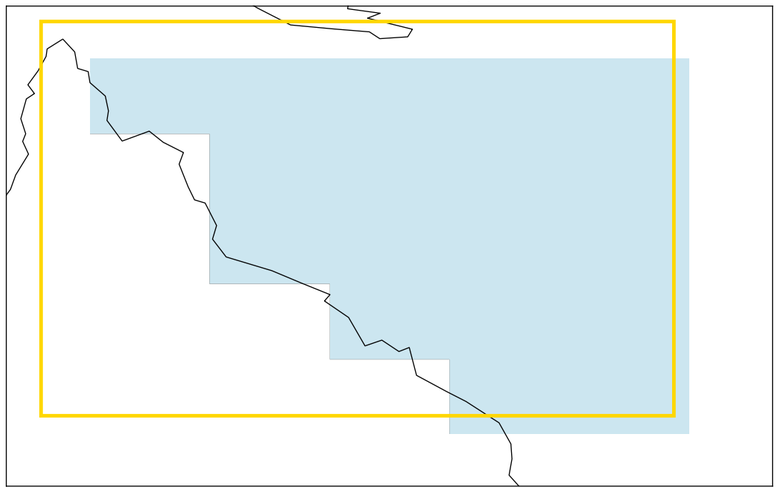
Predicted DHW - Step 46



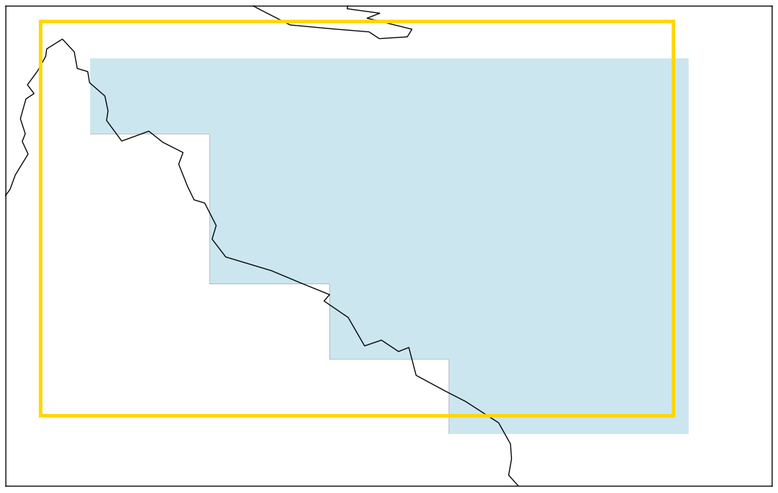
Error (Pred - Actual) - Step 46 [scale ±8]



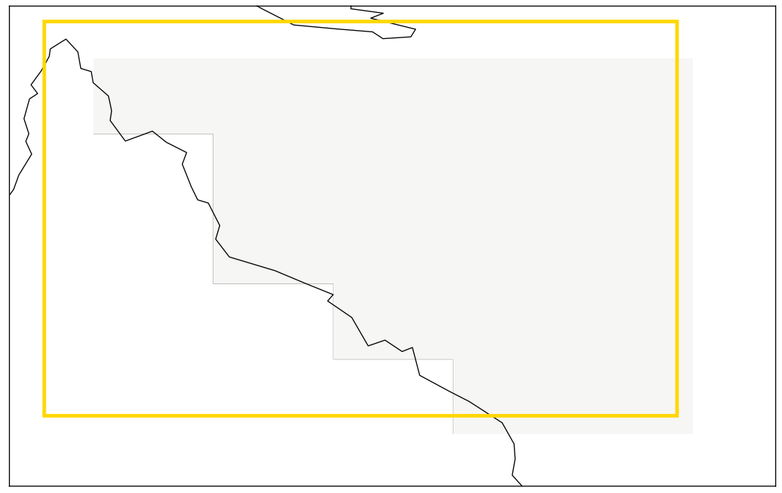
Actual DHW - Step 90

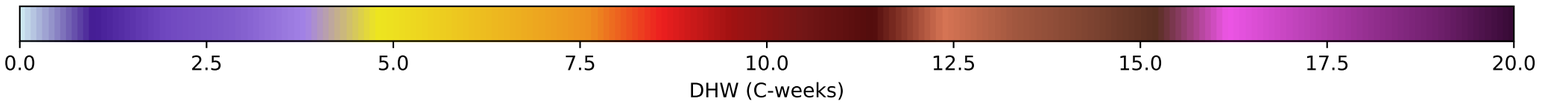


Predicted DHW - Step 90



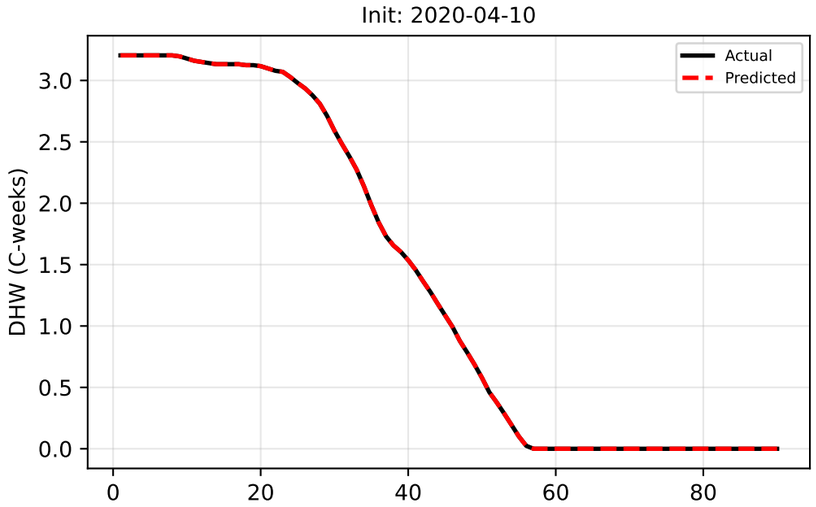
Error (Pred - Actual) - Step 90 [scale ±8]

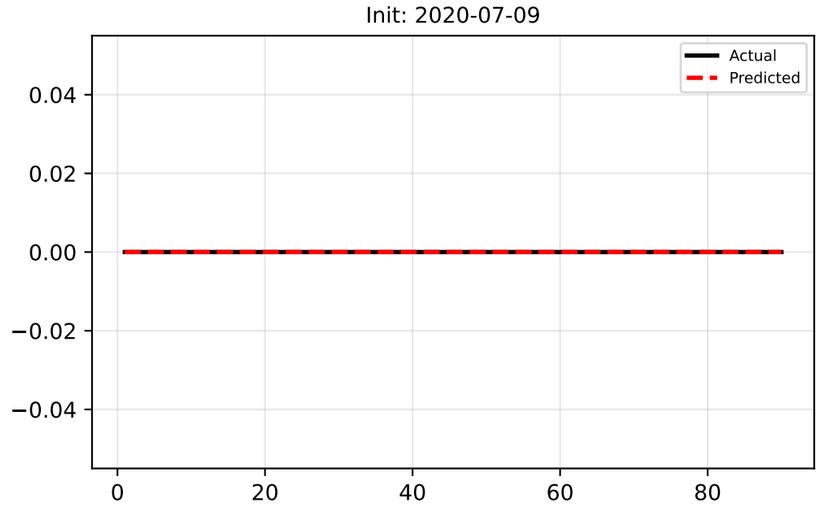


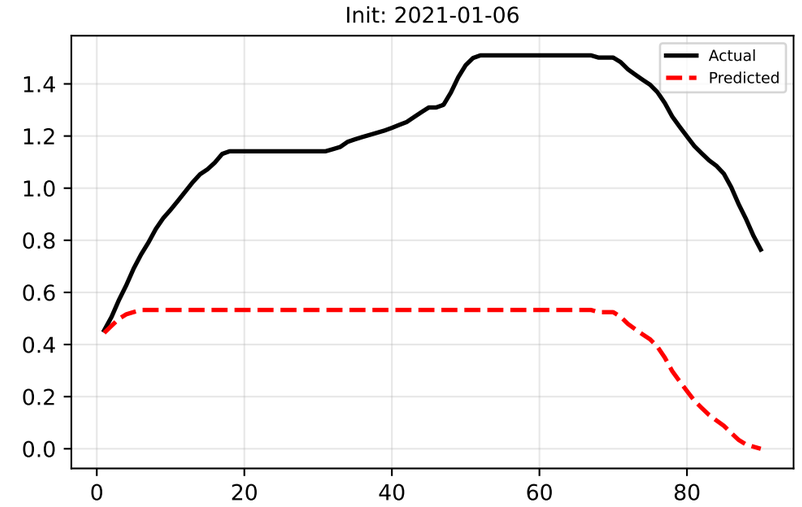


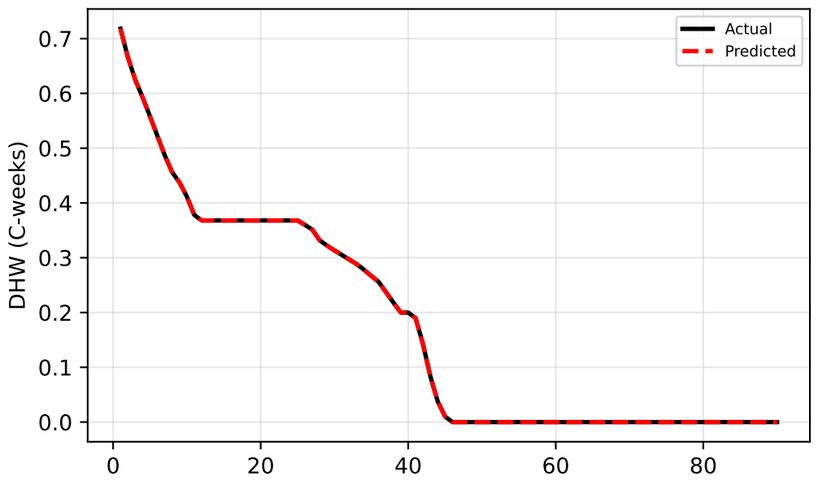


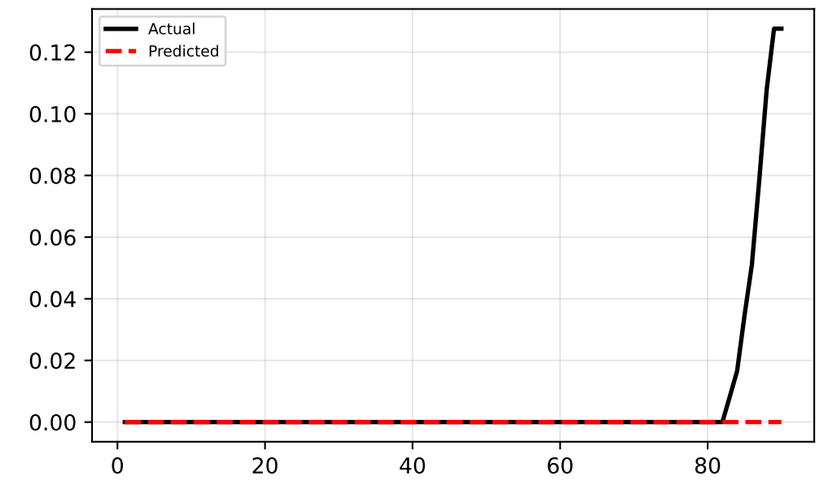
Global-Mean DHW - Test - GBR (C-weeks)

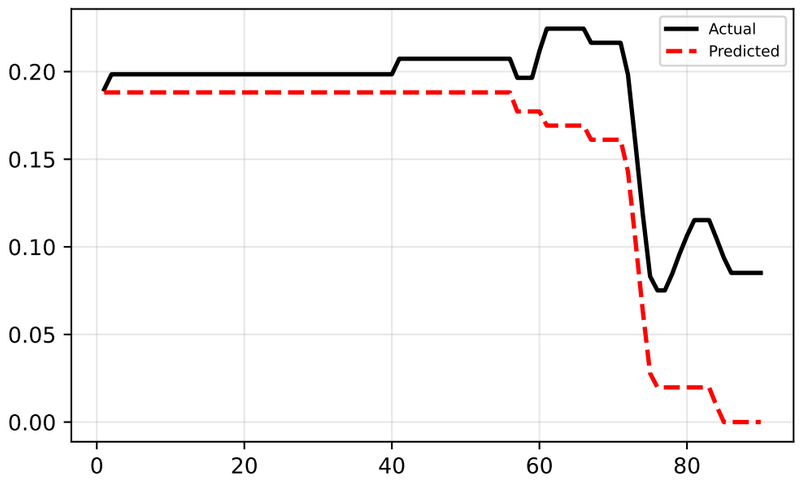


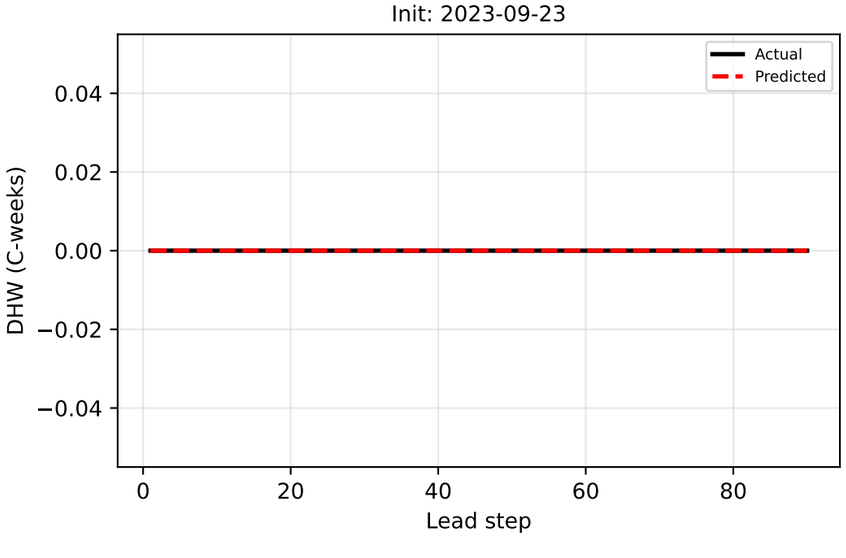


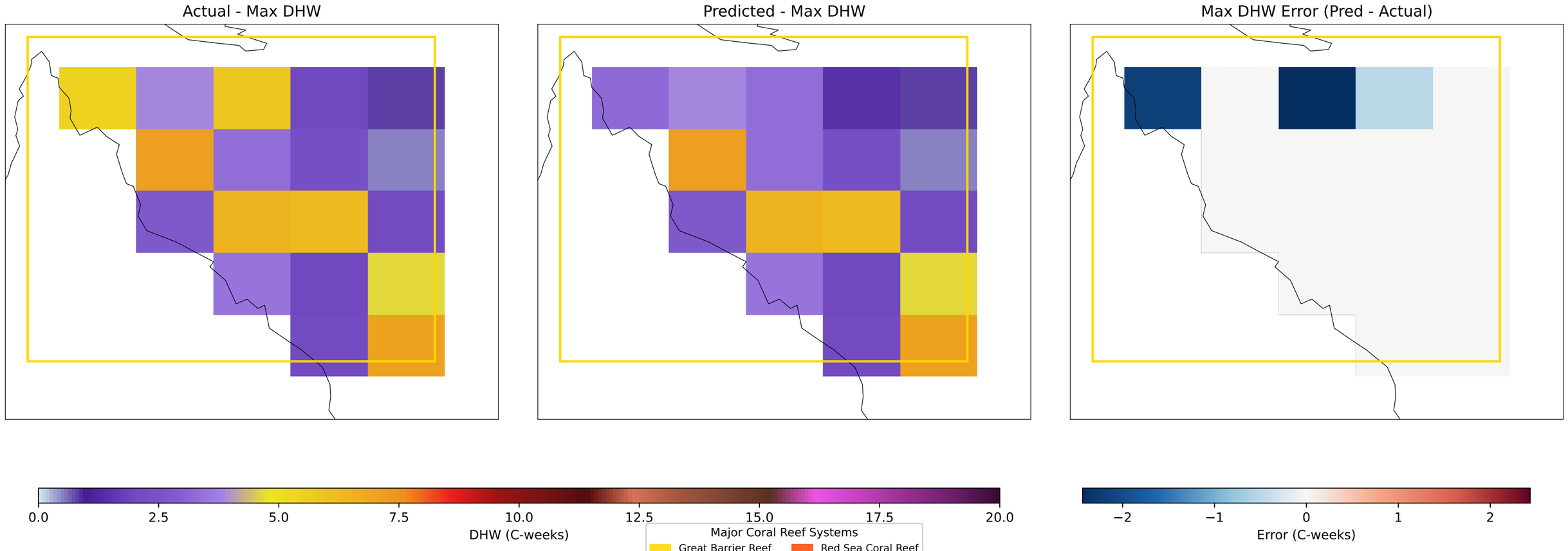


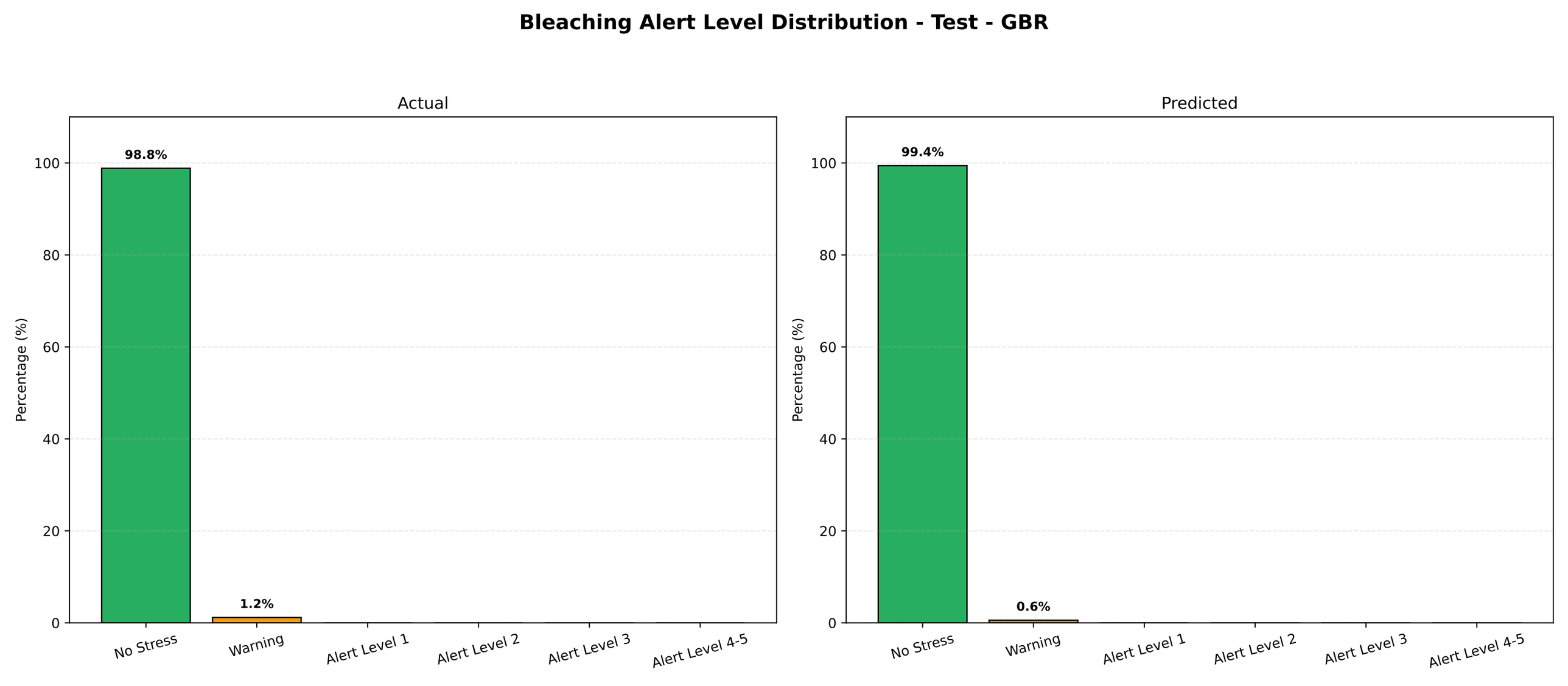






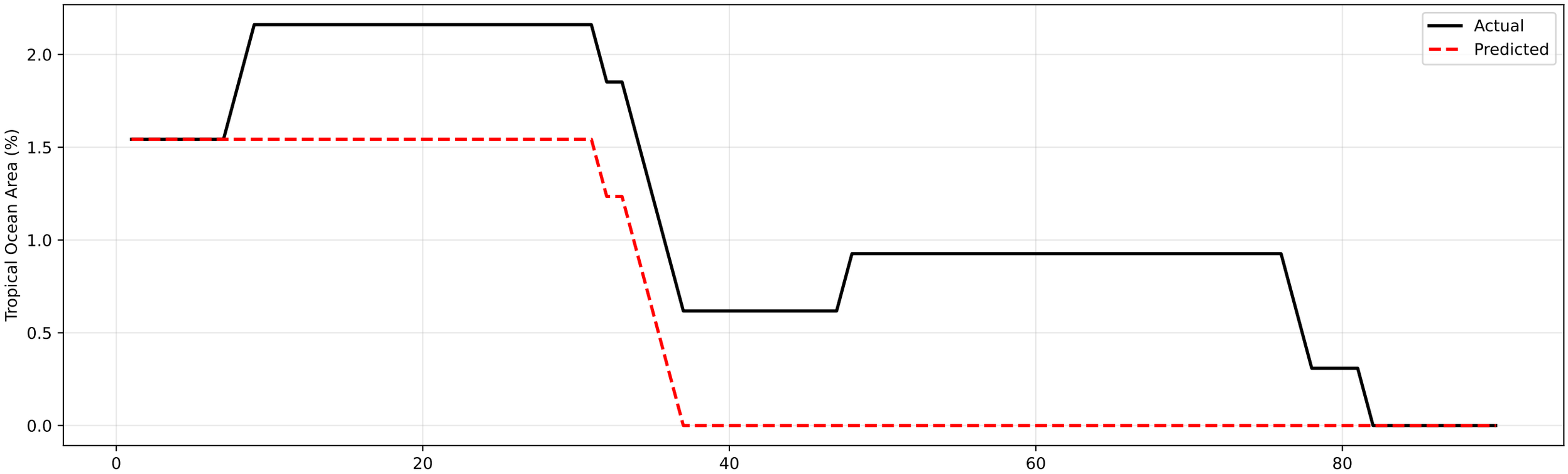



Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−2
−1
0
1
2
Error (C-weeks)

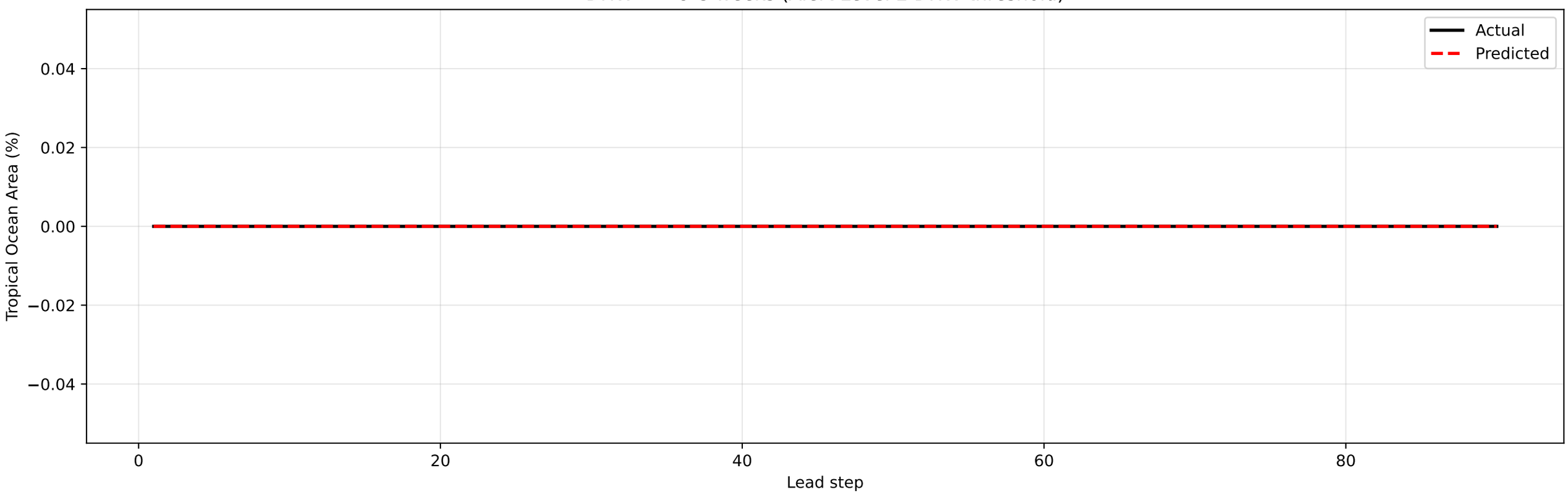

Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.8%
1.2%
99.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



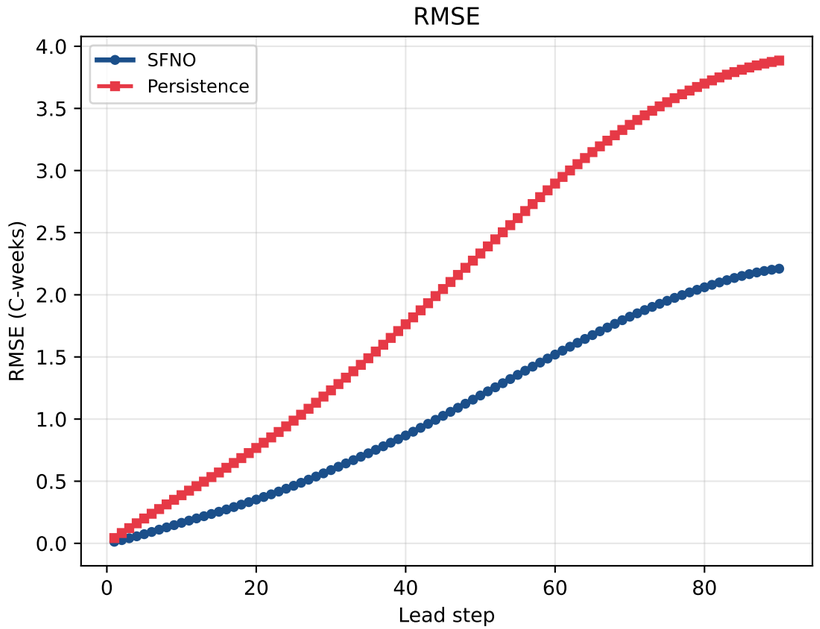
DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

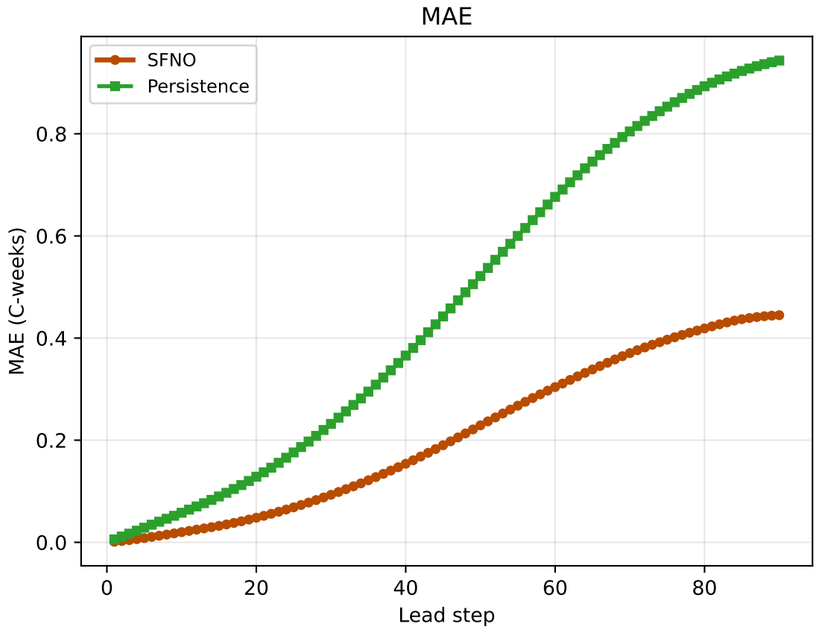


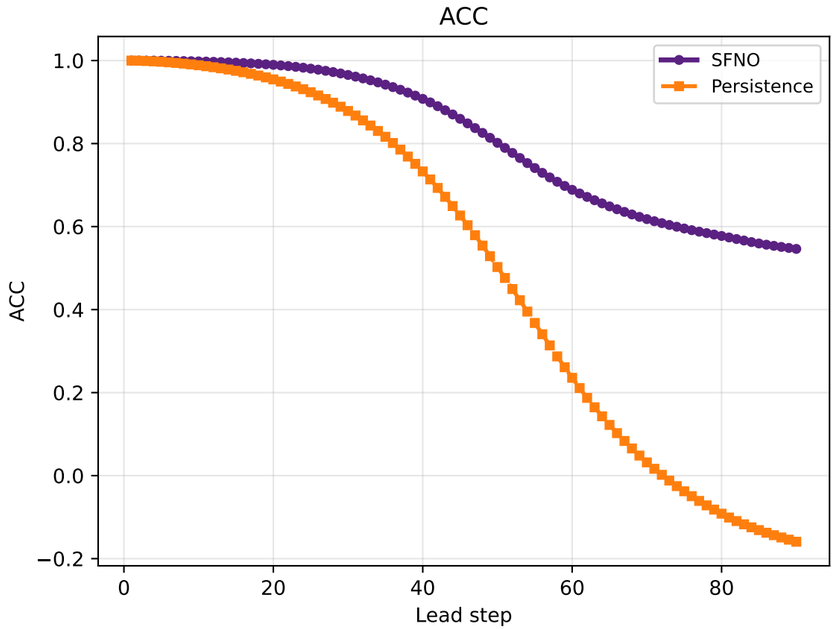
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

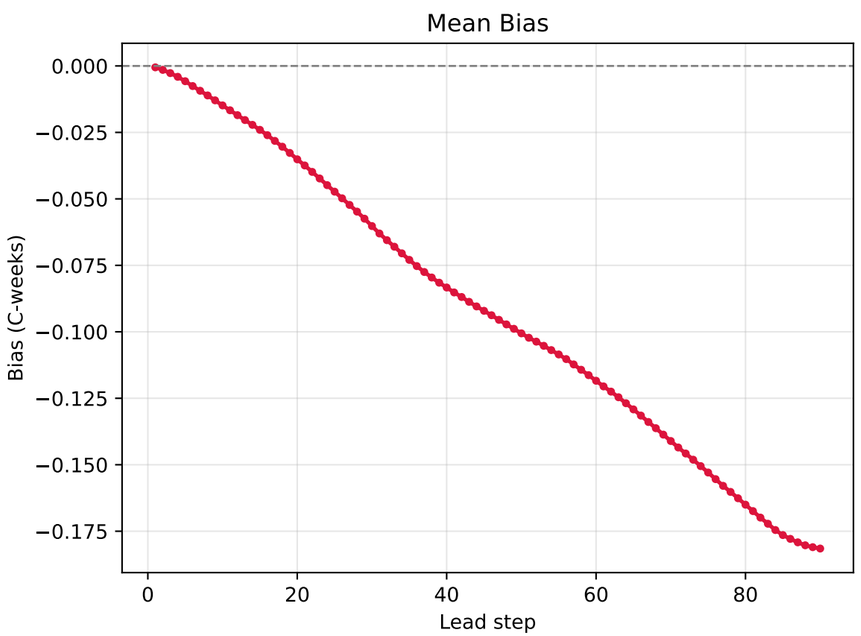
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0129	0.0425	69.6%	0.0011	0.0057	-0.0005	1.0000	0.9999	1.0000
2	0.0261	0.0829	68.5%	0.0026	0.0114	-0.0015	0.9999	0.9995	0.9999
3	0.0412	0.1222	66.3%	0.0044	0.0170	-0.0027	0.9999	0.9989	0.9998
4	0.0568	0.1613	64.8%	0.0062	0.0227	-0.0041	0.9998	0.9980	0.9996
5	0.0736	0.2000	63.2%	0.0083	0.0286	-0.0057	0.9996	0.9970	0.9993
6	0.0916	0.2381	61.5%	0.0106	0.0346	-0.0076	0.9994	0.9957	0.9990
7	0.1101	0.2758	60.1%	0.0129	0.0404	-0.0093	0.9991	0.9943	0.9985
8	0.1284	0.3133	59.0%	0.0152	0.0461	-0.0111	0.9988	0.9926	0.9980
9	0.1465	0.3504	58.2%	0.0175	0.0520	-0.0129	0.9984	0.9907	0.9973
10	0.1650	0.3874	57.4%	0.0199	0.0579	-0.0148	0.9979	0.9886	0.9966
11	0.1829	0.4241	56.9%	0.0224	0.0641	-0.0167	0.9975	0.9863	0.9958
12	0.2007	0.4601	56.4%	0.0248	0.0702	-0.0185	0.9969	0.9839	0.9949
13	0.2184	0.4965	56.0%	0.0272	0.0766	-0.0204	0.9963	0.9812	0.9939
14	0.2362	0.5332	55.7%	0.0297	0.0831	-0.0222	0.9957	0.9783	0.9929
15	0.2541	0.5701	55.4%	0.0322	0.0898	-0.0240	0.9949	0.9752	0.9917
16	0.2727	0.6080	55.1%	0.0350	0.0970	-0.0260	0.9941	0.9717	0.9903
17	0.2920	0.6467	54.8%	0.0380	0.1044	-0.0282	0.9932	0.9680	0.9888
18	0.3115	0.6861	54.6%	0.0412	0.1121	-0.0304	0.9921	0.9639	0.9872
19	0.3317	0.7264	54.3%	0.0446	0.1201	-0.0327	0.9909	0.9594	0.9853
20	0.3523	0.7676	54.1%	0.0482	0.1285	-0.0351	0.9896	0.9546	0.9832
21	0.3731	0.8096	53.9%	0.0519	0.1371	-0.0374	0.9882	0.9494	0.9810
22	0.3944	0.8524	53.7%	0.0558	0.1462	-0.0399	0.9866	0.9437	0.9785
23	0.4165	0.8961	53.5%	0.0599	0.1557	-0.0423	0.9848	0.9376	0.9757
24	0.4397	0.9412	53.3%	0.0642	0.1657	-0.0448	0.9828	0.9309	0.9726
25	0.4631	0.9875	53.1%	0.0685	0.1760	-0.0473	0.9805	0.9237	0.9691
26	0.4873	1.0350	52.9%	0.0731	0.1866	-0.0498	0.9780	0.9158	0.9653
27	0.5123	1.0830	52.7%	0.0778	0.1974	-0.0523	0.9753	0.9074	0.9612
28	0.5376	1.1316	52.5%	0.0827	0.2084	-0.0548	0.9722	0.8984	0.9566
29	0.5633	1.1811	52.3%	0.0878	0.2200	-0.0575	0.9689	0.8888	0.9517
30	0.5895	1.2311	52.1%	0.0930	0.2319	-0.0602	0.9653	0.8785	0.9464
31	0.6164	1.2818	51.9%	0.0985	0.2441	-0.0630	0.9613	0.8675	0.9405
32	0.6432	1.3332	51.8%	0.1041	0.2564	-0.0655	0.9570	0.8558	0.9343
33	0.6697	1.3846	51.6%	0.1097	0.2690	-0.0680	0.9524	0.8434	0.9278
34	0.6967	1.4364	51.5%	0.1154	0.2818	-0.0705	0.9474	0.8304	0.9207
35	0.7247	1.4894	51.3%	0.1215	0.2951	-0.0729	0.9419	0.8164	0.9130
36	0.7527	1.5434	51.2%	0.1278	0.3088	-0.0753	0.9360	0.8014	0.9048
37	0.7811	1.5981	51.1%	0.1341	0.3227	-0.0775	0.9297	0.7854	0.8960
38	0.8097	1.6531	51.0%	0.1405	0.3368	-0.0796	0.9228	0.7687	0.8867
39	0.8385	1.7081	50.9%	0.1471	0.3512	-0.0815	0.9155	0.7511	0.8768
40	0.8679	1.7629	50.8%	0.1538	0.3657	-0.0833	0.9077	0.7329	0.8664
41	0.8982	1.8185	50.6%	0.1608	0.3806	-0.0852	0.8992	0.7136	0.8552
42	0.9296	1.8750	50.4%	0.1680	0.3959	-0.0869	0.8900	0.6932	0.8432
43	0.9617	1.9323	50.2%	0.1753	0.4113	-0.0887	0.8803	0.6717	0.8306
44	0.9938	1.9898	50.1%	0.1826	0.4269	-0.0905	0.8701	0.6495	0.8176
45	1.0261	2.0466	49.9%	0.1901	0.4424	-0.0921	0.8596	0.6267	0.8042
46	1.0583	2.1032	49.7%	0.1977	0.4581	-0.0938	0.8486	0.6033	0.7904
47	1.0909	2.1599	49.5%	0.2054	0.4738	-0.0955	0.8373	0.5792	0.7761
48	1.1235	2.2173	49.3%	0.2132	0.4897	-0.0972	0.8257	0.5542	0.7616
49	1.1565	2.2749	49.2%	0.2210	0.5055	-0.0989	0.8139	0.5286	0.7468
50	1.1898	2.3325	49.0%	0.2290	0.5215	-0.1006	0.8018	0.5026	0.7317
51	1.2229	2.3896	48.8%	0.2368	0.5374	-0.1022	0.7896	0.4762	0.7163
52	1.2557	2.4464	48.7%	0.2446	0.5532	-0.1037	0.7774	0.4495	0.7011
53	1.2887	2.5032	48.5%	0.2523	0.5689	-0.1053	0.7652	0.4226	0.6857
54	1.3221	2.5604	48.4%	0.2599	0.5845	-0.1069	0.7530	0.3952	0.6702
55	1.3558	2.6175	48.2%	0.2676	0.6001	-0.1085	0.7410	0.3679	0.6550
56	1.3892	2.6741	48.0%	0.2751	0.6155	-0.1103	0.7294	0.3407	0.6402
57	1.4222	2.7302	47.9%	0.2825	0.6310	-0.1123	0.7184	0.3138	0.6261
58	1.4547	2.7859	47.8%	0.2898	0.6464	-0.1143	0.7079	0.2872	0.6125
59	1.4870	2.8409	47.7%	0.2970	0.6616	-0.1163	0.6979	0.2611	0.5994
60	1.5190	2.8953	47.5%	0.3040	0.6764	-0.1184	0.6886	0.2357	0.5869
61	1.5507	2.9487	47.4%	0.3110	0.6909	-0.1205	0.6798	0.2110	0.5751
62	1.5823	3.0007	47.3%	0.3180	0.7050	-0.1225	0.6715	0.1873	0.5636
63	1.6137	3.0513	47.1%	0.3249	0.7188	-0.1246	0.6635	0.1647	0.5526
64	1.6449	3.1006	47.0%	0.3318	0.7323	-0.1269	0.6559	0.1429	0.5419
65	1.6759	3.1484	46.8%	0.3385	0.7456	-0.1292	0.6486	0.1222	0.5315
66	1.7064	3.1948	46.6%	0.3452	0.7582	-0.1315	0.6418	0.1024	0.5216
67	1.7365	3.2402	46.4%	0.3516	0.7704	-0.1339	0.6354	0.0835	0.5121
68	1.7665	3.2842	46.2%	0.3581	0.7822	-0.1363	0.6292	0.0655	0.5027
69	1.7957	3.3266	46.0%	0.3645	0.7936	-0.1387	0.6234	0.0483	0.4937
70	1.8238	3.3677	45.8%	0.3705	0.8047	-0.1411	0.6180	0.0319	0.4853
71	1.8509	3.4069	45.7%	0.3762	0.8153	-0.1435	0.6129	0.0165	0.4773
72	1.8772	3.4445	45.5%	0.3817	0.8253	-0.1458	0.6083	0.0020	0.4699
73	1.9028	3.4811	45.3%	0.3869	0.8349	-0.1481	0.6039	-0.0119	0.4629
74	1.9277	3.5164	45.2%	0.3919	0.8443	-0.1505	0.5996	-0.0252	0.4561
75	1.9518	3.5499	45.0%	0.3968	0.8532	-0.1529	0.5954	-0.0377	0.4495
76	1.9754	3.5823	44.9%	0.4016	0.8619	-0.1554	0.5913	-0.0496	0.4432
77	1.9979	3.6139	44.7%	0.4061	0.8703	-0.1579	0.5878	-0.0609	0.4377
78	2.0194	3.6437	44.6%	0.4105	0.8782	-0.1602	0.5843	-0.0716	0.4321
79	2.0403	3.6724	44.4%	0.4147	0.8858	-0.1626	0.5808	-0.0819	0.4268
80	2.0606	3.6998	44.3%	0.4189	0.8931	-0.1650	0.5774	-0.0917	0.4215
81	2.0802	3.7255	44.2%	0.4229	0.9000	-0.1674	0.5738	-0.1009	0.4161
82	2.0995	3.7494	44.0%	0.4269	0.9064	-0.1699	0.5700	-0.1095	0.4105
83	2.1178	3.7716	43.8%	0.4306	0.9123	-0.1722	0.5664	-0.1173	0.4053
84	2.1355	3.7922	43.7%	0.4342	0.9179	-0.1746	0.5627	-0.1245	0.4001
85	2.1517	3.8114	43.5%	0.4370	0.9231	-0.1765	0.5593	-0.1313	0.3954
86	2.1664	3.8295	43.4%	0.4392	0.9278	-0.1779	0.5564	-0.1376	0.3914
87	2.1799	3.8461	43.3%	0.4412	0.9323	-0.1792	0.5536	-0.1434	0.3877
88	2.1917	3.8609	43.2%	0.4428	0.9365	-0.1803	0.5510	-0.1489	0.3841
89	2.2017	3.8738	43.2%	0.4439	0.9402	-0.1810	0.5485	-0.1542	0.3807
90	2.2102	3.8852	43.1%	0.4448	0.9435	-0.1815	0.5462	-0.1593	0.3774

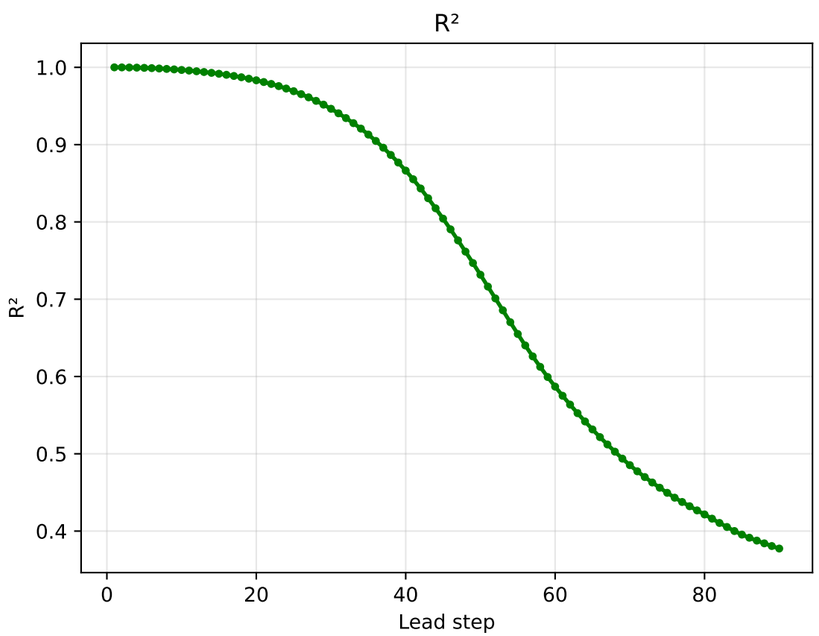
DHW Forecast Skill vs Lead Step - Val

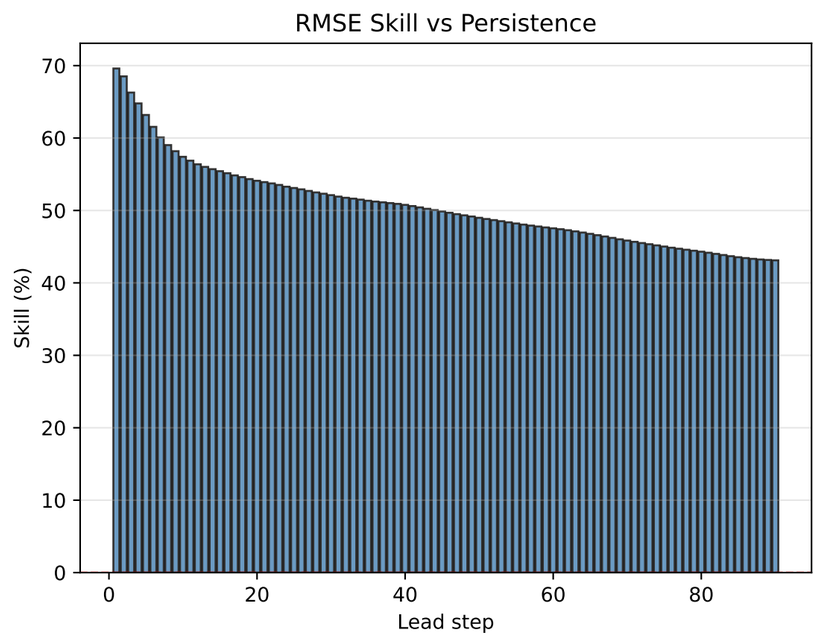


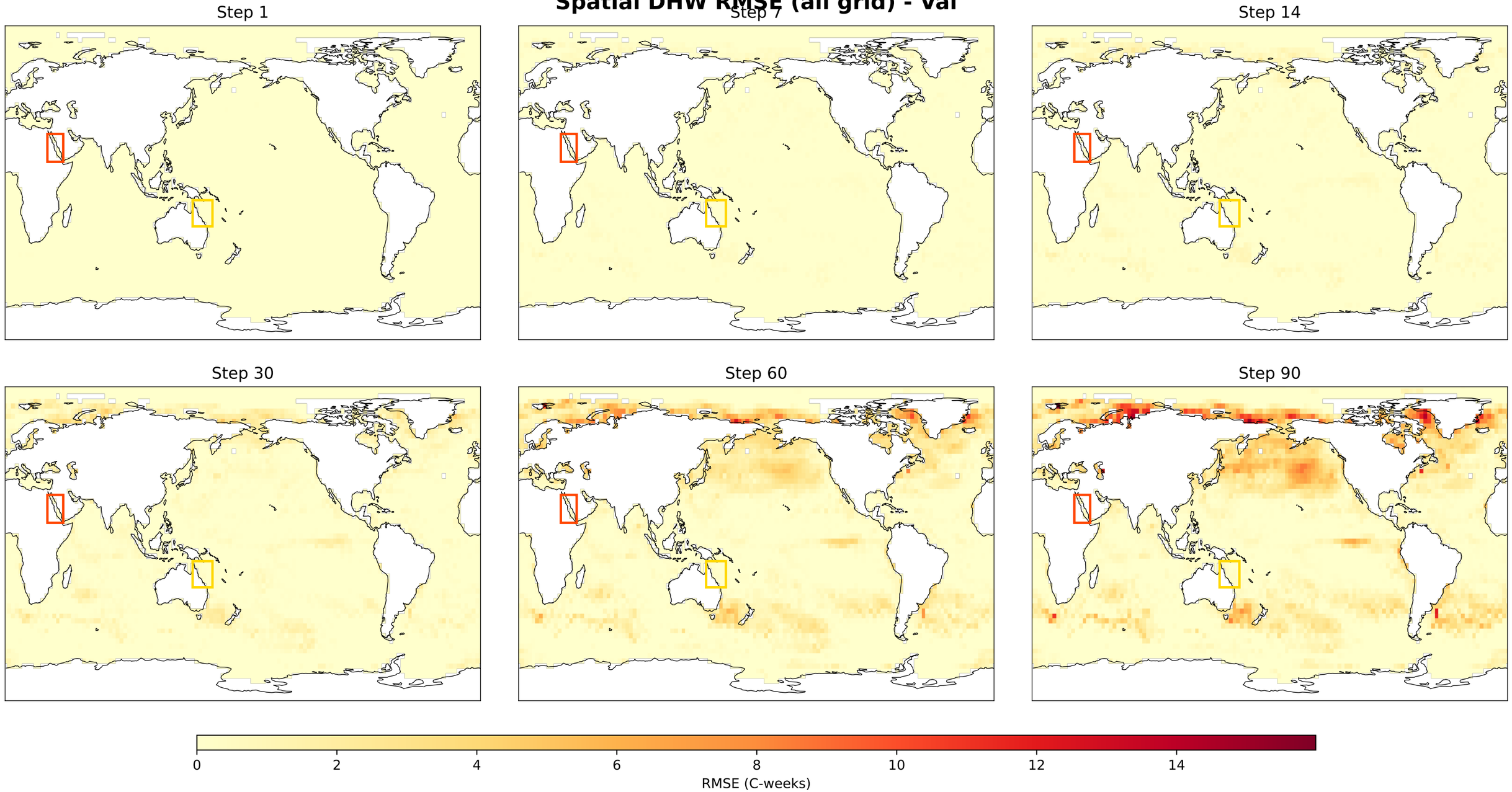


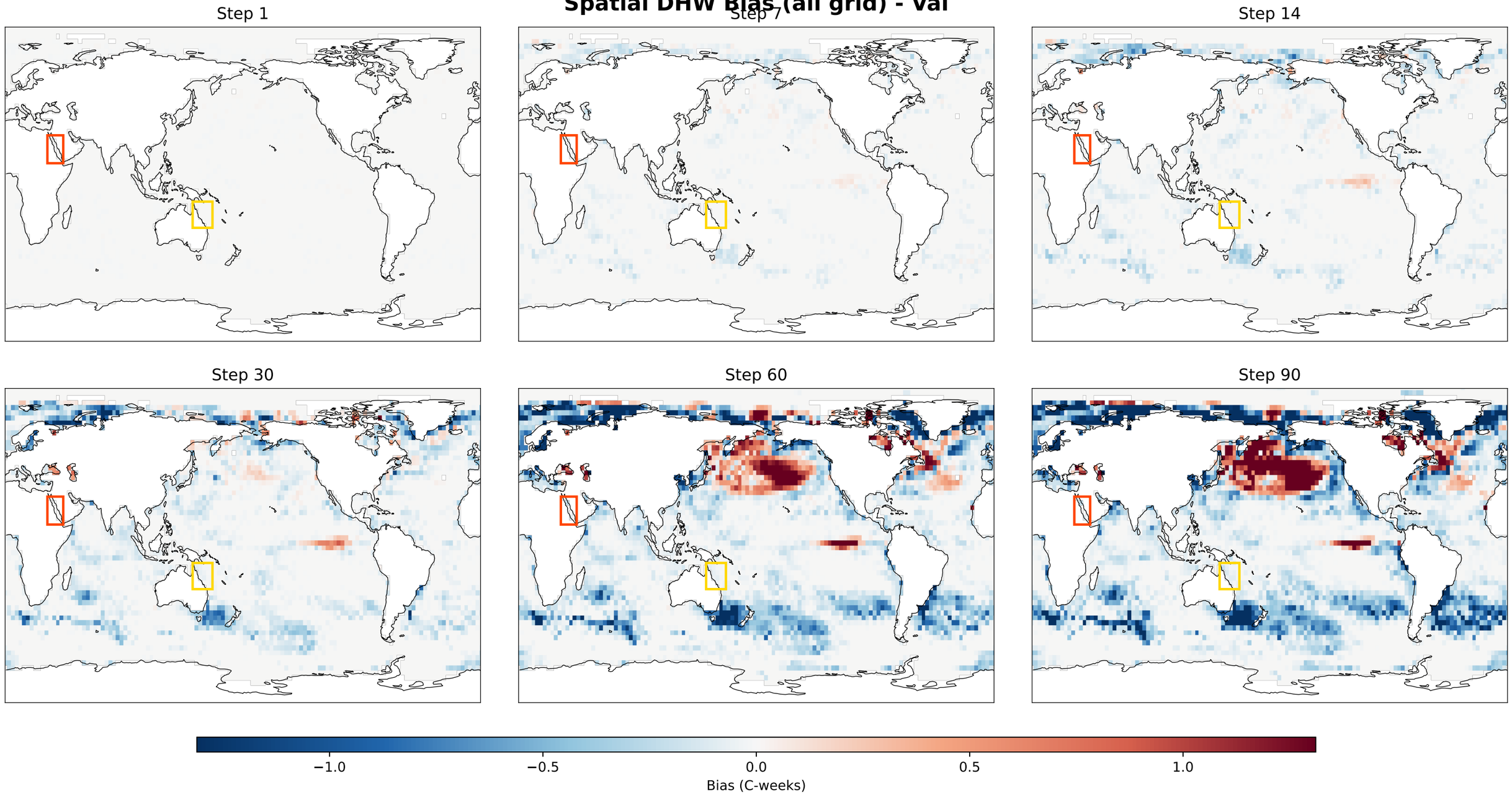


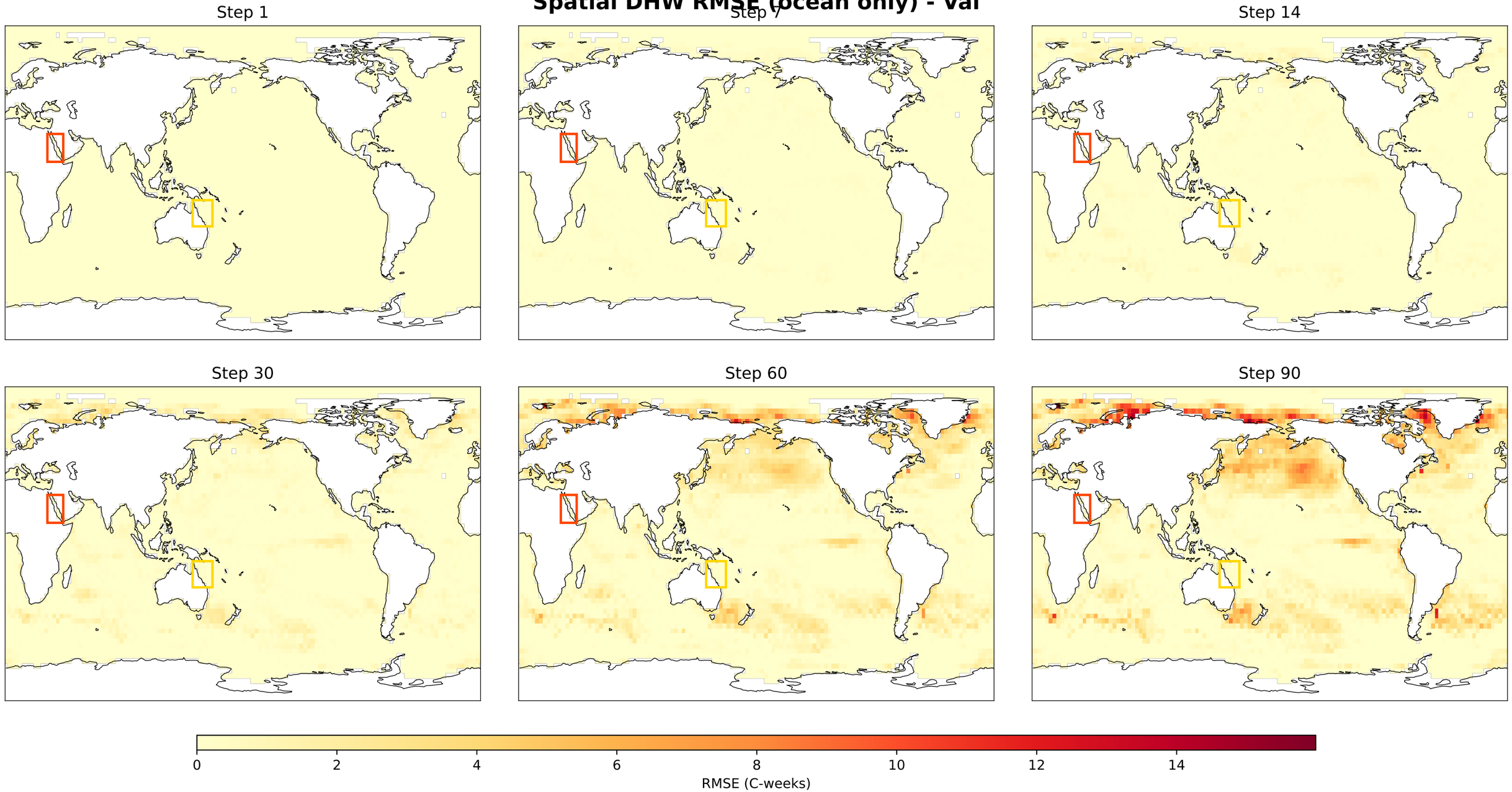


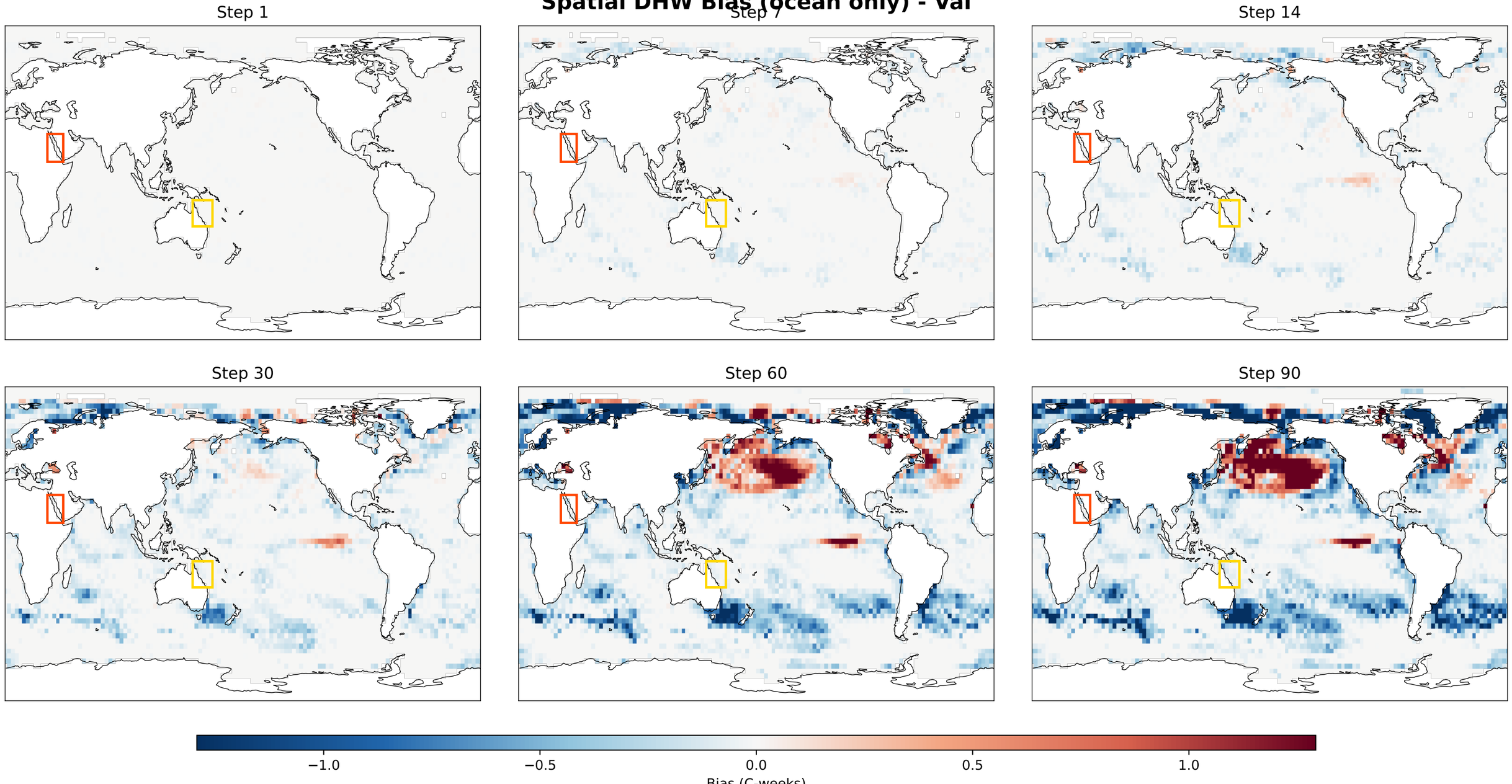





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
RMSE (C-weeks)

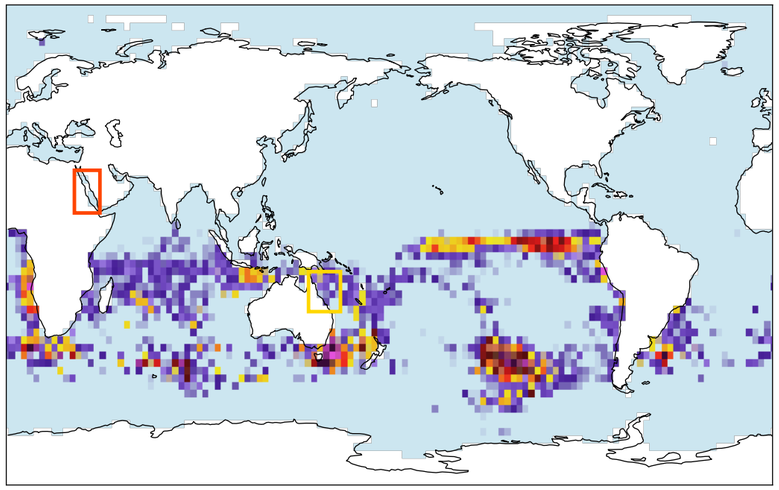

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
RMSE (C-weeks)

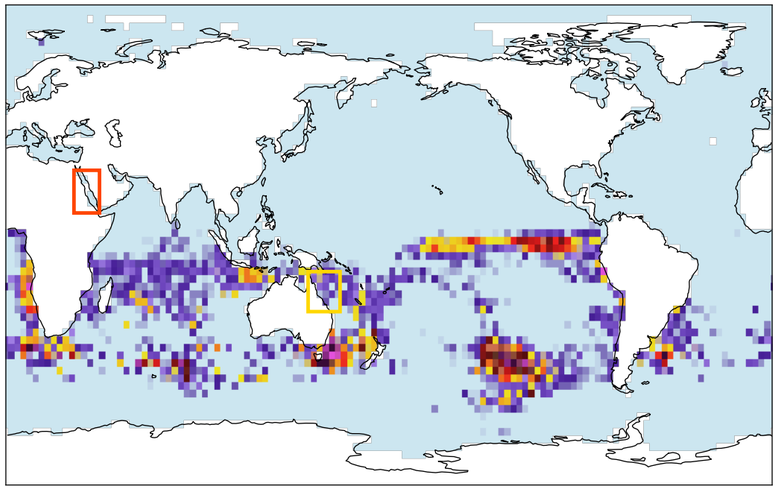

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

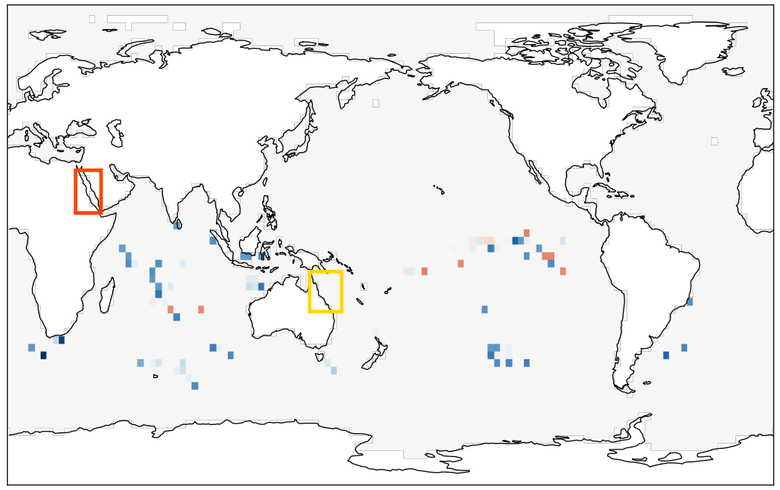
Actual DHW - Step 1



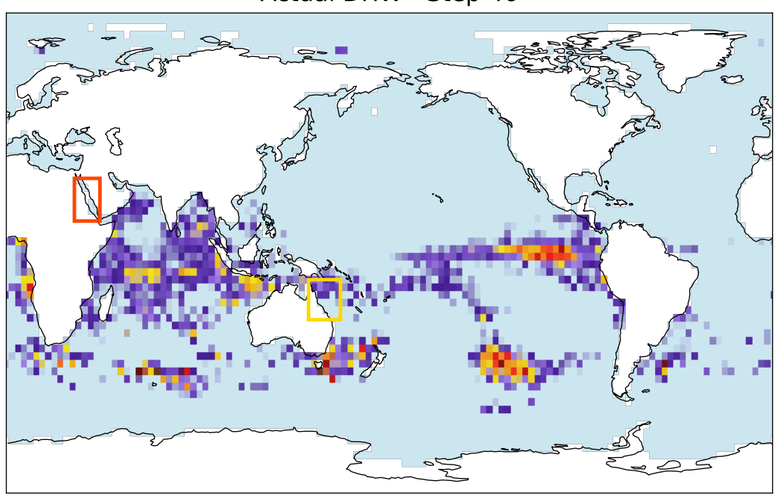
Predicted DHW - Step 1



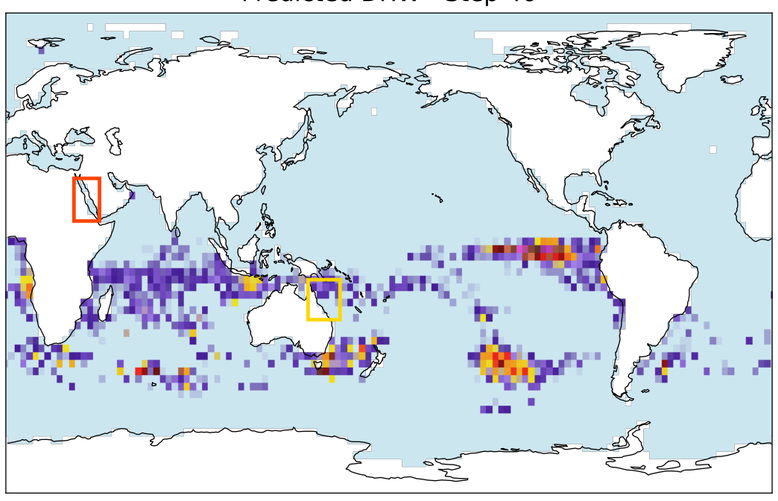
Error (Pred - Actual) - Step 1 [scale ±0.31]



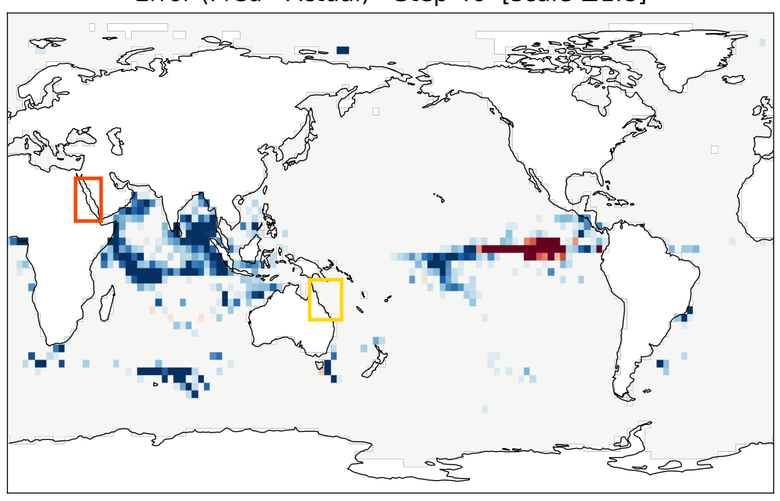
Actual DHW - Step 46



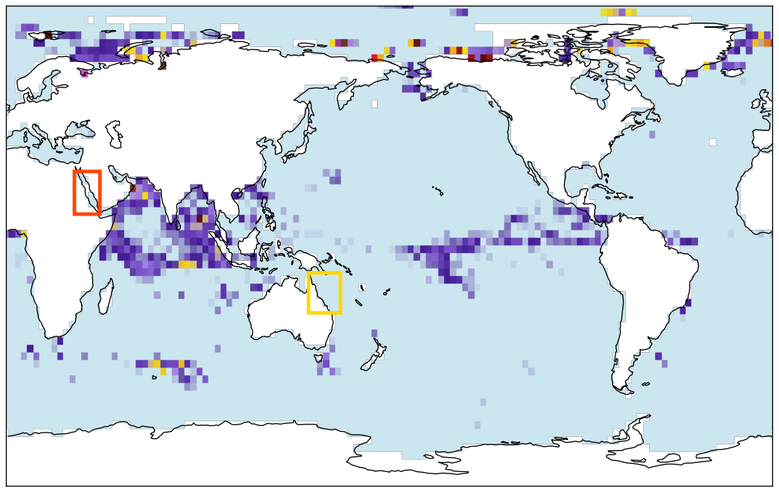
Predicted DHW - Step 46



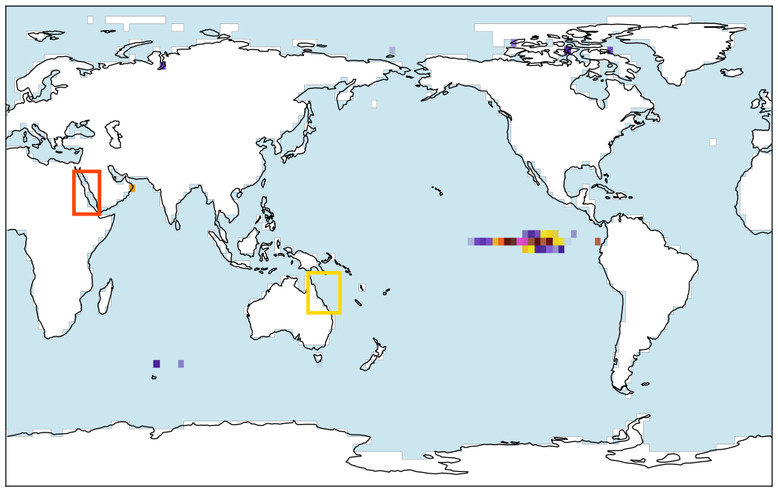
Error (Pred - Actual) - Step 46 [scale ±1.8]



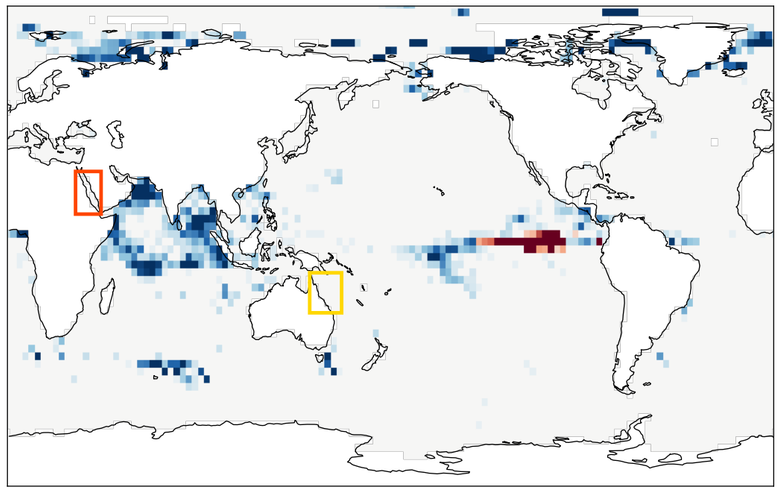
Actual DHW - Step 90

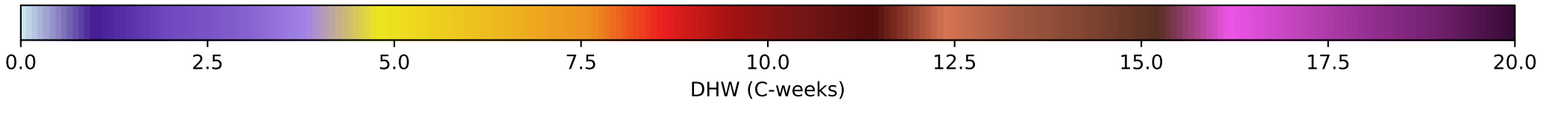


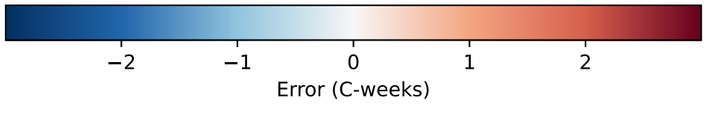
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3]

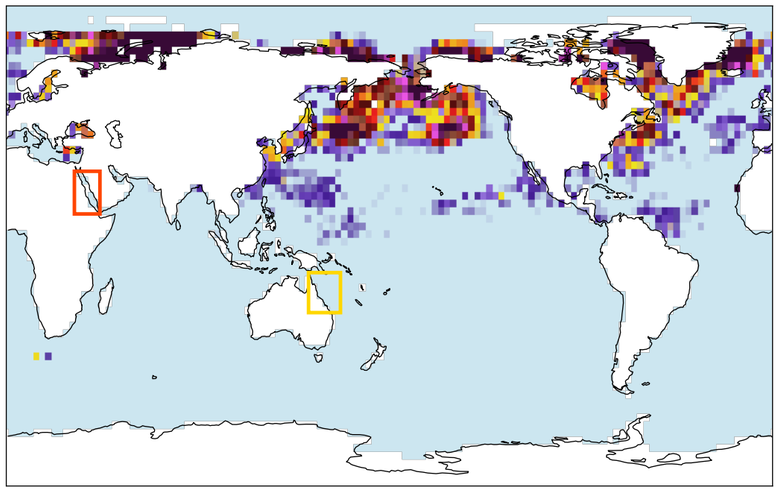




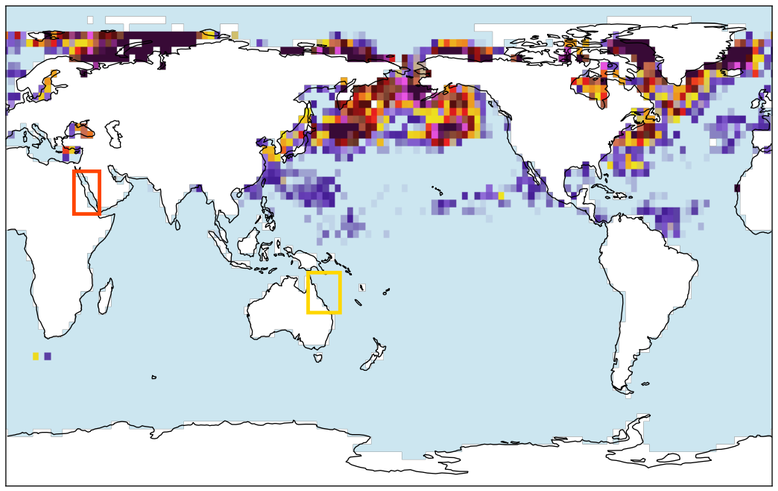


DHW Chain init: 2016-10-07 - Val (C-weeks)

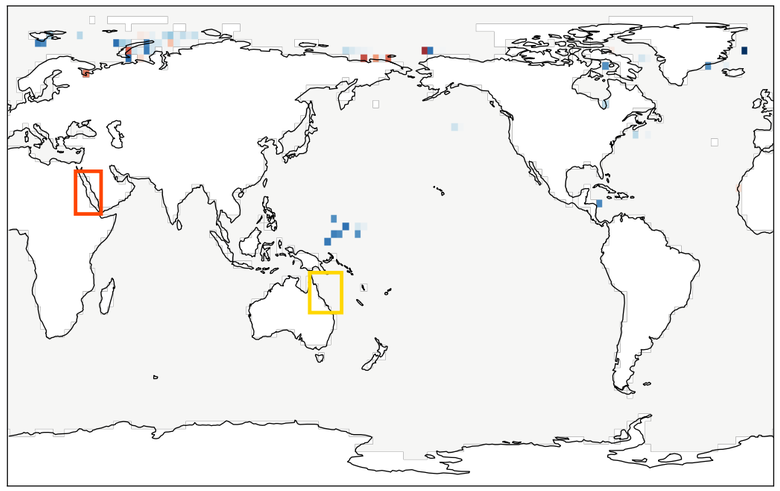
Actual DHW - Step 1



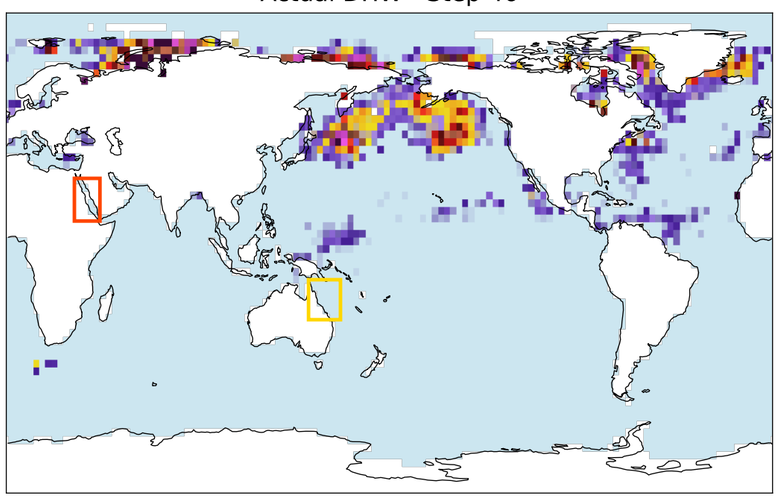
Predicted DHW - Step 1



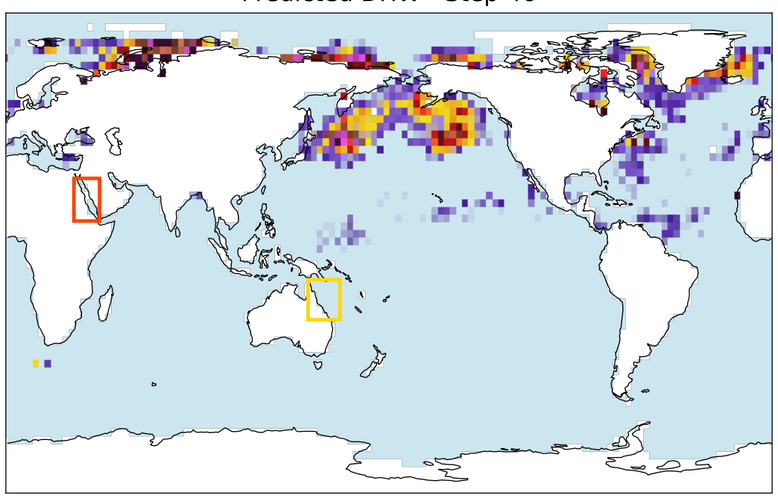
Error (Pred - Actual) - Step 1 [scale ±0.28]



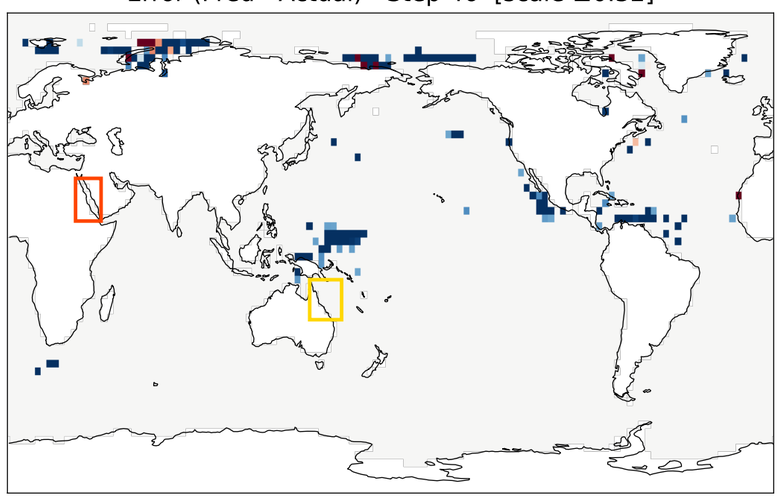
Actual DHW - Step 46



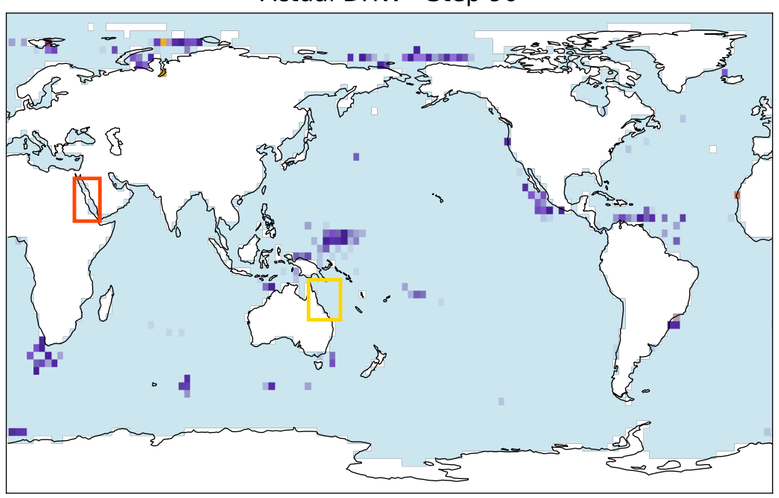
Predicted DHW - Step 46



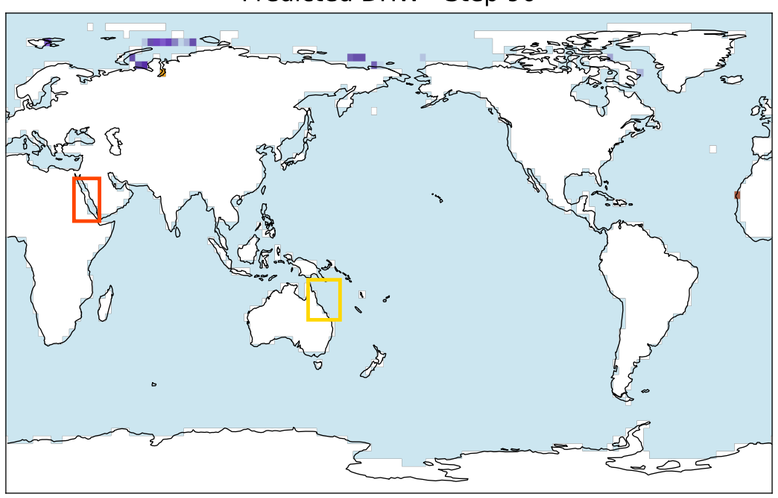
Error (Pred - Actual) - Step 46 [scale ±0.32]



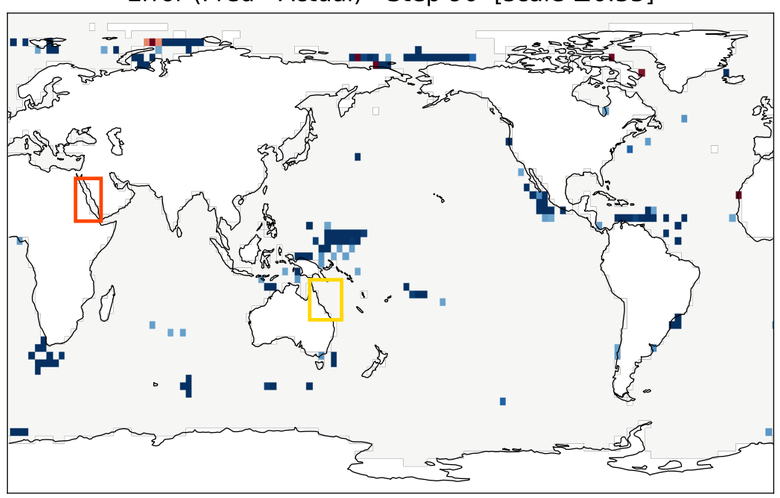
Actual DHW - Step 90

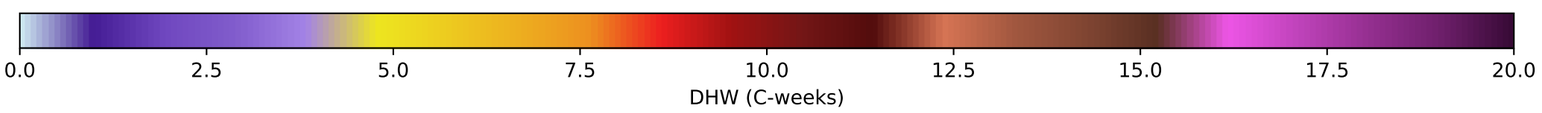


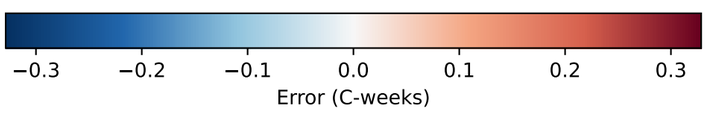
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

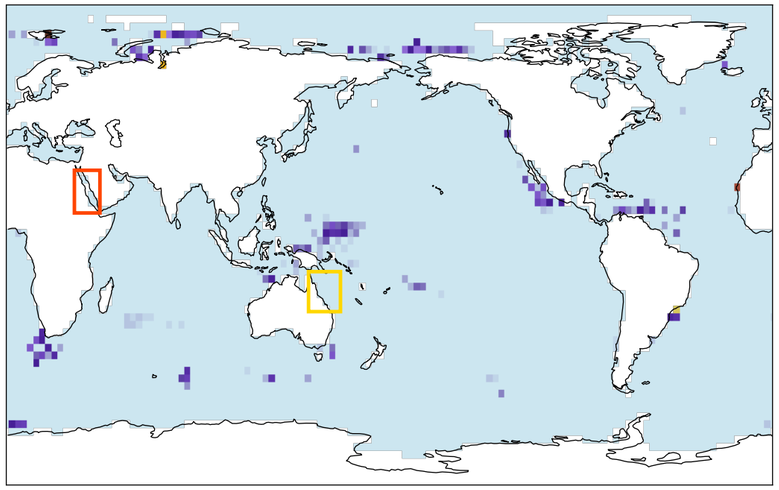




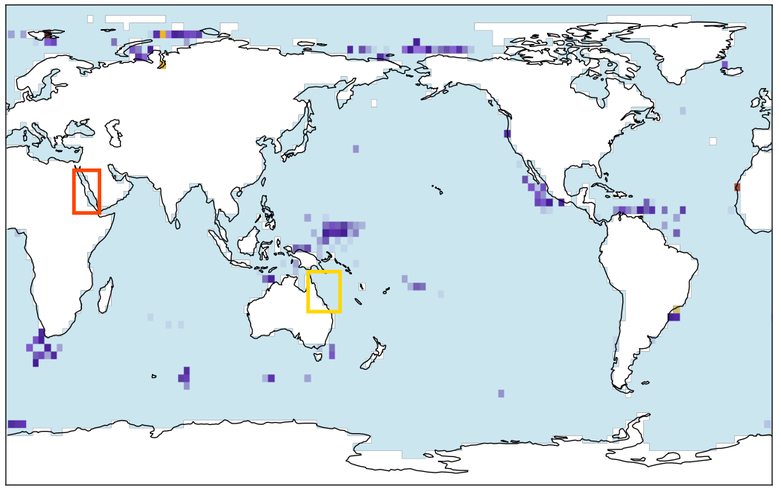


DHW Chain init: 2017-01-05 - Val (C-weeks)

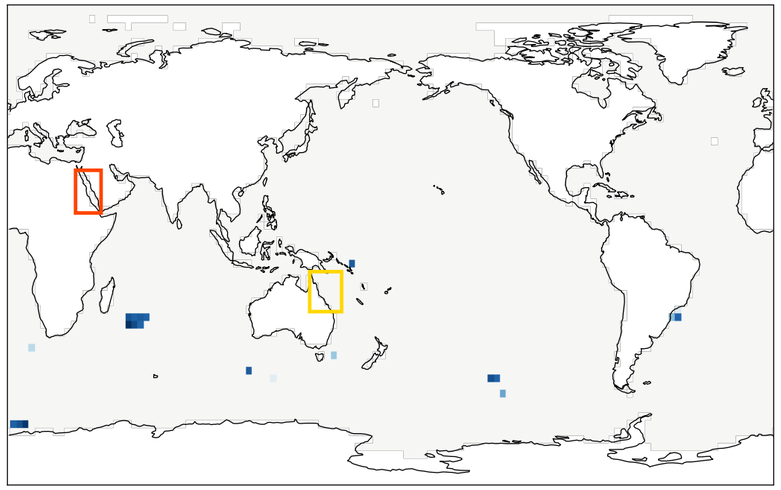
Actual DHW - Step 1



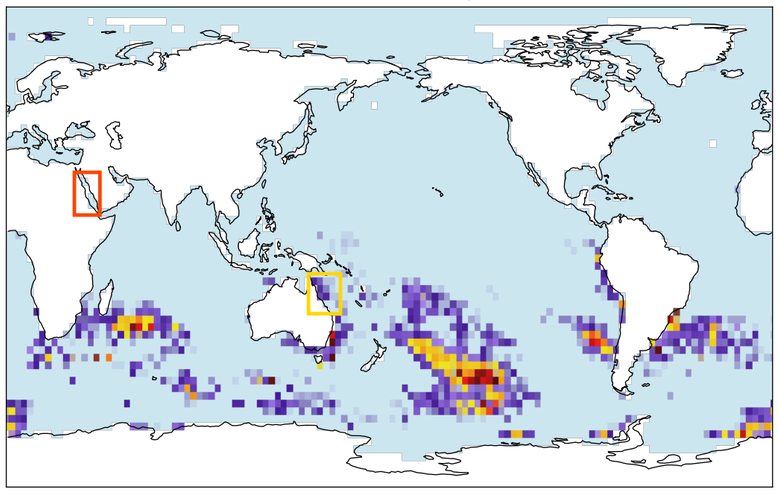
Predicted DHW - Step 1



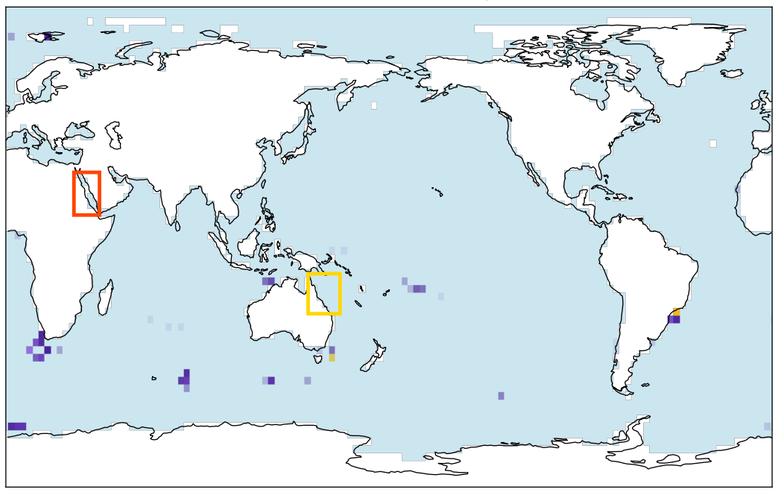
Error (Pred - Actual) - Step 1 [scale ±0.22]



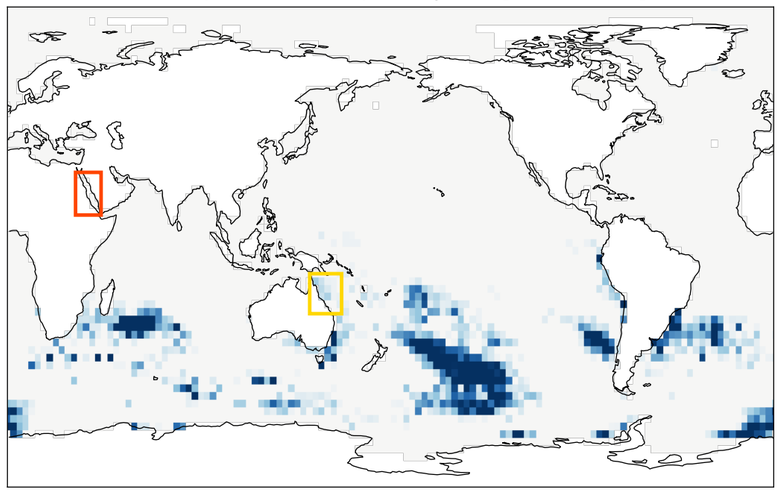
Actual DHW - Step 46



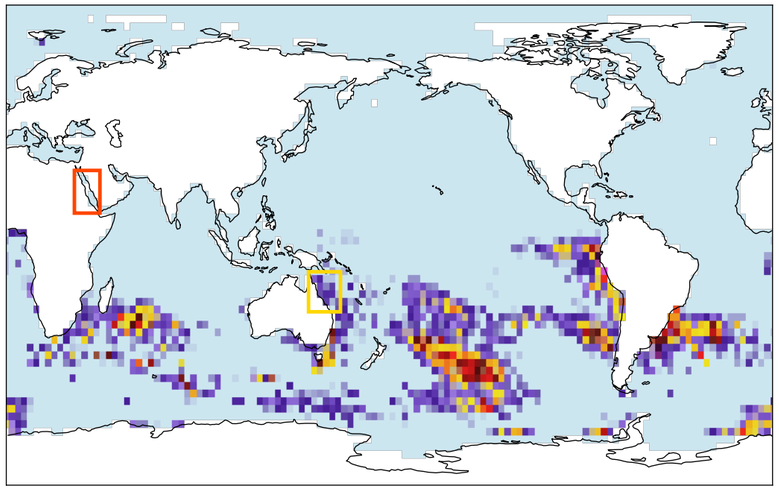
Predicted DHW - Step 46



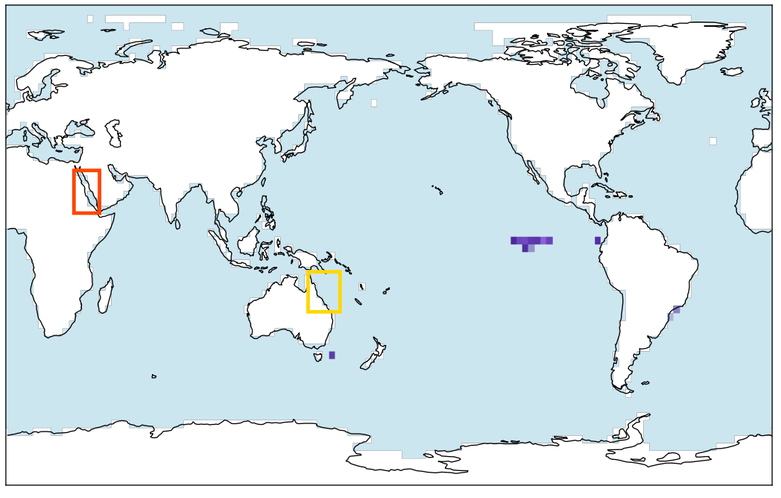
Error (Pred - Actual) - Step 46 [scale ±4.8]



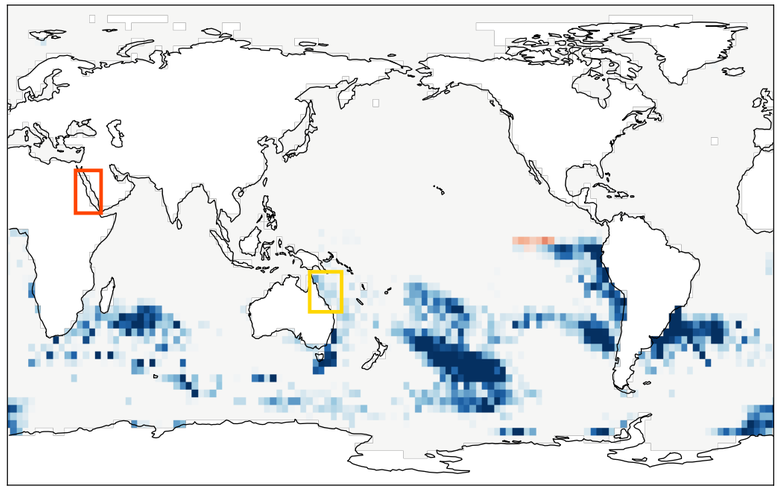
Actual DHW - Step 90

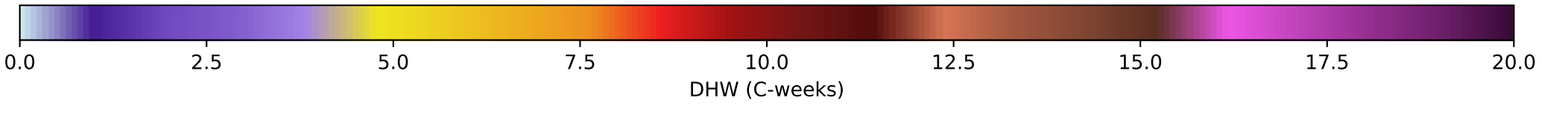


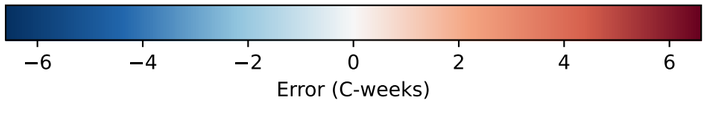
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

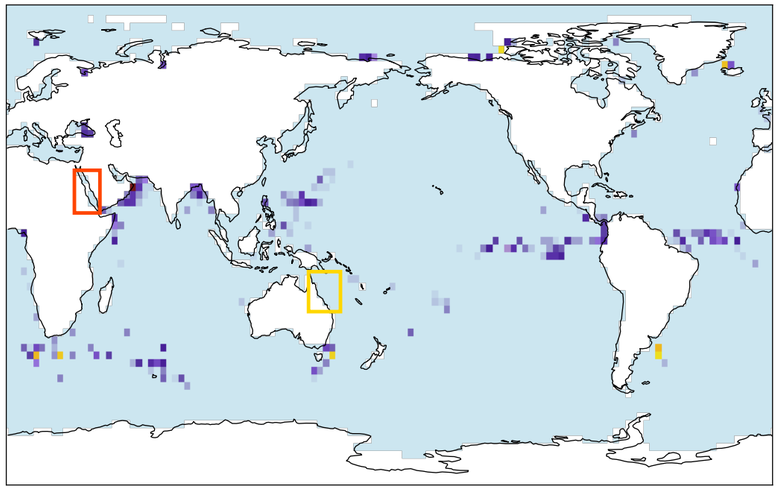




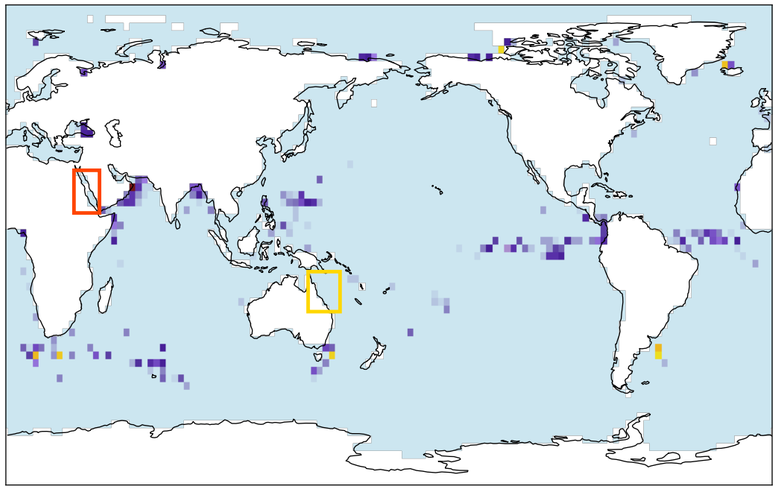


DHW Chain init: 2017-07-04 - Val (C-weeks)

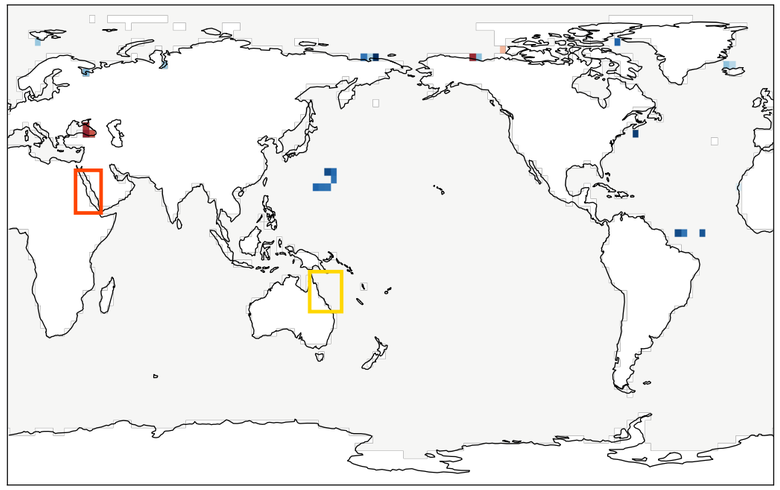
Actual DHW - Step 1



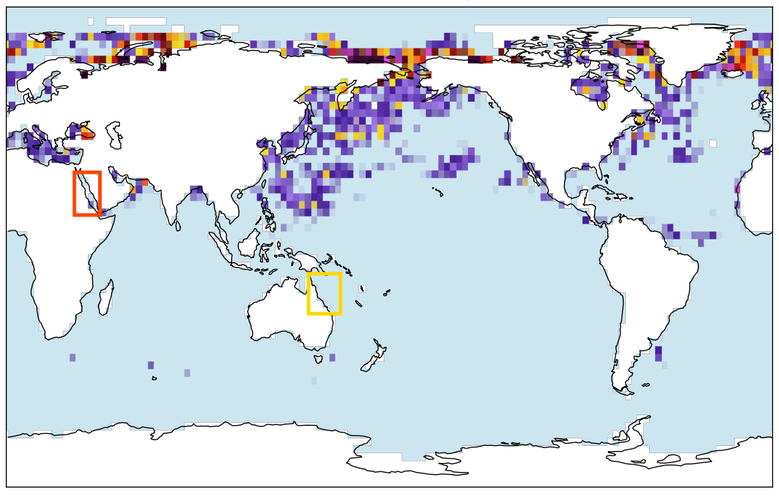
Predicted DHW - Step 1



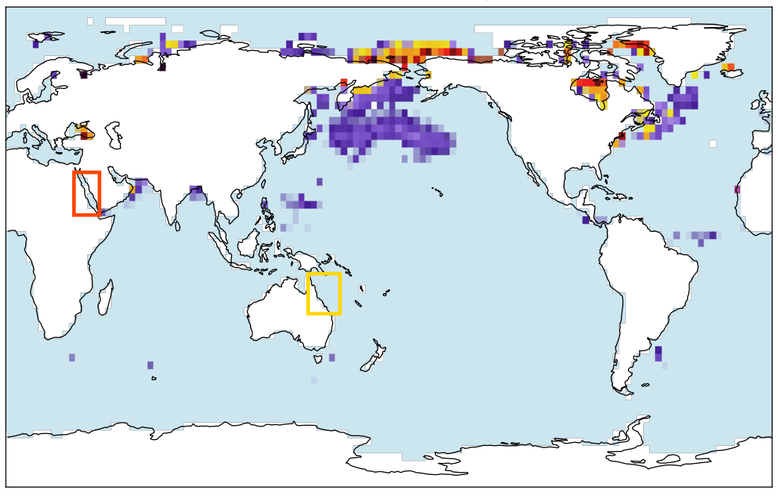
Error (Pred - Actual) - Step 1 [scale ±0.24]



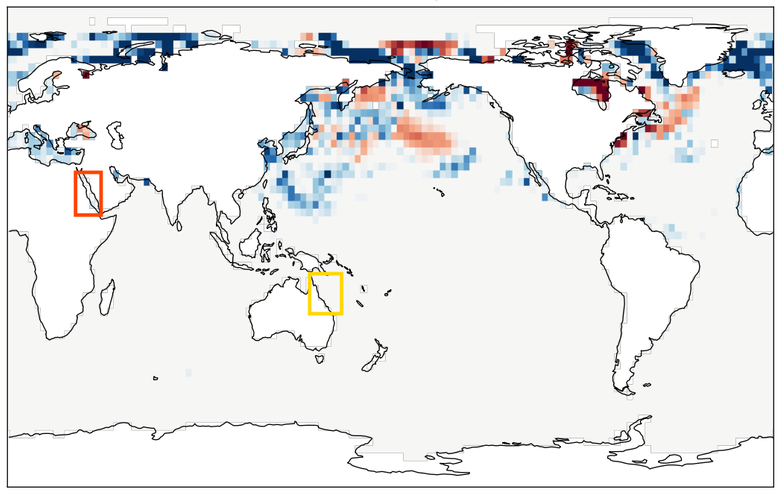
Actual DHW - Step 46



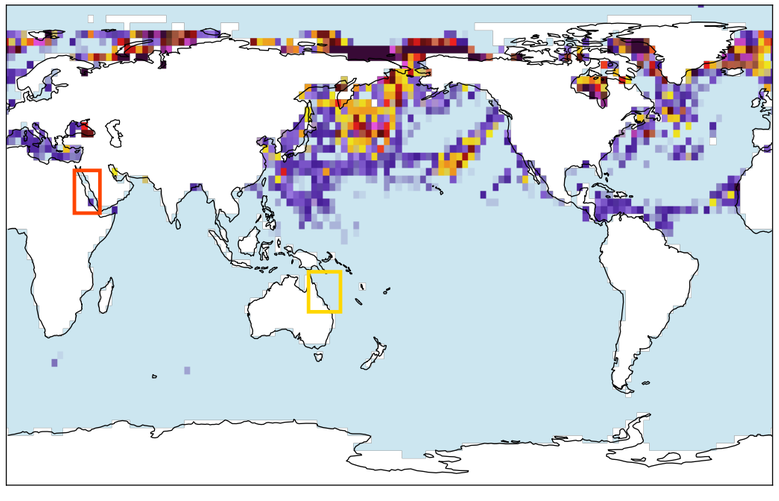
Predicted DHW - Step 46



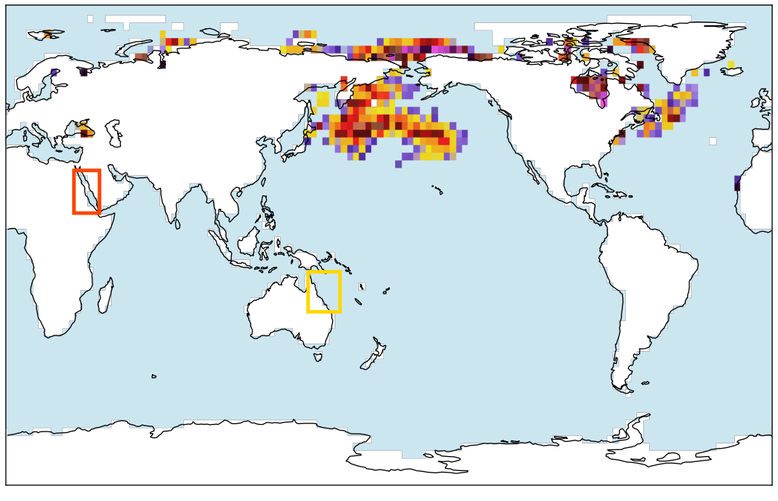
Error (Pred - Actual) - Step 46 [scale ±5.5]



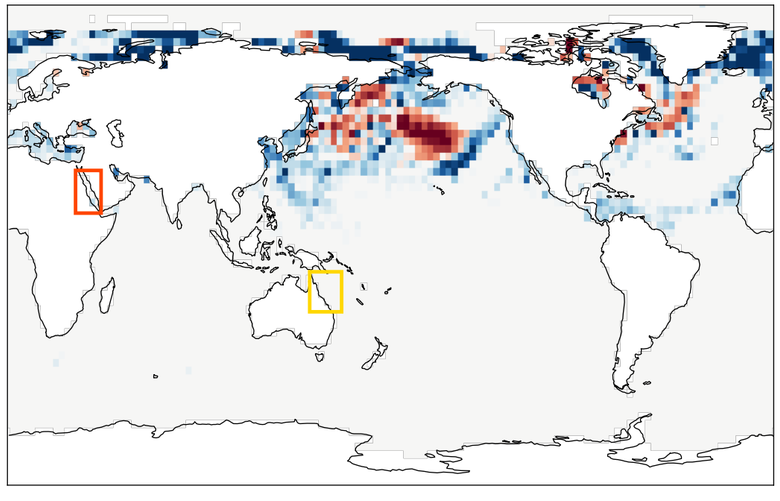
Actual DHW - Step 90

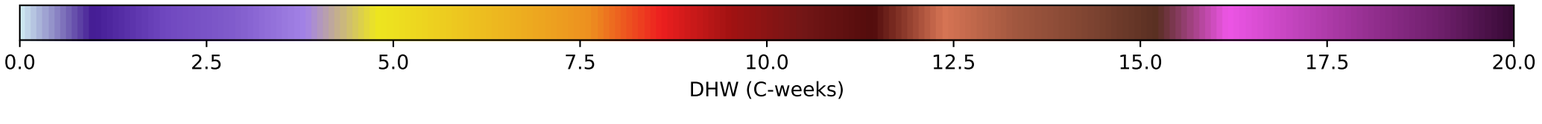


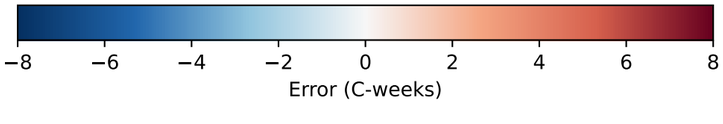
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

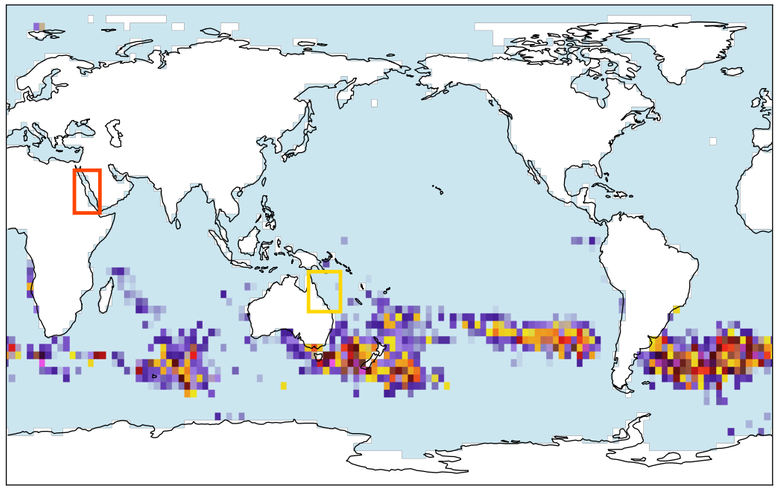




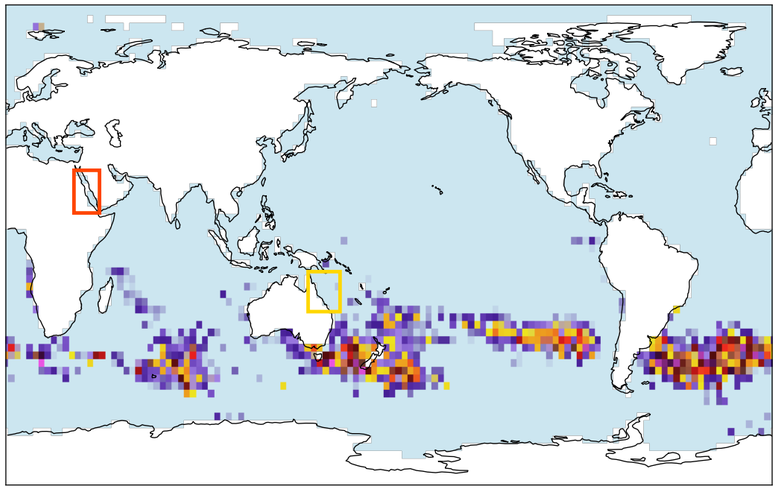


DHW Chain init: 2018-03-31 - Val (C-weeks)

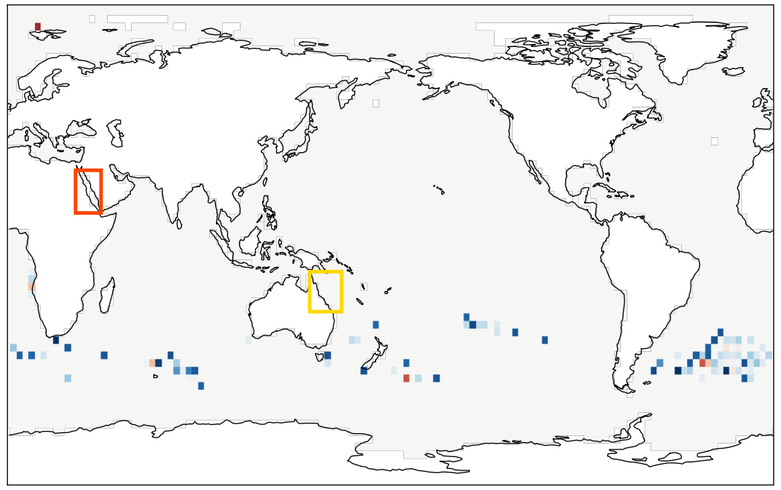
Actual DHW - Step 1



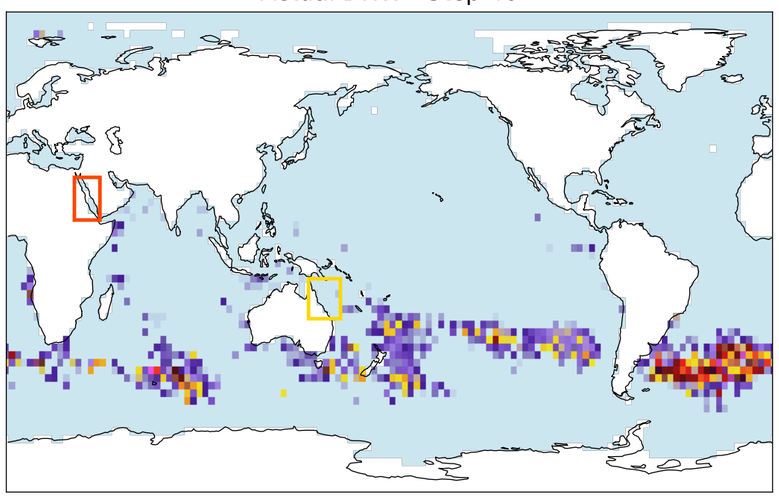
Predicted DHW - Step 1



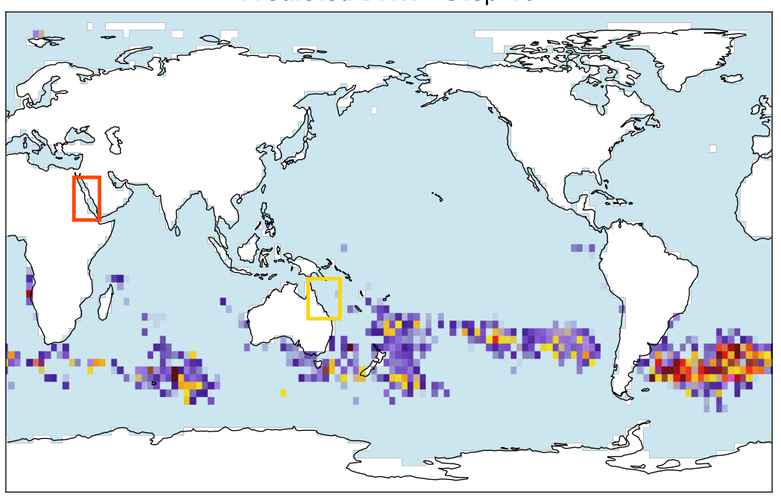
Error (Pred - Actual) - Step 1 [scale ±0.21]



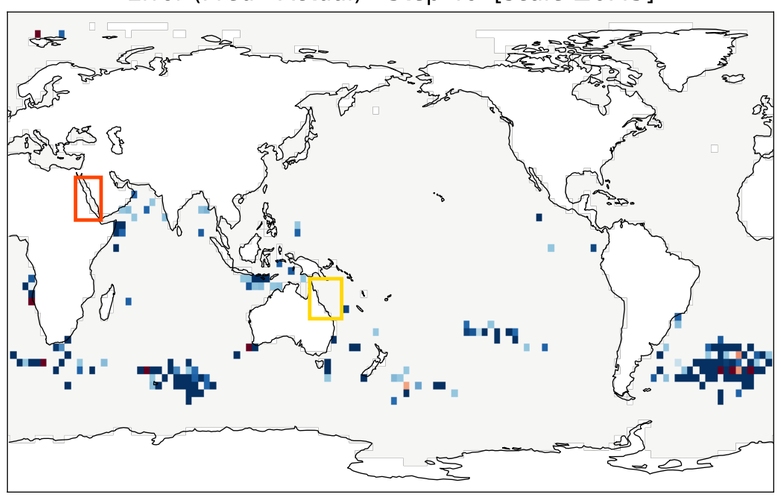
Actual DHW - Step 46



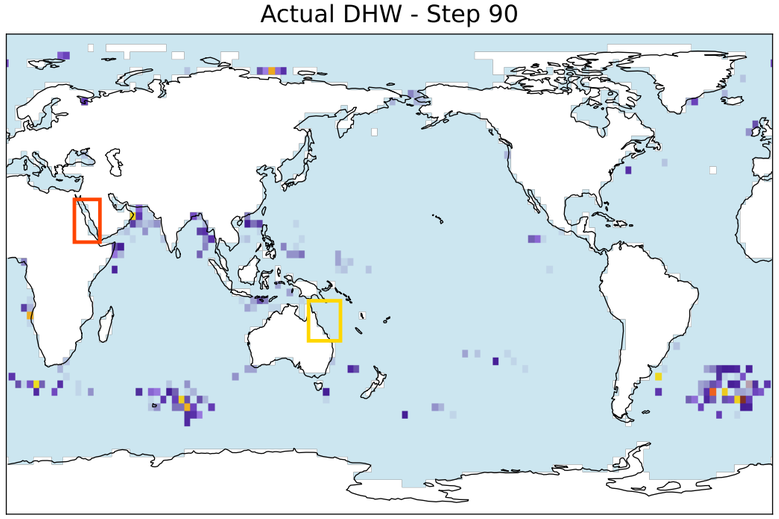
Predicted DHW - Step 46



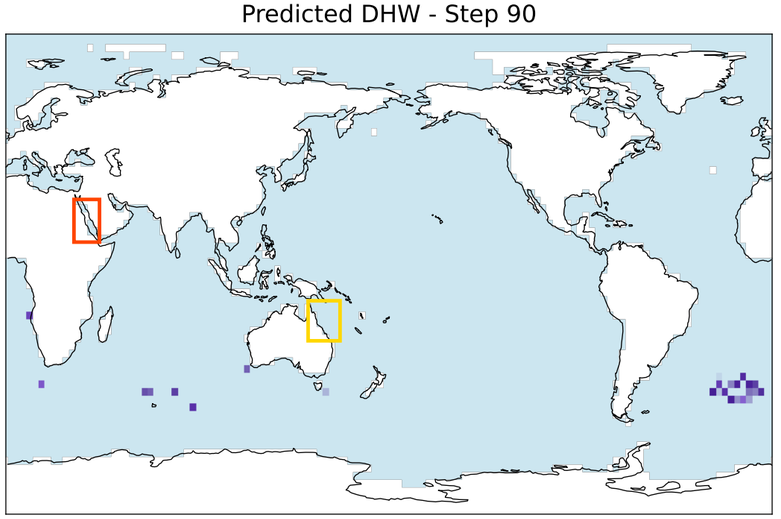
Error (Pred - Actual) - Step 46 [scale ±0.45]



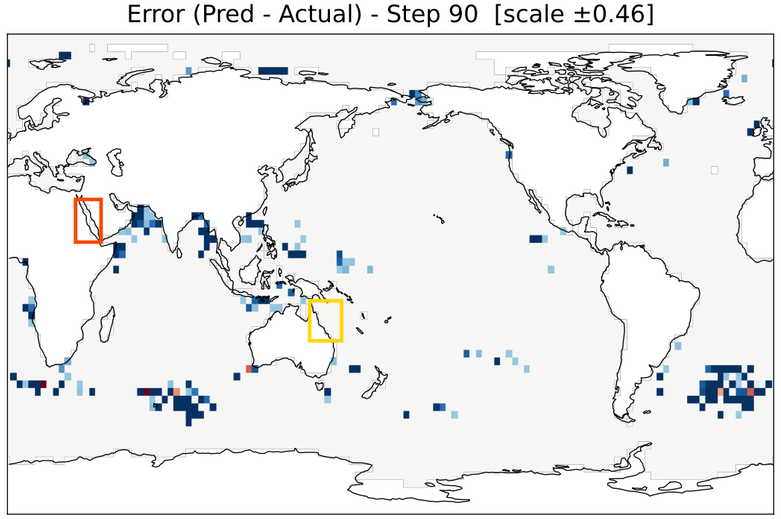
Actual DHW - Step 90

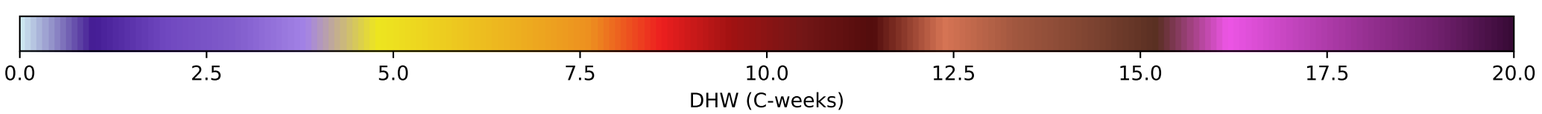


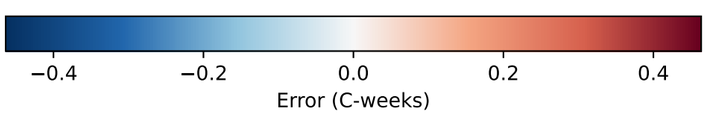
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.46]

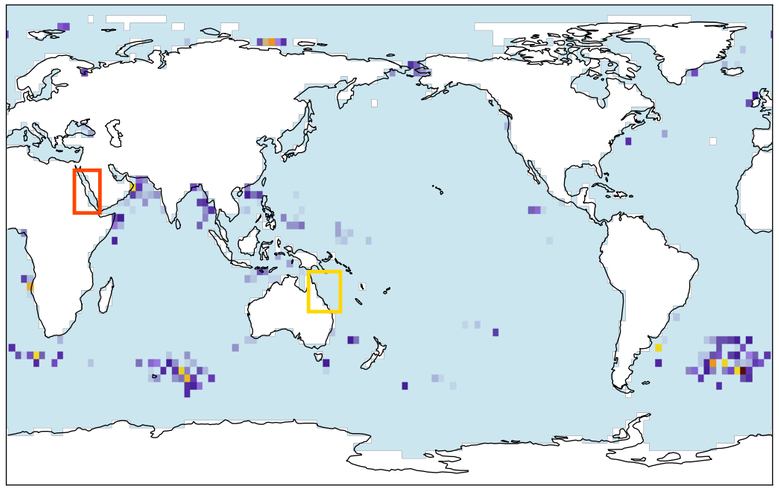




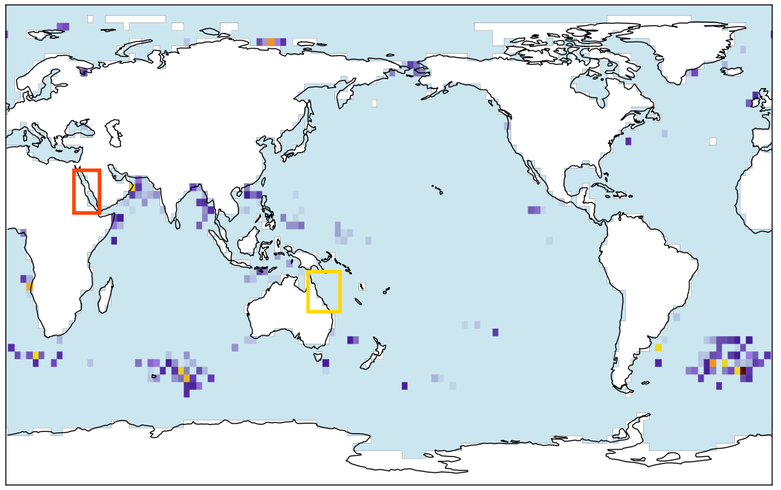


DHW Chain init: 2018-06-29 - Val (C-weeks)

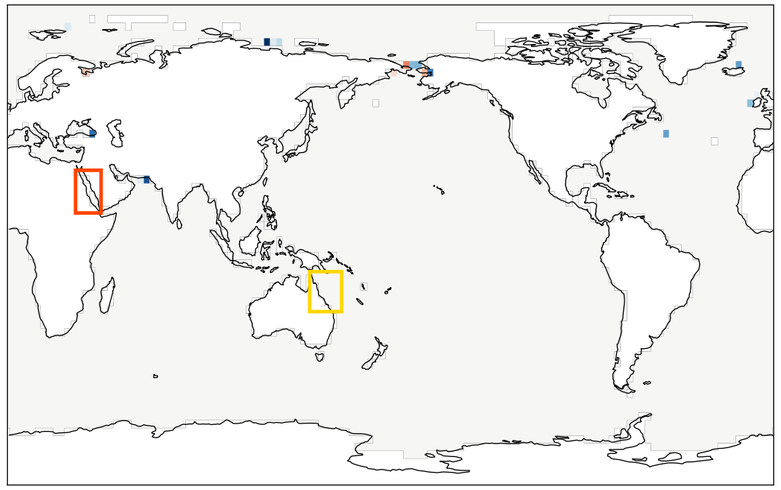
Actual DHW - Step 1



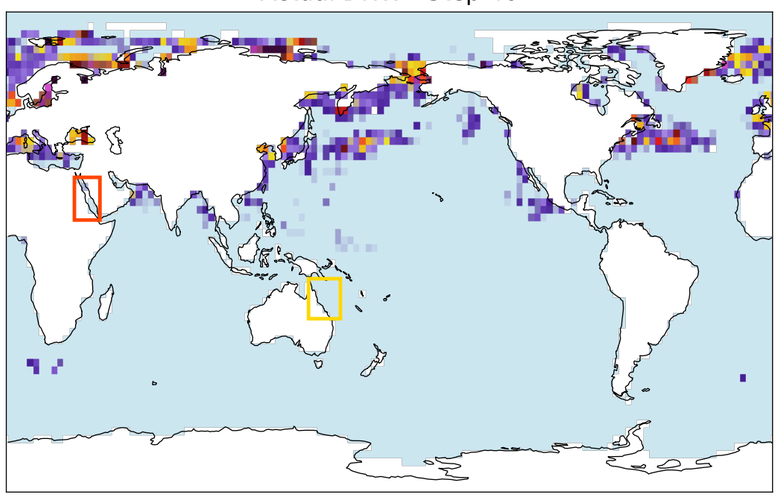
Predicted DHW - Step 1



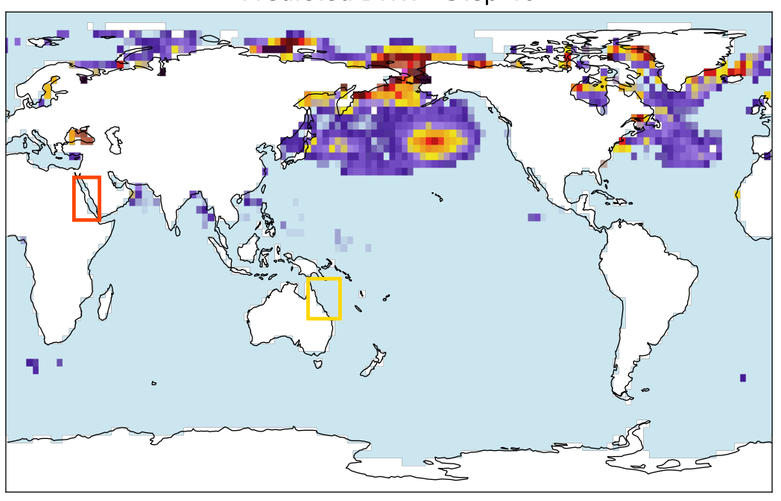
Error (Pred - Actual) - Step 1 [scale ±0.31]



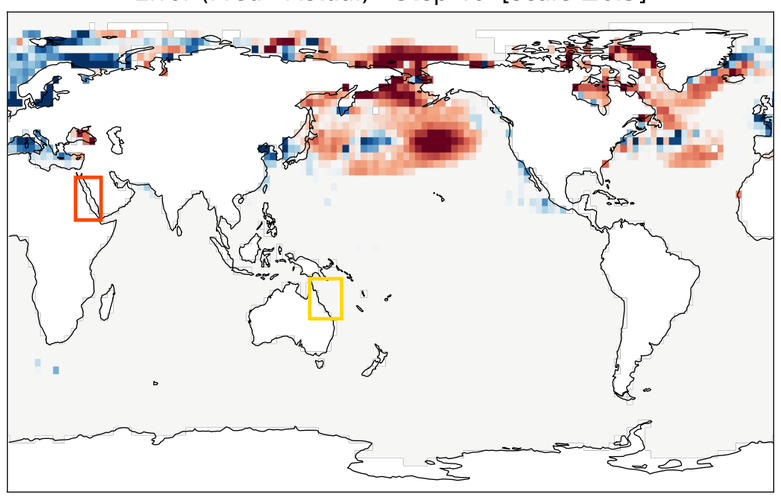
Actual DHW - Step 46



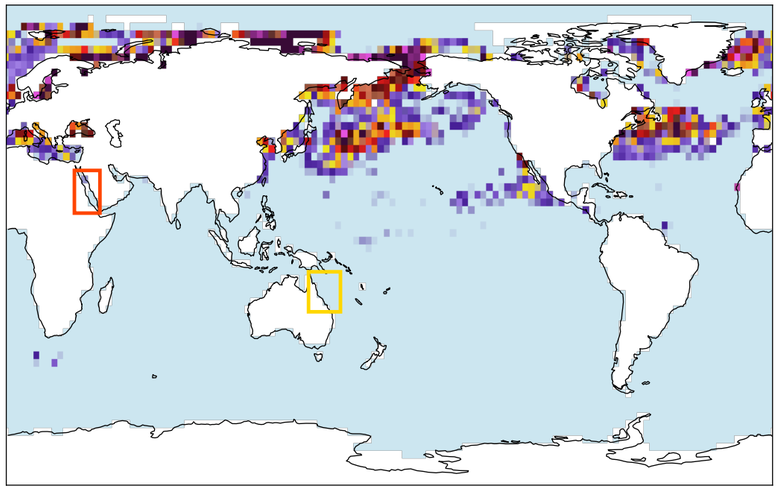
Predicted DHW - Step 46



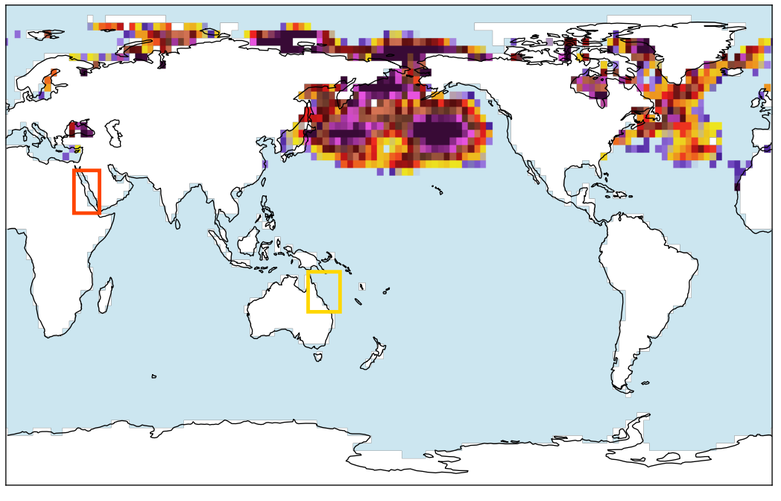
Error (Pred - Actual) - Step 46 [scale ±6.5]



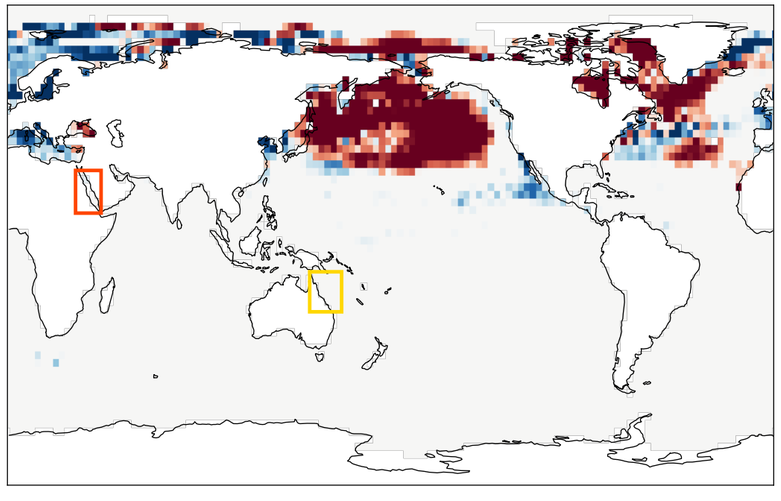
Actual DHW - Step 90

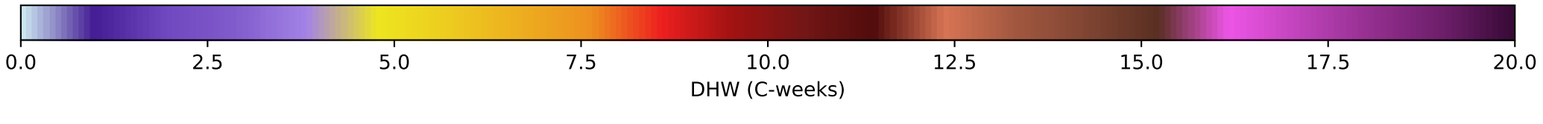


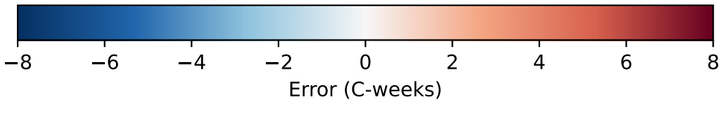
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

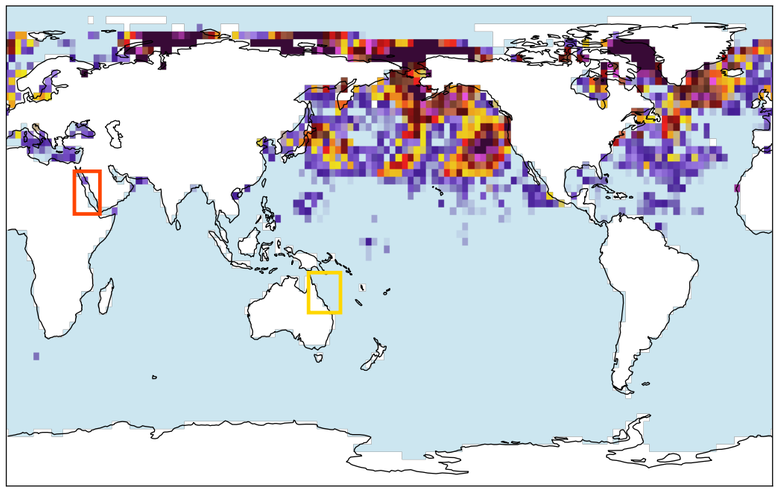




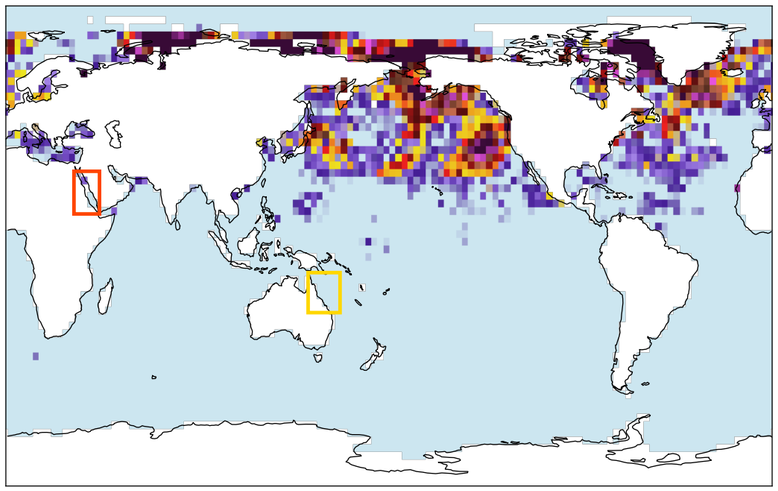


DHW Chain init: 2019-09-22 - Val (C-weeks)

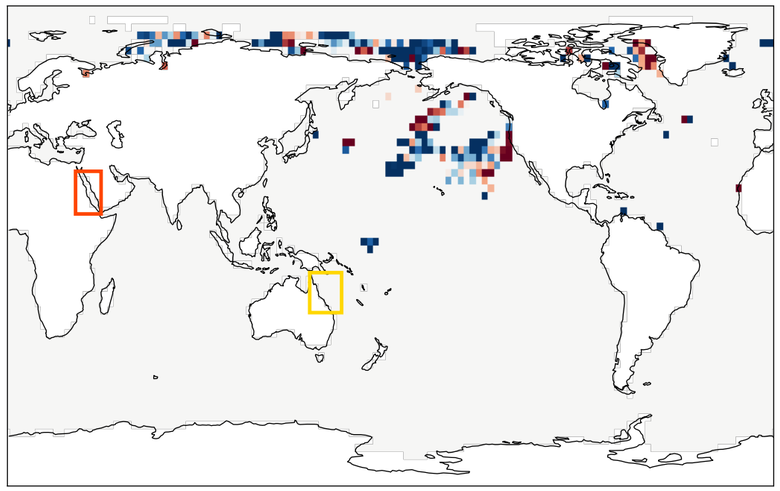
Actual DHW - Step 1



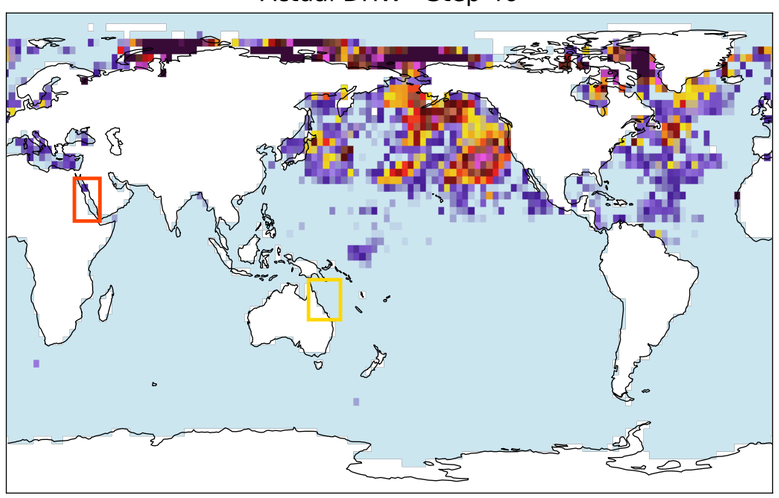
Predicted DHW - Step 1



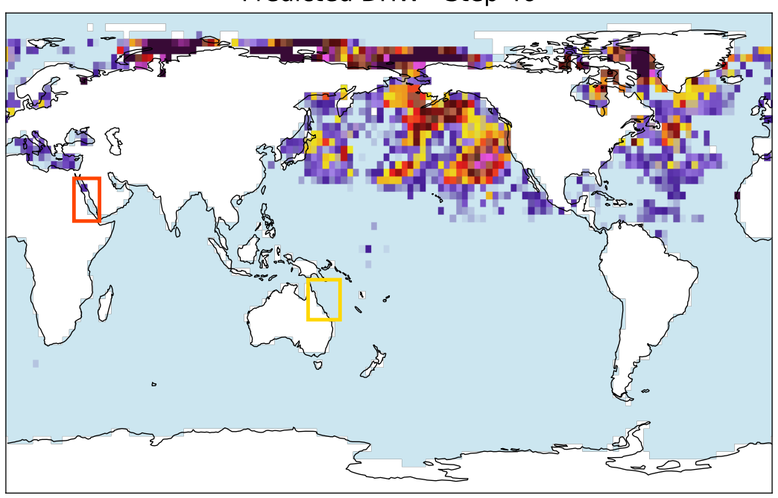
Error (Pred - Actual) - Step 1 [scale ±0.031]



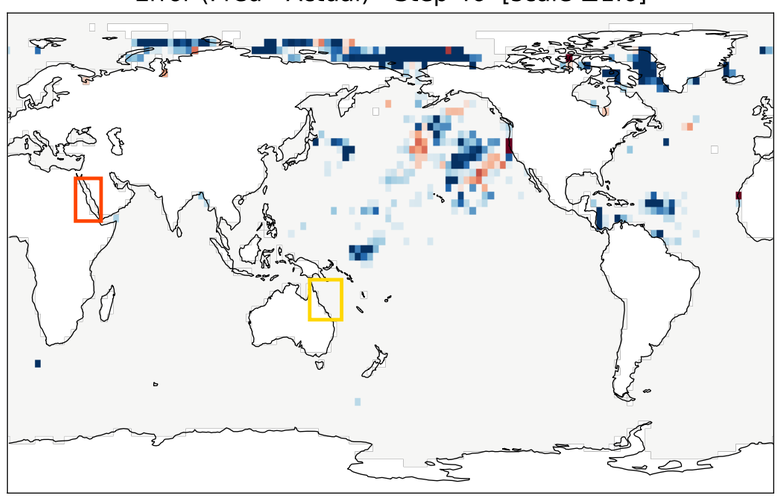
Actual DHW - Step 46



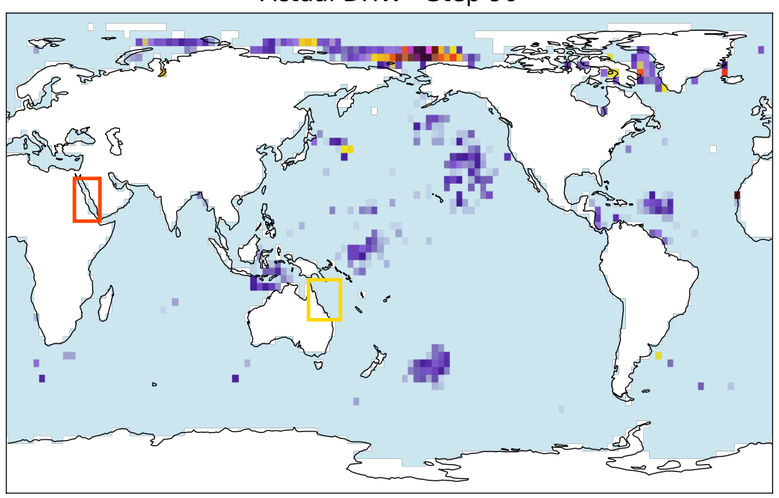
Predicted DHW - Step 46



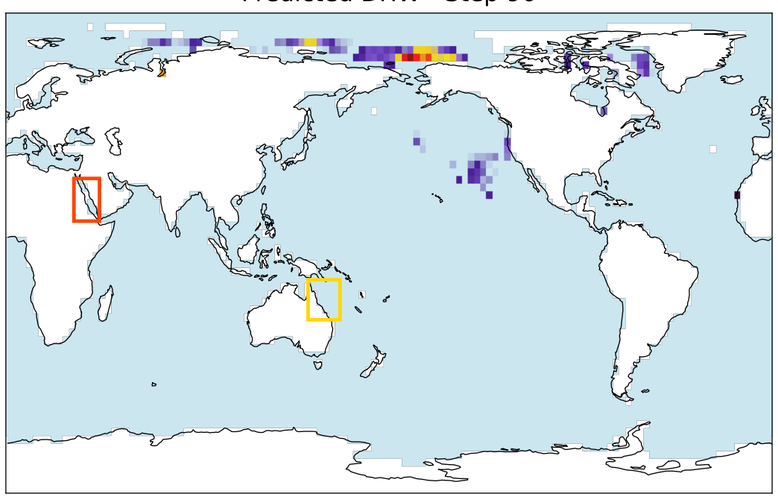
Error (Pred - Actual) - Step 46 [scale ±1.6]



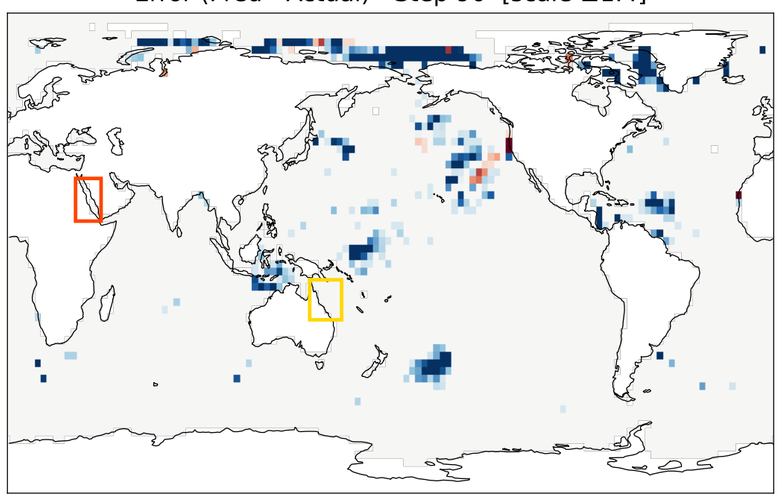
Actual DHW - Step 90

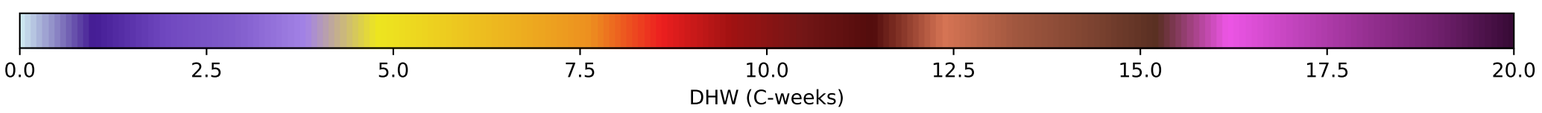


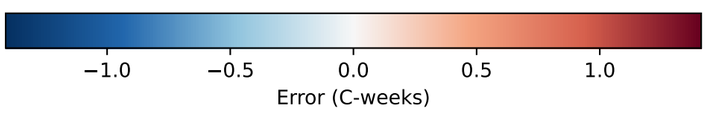
Predicted DHW - Step 90



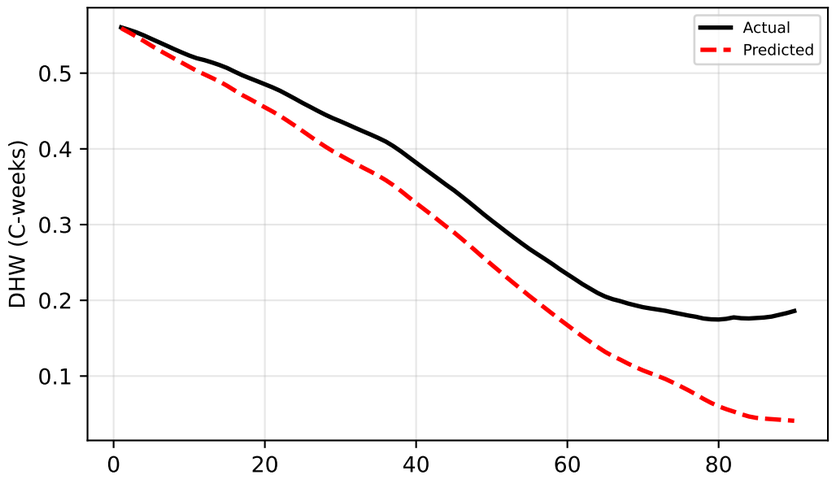
Error (Pred - Actual) - Step 90 [scale ±1.4]

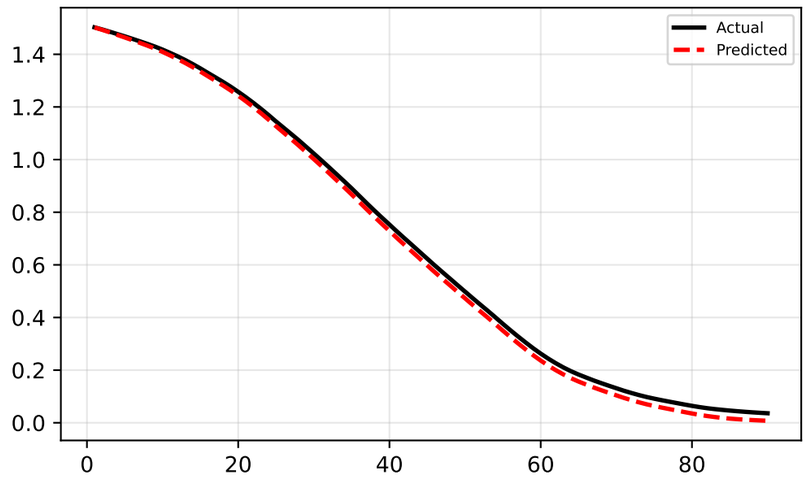


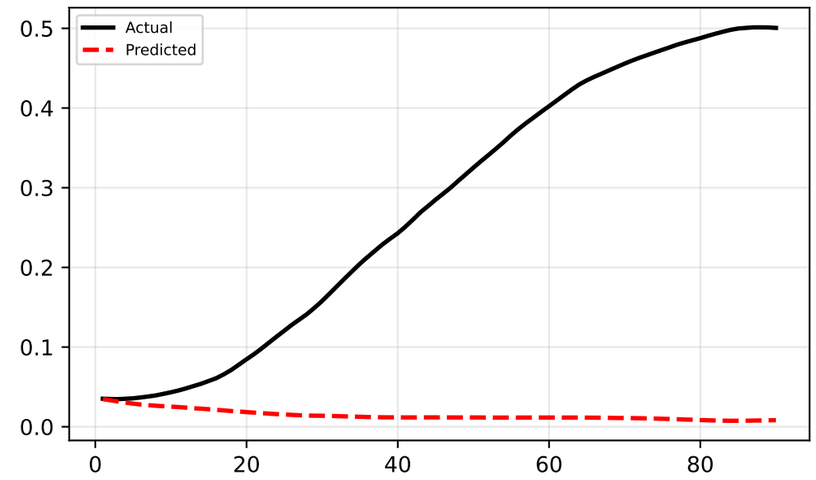


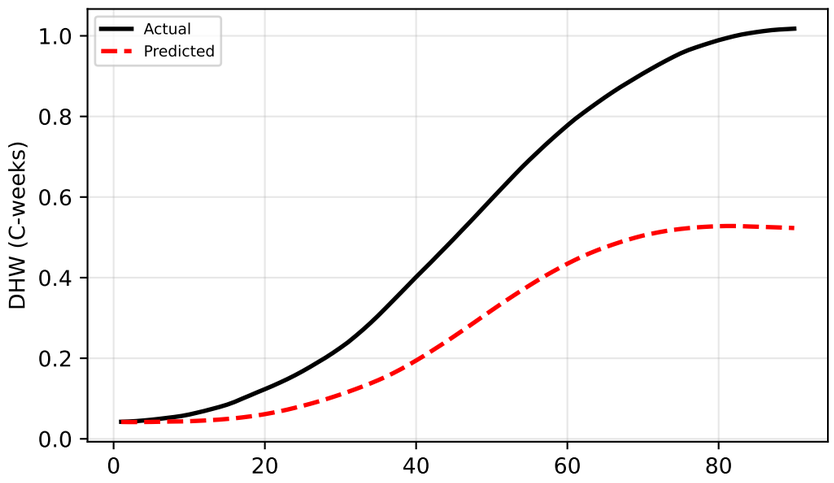


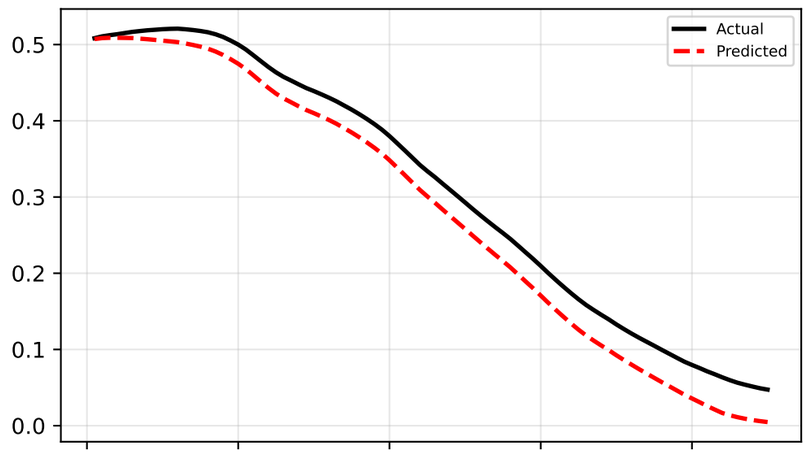
Global-Mean DHW - Val (C-weeks)

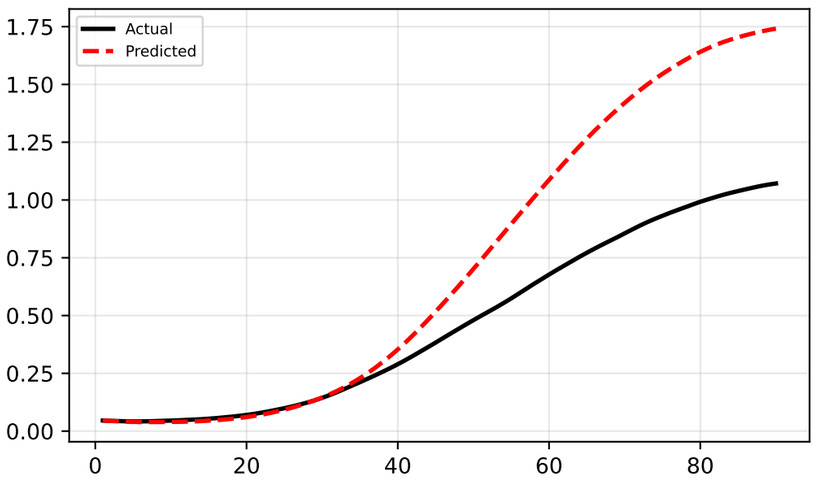


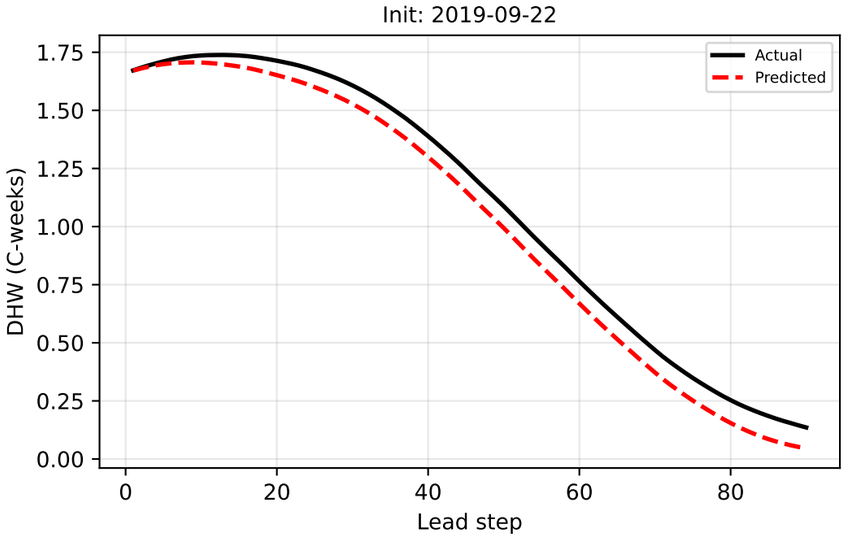




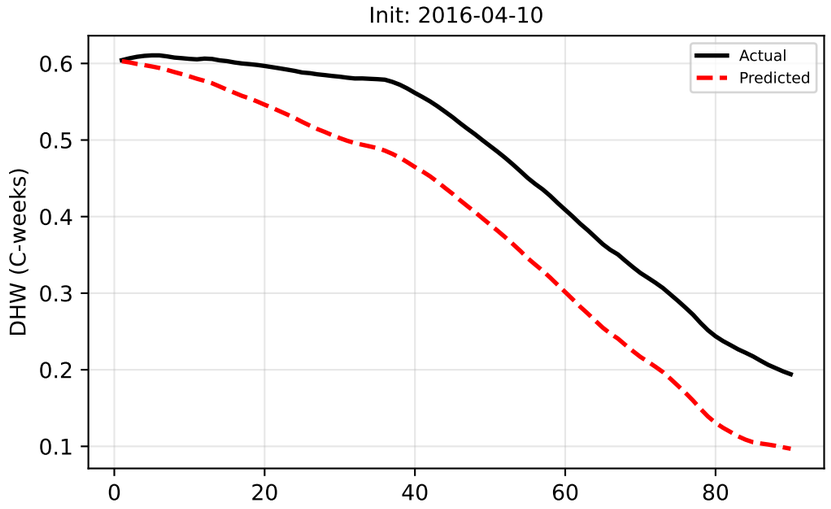


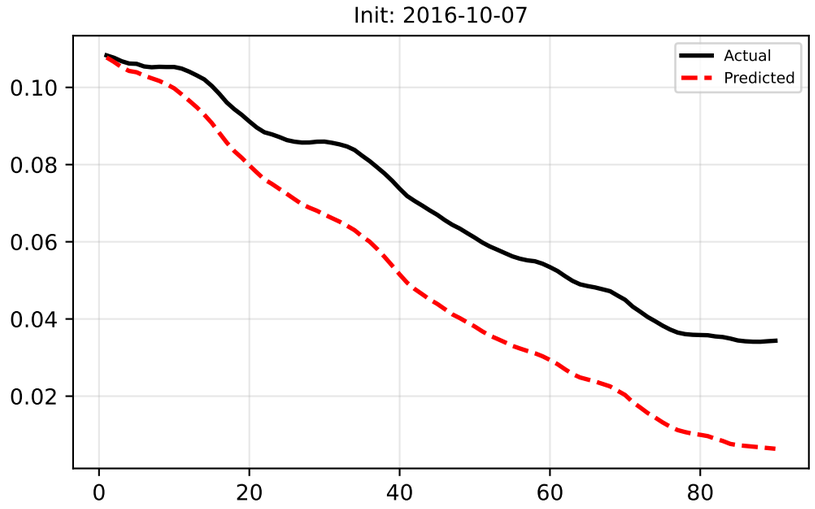


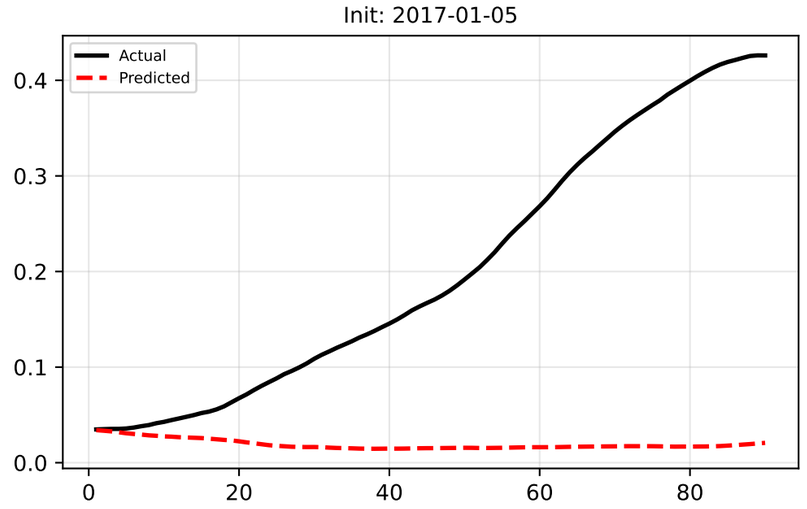


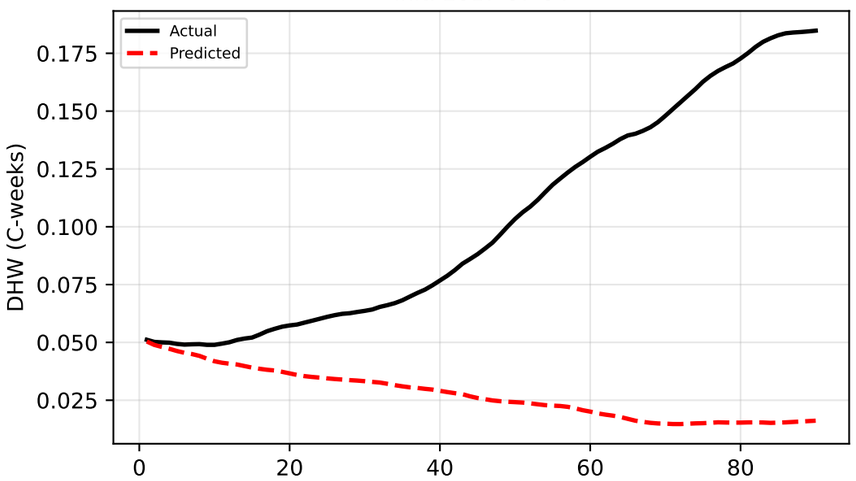


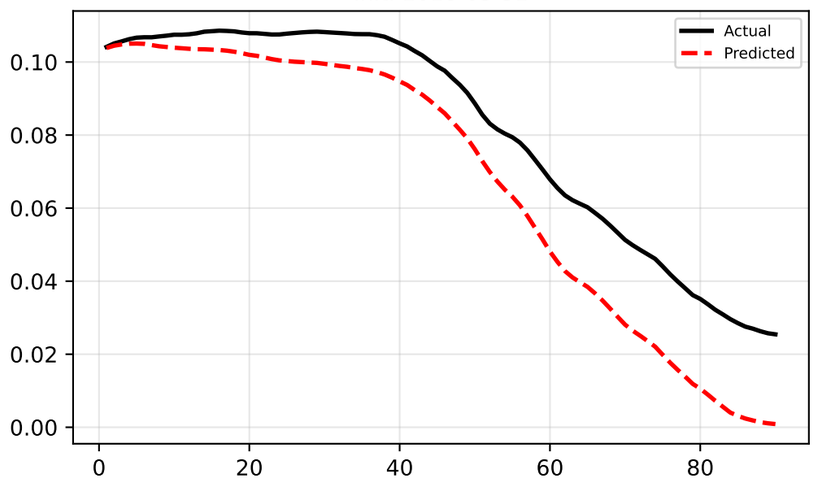
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

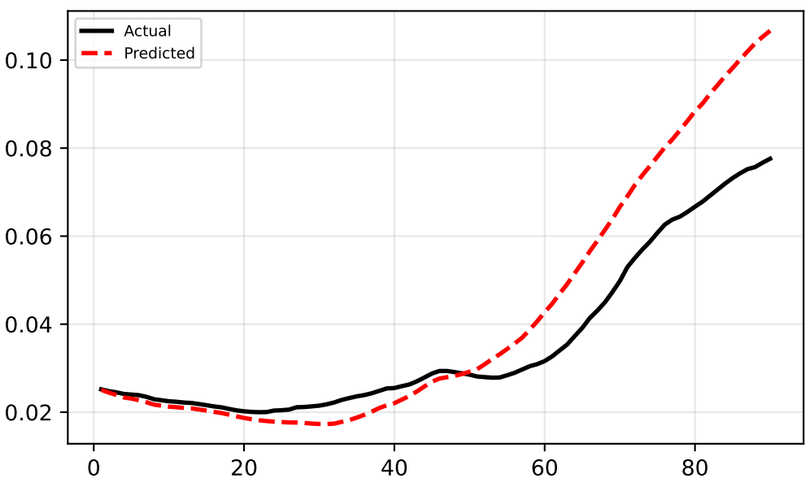


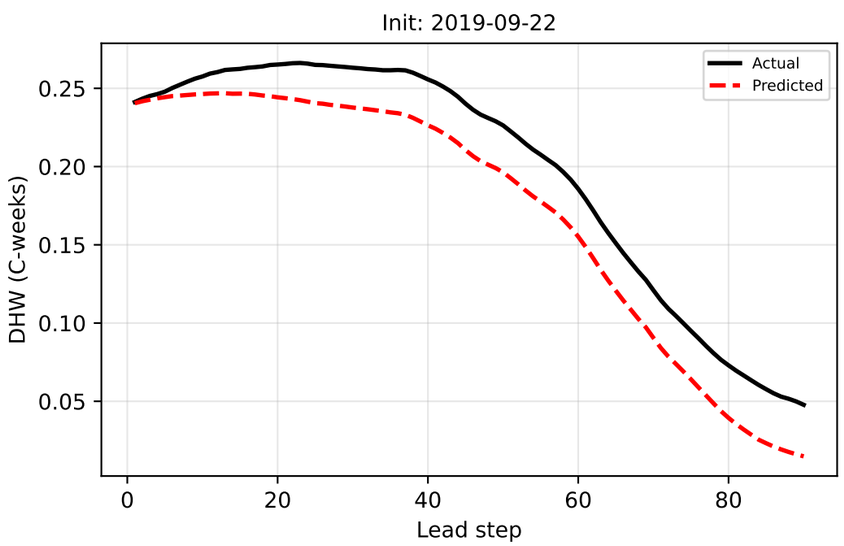




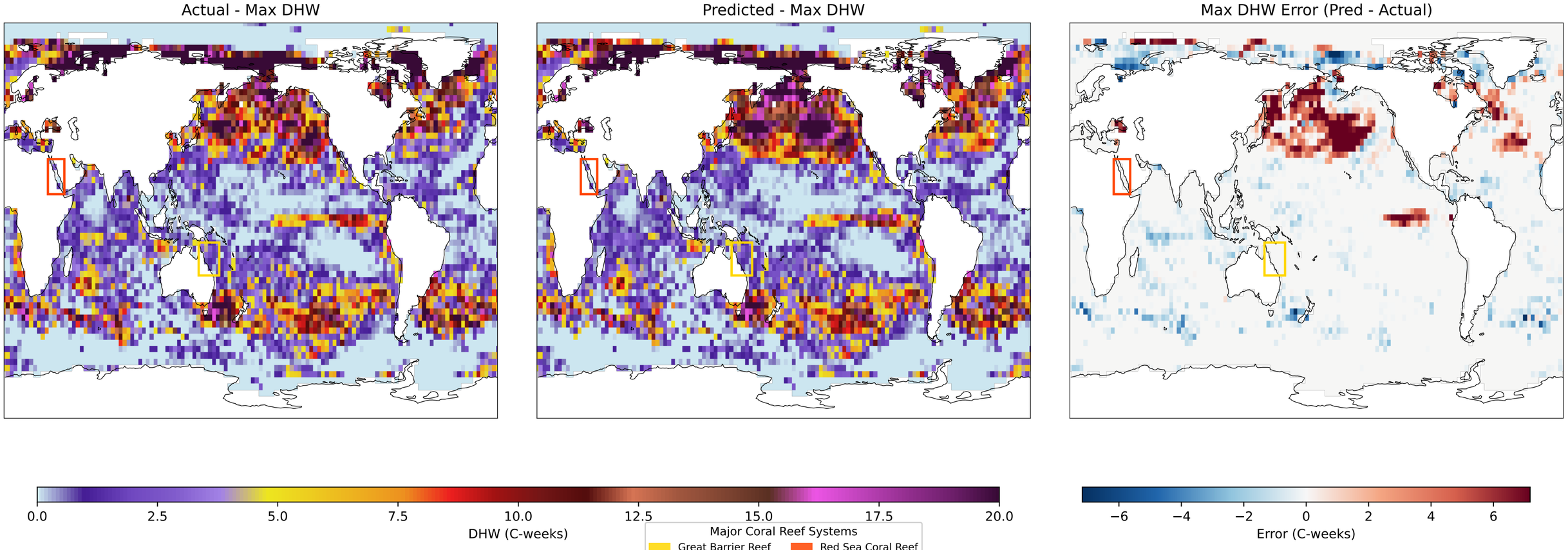


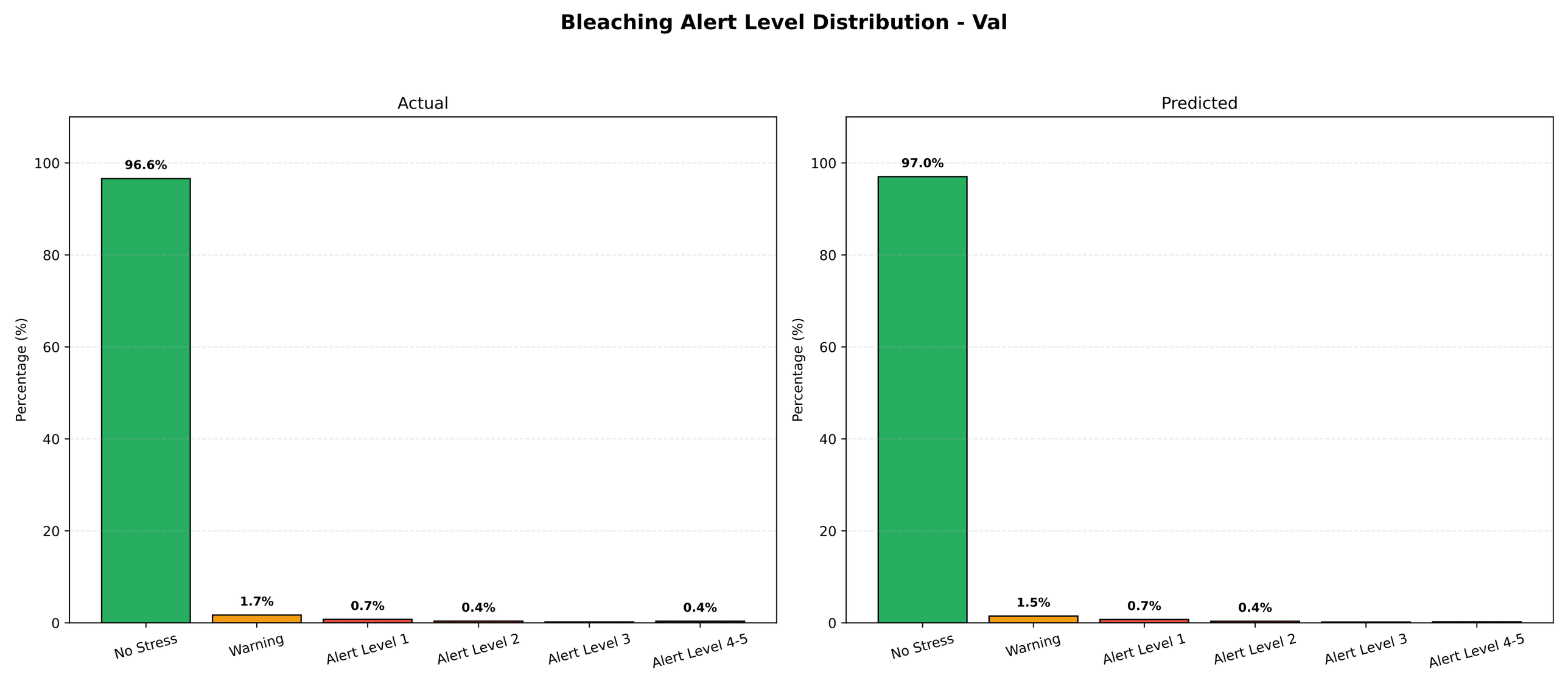






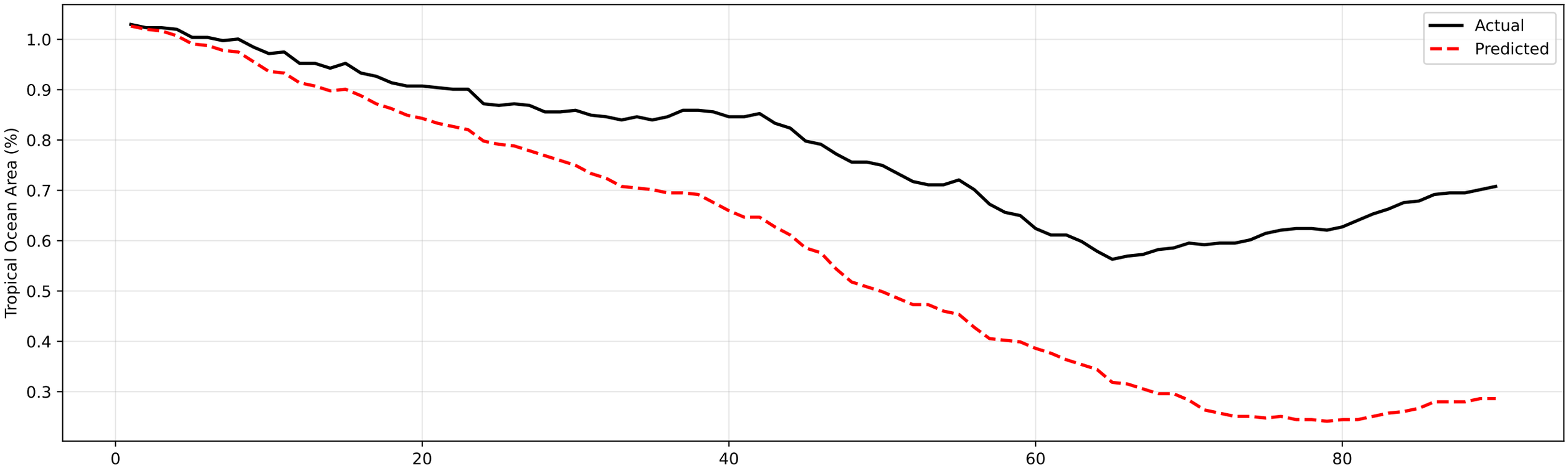
Peak DHW (max over all chains & leads) - Val (C-weeks)



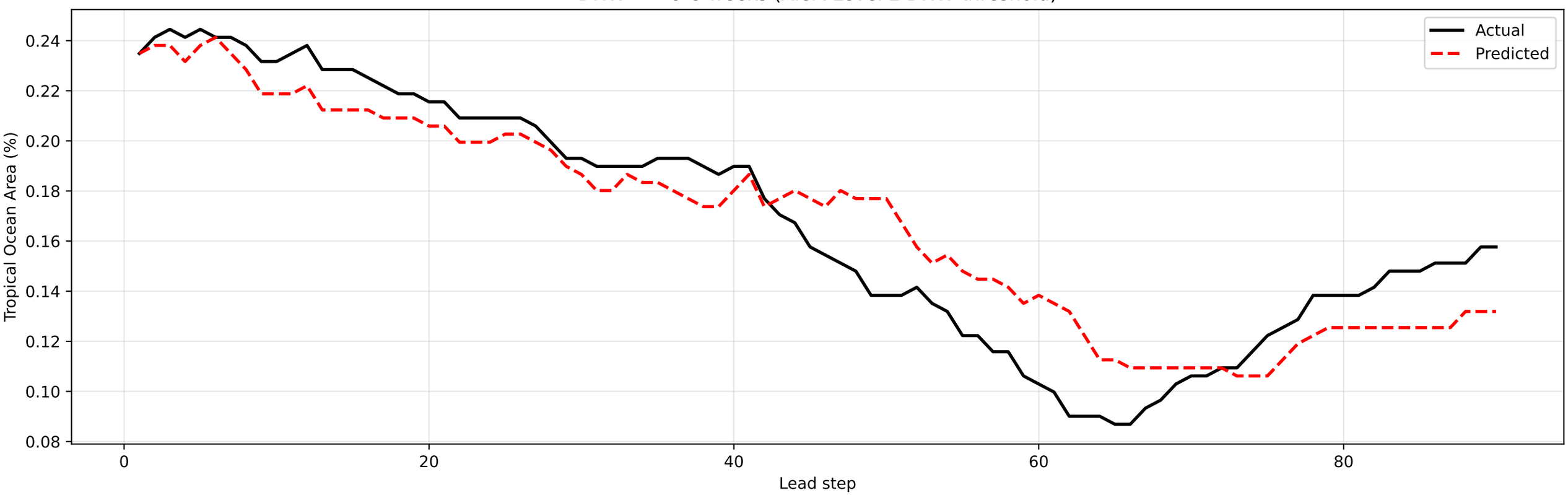

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.7%
0.4%
0.4%
97.0%
1.5%
0.7%
0.4%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



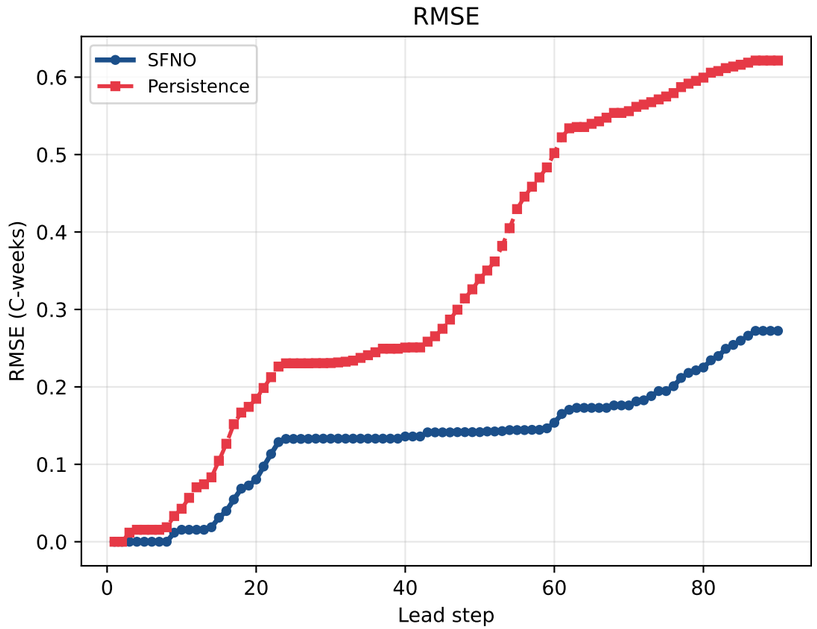
Great Barrier Reef Analysis - Val

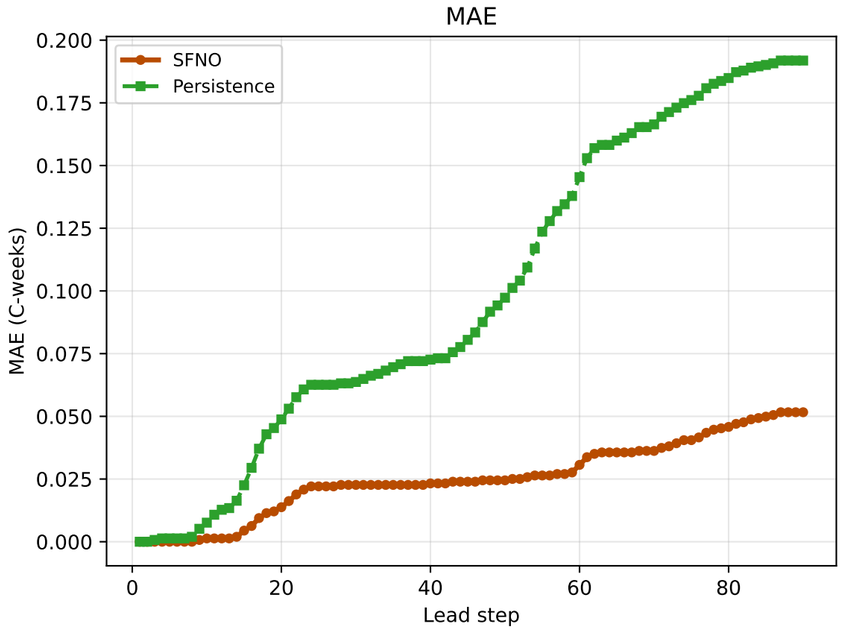
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 5 lat x 5 lon points
```

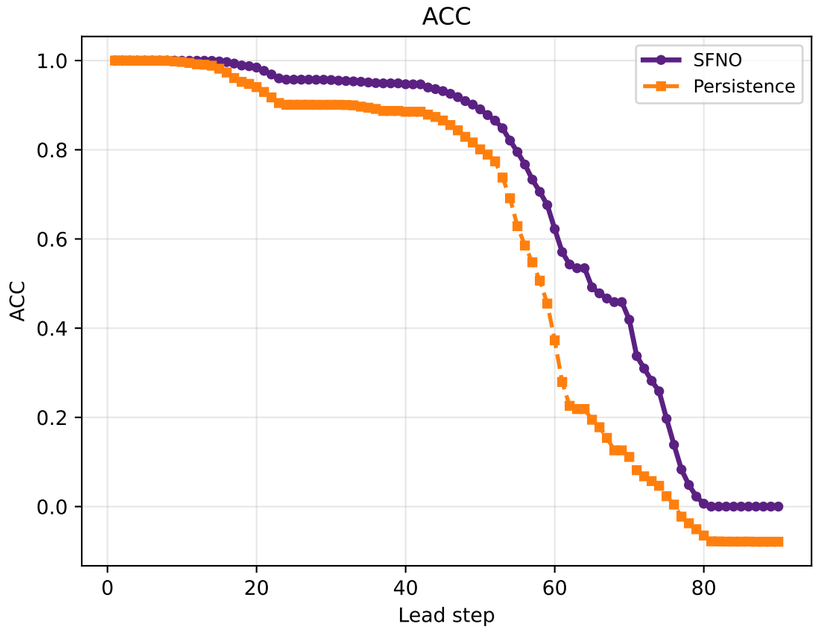
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

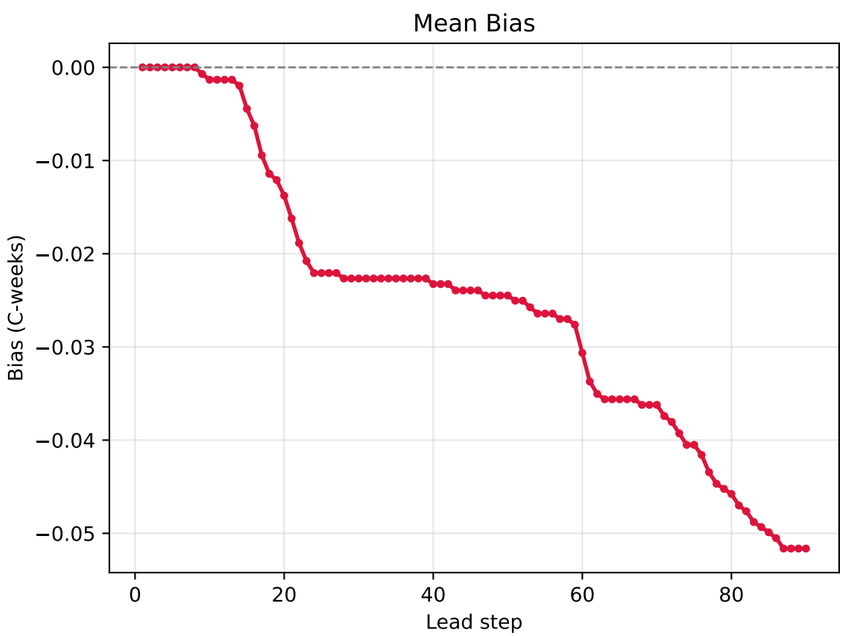
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
2	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
3	0.0000	0.0117	100.0%	0.0000	0.0007	0.0000	1.0000	0.9998	1.0000
4	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
5	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
6	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
7	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
8	0.0000	0.0187	100.0%	0.0000	0.0020	0.0000	1.0000	0.9994	1.0000
9	0.0117	0.0332	64.6%	0.0007	0.0052	-0.0007	0.9997	0.9981	0.9995
10	0.0155	0.0426	63.7%	0.0013	0.0076	-0.0013	0.9996	0.9969	0.9991
11	0.0155	0.0567	72.7%	0.0013	0.0108	-0.0013	0.9995	0.9947	0.9991
12	0.0155	0.0703	78.0%	0.0013	0.0127	-0.0013	0.9995	0.9917	0.9991
13	0.0155	0.0742	79.2%	0.0013	0.0134	-0.0013	0.9995	0.9907	0.9991
14	0.0187	0.0832	77.5%	0.0020	0.0163	-0.0020	0.9993	0.9884	0.9987
15	0.0311	0.1046	70.3%	0.0044	0.0225	-0.0044	0.9980	0.9817	0.9962
16	0.0397	0.1265	68.6%	0.0063	0.0295	-0.0063	0.9966	0.9732	0.9935
17	0.0545	0.1518	64.1%	0.0094	0.0371	-0.0094	0.9932	0.9608	0.9871
18	0.0686	0.1668	58.9%	0.0114	0.0429	-0.0114	0.9889	0.9523	0.9789
19	0.0726	0.1742	58.3%	0.0121	0.0453	-0.0121	0.9874	0.9478	0.9761
20	0.0804	0.1846	56.4%	0.0138	0.0488	-0.0138	0.9843	0.9404	0.9703
21	0.0971	0.1986	51.1%	0.0162	0.0531	-0.0162	0.9770	0.9296	0.9566
22	0.1133	0.2125	46.7%	0.0189	0.0576	-0.0189	0.9688	0.9174	0.9416
23	0.1287	0.2262	43.1%	0.0208	0.0607	-0.0208	0.9600	0.9047	0.9259
24	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
25	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
26	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
27	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
28	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
29	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
30	0.1332	0.2308	42.3%	0.0227	0.0637	-0.0227	0.9564	0.9008	0.9192
31	0.1332	0.2316	42.5%	0.0227	0.0649	-0.0227	0.9550	0.9008	0.9167
32	0.1332	0.2325	42.7%	0.0227	0.0662	-0.0227	0.9542	0.9004	0.9152
33	0.1332	0.2339	43.1%	0.0227	0.0669	-0.0227	0.9533	0.8996	0.9136
34	0.1332	0.2374	43.9%	0.0227	0.0683	-0.0227	0.9525	0.8967	0.9121
35	0.1332	0.2409	44.7%	0.0227	0.0696	-0.0227	0.9509	0.8944	0.9093
36	0.1332	0.2447	45.6%	0.0227	0.0708	-0.0227	0.9496	0.8915	0.9068
37	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
38	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
39	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
40	0.1359	0.2508	45.8%	0.0233	0.0726	-0.0233	0.9469	0.8852	0.9019
41	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
42	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
43	0.1413	0.2584	45.3%	0.0239	0.0756	-0.0239	0.9396	0.8788	0.8887
44	0.1413	0.2654	46.8%	0.0239	0.0776	-0.0239	0.9361	0.8735	0.8822
45	0.1413	0.2750	48.6%	0.0239	0.0805	-0.0239	0.9315	0.8656	0.8740
46	0.1413	0.2869	50.8%	0.0239	0.0834	-0.0239	0.9255	0.8554	0.8631
47	0.1415	0.2997	52.8%	0.0245	0.0876	-0.0245	0.9180	0.8435	0.8499
48	0.1415	0.3142	55.0%	0.0245	0.0917	-0.0245	0.9092	0.8291	0.8344
49	0.1415	0.3260	56.6%	0.0245	0.0942	-0.0245	0.9013	0.8163	0.8207
50	0.1415	0.3395	58.3%	0.0245	0.0973	-0.0245	0.8907	0.8009	0.8025
51	0.1424	0.3503	59.3%	0.0250	0.1012	-0.0250	0.8778	0.7892	0.7809
52	0.1424	0.3620	60.7%	0.0250	0.1041	-0.0250	0.8653	0.7741	0.7605
53	0.1429	0.3821	62.6%	0.0257	0.1093	-0.0257	0.8482	0.7377	0.7322
54	0.1442	0.4049	64.4%	0.0264	0.1169	-0.0264	0.8206	0.6912	0.6890
55	0.1442	0.4296	66.4%	0.0264	0.1236	-0.0264	0.7950	0.6287	0.6501
56	0.1442	0.4457	67.6%	0.0264	0.1278	-0.0264	0.7668	0.5854	0.6088
57	0.1445	0.4585	68.5%	0.0270	0.1318	-0.0270	0.7330	0.5477	0.5614
58	0.1445	0.4705	69.3%	0.0270	0.1346	-0.0270	0.7057	0.5066	0.5248
59	0.1465	0.4834	69.7%	0.0276	0.1379	-0.0276	0.6760	0.4551	0.4861
60	0.1537	0.5019	69.4%	0.0306	0.1454	-0.0306	0.6222	0.3728	0.4186
61	0.1651	0.5223	68.4%	0.0337	0.1529	-0.0337	0.5707	0.2794	0.3594
62	0.1704	0.5339	68.1%	0.0350	0.1569	-0.0350	0.5428	0.2259	0.3295
63	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
64	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
65	0.1729	0.5397	68.0%	0.0356	0.1599	-0.0356	0.4915	0.1947	0.2746
66	0.1729	0.5429	68.2%	0.0356	0.1611	-0.0356	0.4781	0.1779	0.2597
67	0.1729	0.5476	68.4%	0.0356	0.1629	-0.0356	0.4666	0.1539	0.2469
68	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
69	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
70	0.1760	0.5561	68.3%	0.0362	0.1664	-0.0362	0.4189	0.1113	0.2001
71	0.1812	0.5617	67.7%	0.0374	0.1695	-0.0374	0.3375	0.0816	0.1329
72	0.1827	0.5646	67.6%	0.0380	0.1713	-0.0380	0.3096	0.0678	0.1116
73	0.1881	0.5677	66.9%	0.0393	0.1731	-0.0393	0.2819	0.0571	0.0915
74	0.1946	0.5713	65.9%	0.0405	0.1749	-0.0405	0.2585	0.0465	0.0756
75	0.1946	0.5750	66.2%	0.0405	0.1761	-0.0405	0.1969	0.0233	0.0384
76	0.2009	0.5794	65.3%	0.0416	0.1778	-0.0416	0.1385	0.0044	0.0092
77	0.2116	0.5871	64.0%	0.0434	0.1808	-0.0434	0.0831	-0.0226	-0.0142
78	0.2180	0.5916	63.1%	0.0447	0.1826	-0.0447	0.0485	-0.0374	-0.0277
79	0.2214	0.5952	62.8%	0.0452	0.1837	-0.0452	0.0225	-0.0507	-0.0367
80	0.2250	0.5994	62.5%	0.0458	0.1849	-0.0458	0.0064	-0.0649	-0.0418
81	0.2344	0.6058	61.3%	0.0470	0.1872	-0.0470	0.0000	-0.0779	-0.0419
82	0.2398	0.6079	60.6%	0.0476	0.1879	-0.0476	0.0000	-0.0782	-0.0411
83	0.2492	0.6117	59.3%	0.0488	0.1890	-0.0488	0.0000	-0.0785	-0.0399
84	0.2542	0.6137	58.6%	0.0493	0.1896	-0.0493	0.0000	-0.0786	-0.0391
85	0.2597	0.6160	57.8%	0.0499	0.1901	-0.0499	0.0000	-0.0786	-0.0383
86	0.2661	0.6187	57.0%	0.0505	0.1907	-0.0505	0.0000	-0.0785	-0.0374
87	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
88	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
89	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
90	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373

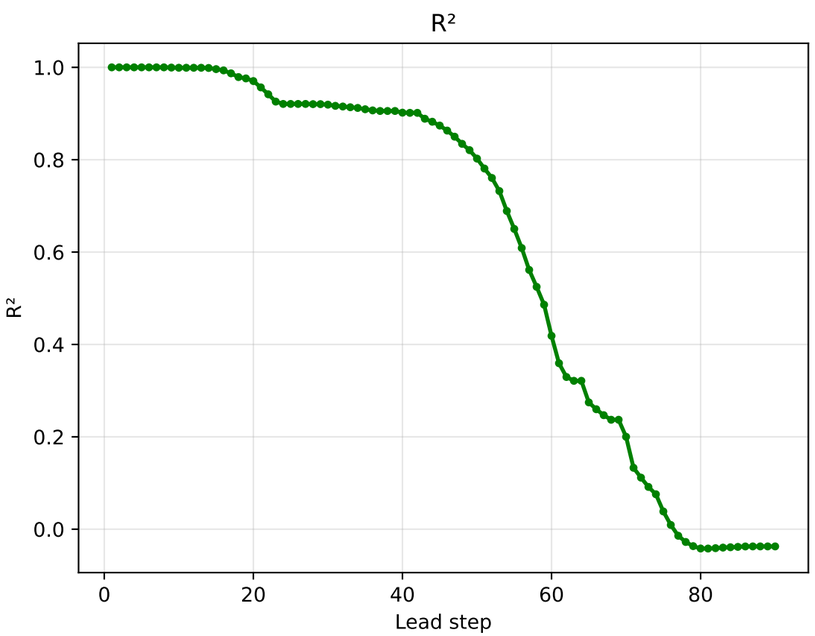
DHW Forecast Skill vs Lead Step - Val - GBR

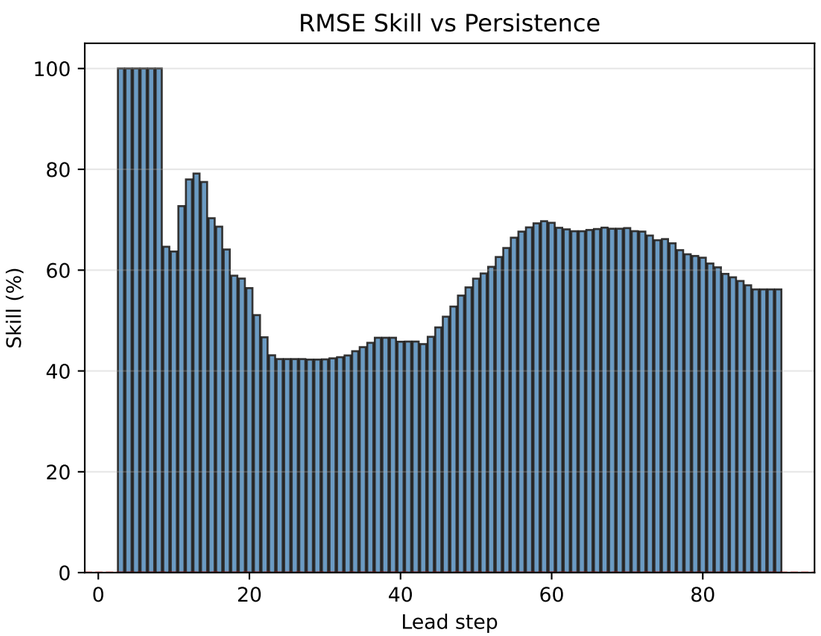


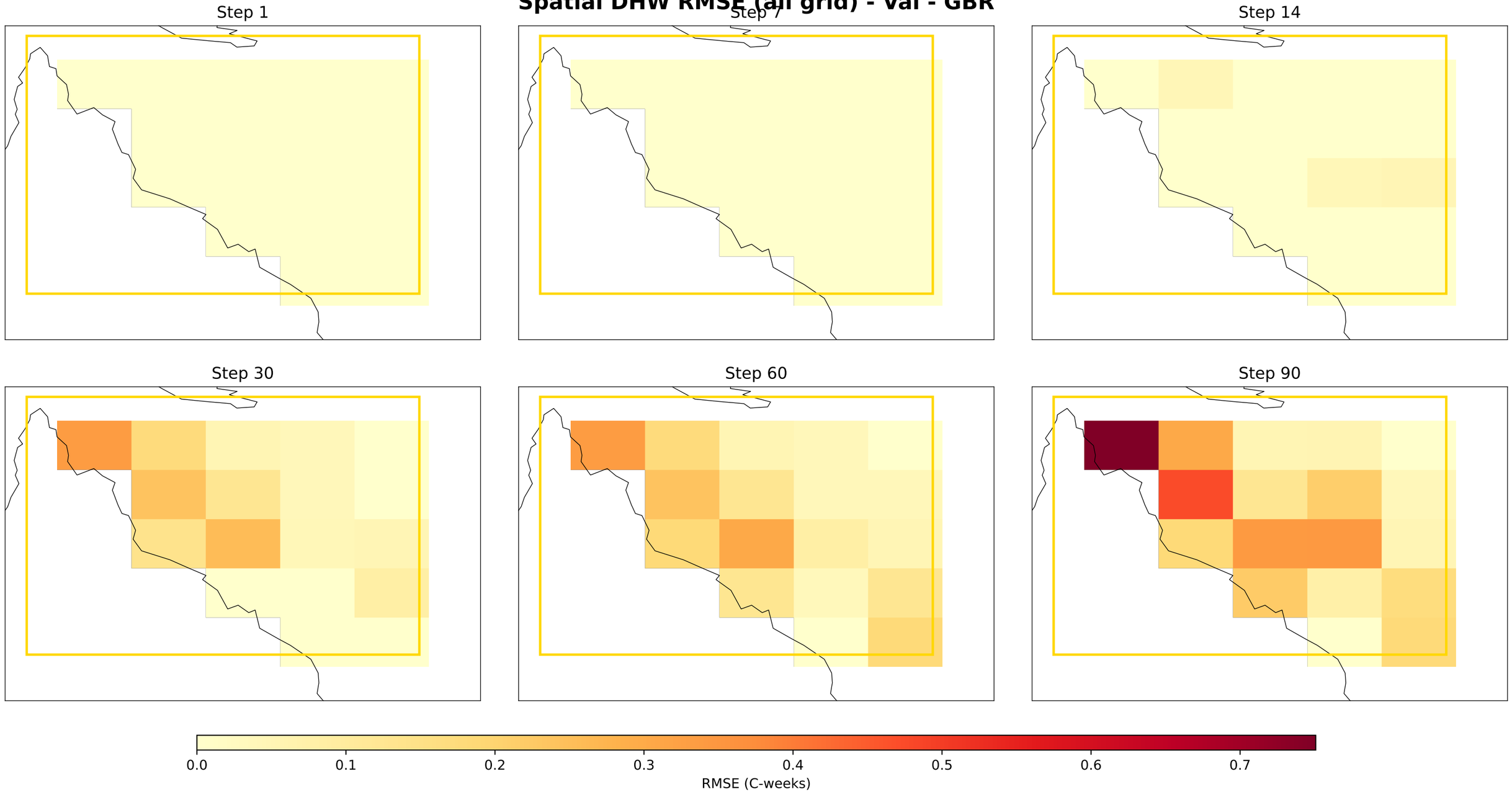


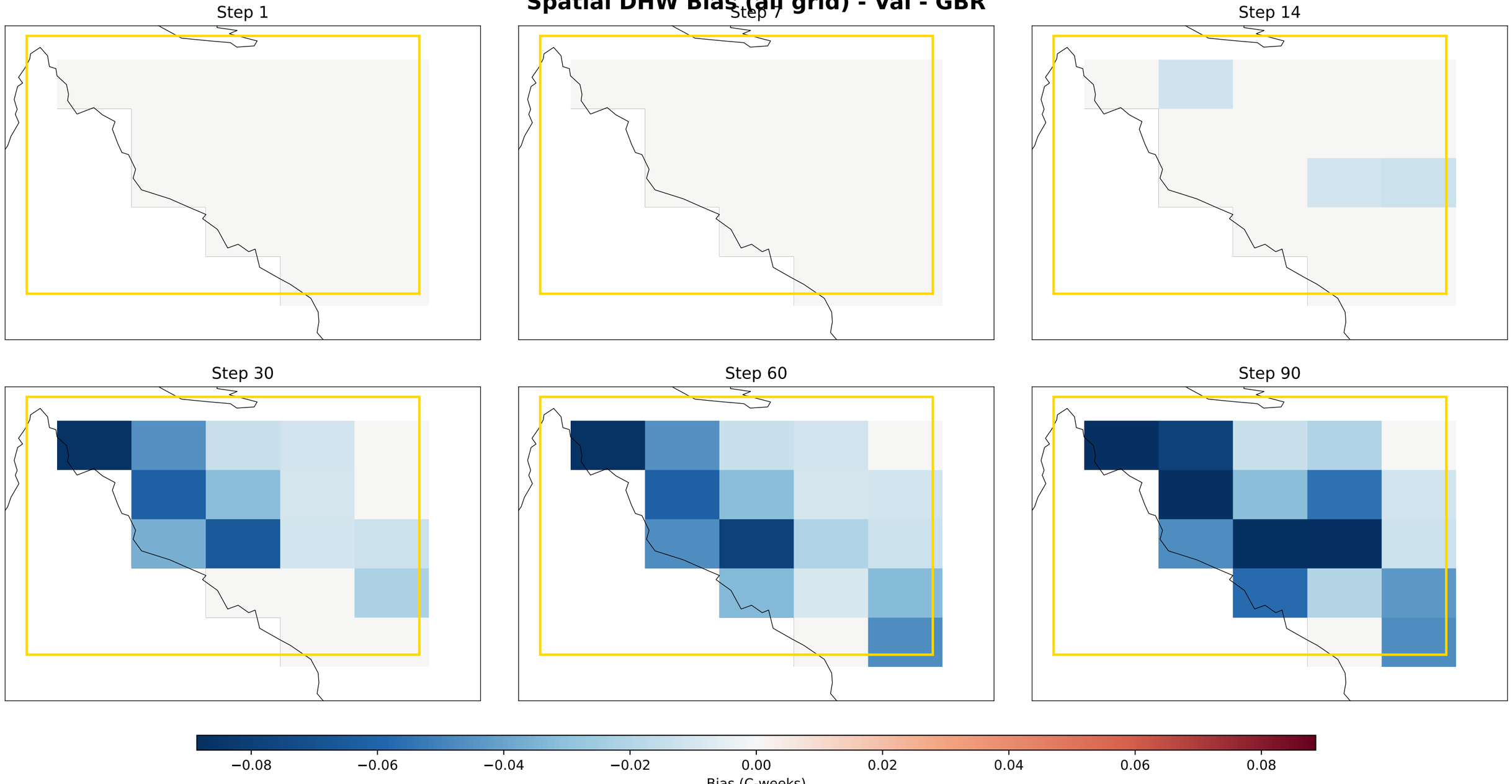


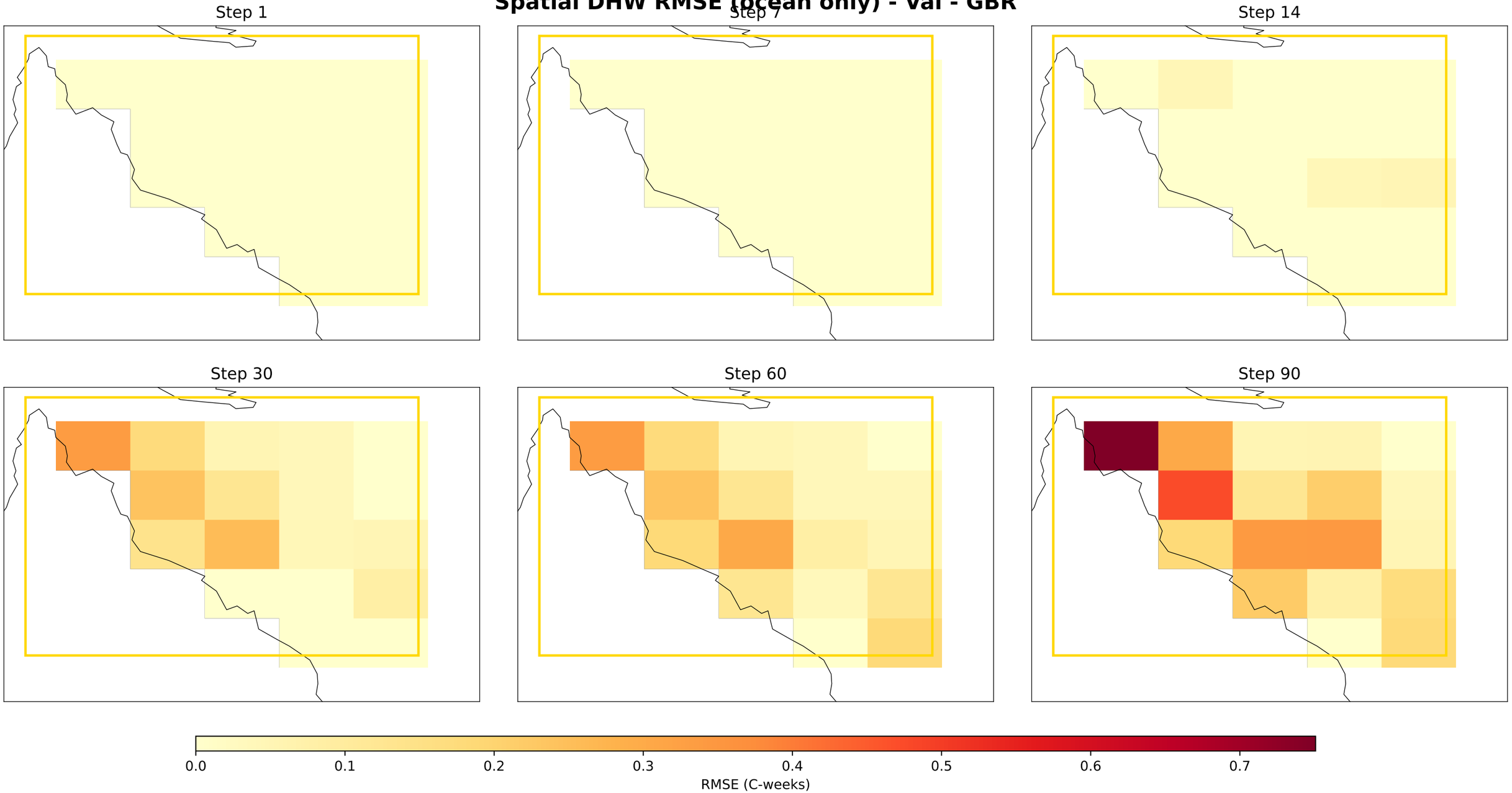


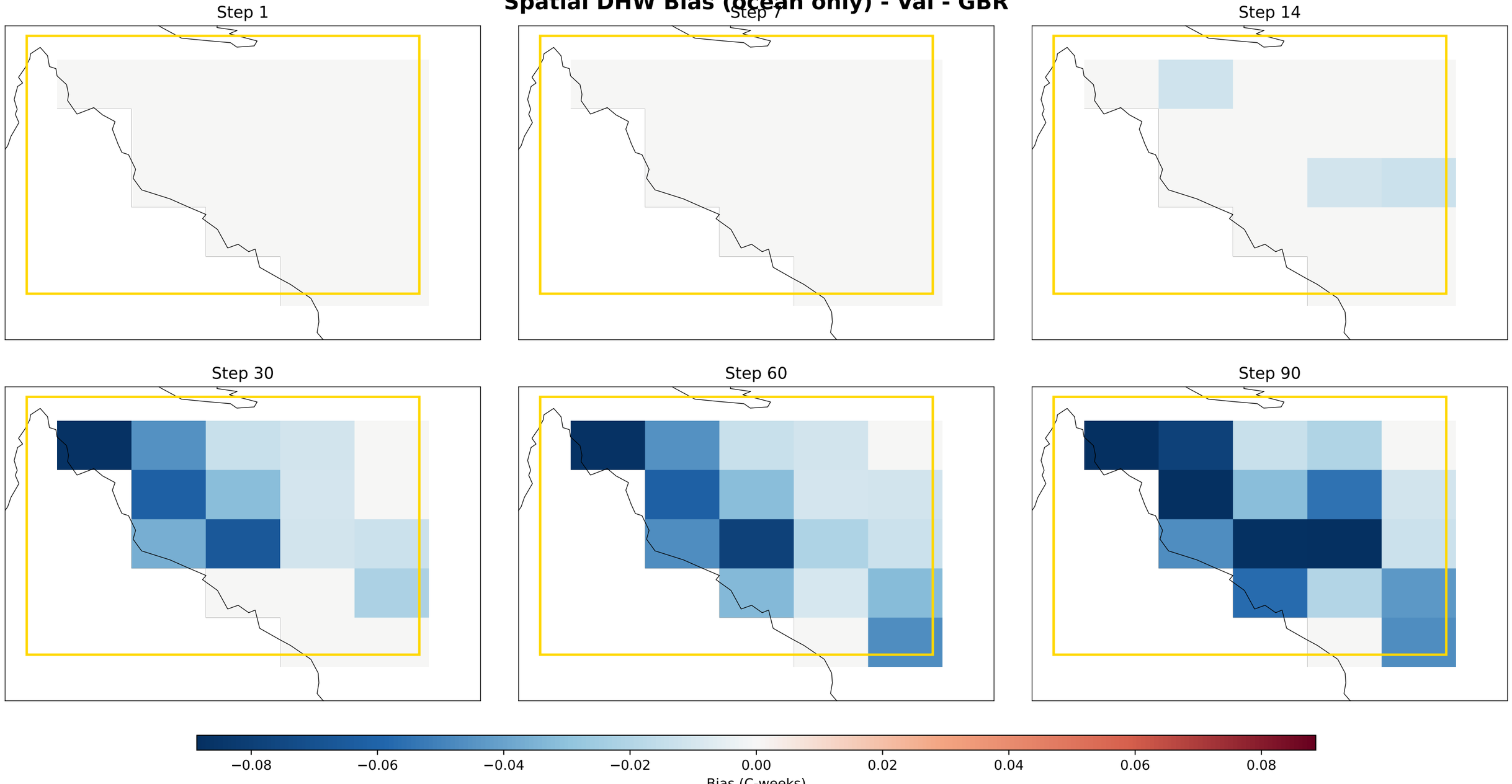





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

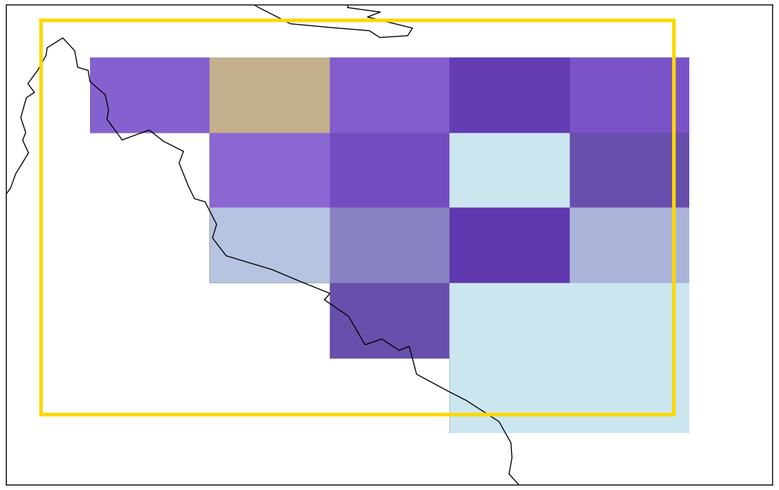

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

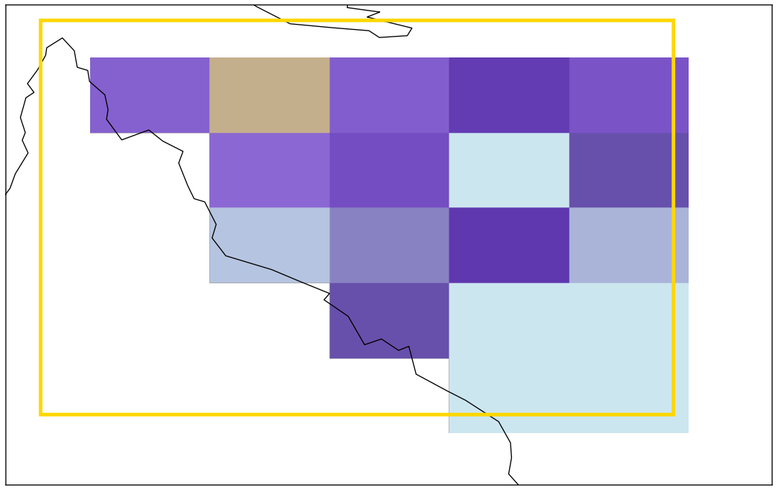

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

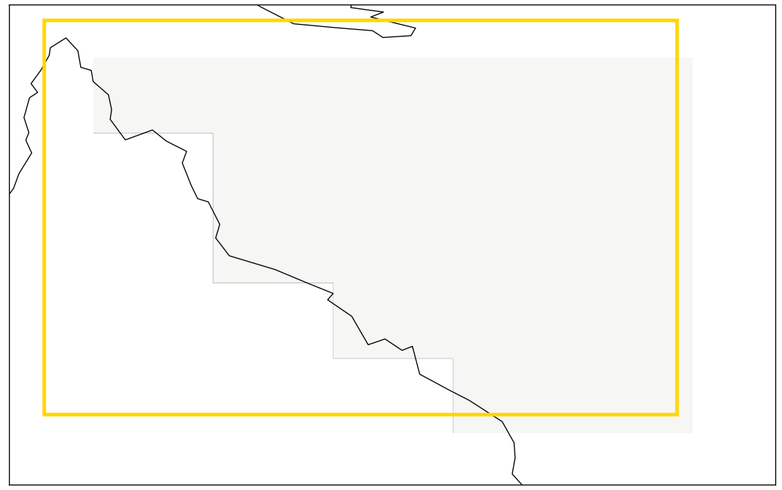
Actual DHW - Step 1



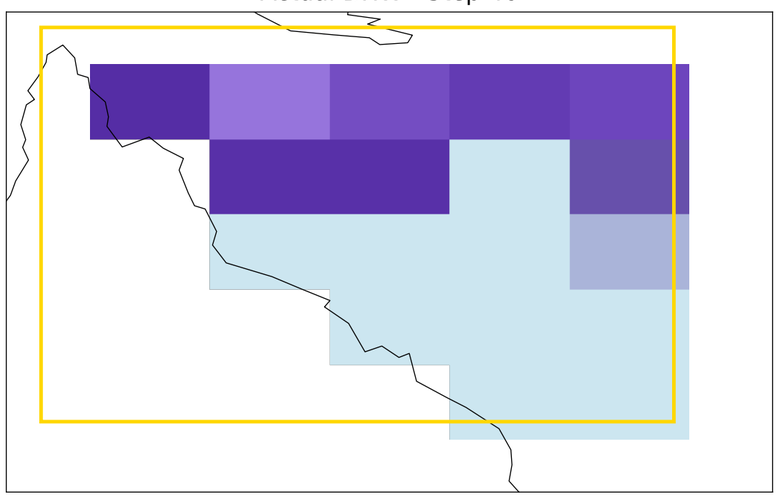
Predicted DHW - Step 1



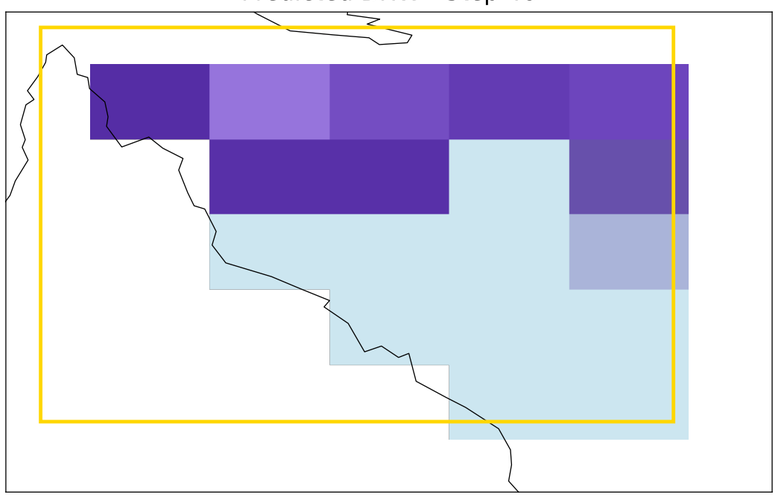
Error (Pred - Actual) - Step 1 [scale ±8]



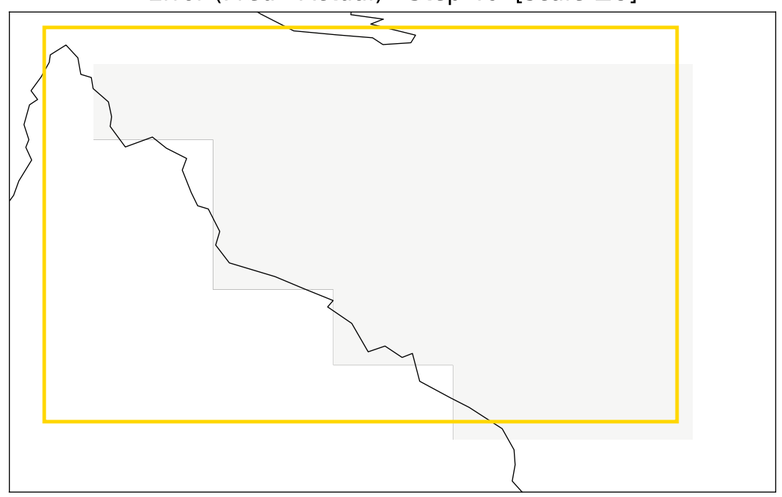
Actual DHW - Step 46



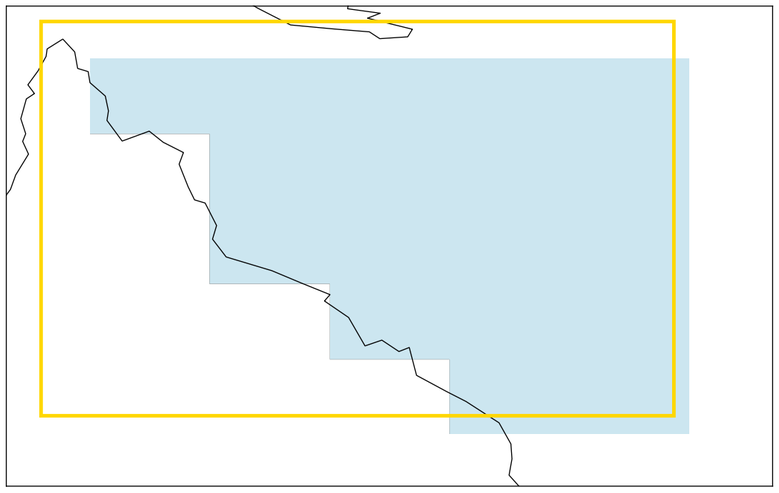
Predicted DHW - Step 46



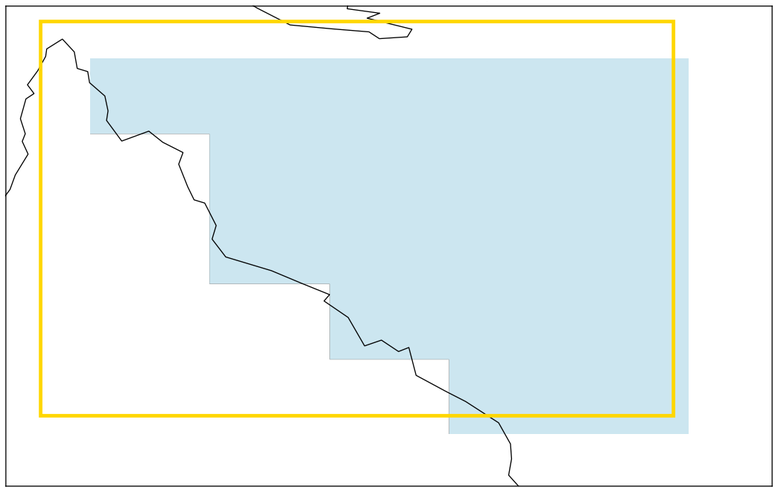
Error (Pred - Actual) - Step 46 [scale ±8]



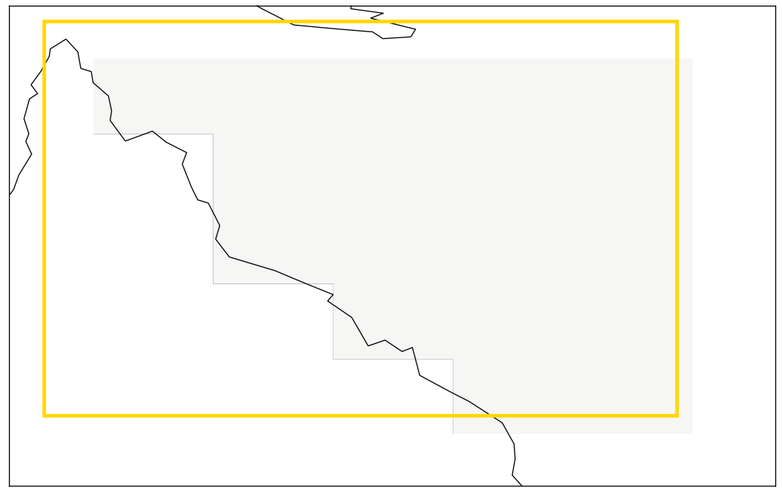
Actual DHW - Step 90

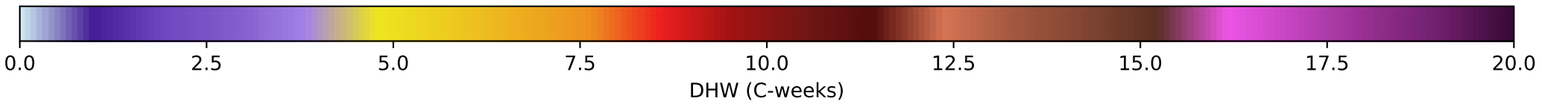


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

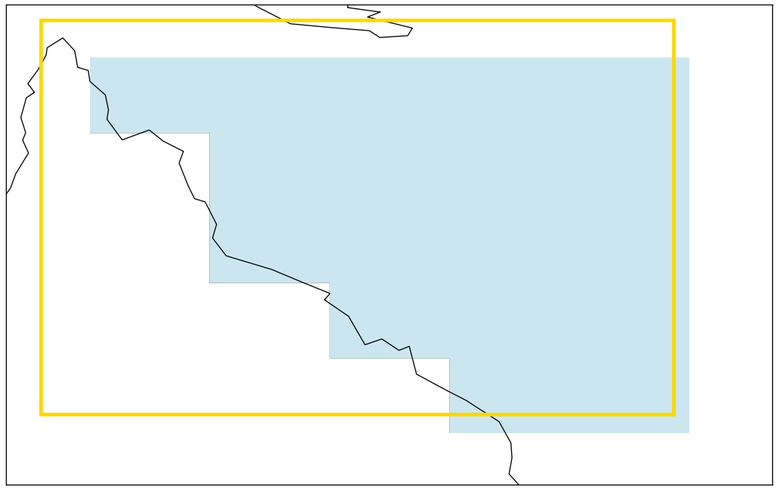




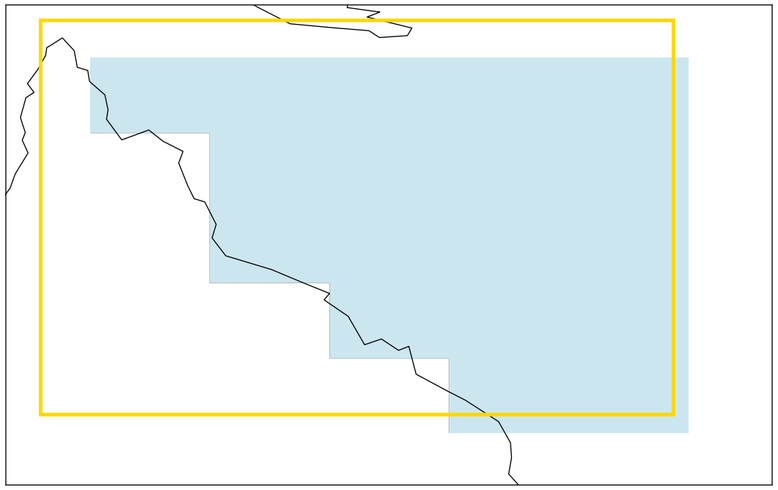


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

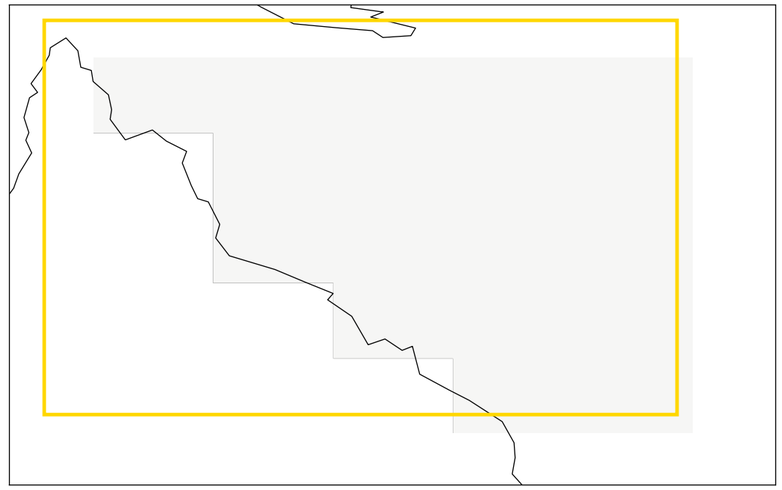
Actual DHW - Step 1



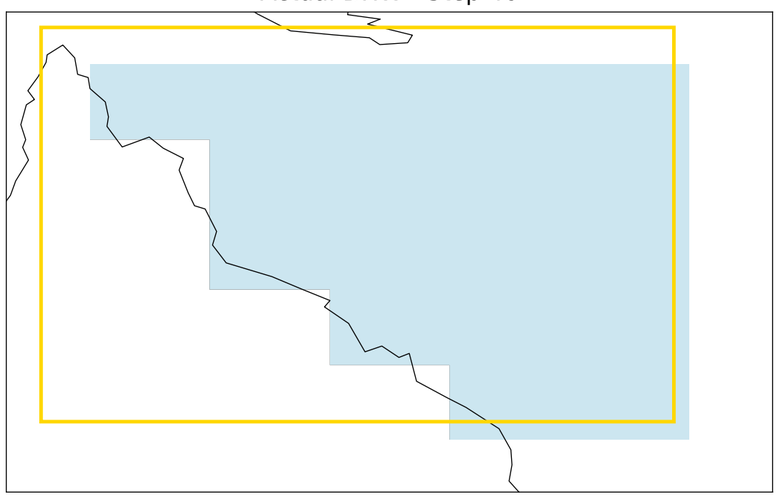
Predicted DHW - Step 1



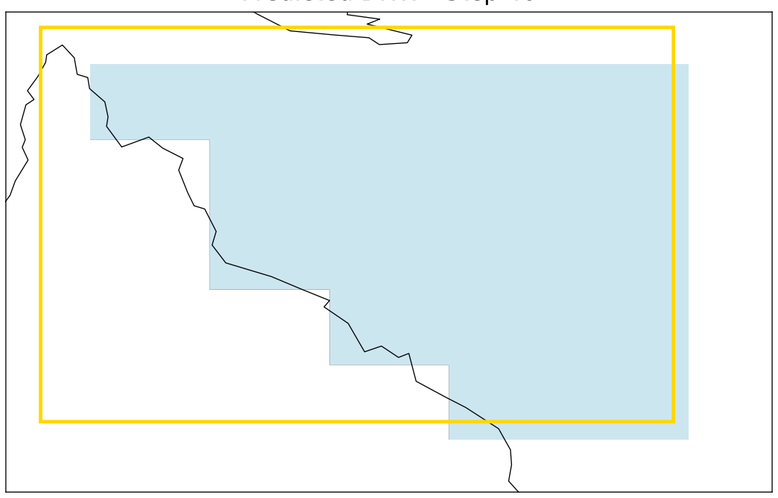
Error (Pred - Actual) - Step 1 [scale ±8]



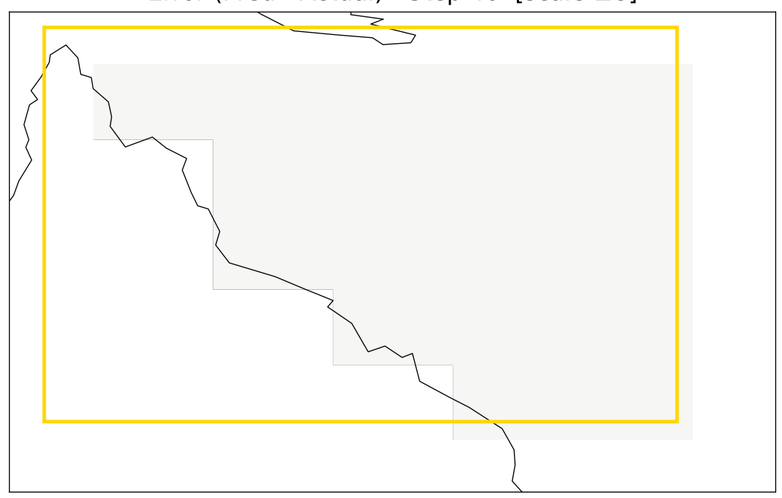
Actual DHW - Step 46



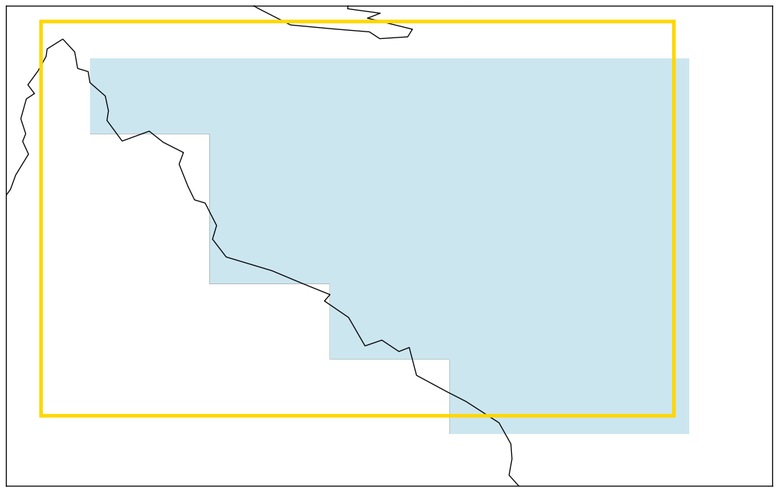
Predicted DHW - Step 46



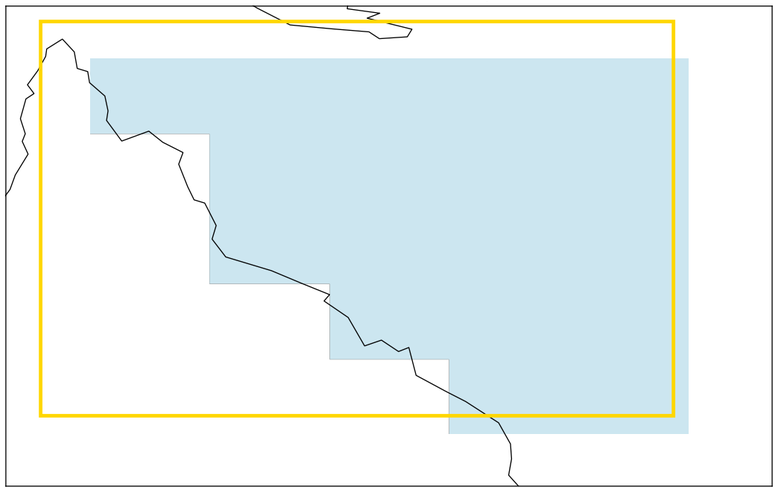
Error (Pred - Actual) - Step 46 [scale ±8]



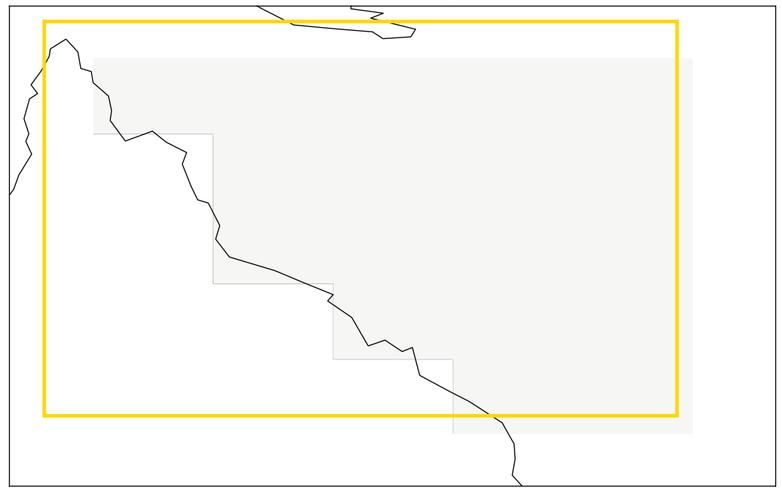
Actual DHW - Step 90

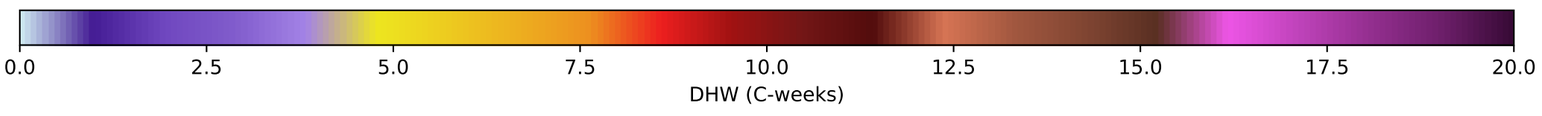


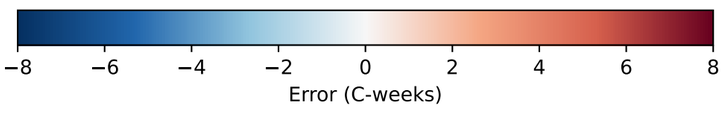
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

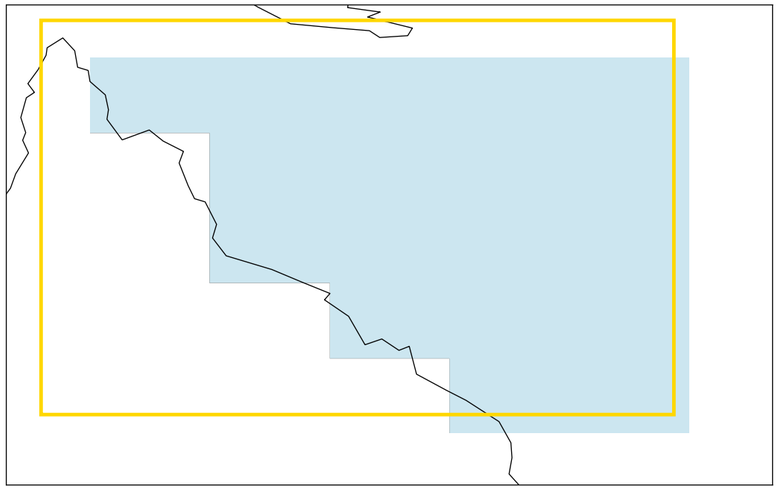




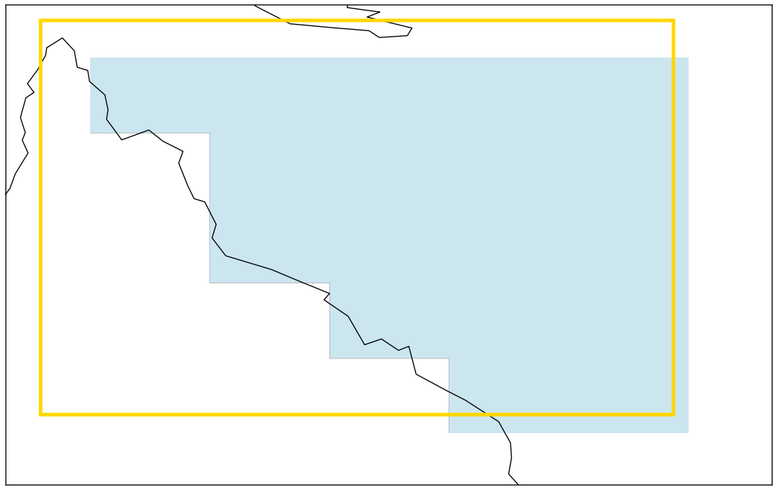


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

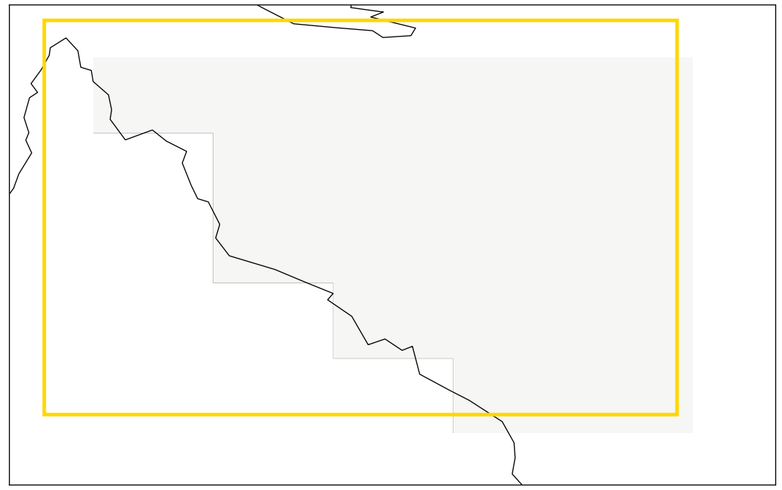
Actual DHW - Step 1



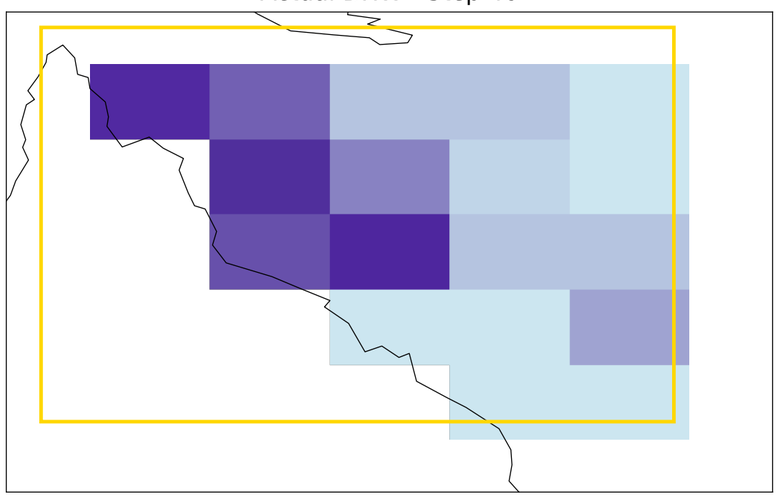
Predicted DHW - Step 1



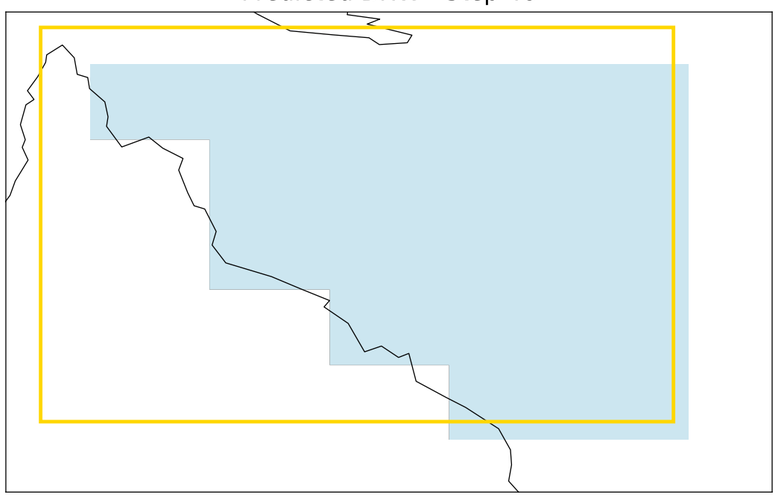
Error (Pred - Actual) - Step 1 [scale ±8]



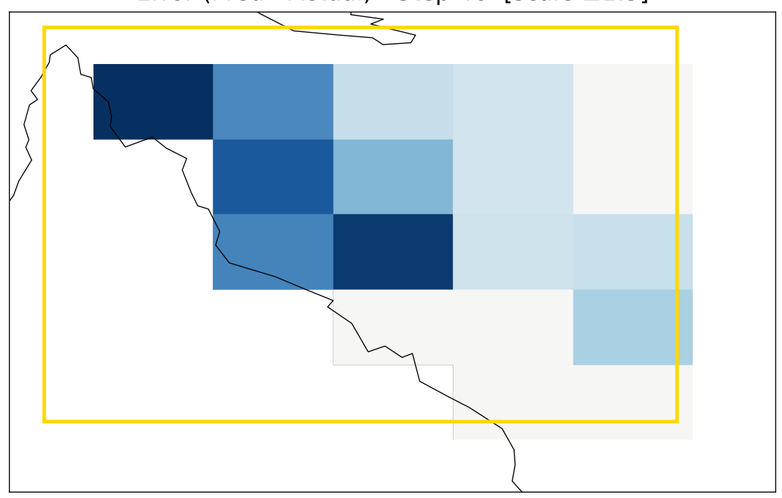
Actual DHW - Step 46



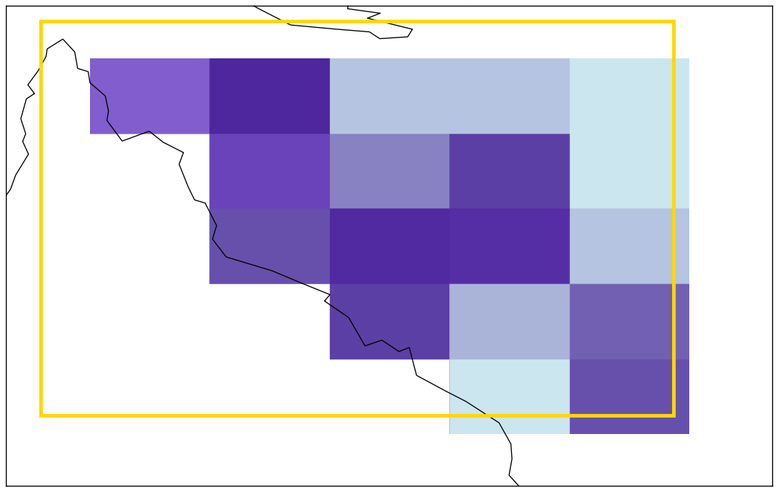
Predicted DHW - Step 46



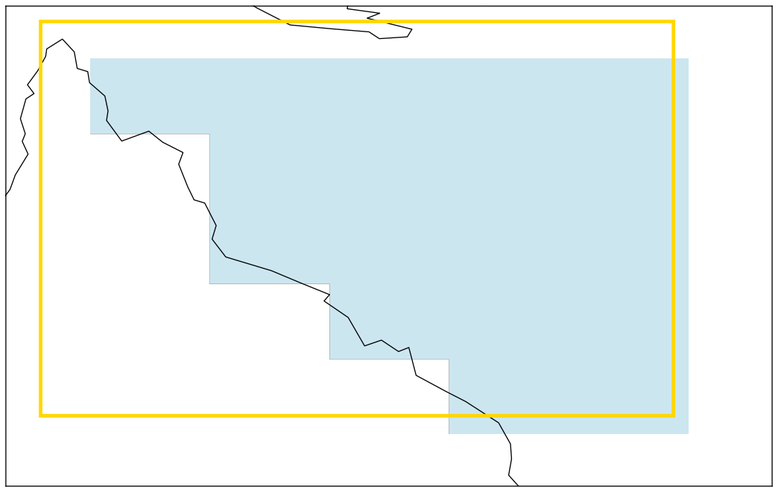
Error (Pred - Actual) - Step 46 [scale ±1.3]



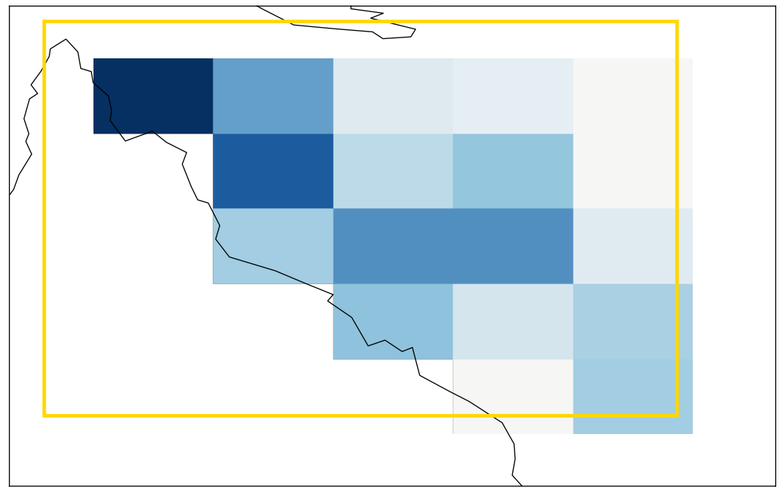
Actual DHW - Step 90

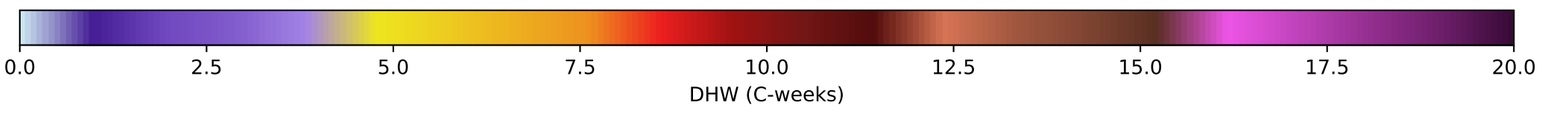


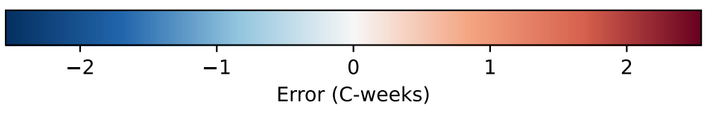
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.5]

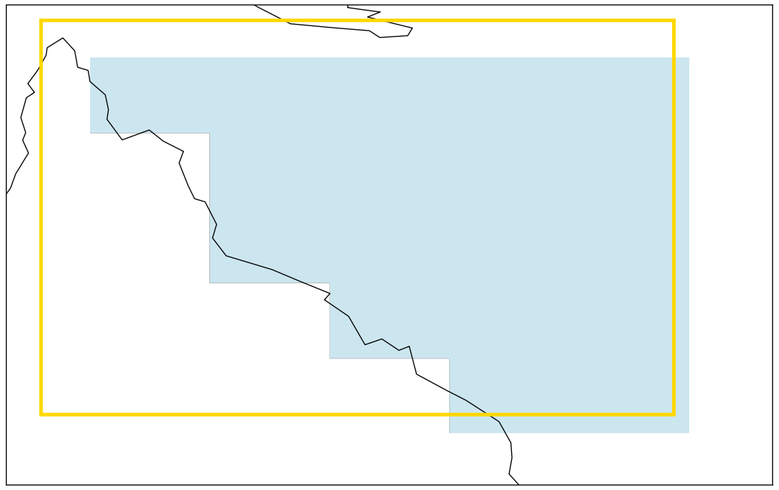




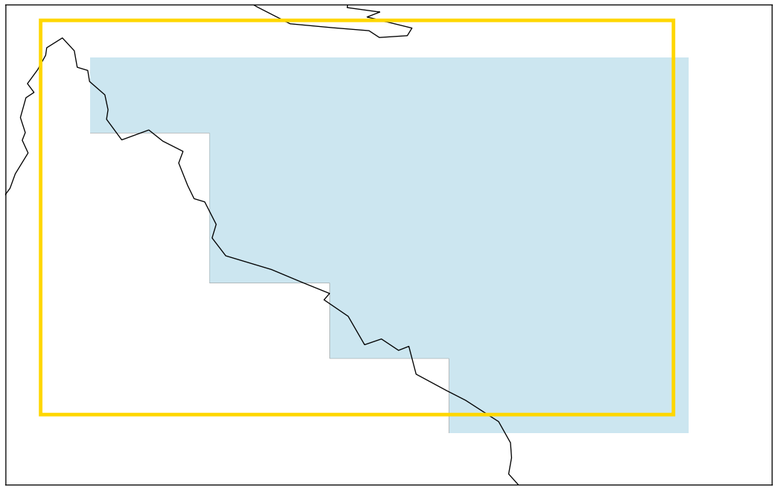


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

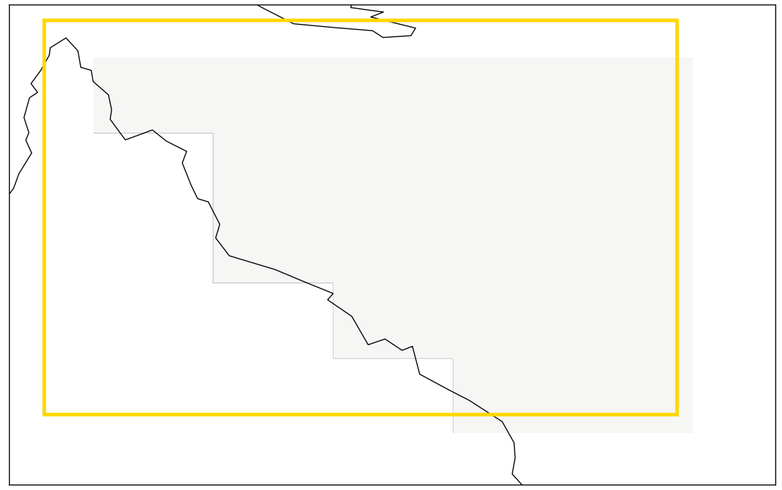
Actual DHW - Step 1



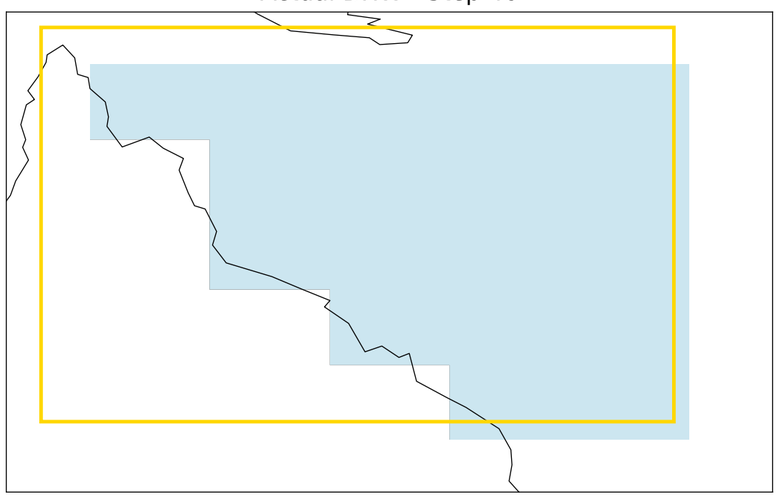
Predicted DHW - Step 1



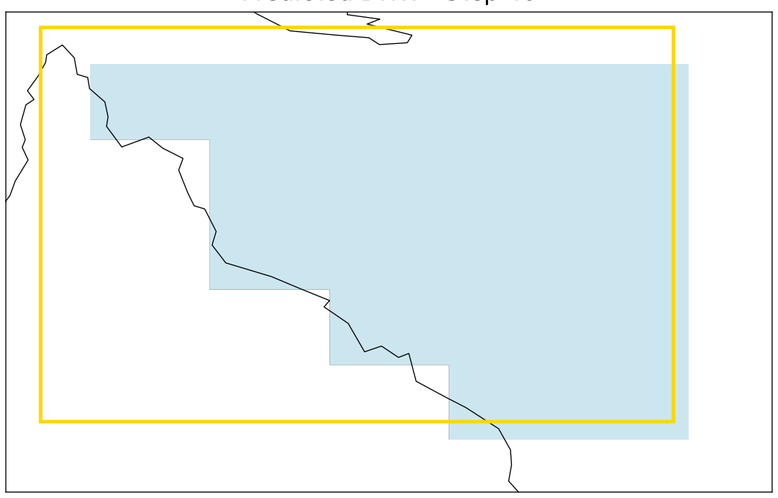
Error (Pred - Actual) - Step 1 [scale ±8]



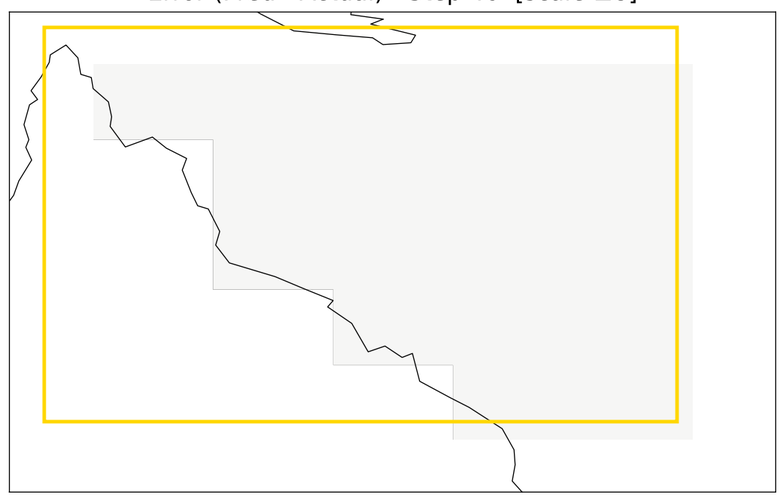
Actual DHW - Step 46



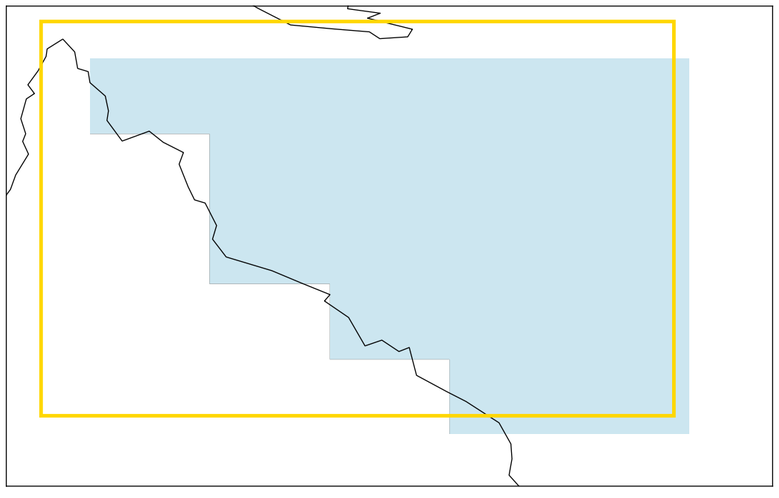
Predicted DHW - Step 46



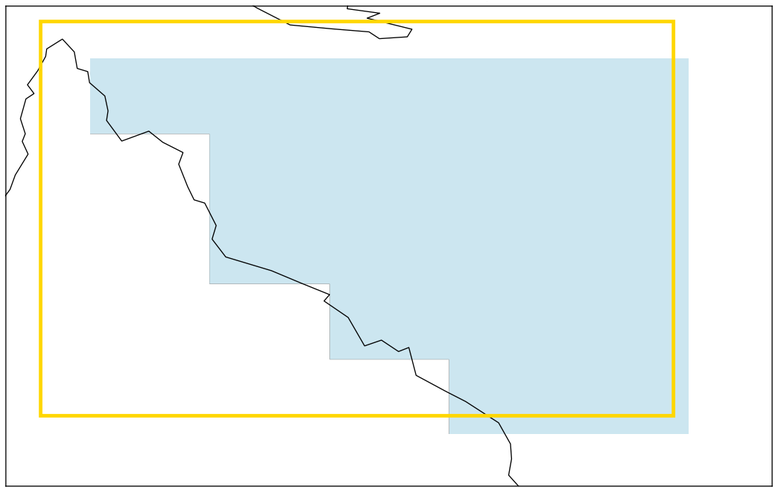
Error (Pred - Actual) - Step 46 [scale ±8]



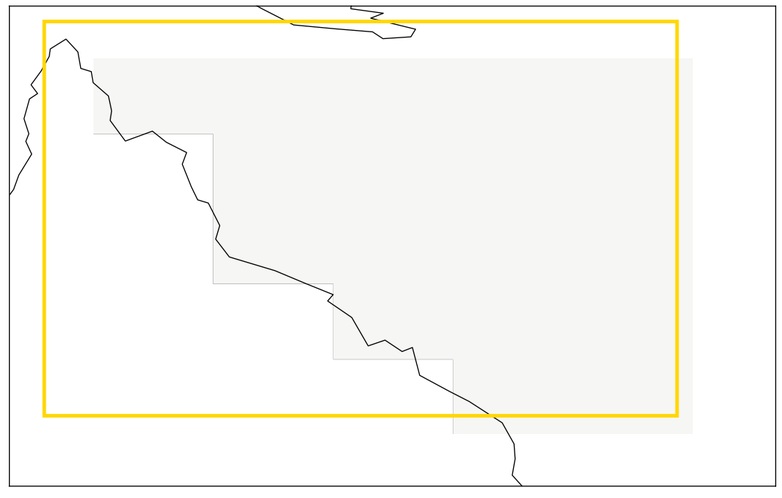
Actual DHW - Step 90

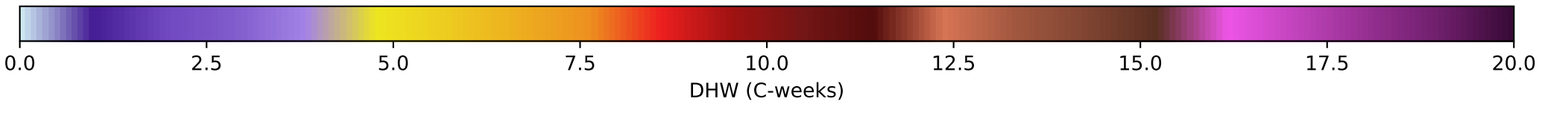


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

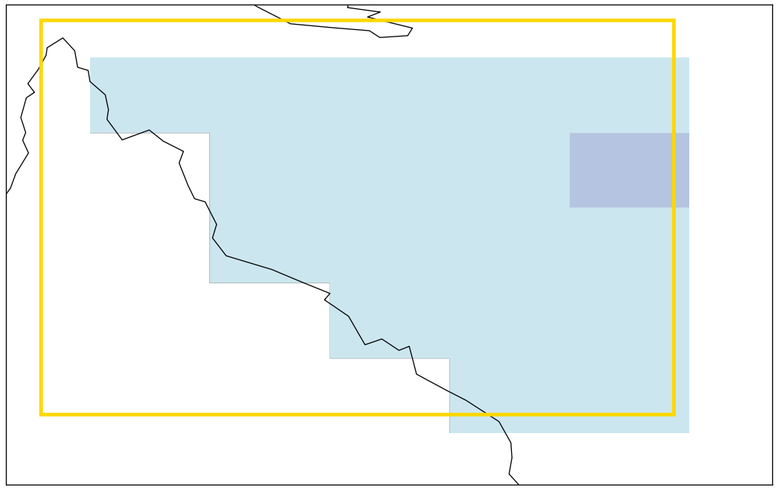




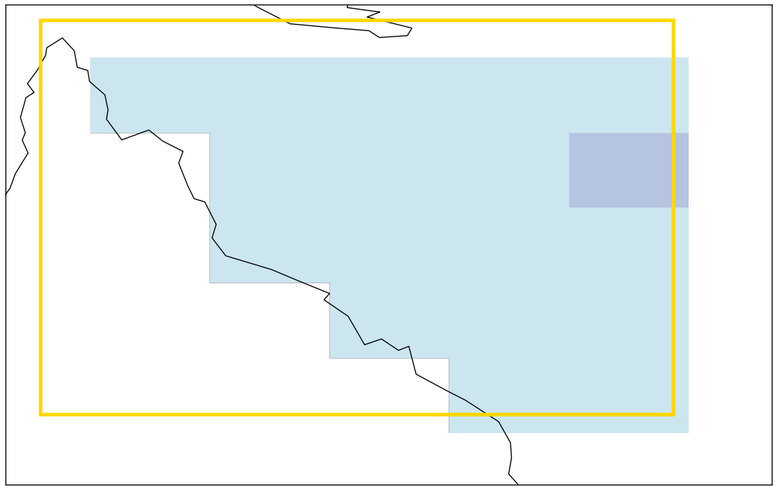


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

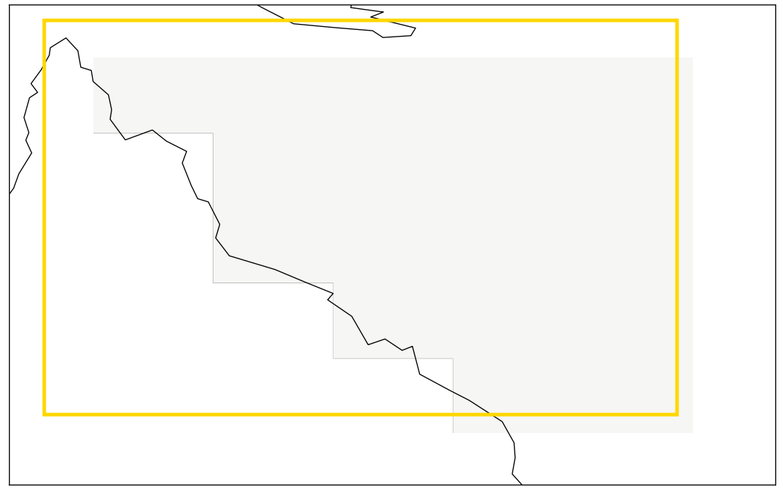
Actual DHW - Step 1



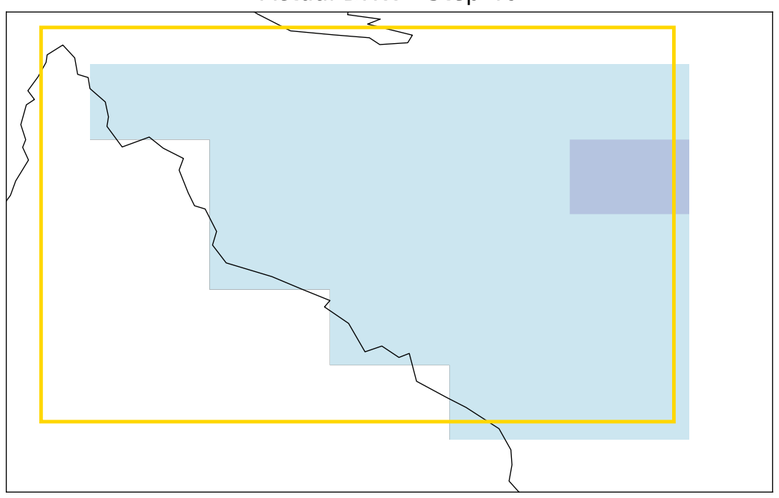
Predicted DHW - Step 1



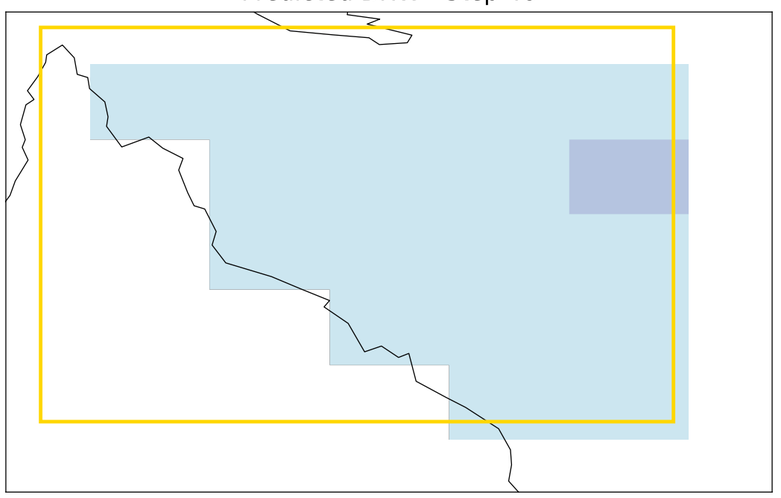
Error (Pred - Actual) - Step 1 [scale ±8]



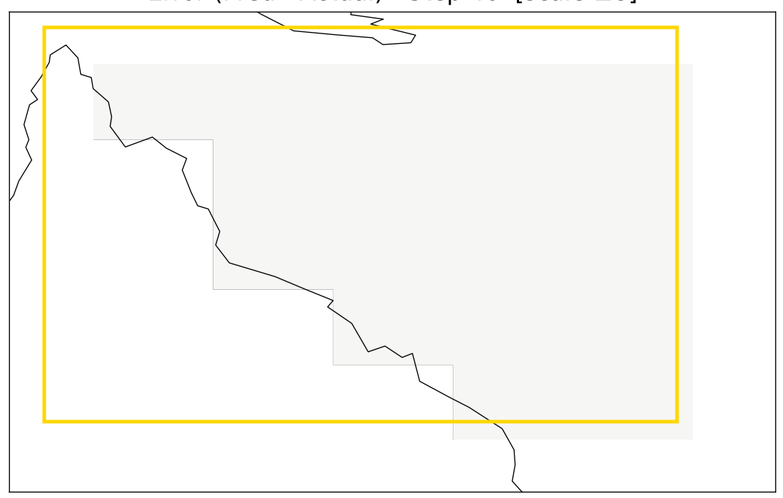
Actual DHW - Step 46



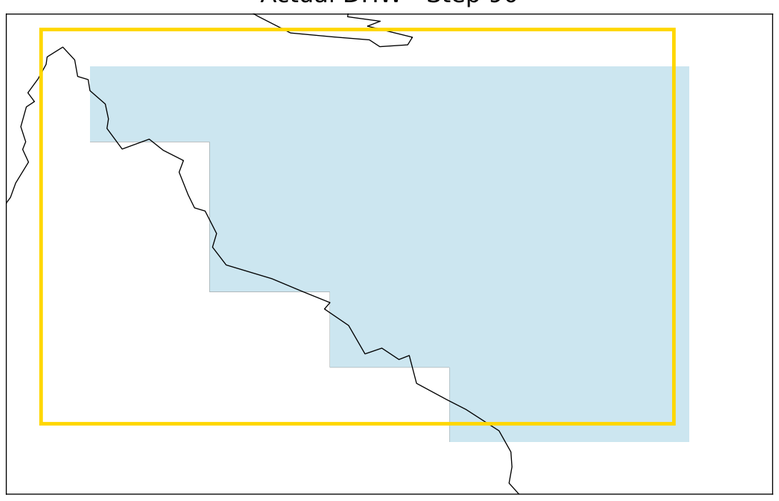
Predicted DHW - Step 46



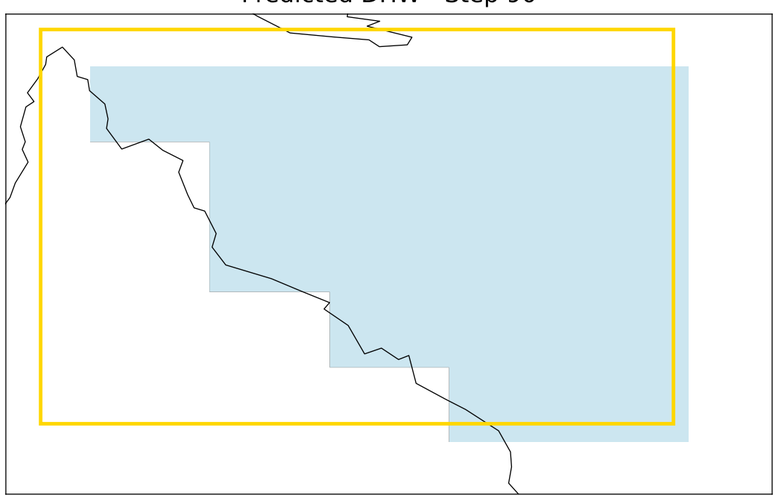
Error (Pred - Actual) - Step 46 [scale ±8]



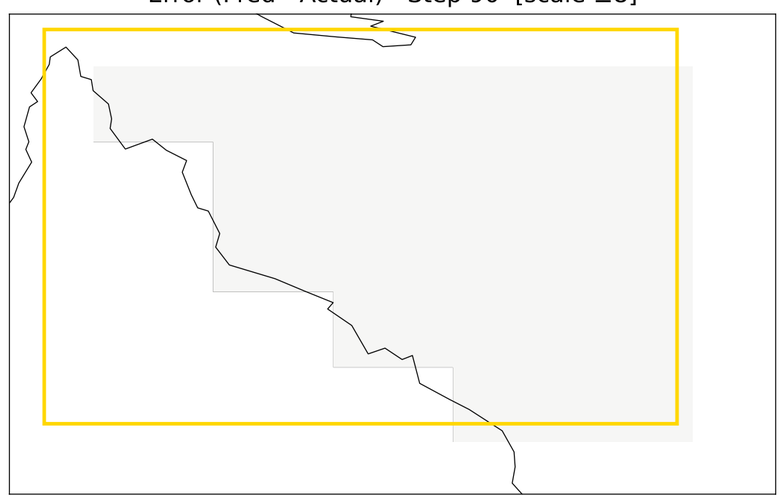
Actual DHW - Step 90

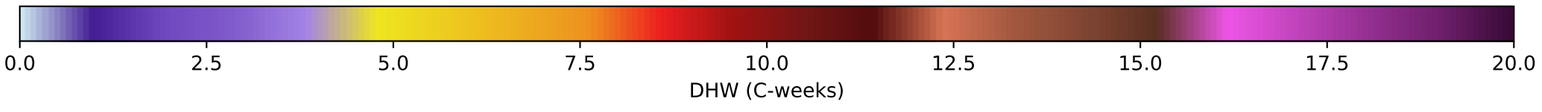


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

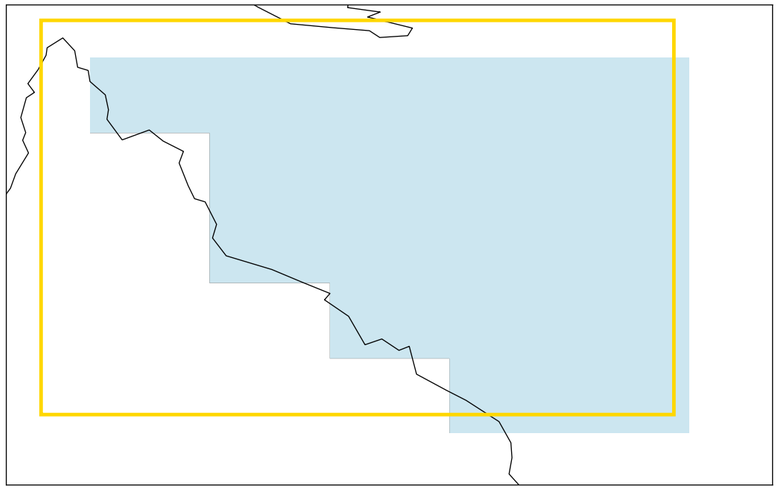




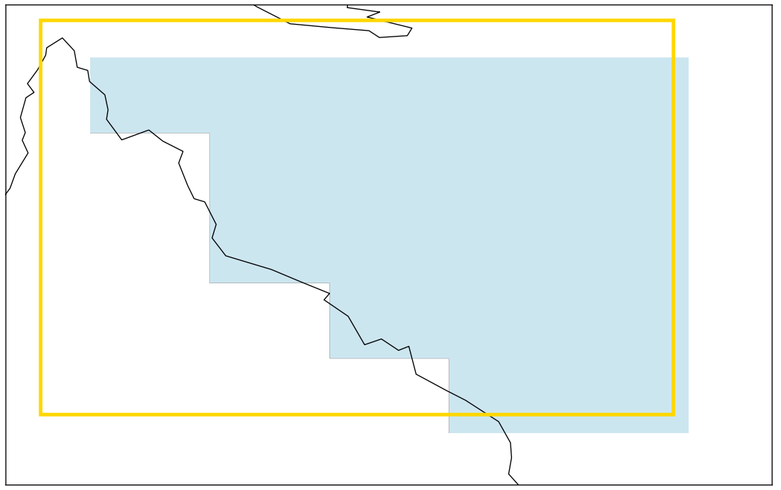


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

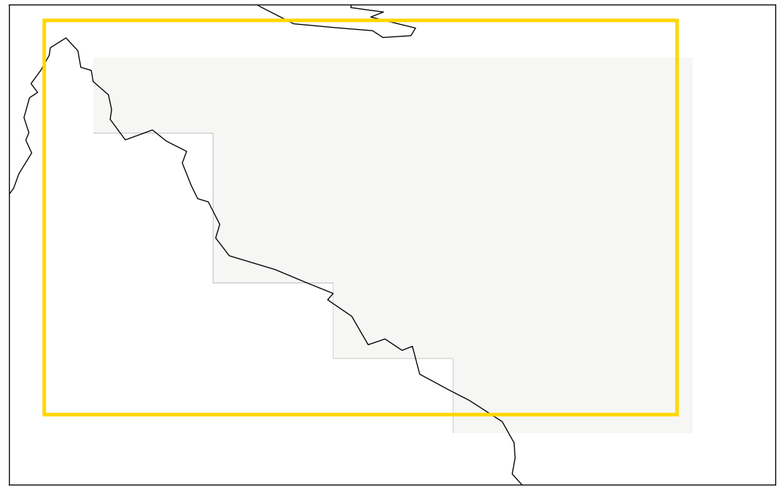
Actual DHW - Step 1



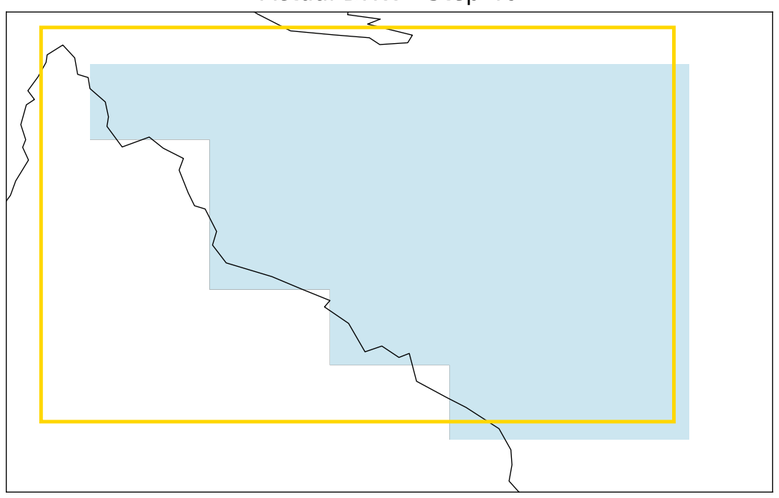
Predicted DHW - Step 1



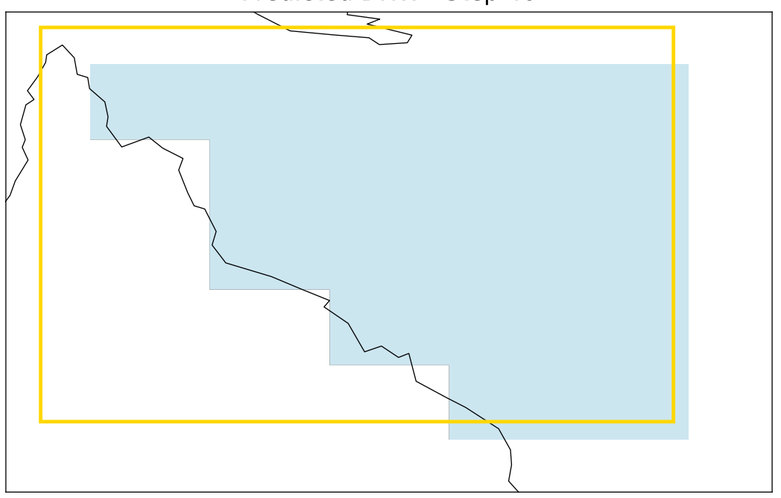
Error (Pred - Actual) - Step 1 [scale ±8]



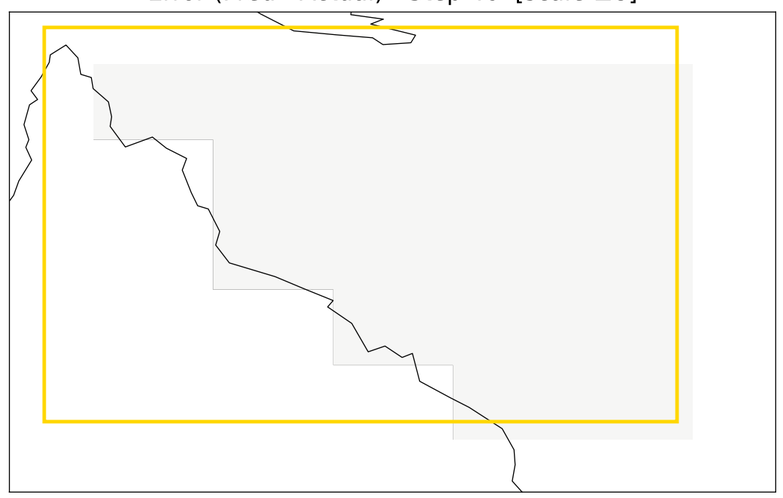
Actual DHW - Step 46



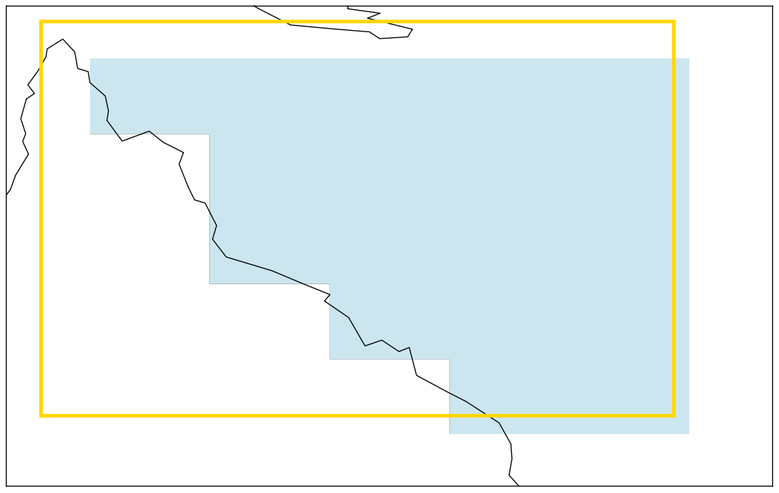
Predicted DHW - Step 46



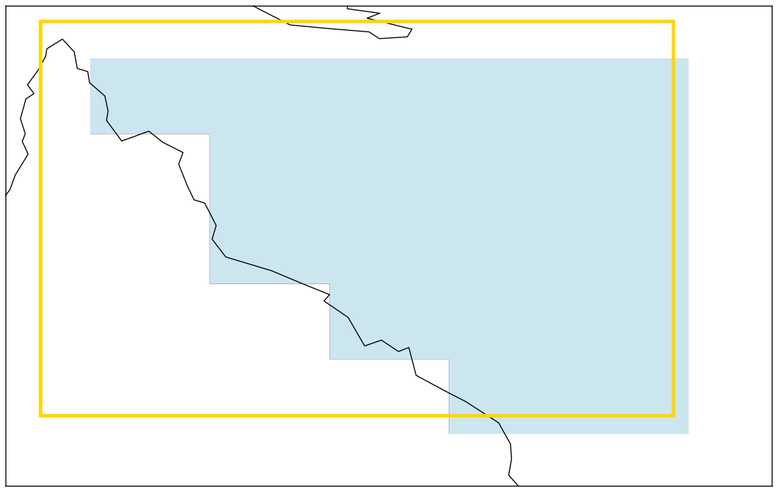
Error (Pred - Actual) - Step 46 [scale ±8]



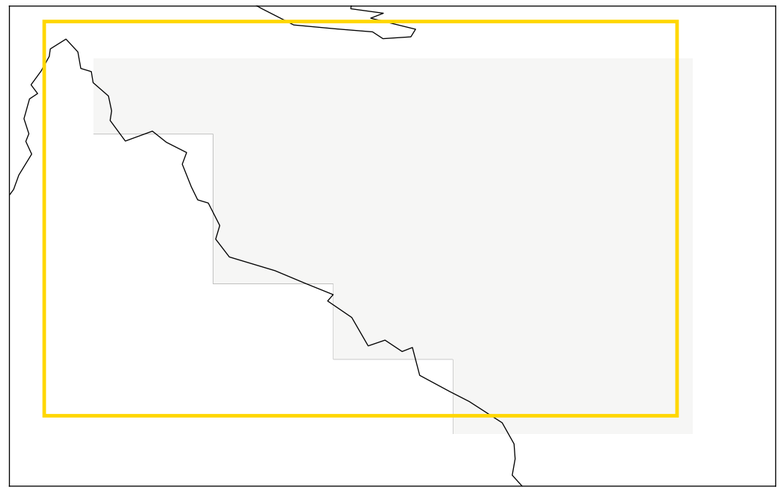
Actual DHW - Step 90

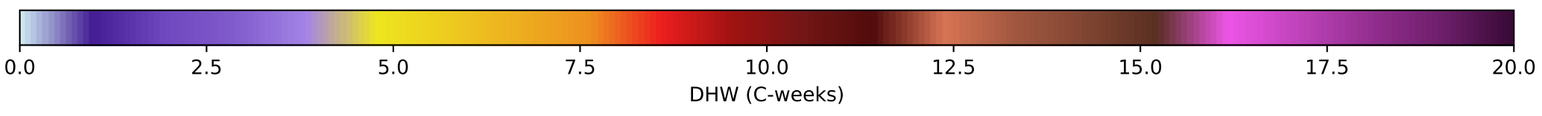


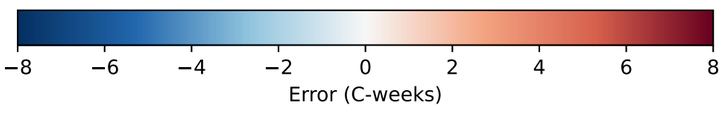
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

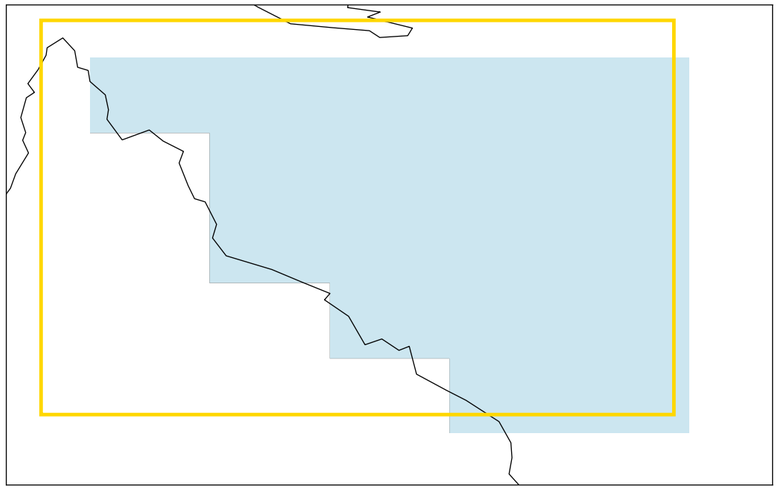




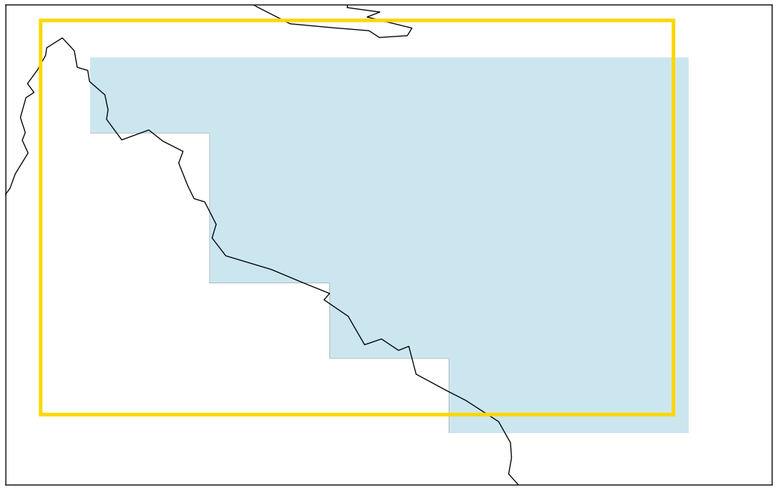


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

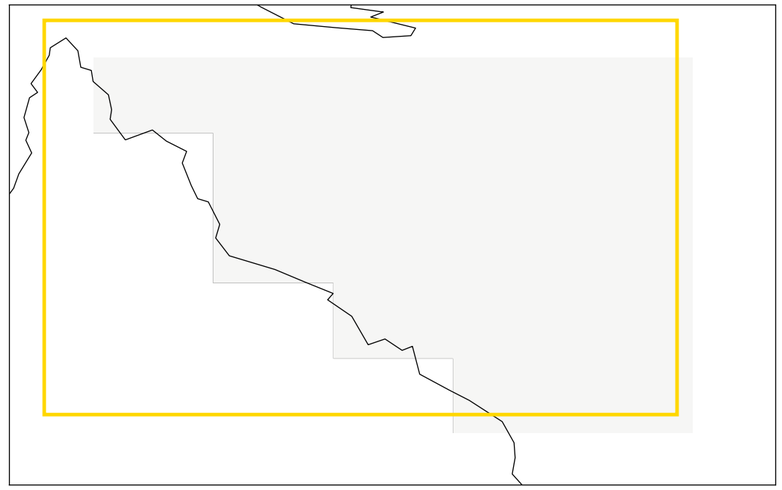
Actual DHW - Step 1



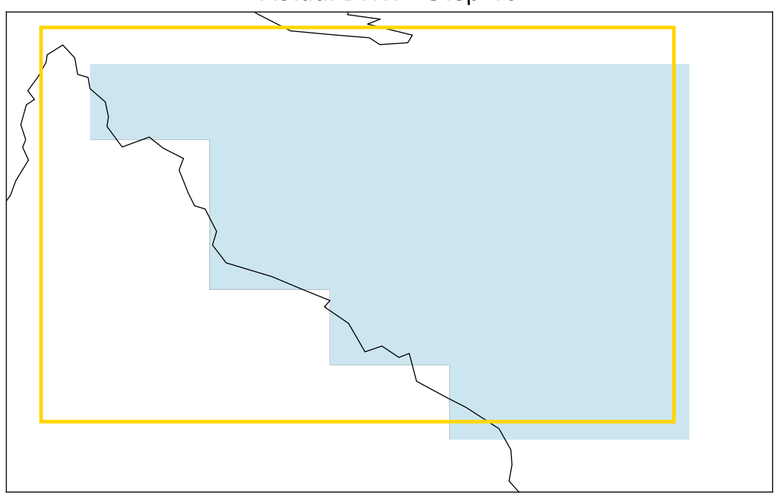
Predicted DHW - Step 1



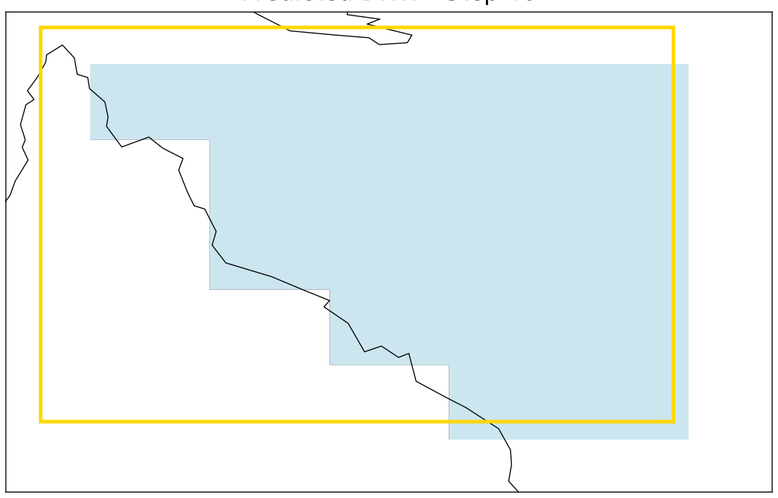
Error (Pred - Actual) - Step 1 [scale ±8]



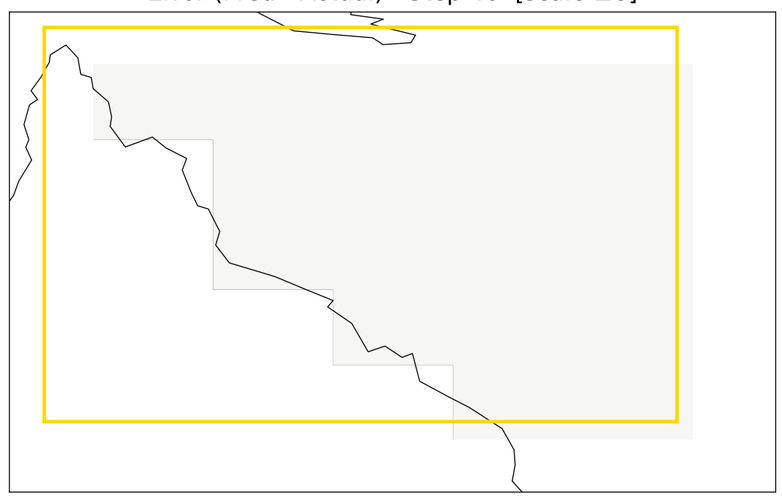
Actual DHW - Step 46



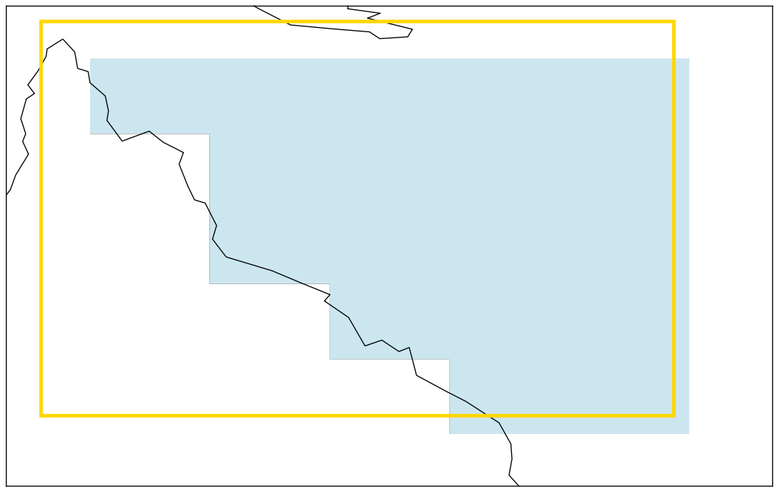
Predicted DHW - Step 46



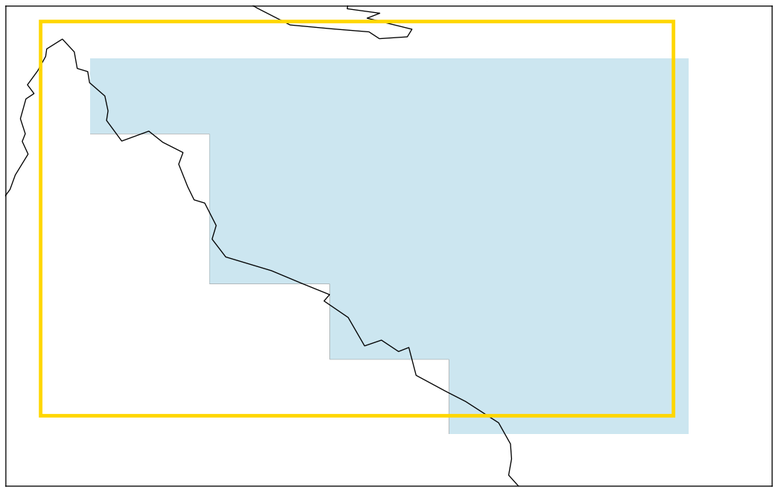
Error (Pred - Actual) - Step 46 [scale ±8]



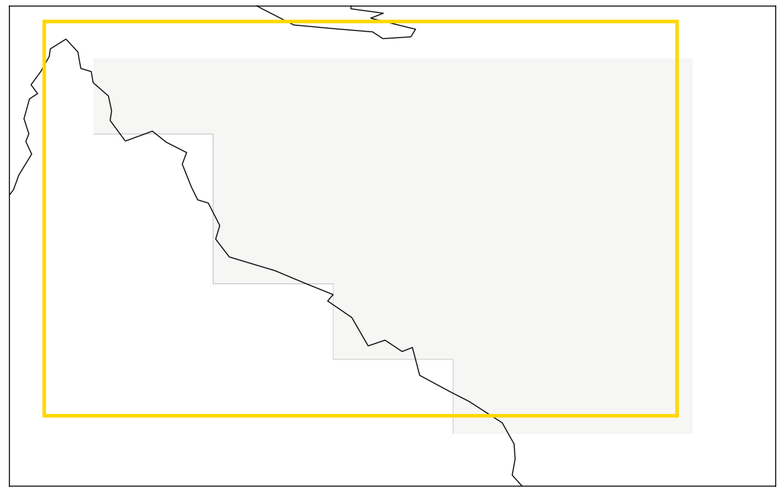
Actual DHW - Step 90

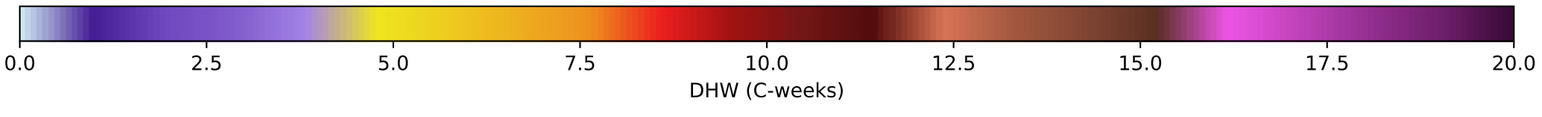


Predicted DHW - Step 90



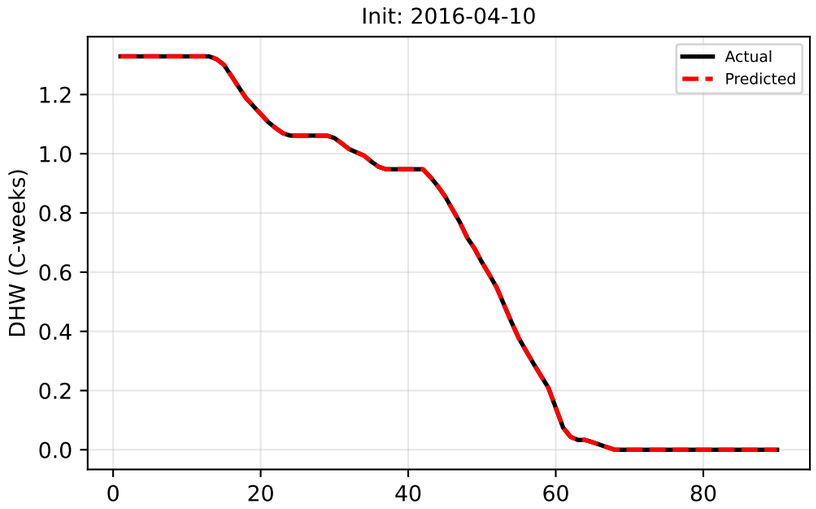
Error (Pred - Actual) - Step 90 [scale ±8]

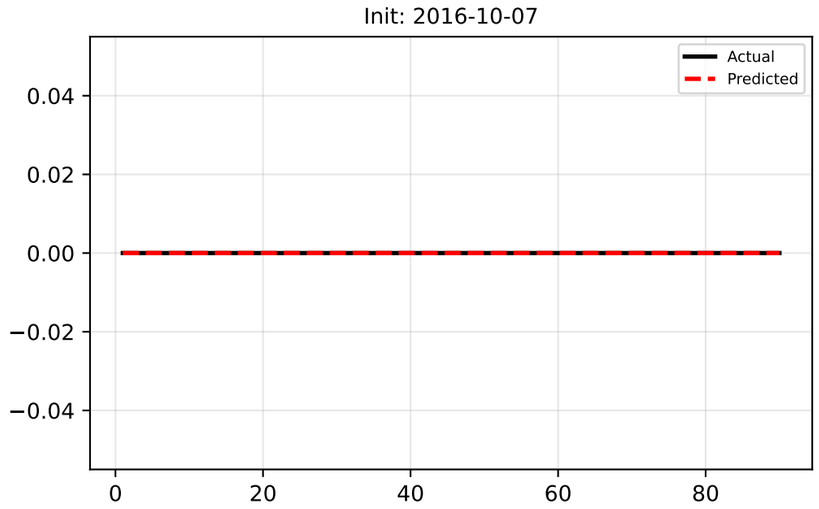


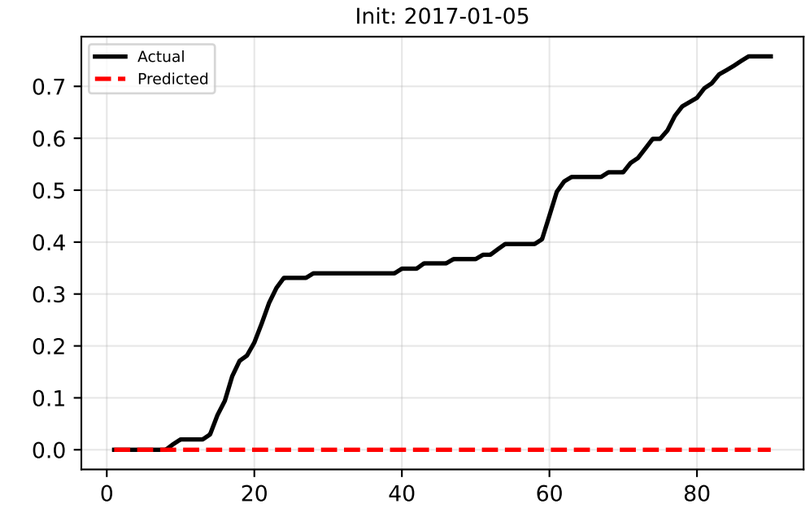


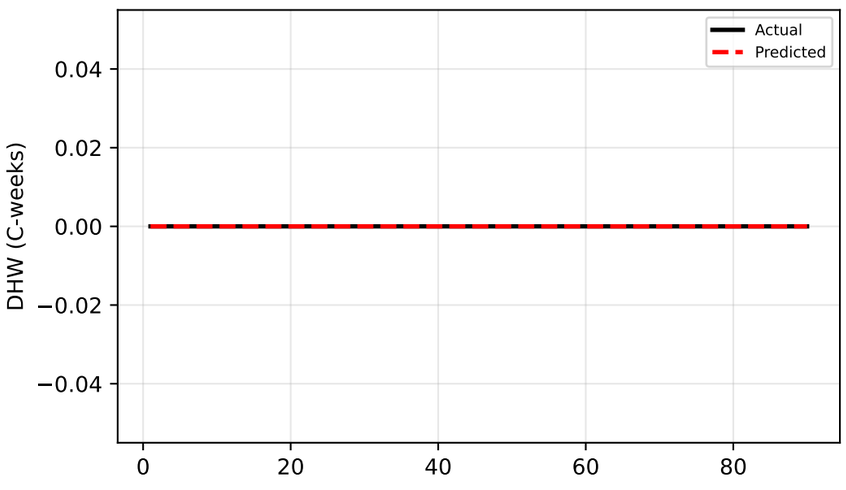


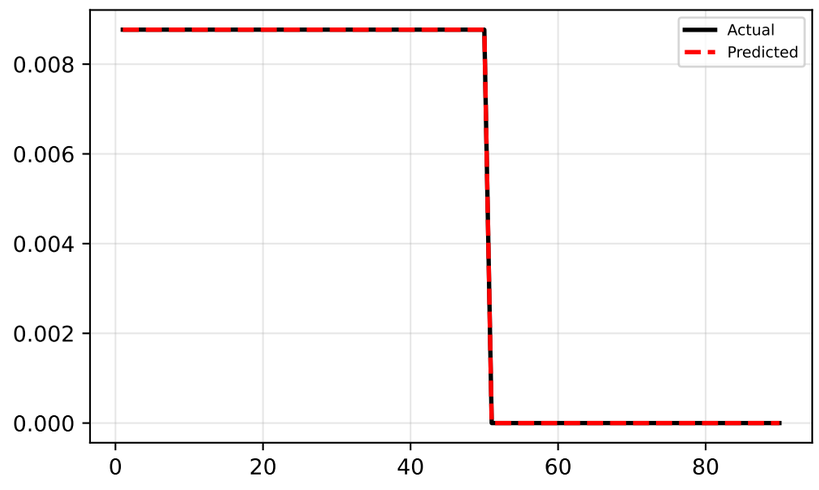
Global-Mean DHW - Val - GBR (C-weeks)

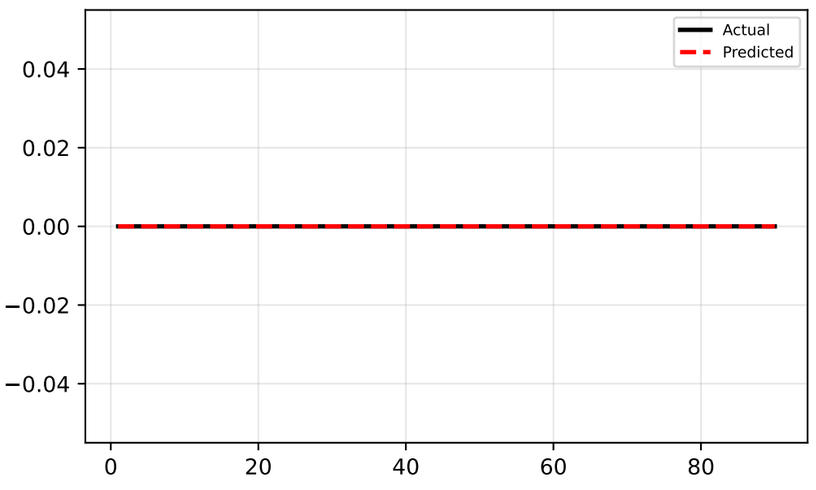


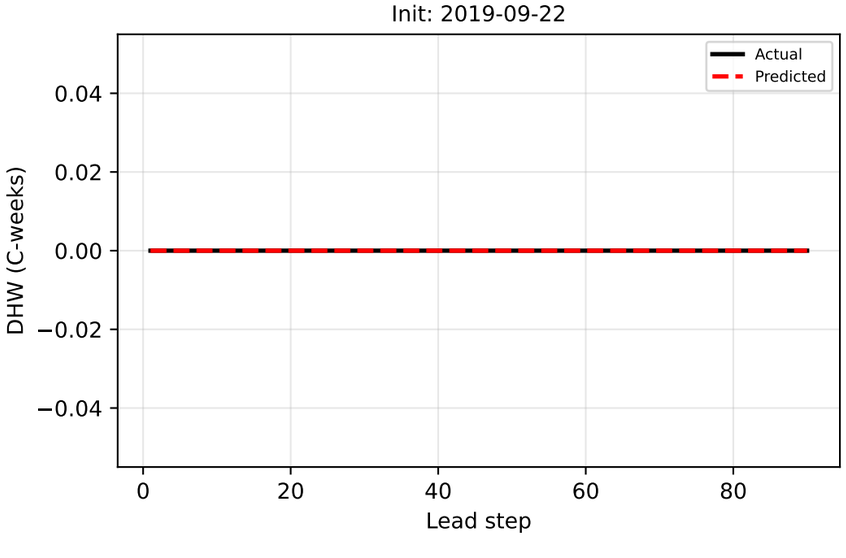


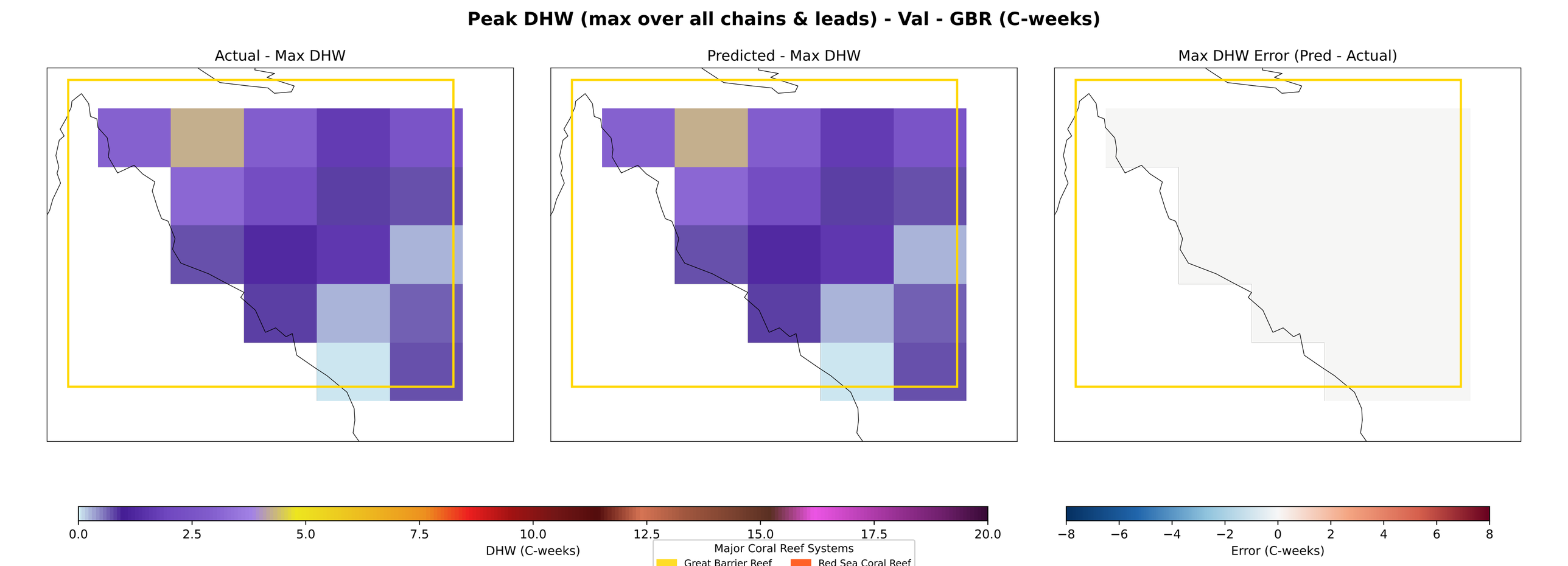


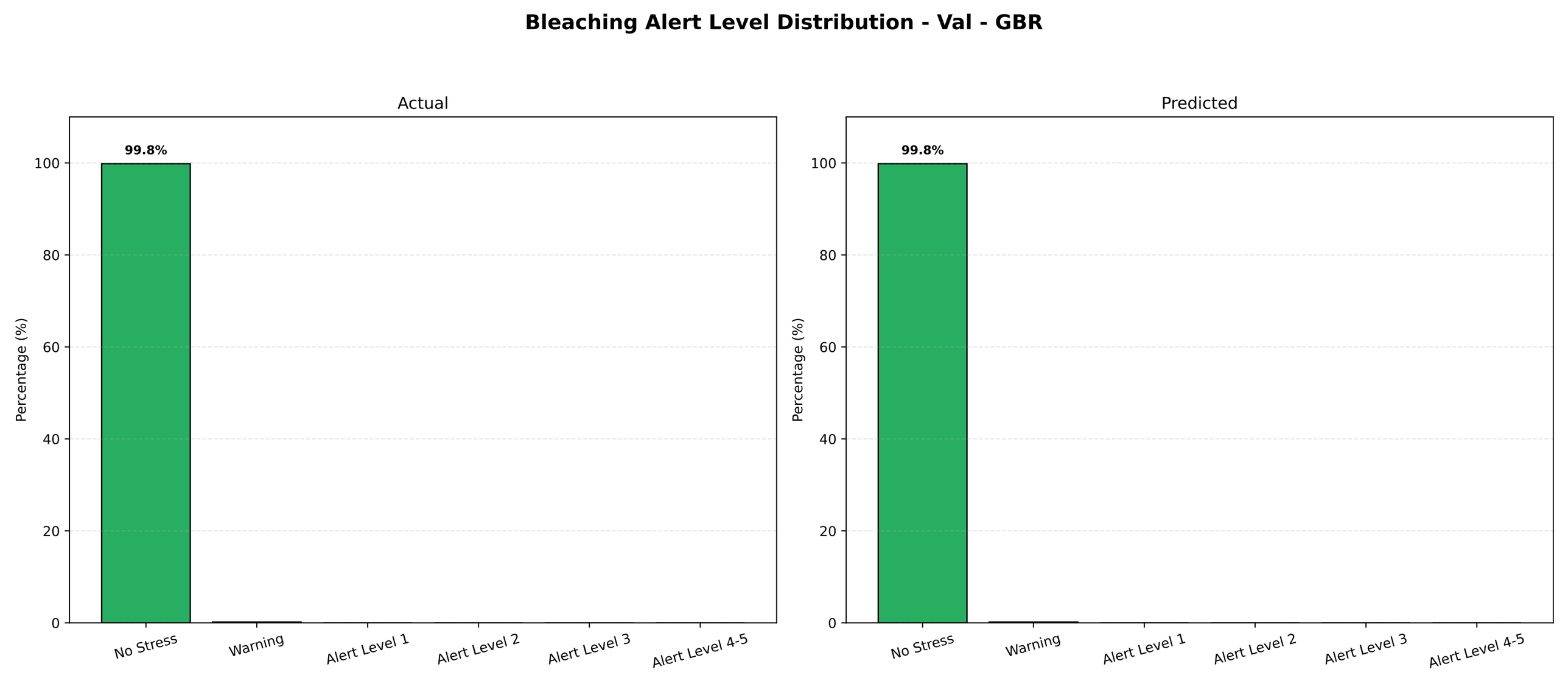






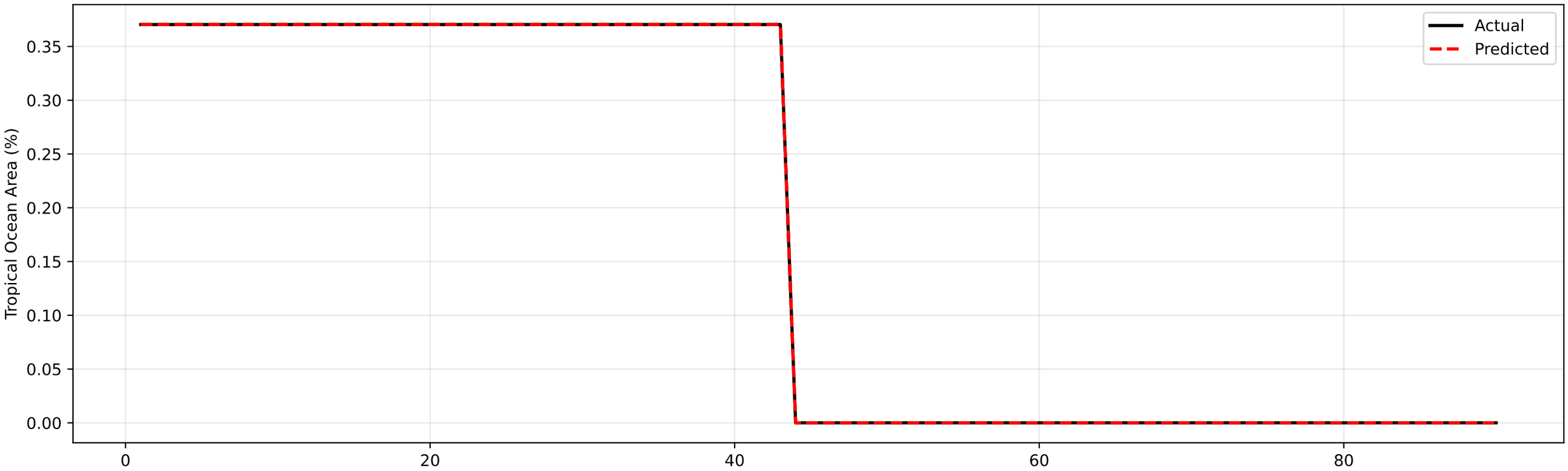



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

